

My Math Roots

Complete Question Bank Review

2670 Questions across **10** Units
Easy | Medium | Hard for every lesson and test bank

Grade 2 | Texas TEKS Aligned

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Unit 1: Basic Fact Strategies

Lesson 1: Count Up & Count Back

Easy (20 questions)

1. Start at 3. Count up 2. What number do you land on?

- A. 4
- B. 5 ✓**
- C. 6
- D. 3

Explanation: Count: 3... 4, 5. You land on 5.

2. What is $2 + 1$?

- A. 3 ✓**
- B. 4
- C. 2
- D. 1

Explanation: Start at 2, count up 1. You get 3.

3. What is $5 + 2$?

- A. 6
- B. 8
- C. 7 ✓**
- D. 5

Explanation: Count up from 5: 6, 7. The answer is 7.

4. What is $4 + 3$?

- A. 8
- B. 6
- C. 7 ✓**
- D. 5

Explanation: Start at 4 and count up 3: 5, 6, 7.

5. Start at 6. Count up 1. What number do you get?

- A. 5
- B. 8
- C. 6
- D. 7 ✓**

Explanation: 6... 7. You land on 7.

6. What is $1 + 3$?

- A. 3
- B. 5
- C. 4 ✓**
- D. 2

Explanation: Start at 1, count up 3: 2, 3, 4. The answer is 4.

7. Count back 1 from 5. What number do you get?

- A. 6
- B. 3
- C. 5
- D. 4 ✓**

Explanation: Start at 5, count back 1: 4.

8. What is $7 - 1$?

- A. 8
- B. 7
- C. 5
- D. 6 ✓**

Explanation: Count back 1 from 7. You get 6.

9. Count back 2 from 6. Where do you land?

- A. 5
- B. 4 ✓**
- C. 3
- D. 8

Explanation: Start at 6, count back: 5, 4. You land on 4.

10. What is $5 - 2$?

- A. 4
- B. 2
- C. 3 ✓**
- D. 7

Explanation: Count back 2 from 5: 4, 3. The answer is 3.

11. What is $3 + 1$?

- A. 2
- B. 5
- C. 3
- D. 4 ✓**

Explanation: Start at 3, count up 1: 4.

12. Count up 3 from 2. What number do you reach?

- A. 4
- B. 6
- C. 5 ✓**
- D. 3

Explanation: Start at 2, count: 3, 4, 5. You reach 5.

13. What is $8 - 2$?

- A. 5
- B. 7
- C. 6 ✓**
- D. 10

Explanation: Count back 2 from 8: 7, 6. The answer is 6.

14. What is $6 - 1$?

- A. 7
- B. 4
- C. 6
- D. 5 ✓**

Explanation: Count back 1 from 6. You get 5.

15. Start at 1. Count up 2. Where do you land?

- A. 2
- B. 4
- C. 1
- D. 3 ✓**

Explanation: Count from 1: 2, 3. You land on 3.

16. What is $9 - 2$?

- A. 8
- B. 6
- C. 11
- D. 7 ✓**

Explanation: Count back 2 from 9: 8, 7. The answer is 7.

17. What is $4 + 1$?

- A. 3
- B. 6
- C. 4
- D. 5 ✓**

Explanation: Start at 4, count up 1: 5.

18. Count back 3 from 7. What number do you get?

- A. 5
- B. 3
- C. 10
- D. 4 ✓**

Explanation: Start at 7, count back: 6, 5, 4. You get 4.

19. What is $2 + 3$?

- A. 4
- B. 6
- C. 5 ✓**
- D. 3

Explanation: Start at 2, count up 3: 3, 4, 5. The answer is 5.

20. What is $10 - 1$?

- A. 11
- B. 8
- C. 9 ✓**
- D. 10

Explanation: Count back 1 from 10. You get 9.

Medium (20 questions)

1. Jada has 8 crayons. She gets 3 more. How many crayons does she have now?

- A. 10
- B. 12
- C. 11 ✓**
- D. 5

Explanation: Count up 3 from 8: 9, 10, 11. Jada has 11 crayons.

2. $__ + 3 = 7$. What is the missing number?

- A. 3
- B. 10
- C. 5
- D. 4 ✓**

Explanation: Think: what plus 3 equals 7? Count back 3 from 7: 6, 5, 4. The missing number is 4.

3. Which way helps you solve $9 + 3$?

- A. Count back 3 from 9
- B. Skip count by 5s
- C. Count up 3 from 9 ✓**
- D. Subtract 3 from 9

Explanation: To add, count up from the bigger number. Start at 9 and count up 3: 10, 11, 12.

4. Leo had 12 grapes. He ate 3. How many grapes are left?

- A. 8
- B. 15
- C. 10
- D. 9 ✓**

Explanation: Count back 3 from 12: 11, 10, 9. Leo has 9 grapes left.

5. Count up 4 from 7. What number do you reach?

- A. 12
- B. 10
- C. 3
- D. 11 ✓**

Explanation: Start at 7, count up: 8, 9, 10, 11. You reach 11.

6. $9 - \underline{\quad} = 6$. What is the missing number?

- A. 2
- B. 15
- C. 4
- D. 3 ✓**

Explanation: Think: 9 minus what gives 6? Count back from 9 to 6: 8, 7, 6. That is 3 steps. The answer is 3.

7. A frog is on lily pad 5. It jumps forward 4 pads. Which pad does it land on?

- A. 10
- B. 1
- C. 8
- D. 9 ✓**

Explanation: Count up 4 from 5: 6, 7, 8, 9. The frog lands on pad 9.

8. Which number sentence can you solve by counting back 2?

- A. $5 + 2$
- B. $7 - 2$ ✓**
- C. $3 + 3$
- D. $2 + 2$

Explanation: Counting back is for subtraction. $7 - 2$ means count back 2 from 7: 6, 5.

9. There are 11 birds on a fence. 4 fly away. How many are left?

- A. 8
- B. 6
- C. 15
- D. 7 ✓**

Explanation: Count back 4 from 11: 10, 9, 8, 7. There are 7 birds left.

10. $6 + \underline{\quad} = 9$. What is the missing number?

- A. 2
- B. 4
- C. 15
- D. 3 ✓**

Explanation: Count up from 6 to 9: 7, 8, 9. That is 3 steps. The missing number is 3.

11. Kai has 7 toy cars. His friend gives him some more. Now Kai has 10. How many did his friend give him?

- A. 2
- B. 4
- C. 17
- D. 3 ✓**

Explanation: Count up from 7 to 10: 8, 9, 10. His friend gave him 3 cars.

12. Which is the best way to solve $11 - 3$?

- A. Count up 3 from 11
- B. Count back 3 from 11 ✓**
- C. Add 3 to 11
- D. Skip count by 3s

Explanation: To subtract, count back. Start at 11, count back 3: 10, 9, 8.

13. $__ - 2 = 8$. What is the missing number?

- A. 6
- B. 11
- C. 9
- D. 10 ✓**

Explanation: Think: what minus 2 equals 8? Count up 2 from 8: 9, 10. The missing number is 10.

14. A bunny hops 3 spaces from 6 on a number line. Where does it land?

- A. 3
- B. 10
- C. 8
- D. 9 ✓**

Explanation: Count up 3 from 6: 7, 8, 9. The bunny lands on 9.

15. There are 14 apples. 4 are eaten. How many are left?

- A. 11
- B. 10 ✓**
- C. 18
- D. 9

Explanation: Count back 4 from 14: 13, 12, 11, 10. There are 10 apples left.

16. Which problem can be solved by counting up from 8?

- A. $8 - 3$
- B. $3 - 8$
- C. $8 + 2$ ✓**
- D. $2 - 8$

Explanation: Count up is for adding. For $8 + 2$, start at 8 and count up 2: 9, 10.

17. Maya picks 6 flowers. Then she picks 5 more. How many flowers does she have?

- A. 12
- B. 1
- C. 10
- D. 11 ✓**

Explanation: Count up 5 from 6: 7, 8, 9, 10, 11. Maya has 11 flowers.

18. $7 + __ = 11$. What is the missing number?

- A. 3
- B. 5
- C. 18
- D. 4 ✓**

Explanation: Count up from 7 to 11: 8, 9, 10, 11. That is 4 steps. The missing number is 4.

19. Start at 15. Count back 3. What number do you land on?

- A. 11
- B. 18
- C. 13
- D. 12 ✓**

Explanation: Count back from 15: 14, 13, 12. You land on 12.

20. A dog has 9 treats. It eats some. Now it has 6. How many did it eat?

- A. 4
- B. 2
- C. 15
- D. 3 ✓**

Explanation: Count back from 9 to 6: 8, 7, 6. That is 3 steps. The dog ate 3 treats.

Hard (20 questions)

1. Mia had 14 stickers. She gave 3 to Kai and 4 to Jada. How many stickers does Mia have left?

- A. 8
- B. 6
- C. 9
- D. 7 ✓**

Explanation: Two steps: $14 - 3 = 11$, then $11 - 4 = 7$. Count back each time. Mia has 7 left.

2. A student wrote: $11 - 3 = 9$. What mistake did the student make?

- A. They added instead
- B. They counted back only 2, not 3 ✓**
- C. They started at 12
- D. They counted up

Explanation: Error analysis: $11 - 3$ should be 8. The student only counted back 2 (to 9) instead of 3.

3. Leo says counting back 4 from 13 gives 8. Is Leo correct?

- A. No, it should be 10
- B. No, it should be 9 ✓**
- C. Yes, Leo is correct
- D. No, it should be 7

Explanation: Check: 13, 12, 11, 10, 9. Counting back 4 from 13 gives 9, not 8. Leo counted back 5 by mistake.

4. $__ + __ = 11$. Both numbers are more than 3. What could they be?

- A. 2 and 9
- B. 4 and 7 ✓**
- C. 3 and 8
- D. 1 and 10

Explanation: Open reasoning: both must be more than 3. $4 + 7 = 11$ and both 4 and 7 are more than 3.

5. Sam had 8 marbles. He found some more and now has 15. Then he lost 3. How many does Sam have now?

- A. 10
- B. 13
- C. 11
- D. 12 ✓**

Explanation: Two steps: Sam went from 8 to 15 marbles. Then $15 - 3 = 12$. Sam has 12 marbles now.

6. Which number sentence is NOT solved by counting back?

- A. $10 - 2$
- B. $13 - 4$
- C. $7 + 3$ ✓**
- D. $9 - 1$

Explanation: Counting back is for subtraction. $7 + 3$ is addition, so you count up, not back.

7. A student wrote: $7 + 4 = 12$. What went wrong?

- A. They subtracted instead
- B. They counted up 5 instead of 4 ✓**
- C. They started at 8 instead of 7
- D. They counted up from 6

Explanation: Error analysis: $7 + 4 = 11$, not 12. The student counted too many: 8, 9, 10, 11, 12 (5 counts, not 4).

8. Zoe is on step 6. She goes up 3 steps, then back down 2 steps. What step is she on now?

- A. 5
- B. 8
- C. 6
- D. 7 ✓**

Explanation: Two steps: $6 + 3 = 9$, then $9 - 2 = 7$. Zoe is on step 7.

9. I am a number. If you count back 3 from me, you get 9. What number am I?

- A. 6
- B. 11
- C. 13
- D. 12 ✓**

Explanation: Think backwards: $__ - 3 = 9$. Count up 3 from 9: 10, 11, 12. The number is 12.

10. True or false: You can solve $6 + 3$ by counting back 3 from 6.

- A. True, counting back works for addition
- B. False, you should count up 3 from 6 ✓**
- C. True, it gives the same answer
- D. False, you should count back from 3

Explanation: Count back is for subtraction. For $6 + 3$, count up from 6: 7, 8, 9. The answer is 9.

11. There were 16 kids at the park. 5 went home, then 3 more went home. How many are still at the park?

- A. 7
- B. 9
- C. 10
- D. 8 ✓**

Explanation: Two steps: $16 - 5 = 11$, then $11 - 3 = 8$. Count back each time. 8 kids are still at the park.

12. Kim says $8 + 3 = 10$. What should Kim do to fix her work?

- A. Count back 3 from 8
- B. Count up one more from 10 ✓**
- C. Start over at 3
- D. Subtract instead of add

Explanation: Error analysis: $8 + 3 = 11$. Kim stopped one count too early. She should count one more: 9, 10, 11.

13. $__ - 4 = 7$. Then $7 + 2 = __$. What are both missing numbers?

- A. 10 and 9
- B. 12 and 9
- C. 11 and 9 ✓**
- D. 11 and 10

Explanation: First: count up 4 from 7 to get 11. Then $7 + 2 = 9$. The answers are 11 and 9.

14. Which problem gives the BIGGEST answer?

- A. $9 + 3$ ✓**
- B. $8 + 2$
- C. $7 + 4$
- D. $6 + 5$

Explanation: Compare: $9 + 3 = 12$, $8 + 2 = 10$, $7 + 4 = 11$, $6 + 5 = 11$. The biggest answer is 12.

15. A number line goes from 0 to 20. You start at 5, jump forward 4, then jump forward 3. Where are you?

- A. 11
- B. 13
- C. 10
- D. 12 ✓**

Explanation: Two jumps: $5 + 4 = 9$, then $9 + 3 = 12$. You are at 12.

16. Eli has 13 cards. He gives away some and has 9 left. Then he finds 4 more. How many does Eli have now?

- A. 12
- B. 14
- C. 8
- D. 13 ✓**

Explanation: First: $13 - 9 = 4$ given away. Then $9 + 4 = 13$. Eli has 13 cards now.

17. A student solved $15 - 4$ and got 10. Explain what went wrong.

- A. They counted back 5 instead of 4 ✓**
- B. They counted back 4 correctly
- C. They added 4 instead
- D. They started at 14

Explanation: Error analysis: $15 - 4 = 11$. The student got 10, meaning they counted back 5 instead of 4.

18. I am thinking of two numbers that add up to 13. Both are greater than 5. What could they be?

- A. 4 and 9
- B. 5 and 8
- C. 6 and 7 ✓**
- D. 3 and 10

Explanation: Open reasoning: both must be greater than 5. $6 + 7 = 13$ and both 6 and 7 are greater than 5.

19. Count back 3 from 12, then count back 2 more. What number do you get?

- A. 8
- B. 6
- C. 9
- D. 7 ✓**

Explanation: Two steps: $12 - 3 = 9$ (count back: 11, 10, 9). Then $9 - 2 = 7$ (count back: 8, 7). You get 7.

20. Is this true or false? $14 - 3$ and $13 - 2$ give the same answer.

- A. True, both equal 11 ✓**
- B. False, $14 - 3 = 10$
- C. False, $13 - 2 = 12$
- D. False, they give different answers

Explanation: Check both: $14 - 3 = 11$ and $13 - 2 = 11$. They are the same! Both equal 11.

Lesson 2: Doubles!

Easy (20 questions)

1. What is $2 + 2$?

- A. 3
- B. 5
- C. 4 ✓**
- D. 2

Explanation: Use doubles: $2 + 2 = 4$.

2. What is $3 + 3$?

- A. 5
- B. 7
- C. 6 ✓**
- D. 3

Explanation: Use doubles: $3 + 3 = 6$.

3. What is $4 + 4$?

- A. 6
- B. 9
- C. 8 ✓**
- D. 7

Explanation: Use doubles: $4 + 4 = 8$.

4. What is $5 + 5$?

- A. 11
- B. 8
- C. 9
- D. 10 ✓**

Explanation: Use doubles: $5 + 5 = 10$.

5. What is $1 + 1$?

- A. 3
- B. 0
- C. 1
- D. 2 ✓**

Explanation: Use doubles: $1 + 1 = 2$.

6. What is $6 + 6$?

- A. 11
- B. 13
- C. 10
- D. 12 ✓**

Explanation: Use doubles: $6 + 6 = 12$.

7. What is $7 + 7$?

- A. 13
- B. 15
- C. 12
- D. 14 ✓**

Explanation: Use doubles: $7 + 7 = 14$.

8. What is $8 + 8$?

- A. 14
- B. 18
- C. 15
- D. 16 ✓**

Explanation: Use doubles: $8 + 8 = 16$.

9. What is $9 + 9$?

- A. 17
- B. 19
- C. 18 ✓**
- D. 16

Explanation: Use doubles: $9 + 9 = 18$.

10. What is $10 + 10$?

- A. 22
- B. 18
- C. 20 ✓**
- D. 19

Explanation: Use doubles: $10 + 10 = 20$.

11. What is double 3?

- A. 5
- B. 9
- C. 3
- D. 6 ✓**

Explanation: Double means add a number to itself. $3 + 3 = 6$.

12. What is double 5?

- A. 15
- B. 5
- C. 10 ✓**
- D. 8

Explanation: Double means $5 + 5$. That equals 10.

13. What is double 7?

- A. 12
- B. 15
- C. 7
- D. 14 ✓**

Explanation: Double 7 means $7 + 7$. That equals 14.

14. What is double 4?

- A. 6
- B. 4
- C. 10
- D. 8 ✓**

Explanation: Double 4 means $4 + 4 = 8$.

15. What is $0 + 0$?

- A. 2
- B. 1
- C. 0 ✓**
- D. 10

Explanation: Use doubles: $0 + 0 = 0$. Zero plus zero is still zero.

16. What is double 8?

- A. 18
- B. 14
- C. 16 ✓**
- D. 8

Explanation: Double 8 means $8 + 8 = 16$.

17. What is double 2?

- A. 2
- B. 6
- C. 3
- D. 4 ✓**

Explanation: Double 2 means $2 + 2 = 4$.

18. What is double 6?

- A. 8
- B. 6
- C. 10
- D. 12 ✓**

Explanation: Double 6 means $6 + 6 = 12$.

19. What is double 1?

- A. 0
- B. 1
- C. 3
- D. 2 ✓**

Explanation: Double 1 means $1 + 1 = 2$.

20. What is double 9?

- A. 16
- B. 19
- C. 18 ✓**
- D. 17

Explanation: Double 9 means $9 + 9 = 18$.

Medium (20 questions)

1. What is $6 + 7$? Hint: Think of a double you know.

- A. 14
- B. 12
- C. 11
- D. 13 ✓**

Explanation: Use doubles plus 1: $6 + 6 = 12$, then add 1 more. $12 + 1 = 13$.

2. What is $5 + 6$?

- A. 12
- B. 10
- C. 11 ✓**
- D. 9

Explanation: Use doubles plus 1: $5 + 5 = 10$, plus 1 more = 11.

3. Sara has 7 red beads and 7 blue beads. How many beads in all?

- A. 13
- B. 15
- C. 12
- D. 14 ✓**

Explanation: Use doubles: $7 + 7 = 14$. Sara has 14 beads.

4. $\underline{\quad} + \underline{\quad} = 16$. Both numbers are the same. What are they?

- A. 7 and 7
- B. 9 and 9
- C. 6 and 6
- D. 8 and 8 ✓**

Explanation: Think: what double equals 16? $8 + 8 = 16$. Both numbers are 8.

5. Which doubles fact helps you solve $8 + 9$?

- A. $7 + 7$
- B. $9 + 9$
- C. $8 + 8$ ✓**
- D. $6 + 6$

Explanation: Strategy: $8 + 9$ is close to $8 + 8 = 16$. Then add 1 more: 17. Use doubles plus 1.

6. What is $7 + 8$?

- A. 14
- B. 16
- C. 13
- D. 15 ✓**

Explanation: Use doubles plus 1: $7 + 7 = 14$, then add 1 more = 15.

7. A boy has 2 hands. Each hand has 5 fingers. How many fingers in all?

- A. 7
- B. 12
- C. 10 ✓**
- D. 8

Explanation: Use doubles: $5 + 5 = 10$. He has 10 fingers in all.

8. What is $4 + 5$?

- A. 10
- B. 8
- C. 7
- D. 9 ✓**

Explanation: Use doubles plus 1: $4 + 4 = 8$, plus 1 more = 9.

9. $__ + 6 = 12$. What is the missing number?

- A. 5
- B. 7
- C. 8
- D. 6 ✓**

Explanation: Think: what plus 6 equals 12? Use doubles: $6 + 6 = 12$. The missing number is 6.

10. Which strategy helps you solve $3 + 4$?

- A. Make a 10
- B. Doubles plus 1 ✓**
- C. Count by 10s
- D. Subtract

Explanation: $3 + 4$ is close to the double $3 + 3 = 6$. Then add 1 more to get 7. Use doubles plus 1.

11. What is $9 + 8$?

- A. 18
- B. 16
- C. 15
- D. 17 ✓**

Explanation: Use doubles minus 1: $9 + 9 = 18$, minus 1 = 17. Or $8 + 8 = 16$, plus 1 = 17.

12. Two friends each have 6 stickers. How many stickers do they have together?

- A. 10
- B. 14
- C. 11
- D. 12 ✓**

Explanation: Use doubles: $6 + 6 = 12$ stickers together.

13. $__ + __ = 10$. Both numbers are the same. What is each number?

- A. 4
- B. 6
- C. 3
- D. 5 ✓**

Explanation: Think: what double equals 10? $5 + 5 = 10$. Each number is 5.

14. What is $3 + 4$? Use a doubles fact to help.

- A. 8
- B. 6
- C. 5
- D. 7 ✓**

Explanation: Doubles plus 1: $3 + 3 = 6$, plus 1 = 7.

15. Max has 8 toy cars. His brother has 7. How many cars do they have together?

- A. 14
- B. 16
- C. 13
- D. 15 ✓**

Explanation: Use doubles plus 1: $7 + 7 = 14$, then 1 more = 15. They have 15 cars.

16. Which doubles fact helps you solve $5 + 6$?

- A. $6 + 6$
- B. $4 + 4$
- C. $5 + 5$ ✓**
- D. $3 + 3$

Explanation: $5 + 6$ is close to $5 + 5 = 10$. Add 1 more to get 11.

17. $__ + 9 = 18$. What is the missing number?

- A. 8
- B. 10
- C. 7
- D. 9 ✓**

Explanation: Think: what plus 9 equals 18? Use doubles: $9 + 9 = 18$. The missing number is 9.

18. What is $8 + 7$?

- A. 14
- B. 16
- C. 13
- D. 15 ✓**

Explanation: Use doubles plus 1: $7 + 7 = 14$, add 1 more = 15. Or $8 + 8 = 16$, minus 1 = 15.

19. Ava and Mia each brought 4 books. How many books are there in all?

- A. 6
- B. 9
- C. 7
- D. 8 ✓**

Explanation: Use doubles: $4 + 4 = 8$ books in all.

20. What is $2 + 3$? Use a doubles fact.

- A. 4
- B. 6
- C. 3
- D. 5 ✓**

Explanation: Doubles plus 1: $2 + 2 = 4$, plus 1 = 5. The answer is 5.

Hard (20 questions)

1. Emma has 7 red marbles and 8 blue marbles. She gives away 3 marbles. How many does she have left?

- A. 11
- B. 13
- C. 10
- D. 12 ✓**

Explanation: Two steps: $7 + 8 = 15$ (use doubles plus 1: $7 + 7 = 14$, $+ 1 = 15$). Then $15 - 3 = 12$.

2. A student said double 7 is 15. What mistake did the student make?

- A. They added $8 + 7$ by mistake ✓**
- B. They subtracted instead
- C. They used the wrong double
- D. They forgot to count

Explanation: Error analysis: double 7 is $7 + 7 = 14$, not 15. The student may have confused it with $7 + 8 = 15$.

3. Jake says $6 + 7 = 14$ because double 7 is 14. Is Jake correct?

A. Yes, $6 + 7 = 14$

B. No, it should be 13 ✓

C. No, it should be 12

D. No, it should be 15

Explanation: Jake used the wrong double. $6 + 7$: use double $6 = 12$, plus $1 = 13$. Or double $7 = 14$, minus $1 = 13$.

4. $__ + __ = 18$. Both numbers are the same. What are they? Then subtract 5 from your answer. What do you get?

A. 9 and 13 ✓

B. 8 and 13

C. 9 and 14

D. 8 and 11

Explanation: Double: $9 + 9 = 18$. Then $18 - 5 = 13$. The numbers are 9, and you get 13.

5. Which problem can NOT be solved with doubles plus 1?

A. $6 + 7$

B. $5 + 6$

C. $4 + 4$ ✓

D. $3 + 4$

Explanation: $4 + 4$ is an exact double, not doubles plus 1. Doubles plus 1 works when numbers are 1 apart.

6. Two teams each score 8 points. Then one team scores 3 more. What is the winning team's total?

A. 16

B. 19

C. 10

D. 11 ✓

Explanation: Two steps: doubles $8 + 8 = 16$ is the tie. Winning team: $8 + 3 = 11$. The winning team has 11.

7. A student wrote $8 + 9 = 16$. What should the student do to fix it?

A. Subtract 1 from 16

B. Add 1 more to get 17 ✓

C. Start over with $8 + 8$

D. Change to $9 + 9$

Explanation: Error analysis: $8 + 9 = 17$. The student got 16 (which is $8 + 8$). They need to add 1 more.

8. I am a doubles fact. My answer is between 10 and 15. What could I be?

A. $5 + 5$ or $6 + 6$

B. $4 + 4$ or $8 + 8$

C. $6 + 6$ or $7 + 7$ ✓

D. $3 + 3$ or $9 + 9$

Explanation: Check: $6 + 6 = 12$ (between 10 and 15). $7 + 7 = 14$ (between 10 and 15). Both work!

9. Lily had 6 coins. She found 6 more. Then she spent 4 coins. How many does she have now?

A. 10

B. 7

C. 6

D. 8 ✓

Explanation: Two steps: $6 + 6 = 12$ (doubles!). Then $12 - 4 = 8$. Lily has 8 coins.

10. True or false: $5 + 6$ and $6 + 5$ give different answers because the numbers switched.

A. True, switching changes the answer

B. False, both equal 11 ✓

C. True, $5 + 6 = 12$

D. False, both equal 12

Explanation: Addition can be done in any order. $5 + 6 = 11$ and $6 + 5 = 11$. Same answer!

11. Which two doubles facts have answers that are 4 apart?

A. **3 + 3 and 5 + 5 ✓**

B. 4 + 4 and 5 + 5

C. 2 + 2 and 3 + 3

D. 6 + 6 and 7 + 7

Explanation: Check: $3 + 3 = 6$ and $5 + 5 = 10$. $10 - 6 = 4$. They are 4 apart!

12. Ben says $5 + 4 = 10$ because double 5 is 10. What did Ben forget?

A. **He should subtract 1 since 4 is one less than 5 ✓**

B. He should add 1 more

C. He used the wrong doubles fact

D. He should use make a 10

Explanation: Error analysis: $5 + 4$ is doubles minus 1. Double 5 = 10, then subtract 1 = 9. Ben forgot to subtract 1.

13. Each of 3 kids has 5 crayons. Is this a doubles problem? Why or why not?

A. Yes, because $5 + 5 = 10$

B. **No, there are 3 groups, not 2 ✓**

C. Yes, because $3 + 3 = 6$

D. No, because 5 is odd

Explanation: Doubles means 2 equal groups. 3 kids with 5 each is 3 groups, so it is not a doubles problem.

14. What is the missing number? Double $\underline{\quad} = 14$. Then add 3 to that number. What do you get?

A. **7 and 10 ✓**

B. 6 and 9

C. 8 and 11

D. 7 and 9

Explanation: Double 7 = 14. Then $7 + 3 = 10$. The answers are 7 and 10.

15. Ava solved $7 + 8$ by thinking $7 + 7 + 1$. Nora solved it by thinking $8 + 8 - 1$. Who is correct?

A. Only Ava

B. Only Nora

C. **Both are correct ✓**

D. Neither is correct

Explanation: Both strategies work! $7 + 7 + 1 = 15$ and $8 + 8 - 1 = 15$. Doubles plus 1 and doubles minus 1 both give 15.

16. A student wrote: double 6 = 11. Then they wrote $6 + 7 = 12$. How many errors did they make?

A. 0 errors

B. 1 error

C. **2 errors ✓**

D. 3 errors

Explanation: Error analysis: double 6 = 12 (not 11), and $6 + 7 = 13$ (not 12). That is 2 errors.

17. $\underline{\quad} + \underline{\quad} = 12$. The two numbers are 2 apart. What could they be?

A. 4 and 8

B. **5 and 7 ✓**

C. 6 and 6

D. 3 and 9

Explanation: Open reasoning: $5 + 7 = 12$, and $7 - 5 = 2$. They are 2 apart!

18. Sam has some stickers. He gets the same number again and now has 16. How many did he start with?

A. 6

B. 9

C. 7

D. **8 ✓**

Explanation: Think: $\underline{\quad} + \underline{\quad} = 16$, same number. That means double $\underline{\quad} = 16$. $8 + 8 = 16$. Sam started with 8.

19. Which is greater: double 6 plus 3 OR double 4 plus 5?

A. Double 6 plus 3 ✓

B. Double 4 plus 5

C. They are equal

D. Cannot tell

Explanation: Compare: $\text{double } 6 + 3 = 12 + 3 = 15$. $\text{Double } 4 + 5 = 8 + 5 = 13$. $15 > 13$. Double 6 plus 3 is greater.

20. A puzzle says: I am a number. Double me, then take away 2. You get 10. What number am I?

A. 5

B. 7

C. 8

D. 6 ✓

Explanation: Work backwards: $10 + 2 = 12$. Now half of 12 is 6. Double 6 = 12, minus 2 = 10. The number is 6.

Lesson 3: Make a 10

Easy (20 questions)

1. What number goes with 9 to make 10?

A. 2

B. 0

C. 3

D. 1 ✓

Explanation: $9 + 1 = 10$. The number 1 goes with 9 to make 10.

2. What number goes with 8 to make 10?

A. 3

B. 1

C. 2 ✓

D. 4

Explanation: $8 + 2 = 10$. The number 2 goes with 8 to make 10.

3. What number goes with 7 to make 10?

A. 4

B. 2

C. 3 ✓

D. 1

Explanation: $7 + 3 = 10$. The number 3 goes with 7 to make 10.

4. What number goes with 6 to make 10?

A. 3

B. 5

C. 4 ✓

D. 6

Explanation: $6 + 4 = 10$. The number 4 goes with 6 to make 10.

5. What number goes with 5 to make 10?

A. 4

B. 6

C. 5 ✓

D. 3

Explanation: $5 + 5 = 10$. The number 5 goes with 5 to make 10.

6. What is $10 + 3$?

- A. 12
- B. 14
- C. 13 ✓**
- D. 11

Explanation: When you add to 10, just put the number after the 1. $10 + 3 = 13$.

7. What is $10 + 5$?

- A. 14
- B. 16
- C. 50
- D. 15 ✓**

Explanation: $10 + 5 = 15$. Adding to 10 is easy!

8. What is $10 + 7$?

- A. 18
- B. 17 ✓**
- C. 16
- D. 71

Explanation: $10 + 7 = 17$.

9. What is $10 + 1$?

- A. 12
- B. 9
- C. 10
- D. 11 ✓**

Explanation: $10 + 1 = 11$.

10. What is $10 + 9$?

- A. 20
- B. 18
- C. 91
- D. 19 ✓**

Explanation: $10 + 9 = 19$.

11. What number goes with 4 to make 10?

- A. 5
- B. 7
- C. 4
- D. 6 ✓**

Explanation: $4 + 6 = 10$. The number 6 goes with 4.

12. What number goes with 3 to make 10?

- A. 6
- B. 8
- C. 7 ✓**
- D. 5

Explanation: $3 + 7 = 10$. The number 7 goes with 3.

13. What number goes with 2 to make 10?

- A. 7
- B. 9
- C. 6
- D. 8 ✓**

Explanation: $2 + 8 = 10$. The number 8 goes with 2.

14. What number goes with 1 to make 10?

- A. 8
- B. 10
- C. 9 ✓
- D. 7

Explanation: $1 + 9 = 10$. The number 9 goes with 1.

15. What is $10 + 2$?

- A. 11
- B. 20
- C. 13
- D. 12 ✓

Explanation: $10 + 2 = 12$.

16. What is $10 + 4$?

- A. 40
- B. 13
- C. 14 ✓
- D. 15

Explanation: $10 + 4 = 14$.

17. What is $10 + 6$?

- A. 60
- B. 17
- C. 15
- D. 16 ✓

Explanation: $10 + 6 = 16$.

18. What is $10 + 8$?

- A. 80
- B. 19
- C. 17
- D. 18 ✓

Explanation: $10 + 8 = 18$.

19. What number goes with 10 to make 10?

- A. 1
- B. 10
- C. 0 ✓
- D. 5

Explanation: $10 + 0 = 10$. You need 0 more to make 10.

20. What is $10 + 0$?

- A. 0
- B. 11
- C. 1
- D. 10 ✓

Explanation: $10 + 0 = 10$. Adding zero changes nothing.

Medium (20 questions)

1. What is $8 + 5$? Think: $8 + 2 = 10$, then $10 + 3 = ?$

- A. 12
- B. 14
- C. 11
- D. 13 ✓

Explanation: Make a 10: Break 5 into 2 and 3. $8 + 2 = 10$. Then $10 + 3 = 13$.

2. What is $9 + 4$?

- A. 14
- B. 12
- C. 11
- D. 13 ✓**

Explanation: Make a 10: 9 needs 1 more to make 10. Break 4 into 1 and 3. $9 + 1 = 10$, then $10 + 3 = 13$.

3. $__ + 6 = 10$. What is the missing number?

- A. 5
- B. 3
- C. 4 ✓**
- D. 6

Explanation: Think: what plus 6 makes 10? $4 + 6 = 10$. The missing number is 4.

4. Which strategy helps solve $7 + 5$?

- A. Count by 2s
- B. Make a 10 first ✓**
- C. Use doubles of 7
- D. Subtract 5 from 7

Explanation: Make a 10: $7 + 3 = 10$, then $10 + 2 = 12$. Break apart 5 into 3 and 2.

5. Kai has 9 toy trucks. He gets 6 more. How many does he have?

- A. 14
- B. 16
- C. 13
- D. 15 ✓**

Explanation: Make a 10: $9 + 1 = 10$, then $10 + 5 = 15$. Break 6 into 1 and 5.

6. What is $7 + 6$? Use make a 10.

- A. 14
- B. 12
- C. 11
- D. 13 ✓**

Explanation: Make a 10: $7 + 3 = 10$, then $10 + 3 = 13$. Break 6 into 3 and 3.

7. $8 + __ = 13$. What is the missing number?

- A. 6
- B. 4
- C. 3
- D. 5 ✓**

Explanation: Make a 10: $8 + 2 = 10$, then $10 + ? = 13$, so $? = 3$. That means $2 + 3 = 5$. The missing number is 5.

8. A jar has 6 red marbles and 8 blue marbles. How many marbles in all?

- A. 15
- B. 12
- C. 13
- D. 14 ✓**

Explanation: Make a 10: $8 + 2 = 10$, then $10 + 4 = 14$. Break 6 into 2 and 4.

9. What is $9 + 7$? Use make a 10.

- A. 15
- B. 17
- C. 14
- D. 16 ✓**

Explanation: Make a 10: $9 + 1 = 10$, then $10 + 6 = 16$. Break 7 into 1 and 6.

10. Which shows how to make a 10 for $8 + 6$?

- A. $8 + 8 = 16$
- B. $8 + 2 + 4 = 14$ ✓
- C. $8 + 6 + 2 = 16$
- D. $6 + 6 = 12$

Explanation: Break 6 into 2 and 4. Now: $8 + 2 = 10$, then $10 + 4 = 14$. That is make a 10!

11. What is $6 + 5$? Use make a 10.

- A. 12
- B. 10
- C. 13
- D. 11 ✓

Explanation: Make a 10: $6 + 4 = 10$, then $10 + 1 = 11$. Break 5 into 4 and 1.

12. Ana has 7 stickers. She finds 4 more. How many does she have now?

- A. 12
- B. 10
- C. 13
- D. 11 ✓

Explanation: Make a 10: $7 + 3 = 10$, then $10 + 1 = 11$. Break 4 into 3 and 1.

13. $9 + \underline{\quad} = 15$. What is the missing number?

- A. 5
- B. 7
- C. 4
- D. 6 ✓

Explanation: Make a 10: $9 + 1 = 10$, then $10 + 5 = 15$. So $1 + 5 = 6$. The missing number is 6.

14. What is $8 + 7$? Use make a 10.

- A. 14
- B. 16
- C. 13
- D. 15 ✓

Explanation: Make a 10: $8 + 2 = 10$, then $10 + 5 = 15$. Break 7 into 2 and 5.

15. Which number is easiest to make a 10 with?

- A. 5
- B. 3
- C. 9 ✓
- D. 6

Explanation: 9 only needs 1 more to make 10. That makes it the easiest to use with make a 10.

16. What is $5 + 8$?

- A. 14
- B. 12
- C. 11
- D. 13 ✓

Explanation: Make a 10: $8 + 2 = 10$, then $10 + 3 = 13$. Break 5 into 2 and 3.

17. $7 + \underline{\quad} = 12$. What is the missing number?

- A. 6
- B. 4
- C. 3
- D. 5 ✓

Explanation: Make a 10: $7 + 3 = 10$, then $10 + 2 = 12$. So $3 + 2 = 5$. The missing number is 5.

18. There are 9 birds in a tree. 5 more land. How many birds now?

- A. 13
- B. 15
- C. 12
- D. 14 ✓**

Explanation: Make a 10: $9 + 1 = 10$, then $10 + 4 = 14$. Break 5 into 1 and 4.

19. What is $6 + 7$? Use make a 10.

- A. 14
- B. 12
- C. 11
- D. 13 ✓**

Explanation: Make a 10: $6 + 4 = 10$, then $10 + 3 = 13$. Break 7 into 4 and 3.

20. $__ + 8 = 10$. What is the missing number?

- A. 3
- B. 1
- C. 4
- D. 2 ✓**

Explanation: Think: what plus 8 makes 10? $2 + 8 = 10$. The missing number is 2.

Hard (20 questions)

1. Mia has 8 grapes. Her mom gives her 7 more. She eats 3. How many does Mia have now?

- A. 11
- B. 13
- C. 10
- D. 12 ✓**

Explanation: Two steps: $8 + 7 = 15$ (make a 10: $8 + 2 = 10$, $+ 5 = 15$). Then $15 - 3 = 12$.

2. A student solved $9 + 5$ and got 13. What mistake did the student make?

- A. They made a 10, then added too many
- B. They forgot to break apart 5
- C. They added $9 + 4$ instead ✓**
- D. They counted wrong by 1

Explanation: Error analysis: $9 + 5 = 14$, not 13. The student may have added $9 + 4 = 13$ instead of $9 + 5$.

3. Tom says you cannot use make a 10 for $6 + 3$ because the answer is less than 10. Is Tom correct?

- A. No, you can always make a 10
- B. Yes, make a 10 works best when the sum is more than 10 ✓**
- C. No, $6 + 3$ equals 10
- D. Yes, but only because 3 is too small

Explanation: Tom is correct! $6 + 3 = 9$, which is less than 10. Make a 10 is most helpful when the sum is over 10.

4. $__ + 7 = 15$. What is the missing number? Then add 3 more to 15. What do you get?

- A. 8 and 18 ✓**
- B. 7 and 18
- C. 8 and 17
- D. 9 and 18

Explanation: Make a 10: $7 + 3 = 10$, $10 + 5 = 15$, so missing is 8. Then $15 + 3 = 18$. Answers: 8 and 18.

5. Which two problems BOTH equal 14?

- A. $8 + 5$ and $9 + 4$
- B. $8 + 6$ and $9 + 5$ ✓**
- C. $7 + 6$ and $8 + 5$
- D. $9 + 6$ and $7 + 7$

Explanation: Check: $8 + 6 = 14$ and $9 + 5 = 14$. Both equal 14! ($8 + 5 = 13$, not 14.)

6. A student wrote: $7 + 6 = 14$. They said $7 + 3 = 10$, then $10 + 4 = 14$. What went wrong?

A. They broke apart 6 into 3 and 4 instead of 3 and 3 ✓

B. They should have started with 6

C. Nothing, the answer is correct

D. They added to 10 wrong

Explanation: Error analysis: 6 = 3 + 3, not 3 + 4. Correct: $7 + 3 = 10$, $10 + 3 = 13$. The answer is 13, not 14.

7. Leo has 9 apples. Kai has 8 apples. They put them together and give 5 to a friend. How many are left?

A. 11

B. 13

C. 10

D. 12 ✓

Explanation: Two steps: $9 + 8 = 17$ (make a 10: $9 + 1 = 10$, $+ 7 = 17$). Then $17 - 5 = 12$.

8. For $8 + 5$, show two different ways to make a 10.

A. $8+2+3$ or $5+5+3$ ✓

B. $8+2+3$ only

C. $5+5+3$ only

D. Neither works

Explanation: Way 1: $8 + 2 = 10$, then $+ 3 = 13$. Way 2: $5 + 5 = 10$, then $+ 3 = 13$. Both make a 10!

9. A puzzle says: I added a number to 9 and got 16. What number did I add?

A. 6

B. 8

C. 5

D. 7 ✓

Explanation: Make a 10: $9 + 1 = 10$, then $10 + ? = 16$, so $? = 6$. Total added: $1 + 6 = 7$.

10. Which problem does NOT need make a 10? Why?

A. $9 + 6$

B. $8 + 5$

C. $10 + 4$ ✓

D. $7 + 8$

Explanation: $10 + 4$ already has a 10! You just add 4 more to get 14. No need to make a 10 first.

11. Sara solved $8 + 6$ using make a 10. She wrote $8 + 2 = 10$, then $10 + 3 = 13$. Did she break apart 6 correctly?

A. Yes, $2 + 3 = 6$

B. No, $2 + 3 = 5$, not 6. She should write $10 + 4 = 14$ ✓

C. No, she should have used $8 + 8$

D. Yes, the answer is 13

Explanation: Error analysis: 6 breaks into 2 and 4, not 2 and 3. Correct: $8 + 2 = 10$, then $10 + 4 = 14$.

12. There are 7 red cubes and 9 blue cubes. Then 8 green cubes are added. How many cubes in all? Use make a 10 to start.

A. 22

B. 25

C. 26

D. 24 ✓

Explanation: First: $7 + 9 = 16$ (make a 10: $9 + 1 = 10$, $+ 6 = 16$). Then $16 + 8 = 24$.

13. What is the same about $9 + 4$ and $8 + 5$?

A. They both use doubles

B. They both equal 13 ✓

C. They both equal 14

D. They both start with make 10 + 3

Explanation: Check: $9 + 4 = 13$ and $8 + 5 = 13$. They both equal 13 even though the numbers are different!

14. $__ + __ = 16$. One number is 2 more than the other. What are they?

- A. 6 and 10
- B. 7 and 9 ✓**
- C. 8 and 8
- D. 5 and 11

Explanation: Open reasoning: $7 + 9 = 16$, and $9 - 7 = 2$. They are 2 apart!

15. A student solved $7 + 8$ and got 16. Explain what went wrong.

- A. They added 1 too many — the answer is 15 ✓**
- B. They subtracted instead
- C. They used the wrong numbers
- D. They forgot to make a 10

Explanation: Error analysis: $7 + 8 = 15$, not 16. The student added 1 too many when making a 10.

16. Nora had 6 pencils. She got 9 more. Then she gave away 4. How many does she have now?

- A. 10
- B. 12
- C. 13
- D. 11 ✓**

Explanation: Two steps: $6 + 9 = 15$ (make a 10: $9 + 1 = 10$, $+ 5 = 15$). Then $15 - 4 = 11$.

17. If $9 + 6 = 15$, what is $9 + 7$?

- A. 15
- B. 17
- C. 14
- D. 16 ✓**

Explanation: Think: related fact. $9 + 7$ is 1 more than $9 + 6$. So $15 + 1 = 16$.

18. Which is greater: $8 + 7$ or $9 + 5$? How do you know?

- A. $8 + 7$, because $15 > 14$ ✓**
- B. $9 + 5$, because 9 is bigger
- C. They are equal, both are 14
- D. They are equal, both are 15

Explanation: Make a 10: $8 + 7 = 15$ and $9 + 5 = 14$. $15 > 14$, so $8 + 7$ is greater.

19. Fill in both blanks: $8 + __ = 10 + __$. The answer is 13.

- A. 5 and 3 ✓**
- B. 4 and 3
- C. 6 and 3
- D. 5 and 4

Explanation: Make a 10: $8 + 5 = 13$. Also $10 + 3 = 13$. So blanks are 5 and 3.

20. Jada says every make-a-10 problem must start with 9 or 8. Is she right?

- A. Yes, only 9 and 8 work
- B. No, you can start with 7 too, like $7 + 5$
- C. No, you can start with any number ✓**
- D. Yes, other numbers are too small

Explanation: Make a 10 works with any number! For example: $6 + 7$: $6 + 4 = 10$, $+ 3 = 13$. Or $5 + 8$: $5 + 5 = 10$, $+ 3 = 13$.

Lesson 4: Number Families

Easy (20 questions)

1. If $3 + 5 = 8$, what is $5 + 3$?

- A. 7
- B. 9
- C. 8 ✓**
- D. 6

Explanation: Fact family: $3 + 5 = 8$ and $5 + 3 = 8$. You can add in any order!

2. If $2 + 6 = 8$, what is $6 + 2$?

- A. 9
- B. 4
- C. 8 ✓**
- D. 7

Explanation: Fact family: $2 + 6 = 8$ and $6 + 2 = 8$. Same numbers, same answer.

3. If $4 + 3 = 7$, what is $7 - 3$?

- A. 3
- B. 5
- C. 4 ✓**
- D. 6

Explanation: Fact family: If $4 + 3 = 7$, then $7 - 3 = 4$. Addition and subtraction are related.

4. If $5 + 4 = 9$, what is $9 - 4$?

- A. 4
- B. 6
- C. 3
- D. 5 ✓**

Explanation: Fact family: If $5 + 4 = 9$, then $9 - 4 = 5$.

5. If $1 + 7 = 8$, what is $8 - 7$?

- A. 2
- B. 0
- C. 7
- D. 1 ✓**

Explanation: Fact family: If $1 + 7 = 8$, then $8 - 7 = 1$.

6. If $6 + 3 = 9$, what is $3 + 6$?

- A. 8
- B. 10
- C. 12
- D. 9 ✓**

Explanation: You can add in any order. $6 + 3 = 9$ and $3 + 6 = 9$.

7. If $2 + 5 = 7$, what is $7 - 2$?

- A. 4
- B. 3
- C. 5 ✓**
- D. 6

Explanation: Fact family: If $2 + 5 = 7$, then $7 - 2 = 5$.

8. If $8 + 2 = 10$, what is $10 - 2$?

- A. 7
- B. 9
- C. 6
- D. 8 ✓**

Explanation: Fact family: If $8 + 2 = 10$, then $10 - 2 = 8$.

9. If $3 + 4 = 7$, what is $7 - 4$?

- A. 4
- B. 2
- C. 3 ✓
- D. 5

Explanation: Fact family: If $3 + 4 = 7$, then $7 - 4 = 3$.

10. If $6 + 4 = 10$, what is $4 + 6$?

- A. 9
- B. 11
- C. 10 ✓
- D. 8

Explanation: Add in any order: $6 + 4 = 10$ and $4 + 6 = 10$.

11. If $9 + 1 = 10$, what is $10 - 9$?

- A. 2
- B. 0
- C. 9
- D. 1 ✓

Explanation: Fact family: If $9 + 1 = 10$, then $10 - 9 = 1$.

12. If $5 + 2 = 7$, what is $7 - 5$?

- A. 3
- B. 1
- C. 4
- D. 2 ✓

Explanation: Fact family: If $5 + 2 = 7$, then $7 - 5 = 2$.

13. If $7 + 3 = 10$, what is $10 - 7$?

- A. 4
- B. 2
- C. 7
- D. 3 ✓

Explanation: Fact family: If $7 + 3 = 10$, then $10 - 7 = 3$.

14. If $1 + 8 = 9$, what is $8 + 1$?

- A. 10
- B. 7
- C. 8
- D. 9 ✓

Explanation: Add in any order: $1 + 8 = 9$ and $8 + 1 = 9$.

15. If $4 + 5 = 9$, what is $9 - 5$?

- A. 3
- B. 5
- C. 6
- D. 4 ✓

Explanation: Fact family: If $4 + 5 = 9$, then $9 - 5 = 4$.

16. If $3 + 7 = 10$, what is $10 - 3$?

- A. 8
- B. 6
- C. 3
- D. 7 ✓

Explanation: Fact family: If $3 + 7 = 10$, then $10 - 3 = 7$.

17. If $2 + 4 = 6$, what is $6 - 4$?

- A. 3
- B. 1
- C. 4
- D. 2 ✓**

Explanation: Fact family: If $2 + 4 = 6$, then $6 - 4 = 2$.

18. If $6 + 2 = 8$, what is $8 - 6$?

- A. 3
- B. 6
- C. 4
- D. 2 ✓**

Explanation: Fact family: If $6 + 2 = 8$, then $8 - 6 = 2$.

19. If $5 + 5 = 10$, what is $10 - 5$?

- A. 6
- B. 4
- C. 3
- D. 5 ✓**

Explanation: Fact family: $5 + 5 = 10$, so $10 - 5 = 5$. Doubles make special fact families!

20. If $1 + 6 = 7$, what is $7 - 1$?

- A. 5
- B. 7
- C. 8
- D. 6 ✓**

Explanation: Fact family: If $1 + 6 = 7$, then $7 - 1 = 6$.

Medium (20 questions)

1. $3 + \underline{\quad} = 10$. What is the missing number? Use a related fact to help.

- A. 8
- B. 6
- C. 5
- D. 7 ✓**

Explanation: Think: related fact. $10 - 3 = 7$. So $3 + 7 = 10$. The missing number is 7.

2. Which number sentence belongs to the same fact family as $6 + 8 = 14$?

- A. $6 + 6 = 12$
- B. $14 - 8 = 6$ ✓**
- C. $8 + 8 = 16$
- D. $14 + 6 = 20$

Explanation: Fact family: $6 + 8 = 14$, $8 + 6 = 14$, $14 - 6 = 8$, and $14 - 8 = 6$. They all use 6, 8, and 14.

3. If $9 + 5 = 14$, which subtraction fact do you also know?

- A. $14 - 9 = 5$ ✓**
- B. $14 - 5 = 8$
- C. $9 - 5 = 14$
- D. $14 + 5 = 19$

Explanation: Fact family: If $9 + 5 = 14$, then $14 - 9 = 5$ and $14 - 5 = 9$.

4. $\underline{\quad} - 4 = 7$. What is the missing number?

- A. 3
- B. 12
- C. 10
- D. 11 ✓**

Explanation: Think: related fact. $7 + 4 = 11$. So $11 - 4 = 7$. The missing number is 11.

5. How many facts are in the fact family for 3, 5, and 8?

- A. 2
- B. 3
- C. 4 ✓**
- D. 5

Explanation: There are 4 facts: $3 + 5 = 8$, $5 + 3 = 8$, $8 - 3 = 5$, $8 - 5 = 3$.

6. Ella has some books. She gives 6 away and has 9 left. How many did she start with?

- A. 3
- B. 16
- C. 14
- D. 15 ✓**

Explanation: Think: related fact. $9 + 6 = 15$. Ella started with 15 books.

7. Which strategy helps solve $13 - 7$?

- A. Count by 10s
- B. Think: $7 + \underline{\quad} = 13$ ✓**
- C. Use doubles of 13
- D. Skip count by 7s

Explanation: Use a related addition fact. Think: $7 + ? = 13$. $7 + 6 = 13$, so $13 - 7 = 6$.

8. If $7 + 6 = 13$, what is $13 - 6$?

- A. 8
- B. 6
- C. 9
- D. 7 ✓**

Explanation: Fact family: If $7 + 6 = 13$, then $13 - 6 = 7$.

9. $8 + \underline{\quad} = 15$. What is the missing number?

- A. 6
- B. 8
- C. 9
- D. 7 ✓**

Explanation: Think: related fact. $15 - 8 = 7$. So $8 + 7 = 15$. The missing number is 7.

10. Which number sentence does NOT belong in the fact family for 5, 9, and 14?

- A. $5 + 9 = 14$
- B. $14 - 5 = 9$
- C. $9 + 5 = 14$
- D. $9 - 5 = 4$ ✓**

Explanation: $9 - 5 = 4$ uses different numbers (4, not 14). It does not belong in the 5, 9, 14 family.

11. A boy had 12 crayons. He lost some and has 5 left. How many did he lose?

- A. 6
- B. 8
- C. 9
- D. 7 ✓**

Explanation: Think: related fact. $5 + ? = 12$. $5 + 7 = 12$. He lost 7 crayons.

12. If $8 + 4 = 12$, which is also true?

- A. $12 - 4 = 8$ ✓**
- B. $12 + 4 = 8$
- C. $8 - 4 = 12$
- D. $4 - 8 = 12$

Explanation: Fact family: $8 + 4 = 12$, so $12 - 4 = 8$ and $12 - 8 = 4$.

13. $14 - \underline{\quad} = 6$. What is the missing number?

- A. 7
- B. 9
- C. 6
- D. 8 ✓**

Explanation: Think: related fact. $6 + ? = 14$. $6 + 8 = 14$. So $14 - 8 = 6$.

14. How many facts are in the fact family for 6, 6, and 12?

- A. 4
- B. 1
- C. 3
- D. 2 ✓**

Explanation: Doubles are special! $6 + 6 = 12$ (only one addition) and $12 - 6 = 6$ (only one subtraction). Just 2 facts!

15. $\underline{\quad} + 9 = 16$. What is the missing number?

- A. 6
- B. 8
- C. 5
- D. 7 ✓**

Explanation: Think: related fact. $16 - 9 = 7$. So $7 + 9 = 16$.

16. Which fact helps you solve $11 - 4$?

- A. $11 + 4 = 15$
- B. $4 + 7 = 11$ ✓**
- C. $4 + 4 = 8$
- D. $11 - 11 = 0$

Explanation: Think: $4 + ? = 11$. $4 + 7 = 11$. So $11 - 4 = 7$.

17. There are 15 kids. 8 are boys. How many are girls?

- A. 6
- B. 8
- C. 9
- D. 7 ✓**

Explanation: Think: related fact. $8 + ? = 15$. $8 + 7 = 15$. There are 7 girls.

18. $5 + \underline{\quad} = 13$. What is the missing number?

- A. 7
- B. 9
- C. 6
- D. 8 ✓**

Explanation: Think: related fact. $13 - 5 = 8$. So $5 + 8 = 13$.

19. If $6 + 7 = 13$, which subtraction fact is in the same family?

- A. $13 - 6 = 7$ ✓**
- B. $13 - 7 = 5$
- C. $7 - 6 = 13$
- D. $6 - 7 = 1$

Explanation: Fact family: $6 + 7 = 13$, $7 + 6 = 13$, $13 - 6 = 7$, $13 - 7 = 6$.

20. Liam has 9 toy cars. His sister gives him more. Now he has 17. How many did she give?

- A. 7
- B. 9
- C. 6
- D. 8 ✓**

Explanation: Think: related fact. $9 + ? = 17$. $9 + 8 = 17$. She gave him 8 cars.

Hard (20 questions)

1. Ana picked 7 flowers. Together with Ben they have 15 flowers. How many did Ben pick? Then 4 flowers wilted. How many are left?

A. 8 and 11 ✓

B. 7 and 11

C. 8 and 12

D. 9 and 11

Explanation: Two steps: $7 + ? = 15$, so Ben picked 8. Then $15 - 4 = 11$ flowers left.

2. A student wrote the fact family for 5, 8, and 13 as: $5 + 8 = 13$, $8 + 5 = 13$, $13 - 5 = 7$. What mistake did they make?

A. $13 - 5 = 8$, not 7 ✓

B. They need more facts

C. $5 + 8$ is wrong

D. The family should use 12

Explanation: Error analysis: $13 - 5 = 8$, not 7. The student subtracted wrong. The correct fact is $13 - 5 = 8$.

3. If $6 + 9 = 15$, which number sentence is NOT in the same fact family?

A. $9 + 6 = 15$

B. $15 - 9 = 6$

C. $15 - 6 = 9$

D. $9 - 6 = 3$ ✓

Explanation: $9 - 6 = 3$ uses different numbers. The fact family for 6, 9, 15 does not include 3.

4. $\underline{\quad} + \underline{\quad} = 14$. Use the fact family to write a subtraction fact. Which pair works?

A. $5 + 9 = 14$, so $14 - 9 = 5$ ✓

B. $5 + 8 = 14$, so $14 - 8 = 6$

C. $6 + 9 = 14$, so $14 - 9 = 6$

D. $4 + 9 = 14$, so $14 - 9 = 5$

Explanation: $5 + 9 = 14$ is correct. The related subtraction fact is $14 - 9 = 5$.

5. Mia says: if I know $8 + 7 = 15$, I also know $15 - 7$ without counting. What is $15 - 7$?

A. 9

B. 7

C. 6

D. 8 ✓

Explanation: Fact family: $8 + 7 = 15$, so $15 - 7 = 8$. Use the related fact — no counting needed!

6. A student wrote: $12 - 5 = 6$. Use a fact family to check. Is this correct?

A. Yes, $5 + 6 = 12$

B. No, $5 + 6 = 11$, not 12. The answer should be 7 ✓

C. No, the answer should be 8

D. Yes, $6 + 6 = 12$

Explanation: Error analysis: check with addition. $5 + 6 = 11$, not 12. So $12 - 5$ is not 6. Correct: $5 + 7 = 12$, so $12 - 5 = 7$.

7. Tom has 16 stickers. He gives some to 2 friends. He gives 7 to one and keeps 4 for himself. How many did the second friend get?

A. 4

B. 6

C. 3

D. 5 ✓

Explanation: Two steps: Tom gave away $16 - 4 = 12$ total. First friend got 7. Second friend: $12 - 7 = 5$.

8. Which fact family has only 2 facts instead of 4?

A. 3, 5, 8

B. 4, 4, 8 ✓

C. 6, 7, 13

D. 5, 9, 14

Explanation: When both addends are the same (doubles), there are only 2 unique facts: $4 + 4 = 8$ and $8 - 4 = 4$.

9. I know $7 + 8 = 15$. Without adding, what is $8 + 7$?

- A. 16
- B. 14
- C. 13
- D. 15 ✓**

Explanation: Addition works in any order. $7 + 8 = 15$, so $8 + 7 = 15$ too. This is the order property.

10. Leo says the fact family for 4, 9, and 13 has 5 facts. Is he right?

- A. Yes, there are 5 facts
- B. No, there are only 4 facts ✓**
- C. No, there are only 3 facts
- D. No, there are only 2 facts

Explanation: A fact family with two different addends always has exactly 4 facts: 2 addition and 2 subtraction.

11. If $6 + \underline{\quad} = 14$, then $14 - 6 = \underline{\quad}$. Fill in both blanks.

- A. 7 and 7
- B. 9 and 9
- C. 8 and 8 ✓**
- D. 7 and 8

Explanation: Think: $6 + 8 = 14$, and $14 - 6 = 8$. Both blanks are 8. That is the fact family connection!

12. A student says $11 - 5 = 7$ because $5 + 7 = 12$. What TWO mistakes did they make?

- A. Wrong addition fact AND wrong subtraction ✓**
- B. Only the subtraction is wrong
- C. Only the addition is wrong
- D. No mistakes were made

Explanation: Error analysis: $5 + 7 = 12$ (not 11), AND $11 - 5 = 6$ (not 7). Two errors: wrong addition fact and wrong answer.

13. Sara had some beads. She lost 8 and has 6 left. She finds 3 more. How many does she have now?

- A. 11
- B. 8
- C. 10
- D. 9 ✓**

Explanation: Step 1: $? - 8 = 6$, so she had 14. Step 2: she has 6 now, finds 3 more: $6 + 3 = 9$.

14. Which number can replace the box? $5 + \text{box} = 12$, and $12 - \text{box} = 5$.

- A. 6
- B. 8
- C. 9
- D. 7 ✓**

Explanation: The box is the same in both! $5 + 7 = 12$, and $12 - 7 = 5$. The box is 7. Fact families show this.

15. Kai wrote four facts: $3 + 9 = 12$, $9 + 3 = 12$, $12 - 3 = 9$, $12 - 9 = 4$. Which fact has an error?

- A. $3 + 9 = 12$
- B. $9 + 3 = 12$
- C. $12 - 3 = 9$
- D. $12 - 9 = 4$ ✓**

Explanation: Error analysis: $12 - 9 = 3$, not 4. The last fact should be $12 - 9 = 3$.

16. If you know $9 + 7 = 16$, name two subtraction facts you also know.

- A. $16 - 9 = 7$ and $16 - 7 = 9$ ✓**
- B. $16 - 9 = 8$ and $16 - 7 = 8$
- C. $9 - 7 = 2$ and $16 - 9 = 6$
- D. $16 - 7 = 8$ and $16 - 9 = 7$

Explanation: Fact family: $9 + 7 = 16$ gives us $16 - 9 = 7$ and $16 - 7 = 9$.

17. ___ - 6 = 9. Then use that answer to find ___ + 3. What are both answers?

A. 15 and 18 ✓

B. 14 and 17

C. 16 and 19

D. 13 and 16

Explanation: Step 1: related fact. $9 + 6 = 15$. Step 2: $15 + 3 = 18$. Answers: 15 and 18.

18. Can 3, 5, and 9 be a fact family? Why or why not?

A. Yes, $3 + 5 = 9$

B. No, because $3 + 5 = 8$, not 9 ✓

C. Yes, $9 - 3 = 5$

D. No, because $9 - 5 = 3$

Explanation: Check: $3 + 5 = 8$, not 9. The three numbers in a fact family must work in addition. 3, 5, and 9 do not.

19. A jar has red and blue cubes. There are 16 cubes total. If you remove all 9 red cubes, 7 blue are left. Write the fact family.

A. $9+7=16$, $7+9=16$, $16-9=7$, $16-7=9$ ✓

B. $9+7=16$, $16-7=8$, $16-9=6$

C. $9+7=15$, $7+9=15$, $15-9=7$

D. Only $9+7=16$

Explanation: The numbers are 9, 7, and 16. Fact family: $9+7=16$, $7+9=16$, $16-9=7$, $16-7=9$.

20. Maya says knowing one addition fact gives you three more facts for free. Is she right?

A. Yes, you get 1 more addition and 2 subtraction facts ✓

B. No, you only get 1 subtraction fact

C. No, you have to memorize each one

D. Yes, but only for doubles

Explanation: Correct! From $6 + 8 = 14$, you also know $8 + 6 = 14$, $14 - 6 = 8$, and $14 - 8 = 6$. Three more facts for free!

Unit Test Bank

Easy (10 questions)

1. Start at 4. Count up 3. What number do you land on?

A. 6

B. 8

C. 7 ✓

D. 5

Explanation: Count up from 4: 5, 6, 7. You land on 7.

2. What is $6 + 6$?

A. 11

B. 13

C. 10

D. 12 ✓

Explanation: Use doubles: $6 + 6 = 12$.

3. What number goes with 8 to make 10?

A. 3

B. 1

C. 2 ✓

D. 4

Explanation: $8 + 2 = 10$. The number 2 goes with 8 to make 10.

4. If $4 + 5 = 9$, what is $5 + 4$?

- A. 8
- B. 10
- C. 9 ✓**
- D. 7

Explanation: Fact family: $4 + 5 = 9$ and $5 + 4 = 9$. Same answer!

5. What is $9 - 3$?

- A. 5
- B. 7
- C. 6 ✓**
- D. 12

Explanation: Count back 3 from 9: 8, 7, 6. The answer is 6.

6. What is double 4?

- A. 6
- B. 4
- C. 10
- D. 8 ✓**

Explanation: Double 4 means $4 + 4 = 8$.

7. What is $10 + 6$?

- A. 60
- B. 17
- C. 15
- D. 16 ✓**

Explanation: $10 + 6 = 16$. Adding to 10 is easy!

8. If $3 + 7 = 10$, what is $10 - 7$?

- A. 4
- B. 2
- C. 7
- D. 3 ✓**

Explanation: Fact family: If $3 + 7 = 10$, then $10 - 7 = 3$.

9. Count back 2 from 8. What number do you get?

- A. 5
- B. 7
- C. 6 ✓**
- D. 10

Explanation: Count back from 8: 7, 6. You get 6.

10. What is $7 + 7$?

- A. 13
- B. 15
- C. 12
- D. 14 ✓**

Explanation: Use doubles: $7 + 7 = 14$.

Medium (10 questions)

1. A cat has 9 treats. It gets 4 more. How many treats does the cat have now?

- A. 14
- B. 12
- C. 11
- D. 13 ✓**

Explanation: Make a 10: $9 + 1 = 10$, then $10 + 3 = 13$. The cat has 13 treats.

2. What is $7 + 8$? Use a doubles fact to help.

- A. 14
- B. 16
- C. 13
- D. 15 ✓**

Explanation: Doubles plus 1: $7 + 7 = 14$, add 1 more = 15.

3. $__ + 5 = 12$. What is the missing number?

- A. 6
- B. 8
- C. 5
- D. 7 ✓**

Explanation: Think: related fact. $12 - 5 = 7$. So $7 + 5 = 12$.

4. Which strategy helps you solve $8 + 5$?

- A. Count by 2s
- B. Use doubles of 8
- C. Make a 10 first ✓**
- D. Subtract 5 from 8

Explanation: Make a 10: $8 + 2 = 10$, then $10 + 3 = 13$.

5. If $9 + 6 = 15$, what is $15 - 9$?

- A. 5
- B. 7
- C. 8
- D. 6 ✓**

Explanation: Fact family: If $9 + 6 = 15$, then $15 - 9 = 6$.

6. There are 13 ducks. 4 swim away. How many are left?

- A. 8
- B. 10
- C. 17
- D. 9 ✓**

Explanation: Count back 4 from 13: 12, 11, 10, 9. There are 9 ducks left.

7. $__ + __ = 14$. Both numbers are the same. What are they?

- A. 6 and 6
- B. 8 and 8
- C. 5 and 5
- D. 7 and 7 ✓**

Explanation: Think: what double equals 14? $7 + 7 = 14$. Both numbers are 7.

8. Which number sentence belongs to the same fact family as $8 + 5 = 13$?

- A. $13 - 5 = 8$ ✓**
- B. $13 - 5 = 7$
- C. $8 + 8 = 16$
- D. $5 + 5 = 10$

Explanation: Fact family: 8, 5, 13. So $13 - 5 = 8$ is in the same family.

9. $9 - __ = 5$. What is the missing number?

- A. 3
- B. 5
- C. 2
- D. 4 ✓**

Explanation: Think: $5 + ? = 9$. Count up: 6, 7, 8, 9. That is 4 steps. The missing number is 4.

10. Ben has 6 red blocks and 7 blue blocks. How many blocks in all?

- A. 14
- B. 12
- C. 11
- D. 13 ✓**

Explanation: Make a 10: $6 + 4 = 10$, then $10 + 3 = 13$. Or doubles plus 1: $6 + 6 = 12$, $+ 1 = 13$.

Hard (10 questions)

1. Kai had 14 toy trucks. He gave 6 to his brother and 3 to his friend. How many does Kai have left?

- A. 4
- B. 6
- C. 3
- D. 5 ✓**

Explanation: Two steps: $14 - 6 = 8$, then $8 - 3 = 5$. Count back each time. Kai has 5 left.

2. A student wrote: double 8 = 15. Then they said $8 + 9 = 16$. How many errors are there?

- A. 0
- B. 1
- C. 2 ✓**
- D. 3

Explanation: Error analysis: double 8 = 16 (not 15), and $8 + 9 = 17$ (not 16). That is 2 errors.

3. If $7 + 9 = 16$, what are the two subtraction facts in the same family?

- A. $16 - 7 = 9$ and $16 - 9 = 7$ ✓**
- B. $16 - 7 = 8$ and $16 - 9 = 6$
- C. $9 - 7 = 2$ and $16 - 7 = 8$
- D. $7 - 9 = 2$ and $16 - 9 = 8$

Explanation: Fact family: $7 + 9 = 16$, $9 + 7 = 16$, $16 - 7 = 9$, $16 - 9 = 7$.

4. Emma has 8 stickers. She finds 7 more, then gives 6 away. How many does she have now?

- A. 10
- B. 8
- C. 11
- D. 9 ✓**

Explanation: Two steps: $8 + 7 = 15$ (doubles plus 1). Then $15 - 6 = 9$. Emma has 9 stickers.

5. A student solved $9 + 4$ using make a 10: $9 + 1 = 10$, then $10 + 4 = 14$. What went wrong?

- A. They should have added 10 + 3, not 10 + 4 ✓**
- B. They started at the wrong number
- C. Nothing, the answer is 14
- D. They should use doubles instead

Explanation: Error analysis: After using 1 from the 4, only 3 is left. Correct: $10 + 3 = 13$, not $10 + 4 = 14$.

6. $\underline{\quad} + \underline{\quad} = 13$. Both numbers are greater than 4. One is 3 more than the other. What are they?

- A. 4 and 9
- B. 6 and 7
- C. 5 and 8 ✓**
- D. 3 and 10

Explanation: Open reasoning: $5 + 8 = 13$, both > 4 , and $8 - 5 = 3$. They are 3 apart!

7. Leo says $15 - 8 = 6$ because $8 + 6 = 14$. What TWO things did Leo get wrong?

- A. Wrong addition fact AND wrong subtraction ✓**
- B. Only the addition is wrong
- C. Only the subtraction is wrong
- D. Nothing is wrong

Explanation: Error analysis: $8 + 6 = 14$ (not 15), so the addition fact is wrong. And $15 - 8 = 7$, not 6.

8. Which is greater: $9 + 6$ or double 7 plus 1?

A. $9 + 6$, because $15 > 14$

B. Double 7 plus 1, because $15 > 14$

C. They are equal, both are 15 ✓

D. They are equal, both are 14

Explanation: Check: $9 + 6 = 15$. Double 7 = 14, plus 1 = 15. They are equal! Both are 15.

9. I am a number. Add me to myself and you get 12. Then subtract 5. What do you get?

A. 6 and 1

B. 5 and 7

C. 7 and 8

D. 6 and 7 ✓

Explanation: Double ? = 12, so the number is 6. Then $12 - 5 = 7$. Answers: 6 and 7.

10. Can 4, 8, and 11 be a fact family? Explain.

A. Yes, $4 + 8 = 11$

B. No, because $4 + 8 = 12$, not 11 ✓

C. Yes, $11 - 8 = 4$

D. No, because $11 - 4 = 8$

Explanation: Check: $4 + 8 = 12$, not 11. The numbers do not work together. 4, 8, and 11 cannot be a fact family.

Unit 2: Place Value

Lesson 1: Big Numbers

Easy (20 questions)

1. What digit is in the tens place of 374?

- A. 3
- B. 7 ✓**
- C. 4
- D. 37

Explanation: In 374, the 7 is in the tens place. It sits between the hundreds and the ones.

2. What digit is in the hundreds place of 851?

- A. 1
- B. 5
- C. 8 ✓**
- D. 85

Explanation: The hundreds place is the first digit on the left. In 851, that digit is 8.

3. What digit is in the ones place of 639?

- A. 6
- B. 3
- C. 9 ✓**
- D. 39

Explanation: The ones place is the last digit on the right. In 639, that digit is 9.

4. How many hundreds are in 400?

- A. 40
- B. 4 ✓**
- C. 0
- D. 400

Explanation: 400 has 4 hundreds. Each hundred is worth 100, and $4 \times 100 = 400$.

5. Which number has a 6 in the tens place?

- A. 602
- B. 260
- C. 326
- D. 463 ✓**

Explanation: In 463, the 6 is in the tens place (the middle digit). Look at the second digit from the right.

6. What digit is in the hundreds place of 207?

- A. 7
- B. 0
- C. 2 ✓**
- D. 20

Explanation: The hundreds digit is the one on the far left. In 207, the hundreds digit is 2.

7. How many tens are in 90?

- A. 90
- B. 0
- C. 9 ✓**
- D. 900

Explanation: 90 has 9 tens. Each ten is worth 10, and $9 \times 10 = 90$.

8. Jake has 3 hundreds blocks, 2 tens blocks, and 5 ones blocks. What number did he make?

- A. 235
- B. 352
- C. 325 ✓**
- D. 532

Explanation: 3 hundreds = 300, 2 tens = 20, 5 ones = 5. Put them together: 325.

9. What digit is in the tens place of 148?

- A. 1
- B. 4 ✓**
- C. 8
- D. 48

Explanation: The tens digit is the middle one. In 148, the digit 4 is in the tens place.

10. Which number has a 9 in the ones place?

- A. 952
- B. 295
- C. 529 ✓**
- D. 590

Explanation: The ones digit is the last digit on the right. In 529, the ones digit is 9.

11. How many ones are in the number 47?

- A. 4
- B. 7 ✓**
- C. 47
- D. 0

Explanation: The ones digit in 47 is 7. There are 7 ones.

12. Maria put 5 hundreds blocks on the table. What number do just those blocks show?

- A. 5
- B. 50
- C. 500 ✓**
- D. 505

Explanation: Each hundreds block is worth 100. Five of them make 500.

13. What digit is in the hundreds place of 999?

- A. 9 ✓**
- B. 99
- C. 999
- D. 90

Explanation: In 999, every digit is 9. The hundreds digit is the first 9 on the left.

14. Which place is the 3 in for the number 735?

- A. Hundreds
- B. Tens ✓**
- C. Ones
- D. Thousands

Explanation: In 735, the 3 is the middle digit. The middle digit is always in the tens place.

15. A number has 0 hundreds, 6 tens, and 2 ones. What is the number?

- A. 620
- B. 602
- C. 62 ✓**
- D. 260

Explanation: 0 hundreds = 0, 6 tens = 60, 2 ones = 2. The number is 62.

16. What digit is in the ones place of 810?

- A. 8
- B. 1
- C. 0 ✓
- D. 10

Explanation: The ones place is the last digit. In 810, the ones digit is 0.

17. How many hundreds are in 700?

- A. 70
- B. 7 ✓
- C. 0
- D. 700

Explanation: $700 = 7$ hundreds. Each hundred is 100, and $7 \times 100 = 700$.

18. Tom has base-ten blocks: 1 hundred, 4 tens, 3 ones. What number is that?

- A. 134
- B. 413
- C. 143 ✓
- D. 314

Explanation: 1 hundred = 100, 4 tens = 40, 3 ones = 3. Together that is 143.

19. Which number has a 2 in the hundreds place?

- A. 832
- B. 328
- C. 283
- D. 238 ✓

Explanation: The hundreds place is the first digit. In 238, the first digit is 2.

20. What digit is in the tens place of 506?

- A. 5
- B. 0 ✓
- C. 6
- D. 50

Explanation: In 506, the tens digit is 0. That means there are zero tens.

Medium (20 questions)

1. In the number 358, what is the VALUE of the digit 5?

- A. 5
- B. 50 ✓
- C. 500
- D. 58

Explanation: The digit 5 is in the tens place. Its value is 50 because 5 tens = 50. The digit tells the place, the value tells how much it is worth.

2. Which number has a 4 in the hundreds place AND a 7 in the ones place?

- A. 470
- B. 407 ✓
- C. 740
- D. 347

Explanation: Look for 4 as the first digit (hundreds) and 7 as the last digit (ones). 407 fits: 4 hundreds, 0 tens, 7 ones.

3. Rosa has 2 hundreds blocks, 14 ones blocks, and no tens blocks. What number did she build?

- A. 214 ✓
- B. 240
- C. 200
- D. 2,014

Explanation: 2 hundreds = 200, 0 tens = 0, 14 ones = 14. Add them: $200 + 14 = 214$.

4. Which shows how to break apart 745 into hundreds, tens, and ones?

- A. **$700 + 40 + 5$** ✓
- B. $7 + 4 + 5$
- C. $74 + 5$
- D. $700 + 45$

Explanation: $745 = 700 + 40 + 5$. The 7 is worth 700, the 4 is worth 40, and the 5 is worth 5.

5. A number has 6 hundreds and 3 ones but NO tens. What is the number?

- A. 630
- B. **603** ✓
- C. 63
- D. 360

Explanation: 6 hundreds = 600, 0 tens = 0, 3 ones = 3. The number is 603. When there are no tens, put a 0 in the tens place.

6. What is the value of the digit 9 in 492?

- A. 9
- B. **90** ✓
- C. 900
- D. 92

Explanation: The 9 is in the tens place, so its value is 90. Remember: digit tells the place, value tells what it is worth.

7. Ben says the number 830 has 83 tens. Is he right?

- A. **Yes, 83 tens = 830** ✓
- B. No, it has 3 tens
- C. No, it has 8 tens
- D. No, it has 30 tens

Explanation: This is tricky! $83 \times 10 = 830$, so yes, 830 does have 83 tens total. You can think of the hundreds as groups of 10 tens.

8. ____ hundreds + 2 tens + 9 ones = 529. What goes in the blank?

- A. 2
- B. **5** ✓
- C. 9
- D. 52

Explanation: $529 = 500 + 20 + 9$. That means 5 hundreds + 2 tens + 9 ones. The blank is 5.

9. Mia says the value of the 6 in 612 is 6. What should the value be?

- A. 6
- B. 60
- C. **600** ✓
- D. 612

Explanation: The 6 is in the hundreds place, so its value is 600, not 6. Mia mixed up the digit with the value.

10. Which number shows 4 hundreds, 0 tens, and 8 ones?

- A. 480
- B. **408** ✓
- C. 48
- D. 840

Explanation: 4 hundreds = 400, 0 tens = 0, 8 ones = 8. The number is 408.

11. Carlos built 562 with base-ten blocks. He lost 1 tens block. What number does he have now?

- A. **552** ✓
- B. 462
- C. 561
- D. 542

Explanation: 562 has 5 hundreds, 6 tens, 2 ones. Take away 1 ten: 5 hundreds, 5 tens, 2 ones = 552.

12. In the number 444, which digit has the GREATEST value?

A. They are all the same

B. The first 4 ✓

C. The middle 4

D. The last 4

Explanation: Even though all digits are 4, their values differ. The first 4 = 400, the middle 4 = 40, the last 4 = 4. The first 4 has the greatest value.

13. What number has 9 in the ones place and 1 in the hundreds place?

A. Only 109

B. Only 119

C. Only 199

D. There are many answers ✓

Explanation: The tens digit could be anything (0-9), so numbers like 109, 119, 129... 199 all work. There are many possible answers.

14. Kira added 1 hundreds block to 348. What is her new number?

A. 349

B. 358

C. 448 ✓

D. 1,348

Explanation: Adding 1 hundreds block means adding 100. $348 + 100 = 448$.

15. Which digit in 276 has a value of 200?

A. 2 ✓

B. 7

C. 6

D. 27

Explanation: The digit 2 is in the hundreds place. 2 hundreds = 200.

16. Omar has 3 hundreds blocks and 15 ones cubes. He trades 10 ones for 1 tens block. What number does he have?

A. 315 ✓

B. 305

C. 3,015

D. 350

Explanation: 3 hundreds = 300. Trading 10 ones for 1 ten does not change the total. $300 + 10 + 5 = 315$.

17. The value of the digit 8 in which number is 800?

A. 380

B. 108

C. 821 ✓

D. 488

Explanation: In 821, the 8 is in the hundreds place, so its value is 800. In the other numbers, 8 is in the tens or ones place.

18. 7 hundreds + ____ tens + 4 ones = 764. What goes in the blank?

A. 4

B. 6 ✓

C. 7

D. 64

Explanation: $764 = 700 + 60 + 4$. So 7 hundreds + 6 tens + 4 ones. The blank is 6.

19. Lucy says 305 has 3 hundreds and 5 ones. Did she describe it fully?

A. Yes, she is correct

B. No, she forgot 0 tens ✓

C. No, she forgot 5 tens

D. No, she forgot 3 tens

Explanation: $305 = 3$ hundreds, 0 tens, 5 ones. Lucy forgot to say 0 tens. Even if there are 0, we still count that place.

20. Which pair of numbers have the SAME digit in the tens place?

- A. 345 and 543
- B. 291 and 198
- C. 672 and 478 ✓**
- D. 456 and 853

Explanation: 672 has 7 in the tens place. 478 also has 7 in the tens place. Both share a tens digit of 7.

Hard (20 questions)

1. What number has 3 hundreds, 15 tens, and 7 ones?

- A. 3,157
- B. 457 ✓**
- C. 307
- D. 1,507

Explanation: 15 tens = 150. But 10 tens make another hundred! So 3 hundreds + 15 tens + 7 ones = 300 + 150 + 7 = 457. You must regroup the extra tens into a hundred.

2. Jada wrote $600 + 40 + 15$ in standard form as 6,415. What mistake did she make?

- A. She added wrong
- B. She put digits side by side instead of adding ✓**
- C. She forgot the zero
- D. She switched hundreds and tens

Explanation: Jada wrote each number's digits next to each other (6-4-15) instead of adding. The correct answer is $600 + 40 + 15 = 655$.

3. Sam says $452 > 459$ because $5 > 4$. Is Sam right?

- A. Yes, 5 is bigger than 4
- B. No, the hundreds are different
- C. No, both start with 45 so check ones: $2 < 9$ ✓**
- D. No, you cannot compare these numbers

Explanation: Sam compared the wrong digits. Both numbers have 4 hundreds and 5 tens. Since the ones are different, compare 2 and 9. Since $2 < 9$, we know $452 < 459$.

4. A mystery number has the same digit in every place. The digit values add up to 666. What is the number?

- A. 222
- B. 666 ✓**
- C. 333
- D. 999

Explanation: If the same digit d is in every place, the values are d hundreds + d tens + d ones. That is $100d + 10d + d = 111 \times d$. $111 \times 6 = 666$. The number is 666.

5. ____ is 100 more than 867 and 10 less than 977. What is ____?

- A. 877
- B. 967 ✓**
- C. 957
- D. 857

Explanation: 100 more than 867 = 967. Check: 10 less than 977 = 967. Both clues give 967, so the answer is 967.

6. Dev says 709 has 7 hundreds and 9 tens. Find his mistake.

- A. The 9 is in the ones place, not tens ✓**
- B. The 7 is in the tens place
- C. 709 has 70 tens
- D. There is no mistake

Explanation: In 709, the 0 is in the tens place and the 9 is in the ones place. Dev thought the 9 was in the tens place, but it is actually in the ones place.

7. I am a 3-digit number. My hundreds digit is double my ones digit. My tens digit is 0. My ones digit is 3. What number am I?

- A. 603 ✓
- B. 306
- C. 630
- D. 360

Explanation: Ones digit = 3. Hundreds digit = double 3 = 6. Tens digit = 0. The number is 603.

8. Ava shows 524 three different ways. Which is NOT correct?

- A. $500 + 20 + 4$
- B. 5 hundreds, 2 tens, 4 ones
- C. 50 tens and 24 ones
- D. $500 + 20 + 40$ ✓

Explanation: $500 + 20 + 40 = 560$, not 524. The other three all equal 524. Check: 50 tens = 500, plus 24 ones = 524. Correct! But $500 + 20 + 40$ adds extra tens.

9. What number has 12 tens, 4 hundreds, and 3 ones?

- A. 1,243
- B. 523 ✓
- C. 4,123
- D. 1,423

Explanation: 12 tens = 120. 4 hundreds = 400. 3 ones = 3. Total: $400 + 120 + 3 = 523$. The 12 tens regroup into 1 extra hundred and 2 tens.

10. Eli says that in 380, the value of 8 is greater than the value of 3. Is he right? Why?

- A. Yes, because $8 > 3$
- B. No, 3 is in hundreds so $300 > 80$ ✓
- C. Yes, because $80 > 30$
- D. No, they are equal

Explanation: Even though 8 is a bigger digit, the 3 is in the hundreds place (value 300) and the 8 is in the tens place (value 80). $300 > 80$, so the 3 has greater value.

11. Two students build the same number. Ana uses 4 hundreds and 23 ones. Ben uses 3 hundreds and 12 tens. Who has a leftover, and how many ones?

- A. Ana, 3 ones left over
- B. Ben, 3 ones left over
- C. Ana has 423, Ben has 420, so Ana has 3 extra ones ✓
- D. They both have 423

Explanation: Ana: $400 + 23 = 423$. Ben: $300 + 120 = 420$. Ana's number is 3 more than Ben's. They did NOT build the same number. Ana has 3 extra ones.

12. I am thinking of a number. It has a 0 in the tens place. The hundreds digit is 2 more than the ones digit. The ones digit is 5. What is my number?

- A. 705 ✓
- B. 507
- C. 750
- D. 570

Explanation: Ones = 5. Hundreds = $5 + 2 = 7$. Tens = 0. The number is 705.

13. Kai wrote $300 + 50 + 17 = 3,517$. What should the answer really be?

- A. 367 ✓
- B. 857
- C. 3,517
- D. 350

Explanation: Add step by step: $300 + 50 = 350$, then $350 + 17 = 367$. Kai placed digits side by side (3-5-17) instead of adding.

14. Which number does NOT have 4 in a place where its value is 40?

- A. 241
- B. 843
- C. 540 ✓**
- D. 142

Explanation: For the value to be 40, the 4 must be in the tens place. 241 has 4 in tens (value 40). 843 has 4 in tens (value 40). 142 has 4 in tens (value 40). But 540 has 4 in the ones place (value 4), not 40.

15. Maya traded 1 hundreds block for 10 tens blocks. She started with 3 hundreds, 2 tens, 5 ones. How many tens blocks does she have now?

- A. 10
- B. 12 ✓**
- C. 102
- D. 32

Explanation: She trades 1 hundred for 10 tens. She had 2 tens + 10 new tens = 12 tens blocks. The total value stays 325: now 2 hundreds + 12 tens + 5 ones = $200 + 120 + 5 = 325$.

16. What 3-digit number has digits that add up to 20, a 9 in the tens place, and an even ones digit?

- A. 596 ✓**
- B. 894
- C. 498
- D. 698

Explanation: Tens = 9, ones must be even. Try 596: $5 + 9 + 6 = 20$. It has 9 in tens and 6 (even) in ones. That works! The others: $894 = 8+9+4=21$, $498 = 4+9+8=21$, $698 = 6+9+8=23$.

17. Zoe built a number with 2 hundreds blocks, 0 tens blocks, and 18 ones cubes. Her teacher said to regroup. What is the standard form?

- A. 218 ✓**
- B. 2,018
- C. 228
- D. 318

Explanation: 2 hundreds = 200, 18 ones = 18. Total = 218. The 18 ones could be regrouped as 1 ten and 8 ones, giving 2 hundreds, 1 ten, 8 ones = 218.

18. A number has more hundreds than tens and more tens than ones. Which could it be?

- A. 321 ✓**
- B. 123
- C. 213
- D. 231

Explanation: We need hundreds digit > tens digit > ones digit. In 321: $3 > 2 > 1$. That works! In 123: $1 < 2$, so no. In 213: $2 > 1$ but $1 < 3$, no. In 231: $2 < 3$, no.

19. Pedro says $500 + 0 + 8 = 58$. What is wrong?

- A. He dropped the hundreds
- B. He forgot the zero in the tens place ✓**
- C. He added wrong
- D. Nothing, he is correct

Explanation: $500 + 0 + 8 = 508$, not 58. Pedro forgot that 0 tens still needs a 0 in the tens place. Without it, 508 becomes 58.

20. Use these clues to find the number: It is between 550 and 600. The tens digit is 3 more than the ones digit. The ones digit is 2.

- A. 552 ✓**
- B. 572
- C. 582
- D. 523

Explanation: Between 550 and 600 means hundreds = 5. Ones = 2. Tens = $2 + 3 = 5$. The number is 552. Check: 552 is between 550 and 600, and $5 - 2 = 3$.

Lesson 2: Different Ways to Write Numbers

Easy (20 questions)

1. How do you write 245 in words?

A. Two hundred forty-five ✓

B. Two hundred fifty-four

C. Two forty-five

D. Twenty-four five

Explanation: $245 = \text{two hundred forty-five}$. Say the hundreds first, then the tens and ones together.

2. What is the standard form of three hundred twelve?

A. 3,012

B. 312 ✓

C. 321

D. 213

Explanation: Three hundred twelve = 312. Standard form uses digits, not words.

3. What is $400 + 30 + 7$?

A. 437 ✓

B. 4,307

C. 473

D. 347

Explanation: Add each part: $400 + 30 + 7 = 437$. This is expanded form turning into standard form.

4. How do you write 860 in expanded form?

A. $800 + 60$

B. $800 + 60 + 0$ ✓

C. $86 + 0$

D. $8 + 6 + 0$

Explanation: $860 = 800 + 60 + 0$. In expanded form, we show the value of each digit. The ones place is 0.

5. Which is the word form of 519?

A. Five nineteen

B. Five hundred nineteen ✓

C. Fifty-one nine

D. Five hundred ninety-one

Explanation: 519 in words is five hundred nineteen. Always say 'hundred' for 3-digit numbers.

6. What is the standard form of $200 + 50 + 3$?

A. 253 ✓

B. 2,503

C. 235

D. 523

Explanation: Just add them up: $200 + 50 + 3 = 253$. Expanded form shows each place value added together.

7. How do you write seven hundred four?

A. 740

B. 704 ✓

C. 74

D. 7,004

Explanation: Seven hundred four = 704. There are 0 tens, so a 0 goes in the tens place.

8. Which shows 651 in expanded form?

A. $600 + 50 + 1$ ✓

B. $6 + 5 + 1$

C. $650 + 1$

D. $65 + 1$

Explanation: $651 = 600 + 50 + 1$. Each digit is written as its full value.

9. Amy wrote five hundred thirty. Which number is that?

- A. 513
- B. 530 ✓**
- C. 503
- D. 5,030

Explanation: Five hundred thirty = 530. Thirty means 3 tens and 0 ones.

10. What is $100 + 90 + 2$?

- A. 912
- B. 192 ✓**
- C. 129
- D. 1,092

Explanation: $100 + 90 + 2 = 192$. The 1 is hundreds, 9 is tens, 2 is ones.

11. How do you write 408 in words?

- A. Four hundred eight ✓**
- B. Four hundred eighty
- C. Forty-eight
- D. Four zero eight

Explanation: 408 = four hundred eight. We do not say the zero -- we skip from hundreds right to ones.

12. Which is the expanded form of 370?

- A. $37 + 0$
- B. $300 + 70 + 0$ ✓**
- C. $300 + 7$
- D. $3 + 7 + 0$

Explanation: $370 = 300 + 70 + 0$. Show each digit as its full value: 3 hundreds, 7 tens, 0 ones.

13. Jake says nine hundred sixteen is 960. Is he right?

- A. Yes
- B. No, it is 916 ✓**
- C. No, it is 906
- D. No, it is 196

Explanation: Nine hundred sixteen = 916. Jake wrote sixty (60) instead of sixteen (16).

14. What is the standard form of $700 + 8$?

- A. 78
- B. 780
- C. 708 ✓**
- D. 7,008

Explanation: $700 + 8 = 708$. There are 0 tens, so we need a 0 in the middle.

15. Which number is written in word form?

- A. 452
- B. $400 + 50 + 2$
- C. Four hundred fifty-two ✓**
- D. 4 hundreds 5 tens 2 ones

Explanation: Word form uses words to write the number. 'Four hundred fifty-two' is word form. The others are standard form, expanded form, and place-value form.

16. How do you write 999 in expanded form?

- A. $900 + 90 + 9$ ✓**
- B. $9 + 9 + 9$
- C. 999
- D. $99 + 9$

Explanation: $999 = 900 + 90 + 9$. Each 9 has a different value based on its place.

17. What is the word form of 110?

- A. One hundred ten ✓
- B. One hundred one
- C. Eleven
- D. One ten

Explanation: 110 = one hundred ten. 1 hundred and 1 ten and 0 ones.

18. Maria wrote $300 + 20 + 6$. What number is that?

- A. 326 ✓
- B. 362
- C. 236
- D. 3,206

Explanation: $300 + 20 + 6 = 326$. Add each part together to get the standard form.

19. Which is the expanded form of 205?

- A. $200 + 5$
- B. $200 + 0 + 5$ ✓
- C. $20 + 5$
- D. $2 + 0 + 5$

Explanation: $205 = 200 + 0 + 5$. Even though there are 0 tens, we show it in expanded form.

20. What number is six hundred one?

- A. 610
- B. 601 ✓
- C. 61
- D. 6,001

Explanation: Six hundred one = 601. Be careful not to write sixty-one (610) by mistake.

Medium (20 questions)

1. What is $600 + 30 + 8$ in standard form?

- A. 6,308
- B. 6,038
- C. 638 ✓
- D. 368

Explanation: Add the parts: $600 + 30 + 8 = 638$. Standard form is just the number written with digits.

2. Which number is shown in expanded form?

- A. 523
- B. $500 + 20 + 3$ ✓
- C. Five hundred twenty-three
- D. 5 hundreds 2 tens 3 ones

Explanation: Expanded form breaks the number into hundreds + tens + ones using addition. $500 + 20 + 3$ is expanded form.

3. Is $400 + 20 + 8$ the same as 428?

- A. Yes, they are equal ✓
- B. No, $400 + 20 + 8 = 4,208$
- C. No, $400 + 20 + 8 = 482$
- D. No, they are close but not the same

Explanation: $400 + 20 + 8 = 428$. Expanded form and standard form are just two ways to write the same number.

4. $300 + \underline{\quad} + 6 = 376$. What goes in the blank?

- A. 7
- B. 70 ✓
- C. 76
- D. 700

Explanation: $376 = 300 + 70 + 6$. The missing part is 70, which is the tens value.

5. Leo writes the word form of 802 as 'eight hundred twenty.' What should it be?

- A. Eight hundred twenty
- B. Eight hundred two ✓**
- C. Eighty-two
- D. Eight hundred twelve

Explanation: 802 has 0 tens and 2 ones. The word form is 'eight hundred two.' Leo confused 02 with twenty.

6. Which two ways show the SAME number?

- A. 306 and $300 + 60$
- B. 490 and $400 + 90 + 0$ ✓**
- C. 715 and $700 + 50 + 1$
- D. 243 and $200 + 40 + 30$

Explanation: $400 + 90 + 0 = 490$. They match! $300 + 60 = 360$ (not 306). $700 + 50 + 1 = 751$ (not 715). $200 + 40 + 30 = 270$ (not 243).

7. Nora wrote 1,052 in expanded form as $1,000 + 52$. Is there a better way?

- A. No, that is correct
- B. Yes: $1,000 + 50 + 2$ ✓**
- C. Yes: $1,000 + 500 + 2$
- D. Yes: $100 + 50 + 2$

Explanation: Expanded form should break apart every place value. $1,052 = 1,000 + 0 + 50 + 2$. The best answer is $1,000 + 50 + 2$.

8. How do you write one thousand two hundred in standard form?

- A. 1,200 ✓**
- B. 1,002
- C. 12,000
- D. 120

Explanation: One thousand two hundred = 1,200. One thousand = 1,000 plus two hundred = 200.

9. $\underline{\quad} + 40 + 5 = 945$. What goes in the blank?

- A. 9
- B. 90
- C. 900 ✓**
- D. 9,000

Explanation: $945 - 40 - 5 = 900$. The missing part is 900, which is the hundreds value.

10. Ella says $700 + 15$ is the same as 715. Is she correct?

- A. Yes, $700 + 15 = 715$ ✓**
- B. No, it equals 7,015
- C. No, it equals 850
- D. No, it equals 7,150

Explanation: $700 + 15 = 715$. Even though 15 is not a 'pure' tens or ones, you can still add it to 700.

11. Which is NOT a correct way to write 358?

- A. $300 + 50 + 8$
- B. Three hundred fifty-eight
- C. 35 tens and 8 ones
- D. $300 + 80 + 5$ ✓**

Explanation: $300 + 80 + 5 = 385$, not 358. The other three all equal 358. Check: 35 tens = 350, plus 8 = 358.

12. Write the number that is the same as nine hundred forty in expanded form.

- A. $900 + 40$ ✓**
- B. $900 + 4$
- C. $90 + 40$
- D. $9 + 4 + 0$

Explanation: Nine hundred forty = 940 = $900 + 40 + 0$. The best expanded form shown is $900 + 40$.

13. Kevin wrote 'five hundred sixty-three' as 500603. What did he do wrong?

A. He wrote each part separately instead of combining ✓

B. He put too many zeros

C. He switched the digits

D. Nothing, he is right

Explanation: Kevin wrote 500, then 60, then 3 side by side. He should add: $500 + 60 + 3 = 563$.

14. What is the expanded form of 1,006?

A. $1,000 + 6$

B. $1,000 + 60$

C. $100 + 6$

D. $1,000 + 0 + 0 + 6$ ✓

Explanation: $1,006 = 1,000 + 0 + 0 + 6$. The fullest expanded form shows every place. $1,000 + 6$ is also acceptable.

15. Sara says 260 in word form is 'two hundred and six.' What is the correct word form?

A. Two hundred six

B. Two hundred sixty ✓

C. Twenty-six

D. Two sixty

Explanation: $260 =$ two hundred sixty. Sara wrote 206 (two hundred six) instead. The 6 is in the tens place, making it sixty.

16. Which expanded form equals 809?

A. $800 + 90$

B. $800 + 0 + 9$ ✓

C. $80 + 9$

D. $800 + 9 + 0$

Explanation: $809 = 800 + 0 + 9$. The zero in the tens place matters -- it shows there are no tens.

17. Tom says $500 + 30 + 12$ is not in proper expanded form. Why?

A. Because 12 has two digits ✓

B. Because you cannot add three numbers

C. Because 500 is too big

D. Tom is wrong, it is fine

Explanation: In expanded form, each part should be a single place value. 12 covers both tens and ones. Proper form: $500 + 40 + 2 = 542$.

18. What number does this show: $1,000 + 100 + 20 + 5$?

A. 1,125 ✓

B. 11,025

C. 1,000,125

D. 10,125

Explanation: Add step by step: $1,000 + 100 = 1,100$, then $+ 20 = 1,120$, then $+ 5 = 1,125$.

19. Which shows the same number as eight hundred five?

A. 850

B. 805 ✓

C. $800 + 50$

D. 8,005

Explanation: Eight hundred five = 805. Be careful: eight hundred fifty = 850, which is different.

20. Fill in the blank: $147 = \underline{\hspace{1cm}} + 40 + 7$

A. 1

B. 10

C. 100 ✓

D. 140

Explanation: $147 = 100 + 40 + 7$. The hundreds value is 100.

Hard (20 questions)

1. Lily wrote $600 + 40 + 15$ in standard form as 6,415. What mistake did she make?

- A. She added wrong
- B. She put digits side by side instead of adding ✓**
- C. She forgot to carry
- D. She switched the tens and ones

Explanation: Lily placed 6, 4, and 15 side by side to get 6,415. She should have added: $600 + 40 + 15 = 655$.

2. Marcus wrote 1,072 three ways. Which one is WRONG?

- A. $1,000 + 70 + 2$
- B. One thousand seventy-two
- C. $1,000 + 700 + 2$ ✓**
- D. 10 hundreds + 7 tens + 2 ones

Explanation: $1,000 + 700 + 2 = 1,702$, not 1,072. Marcus confused 70 with 700. The other three all equal 1,072.

3. Ava wrote 'four hundred sixteen' as 4016. What should the correct number be?

- A. 4,016
- B. 416 ✓**
- C. 4,160
- D. 460

Explanation: Four hundred sixteen = 416. Ava split 'four hundred' as 400 and 'sixteen' as 16, then wrote them side by side: 4016. She should combine: $400 + 16 = 416$.

4. Which number can be written as 11 hundreds + 5 tens + 3 ones?

- A. 1,153 ✓**
- B. 1,503
- C. 11,053
- D. 153

Explanation: 11 hundreds = 1,100. Add $50 + 3 = 53$. Total: $1,100 + 53 = 1,153$. You must regroup: 11 hundreds = 1 thousand + 1 hundred.

5. Two students write 837 in expanded form. Kai writes $800 + 30 + 7$. Zoe writes $800 + 37$. Who is correct?

- A. Only Kai
- B. Only Zoe
- C. Both are correct ✓**
- D. Neither is correct

Explanation: $800 + 30 + 7 = 837$ and $800 + 37 = 837$. Both are valid ways to decompose the number, even though Kai's is the standard expanded form.

6. Write $500 + 130 + 4$ in standard form.

- A. 5,134
- B. 634 ✓**
- C. 504
- D. 513

Explanation: $500 + 130 + 4$. Since $130 = 100 + 30$, this is really $500 + 100 + 30 + 4 = 634$. The 130 carries an extra hundred.

7. Eli says the expanded form of 1,200 is $1,000 + 200$. His teacher says he can also write it as 12 hundreds. Are both right?

- A. Only Eli is right
- B. Only the teacher is right
- C. Both are right ✓**
- D. Neither is right

Explanation: $1,000 + 200 = 1,200$ and $12 \text{ hundreds} = 12 \times 100 = 1,200$. Both ways equal 1,200, so both are correct.

8. Dani writes the word form of 1,108 as 'one thousand one hundred eight.' Her friend writes 'one thousand one hundred and eight.' Who is correct?

- A. Only Dani
- B. Only her friend
- C. Both are correct ✓**
- D. Neither is correct

Explanation: Both ways are accepted. Some people include 'and' before the last part, some do not. Both say the same number: 1,108.

9. $900 + \underline{\quad} + 14 = 964$. What goes in the blank?

- A. 50 ✓
- B. 64
- C. 5
- D. 54

Explanation: $964 - 900 - 14 = 50$. Check: $900 + 50 + 14 = 964$. The blank is 50.

10. Mr. Chen asks: 'Write 708 in expanded form.' Three students answer. Who is WRONG? A) $700 + 8$ B) $700 + 0 + 8$ C) $70 + 8$

- A. A is wrong
- B. B is wrong
- C. C is wrong ✓
- D. They are all correct

Explanation: $70 + 8 = 78$, not 708. The student forgot the hundreds! $700 + 8 = 708$ and $700 + 0 + 8 = 708$, so A and B are both correct.

11. A number in expanded form is $1,000 + 100 + 50 + 12$. What is it in standard form?

- A. 1,1512
- B. 1,162 ✓
- C. 11,512
- D. 1,152

Explanation: Add step by step: $1,000 + 100 = 1,100$. Then $1,100 + 50 = 1,150$. Then $1,150 + 12 = 1,162$. The 12 ones regroup as 1 ten and 2 ones.

12. Rosa writes three hundred nine as 3009. Oscar writes it as 309. Who is correct and why?

- A. Rosa, because three hundred = 300 and nine = 09
- B. Oscar, because $300 + 9 = 309$ ✓
- C. Both are correct
- D. Neither is correct

Explanation: Three hundred nine = 309. Oscar is correct. Rosa put 300 and 9 side by side instead of adding: $300 + 9 = 309$, not 3009.

13. Which expanded form is WRONG for 555?

- A. $500 + 50 + 5$
- B. 5 hundreds + 5 tens + 5 ones
- C. 55 tens + 5 ones
- D. $500 + 55 + 5$ ✓

Explanation: $500 + 55 + 5 = 560$, not 555. The other three all equal 555. Check: 55 tens = 550, $+ 5 = 555$.

14. Tyler says he can write 476 as 4 hundreds + 7 tens + 6 ones OR as 3 hundreds + 17 tens + 6 ones. Is the second way correct?

- A. No, you cannot have 17 tens
- B. Yes, $300 + 170 + 6 = 476$ ✓
- C. No, that equals 376
- D. Yes, but only the first way is proper

Explanation: $300 + 170 + 6 = 476$. Yes! 17 tens = 170, and $300 + 170 + 6 = 476$. You can trade 1 hundred for 10 tens.

15. What is wrong with this: 'Two hundred thirteen = $200 + 30 + 1$ '?

- A. Nothing, it is correct
- B. The tens and ones are switched: should be $200 + 10 + 3$ ✓
- C. It should be 2,013
- D. Thirteen means 30, not 13

Explanation: Thirteen = 13, which is $10 + 3$. So $213 = 200 + 10 + 3$. The student wrote $30 + 1$ instead of $10 + 3$, mixing up the tens and ones of thirteen.

16. Write the number: 'The thousands digit is 1, the hundreds digit is the same as the ones digit, the tens digit is 0, and the ones digit is 3.'

A. 1,303 ✓

B. 1,003

C. 1,030

D. 1,330

Explanation: Thousands = 1, tens = 0, ones = 3, hundreds = same as ones = 3. The number is 1,303.

17. Mia says $200 + 60 + 18 = 278$. Is she right?

A. Yes ✓

B. No, it equals 268

C. No, it equals 2,618

D. No, it equals 878

Explanation: $200 + 60 + 18 = 278$. Check: $200 + 60 = 260$, then $260 + 18 = 278$. Mia is correct!

18. A number has these clues: In expanded form, the hundreds part is 900. The word form ends with 'eleven.' What is the number?

A. 911 ✓

B. 901

C. 9,011

D. 9,110

Explanation: Hundreds = 900, ends with eleven = 11. So the number is $900 + 11 = 911$. Eleven means 1 ten and 1 one.

19. Which is the correct expanded form of 1,050?

A. $1,000 + 50$

B. $1,000 + 500$

C. $100 + 50$

D. $1,000 + 0 + 50 + 0$ ✓

Explanation: $1,050 = 1,000 + 0 + 50 + 0$. The fullest expanded form shows every place. $1,000 + 50$ is also correct but less complete.

20. James says 'twelve hundred' and 'one thousand two hundred' are different numbers. Is he right?

A. Yes, twelve hundred = 1,200 and one thousand two hundred = 12,000

B. Yes, they sound different so they must be different

C. No, both equal 1,200 ✓

D. No, both equal 12,000

Explanation: Twelve hundred = $12 \times 100 = 1,200$. One thousand two hundred = $1,000 + 200 = 1,200$. They are the same number said two different ways!

Lesson 3: Bigger or Smaller?

Easy (20 questions)

1. Which is greater: 47 or 74?

A. 47

B. 74 ✓

C. They are equal

D. You cannot tell

Explanation: Compare the tens first. 74 has 7 tens and 47 has 4 tens. Since $7 > 4$, we know $74 > 47$.

2. Which symbol goes in the blank? 385 ____ 583

A. >

B. < ✓

C. =

D. +

Explanation: Compare hundreds first. 385 has 3 hundreds, 583 has 5 hundreds. $3 < 5$, so $385 < 583$.

3. Which number is the smallest: 216, 621, 162?

- A. 216
- B. 621
- C. 162 ✓
- D. They are all equal

Explanation: Compare hundreds: 162 has 1 hundred, 216 has 2, 621 has 6. 1 is the smallest, so 162 is the smallest number.

4. Is 500 greater than, less than, or equal to 499?

- A. Greater than ✓
- B. Less than
- C. Equal to
- D. You cannot tell

Explanation: 500 has 5 hundreds and 499 has 4 hundreds. Since $5 > 4$, we know $500 > 499$.

5. Which symbol goes in the blank? 250 ____ 250

- A. >
- B. <
- C. = ✓
- D. None

Explanation: Both numbers are exactly the same: $250 = 250$. Use the equal sign.

6. Which is greater: 892 or 829?

- A. 892 ✓
- B. 829
- C. They are equal
- D. You cannot tell

Explanation: Both have 8 hundreds. Compare tens: 892 has 9 tens, 829 has 2 tens. Since $9 > 2$, $892 > 829$.

7. Put in order from least to greatest: 100, 10, 1

- A. 100, 10, 1
- B. 1, 10, 100 ✓
- C. 10, 1, 100
- D. 1, 100, 10

Explanation: From least to greatest: 1 is the smallest, then 10, then 100. Answer: 1, 10, 100.

8. Which number is the greatest: 345, 354, 435?

- A. 345
- B. 354
- C. 435 ✓
- D. They are all equal

Explanation: Compare hundreds first. 435 has 4 hundreds, while 345 and 354 only have 3 hundreds. 435 is the greatest.

9. Which symbol goes in the blank? 607 ____ 670

- A. >
- B. < ✓
- C. =
- D. +

Explanation: Both have 6 hundreds. 607 has 0 tens, 670 has 7 tens. $0 < 7$, so $607 < 670$.

10. Is 1,000 greater than 999?

- A. Yes ✓
- B. No
- C. They are equal
- D. You cannot compare them

Explanation: 1,000 has 4 digits and 999 has 3 digits. A 4-digit number is always greater than a 3-digit number. $1,000 > 999$.

11. Which number is less: 418 or 481?

A. 418 ✓

B. 481

C. They are equal

D. You cannot tell

Explanation: Both have 4 hundreds. Compare tens: 418 has 1 ten, 481 has 8 tens. $1 < 8$, so $418 < 481$.

12. Sara has 312 stickers. Tom has 321 stickers. Who has more?

A. Sara

B. Tom ✓

C. They have the same

D. You cannot tell

Explanation: Both have 3 hundreds. Compare tens: 312 has 1 ten, 321 has 2 tens. $2 > 1$, so Tom has more.

13. Which symbol goes in the blank? 755 ____ 575

A. > ✓

B. <

C. =

D. +

Explanation: Compare hundreds: 755 has 7 hundreds, 575 has 5 hundreds. $7 > 5$, so $755 > 575$.

14. Which is the greatest number: 199, 200, 188?

A. 199

B. 200 ✓

C. 188

D. They are all equal

Explanation: 200 has 2 hundreds while 199 and 188 only have 1 hundred. 200 is the greatest.

15. Is 340 greater than or less than 304?

A. Greater than ✓

B. Less than

C. Equal to

D. You cannot tell

Explanation: Both have 3 hundreds. Compare tens: 340 has 4 tens, 304 has 0 tens. $4 > 0$, so $340 > 304$.

16. Which number comes just BEFORE 600?

A. 601

B. 599 ✓

C. 500

D. 610

Explanation: The number just before 600 is 599. Count: ...598, 599, 600.

17. Which number comes just AFTER 999?

A. 998

B. 1,000 ✓

C. 990

D. 9,910

Explanation: The number just after 999 is 1,000. This is where we go from a 3-digit number to a 4-digit number!

18. Leo picked a number between 450 and 460. Which could it be?

A. 449

B. 455 ✓

C. 461

D. 445

Explanation: Between 450 and 460 means numbers like 451, 452... 459. 455 fits!

19. Which is less: 1,100 or 1,010?

- A. 1,100
- B. 1,010 ✓**
- C. They are equal
- D. You cannot tell

Explanation: Both have 1 thousand. Compare hundreds: 1,100 has 1 hundred, 1,010 has 0 hundreds. $0 < 1$, so $1,010 < 1,100$.

20. Which symbol goes in the blank? 900 ____ 900

- A. >
- B. <
- C. = ✓**
- D. +

Explanation: Both numbers are 900. They are exactly equal, so we use =.

Medium (20 questions)

1. Which symbol goes between 358 ____ 385?

- A. >
- B. < ✓**
- C. =
- D. +

Explanation: Both have 3 hundreds. Compare tens: 358 has 5 tens, 385 has 8 tens. $5 < 8$, so $358 < 385$.

2. Put these in order from least to greatest: 512, 521, 215

- A. 512, 521, 215
- B. 215, 512, 521 ✓**
- C. 521, 512, 215
- D. 215, 521, 512

Explanation: Compare hundreds first: 215 has 2 hundreds (smallest), 512 and 521 both have 5. For those, compare tens: 512 has 1 ten, 521 has 2 tens. Order: 215, 512, 521.

3. Which number is closest to 800?

- A. 750
- B. 810 ✓**
- C. 780
- D. 850

Explanation: Find the distance from 800: 750 is 50 away, 810 is 10 away, 780 is 20 away, 850 is 50 away. 810 is closest (only 10 away).

4. Amy says $609 > 690$ because 6 hundreds is the same and $9 > 0$. What is wrong?

- A. Nothing, she is right
- B. She compared ones before tens
- C. She should compare tens first: $0 < 9$ ✓**
- D. The hundreds are different

Explanation: After hundreds match, compare tens next. 609 has 0 tens, 690 has 9 tens. Since $0 < 9$, $609 < 690$. Amy skipped tens and jumped to ones.

5. Which number is between 475 and 500?

- A. 470
- B. 505
- C. 483 ✓**
- D. 475

Explanation: Between 475 and 500 means greater than 475 and less than 500. 483 fits. 470 is too small, 505 is too big.

6. Write <, >, or = for: 1,100 ____ 1,010

- A. > ✓**
- B. <
- C. =
- D. Cannot compare

Explanation: Both have 1 thousand. Compare hundreds: 1,100 has 1 hundred, 1,010 has 0 hundreds. $1 > 0$, so $1,100 > 1,010$.

7. Jake picked a number. It is greater than 350 but less than 360. The ones digit is 7. What is his number?

- A. 347
- B. 357 ✓**
- C. 367
- D. 307

Explanation: Between 350 and 360 with ones digit 7 means the number is 357.

8. Which list is in order from greatest to least?

- A. 410, 401, 140 ✓**
- B. 140, 401, 410
- C. 401, 410, 140
- D. 140, 410, 401

Explanation: Greatest to least means biggest first. $410 > 401 > 140$. Compare hundreds: 410 and 401 both have 4, but 410 has more tens.

9. Which number makes this true? $628 < \underline{\hspace{1cm}}$

- A. 618
- B. 620
- C. 627
- D. 631 ✓**

Explanation: We need a number greater than 628. Only $631 > 628$. The others (618, 620, 627) are all less than 628.

10. Rosa compared 845 and 854. She said they are equal because they use the same digits. Is she right?

- A. Yes, same digits means same number
- B. No, the digits are in different places so the values differ ✓**
- C. Yes, because both have 8 hundreds
- D. No, because 845 has more hundreds

Explanation: Even though both use 8, 4, and 5, the place of each digit matters. In 845, the 4 is in tens (40). In 854, the 5 is in tens (50). $845 < 854$.

11. Which is greater: 1,200 or 999?

- A. 999, because $9 > 1$
- B. 1,200, because it has more digits ✓**
- C. They are equal
- D. 999, because all digits are bigger

Explanation: 1,200 is a 4-digit number and 999 is a 3-digit number. A number with more digits is always greater (for whole numbers).

12. Sam has 467 baseball cards. He wants to have more than Mia, who has 476. How many more does he need?

- A. At least 9 more
- B. At least 10 more ✓**
- C. At least 1 more
- D. He already has more

Explanation: Sam has 467, Mia has 476. Difference: $476 - 467 = 9$. Sam needs MORE than Mia, so he needs at least 10 more to have 477.

13. Put these numbers on a number line. Which is farthest right? 305, 350, 503

- A. 305
- B. 350
- C. 503 ✓**
- D. They are all the same distance

Explanation: On a number line, greater numbers are farther right. $503 > 350 > 305$, so 503 is farthest right.

14. Which comparison is TRUE?

- A. $312 > 321$
- B. $489 < 498$ ✓**
- C. $756 = 765$
- D. $200 > 1,000$

Explanation: Check each: $312 < 321$ (not $>$). $489 < 498$ is true because same hundreds, same tens, and $8 < 9$. 756 does not equal 765. $200 < 1,000$ (not $>$).

15. What is the greatest 3-digit number you can make with the digits 3, 7, and 1?

- A. 137
- B. 371
- C. 713
- D. 731 ✓**

Explanation: To make the greatest number, put the biggest digit in the hundreds place. $7 > 3 > 1$, so: 731.

16. Which number is 1 less than 800?

- A. 700
- B. 801
- C. 799 ✓**
- D. 810

Explanation: 1 less than 800 = 799. When you go back 1 from 800, you get 799.

17. Eli has these numbers on cards: 490, 409, 940. Put them in order from least to greatest.

- A. 409, 490, 940 ✓**
- B. 490, 409, 940
- C. 940, 490, 409
- D. 409, 940, 490

Explanation: Compare hundreds: 409 has 4, 490 has 4, 940 has 9. Between 409 and 490, compare tens: $0 < 9$. So: $409 < 490 < 940$.

18. Which statement is correct?

- A. $450 > 540$
- B. $199 > 200$
- C. $1,050 > 1,005$ ✓**
- D. $387 = 378$

Explanation: Check: $450 < 540$ (no). $199 < 200$ (no). 1,050 vs 1,005: same thousands, 0 vs 0 hundreds, 5 vs 0 tens. $5 > 0$, so $1,050 > 1,005$ (yes!).

19. Which number could go in the blank to make this true? ____ > 567

- A. 560
- B. 557
- C. 566
- D. 570 ✓**

Explanation: We need a number greater than 567. Only $570 > 567$. The others are all less than or equal to 567.

20. Ava says 100 is greater than 99 because it has more digits. Is her reasoning correct?

- A. Yes, more digits always means greater ✓**
- B. No, you must compare each digit
- C. Yes, but only for these numbers
- D. No, 99 is greater

Explanation: For whole numbers, a number with more digits is always greater. 100 (3 digits) > 99 (2 digits). Ava's reasoning is correct!

Hard (20 questions)

1. Put these in order from least to greatest: 899, 908, 890, 981

- A. 899, 890, 908, 981
- B. 890, 899, 908, 981 ✓**
- C. 908, 899, 890, 981
- D. 981, 908, 899, 890

Explanation: Compare hundreds: all have 8 or 9 hundreds. 890 (8h, 9t) < 899 (8h, 9t, 9o) < 908 (9h) < 981 (9h, 8t). Order: 890, 899, 908, 981.

2. Sam says $452 > 459$ because $5 > 4$. Is Sam right? What did he do wrong?

- A. He is right
- B. He compared hundreds instead of ones
- C. He compared random digits instead of going place by place
- D. He compared tens when he should compare ones ✓**

Explanation: Both numbers have 4 hundreds and 5 tens. The difference is in the ones: 2 vs 9 . Since $2 < 9$, $452 < 459$. Sam should have compared ones since hundreds and tens matched.

3. Nina made the greatest 3-digit number and the least 3-digit number using digits 5, 0, and 8 (each once). What is the difference?

- A. 342 ✓
- B. 750
- C. 392
- D. 508

Explanation: Greatest: put biggest digit in hundreds: 850. Least 3-digit: cannot start with 0, so put 5 in hundreds: 508. Difference: $850 - 508 = 342$.

4. Which shows the numbers in order from greatest to least?

- A. $1,102 > 1,120 > 1,012$
- B. $1,120 > 1,102 > 1,012$ ✓
- C. $1,012 > 1,102 > 1,120$
- D. $1,120 > 1,012 > 1,102$

Explanation: All have 1 thousand and 1 or 0 hundreds. 1,120 has 1 hundred and 2 tens. 1,102 has 1 hundred and 0 tens. 1,012 has 0 hundreds. So: $1,120 > 1,102 > 1,012$.

5. Kai says 399 is greater than 400 because $3 + 9 + 9 = 21$ and $4 + 0 + 0 = 4$. What is wrong with his thinking?

- A. Nothing, he is right
- B. You compare place by place, not add digits ✓
- C. You should multiply digits, not add
- D. The digits are too big to add

Explanation: Adding digits does not tell you which number is greater. Compare place by place: 399 has 3 hundreds, 400 has 4 hundreds. Since $3 < 4$, $399 < 400$.

6. Using the digits 6, 2, and 9 (each used once), what is the LEAST 3-digit number you can make?

- A. 269 ✓
- B. 296
- C. 629
- D. 926

Explanation: For the least number, put the smallest digit in hundreds: 2. Then next smallest in tens: 6. Then 9 in ones. Least = 269.

7. Three friends measured their jumps. Ava: 108 cm, Ben: 180 cm, Cal: 118 cm. Put them in order from shortest to longest jump.

- A. Ava, Cal, Ben ✓
- B. Ben, Cal, Ava
- C. Cal, Ava, Ben
- D. Ava, Ben, Cal

Explanation: $108 < 118 < 180$. Shortest to longest: Ava (108), Cal (118), Ben (180).

8. Mia says that since $59 > 51$, then $598 > 519$ is also true. Is her reasoning correct?

- A. Yes, comparing the first two digits works ✓
- B. No, you must compare all three digits starting from hundreds
- C. No, $598 < 519$
- D. Yes, but only by luck

Explanation: Both have 5 hundreds, so compare tens: $9 > 1$, so $598 > 519$. Mia got the right answer. Comparing the first two digits can work when hundreds match, but the proper method is comparing place by place from left to right.

9. What is the smallest number that is greater than 1,199?

- A. 1,200 ✓
- B. 1,100
- C. 1,198
- D. 1,201

Explanation: The very next number after 1,199 is 1,200. So 1,200 is the smallest number greater than 1,199.

10. Tyler has cards showing 4, 7, and 0. He wants to make a 3-digit number between 400 and 500. Which numbers can he make?

- A. Only 407
- B. Only 470
- C. 407 and 470 ✓**
- D. 400 and 470

Explanation: Between 400 and 500 means hundreds digit = 4. The remaining digits are 7 and 0. He can make 407 or 470. Both are between 400 and 500.

11. Put these in order from least to greatest: 1,200, 1,020, 1,002, 1,100

- A. 1,002, 1,020, 1,100, 1,200 ✓**
- B. 1,200, 1,100, 1,020, 1,002
- C. 1,020, 1,002, 1,100, 1,200
- D. 1,002, 1,100, 1,020, 1,200

Explanation: All have 1 thousand. Compare hundreds: 1,002 (0h), 1,020 (0h), 1,100 (1h), 1,200 (2h). For the two with 0 hundreds, compare tens: 1,002 (0t) < 1,020 (2t). Order: 1,002, 1,020, 1,100, 1,200.

12. Eva says $789 < 798 < 879 < 897$. Is she correct?

- A. Yes, all comparisons are right ✓**
- B. No, $798 > 879$
- C. No, $789 > 798$
- D. No, $879 > 897$

Explanation: Check each pair: $789 < 798$ (same hundreds, $8 < 9$ in tens). $798 < 879$ (7 hundreds < 8 hundreds). $879 < 897$ (same hundreds, $7 < 9$ in tens). All correct!

13. A number is greater than 550 but less than 560. Its ones digit is odd. Its digits add up to 13. What is it?

- A. 551
- B. 553 ✓**
- C. 555
- D. 557

Explanation: Between 550 and 560 with odd ones: 551, 553, 555, 557, 559. Check digit sums: $5+5+1=11$, $5+5+3=13$, $5+5+5=15$, $5+5+7=17$, $5+5+9=19$. Only 553 has digits that add to 13.

14. Leo and Zoe each made a 3-digit number. Leo's number has 8 hundreds and is less than 830. Zoe's has 8 hundreds and is greater than 870. Whose number is greater?

- A. Leo's
- B. Zoe's ✓**
- C. They are equal
- D. You cannot tell

Explanation: Leo's number is less than 830, so it could be 800-829. Zoe's is greater than 870, so it could be 871-899. Any number Zoe could have is greater than any number Leo could have.

15. Which set of numbers is in order from GREATEST to LEAST?

- A. 345, 435, 543
- B. 543, 435, 345 ✓**
- C. 435, 345, 543
- D. 345, 543, 435

Explanation: Greatest to least means biggest first: $543 > 435 > 345$. Compare hundreds: $5 > 4 > 3$.

16. Dana says you only need to look at the hundreds digit to compare 738 and 742. Is she right?

- A. Yes, hundreds tells you everything
- B. No, both have 7 hundreds so you must check tens too ✓**
- C. No, you always check ones first
- D. Yes, because $7 = 7$ means they are equal

Explanation: Both numbers have 7 hundreds, so hundreds alone cannot tell us. We must compare tens: 738 has 3 tens, 742 has 4 tens. Since $3 < 4$, $738 < 742$.

17. What is the greatest 3-digit number where no digit repeats and the hundreds digit is even?

- A. 987
- B. 897 ✓**
- C. 896
- D. 986

Explanation: Hundreds must be even, so the biggest even digit for hundreds is 8. Then tens = 9 (biggest remaining), ones = 7 (next biggest). Answer: 897.

18. Nora wrote: $401 < 410 < 400$. Find the error.

- A. 401 should be greater than 410
- B. 400 should come first since it is least
- C. $400 < 401 < 410$ is the correct order ✓**
- D. There is no error

Explanation: $400 < 401 < 410$. Nora put 400 at the end (as if it were greatest), but 400 is the least of the three.

19. I am between 990 and 1,010. I am an even number. My digits add up to 9. What number am I?

- A. 990
- B. 1,008 ✓**
- C. 1,000
- D. 1,002

Explanation: Even numbers between 990 and 1,010: 992, 994, 996, 998, 1,000, 1,002, 1,004, 1,006, 1,008. Digit sums: $992=20$, $994=22$, $996=24$, $998=26$, $1,000=1$, $1,002=3$, $1,004=5$, $1,006=7$, $1,008=9$. Only 1,008 has digits adding to 9.

20. Three kids ran a race. Times: Ava 1,050 seconds, Ben 1,005 seconds, Cho 1,500 seconds. Who was fastest (least time)?

- A. Ava
- B. Ben ✓**
- C. Cho
- D. They tied

Explanation: Fastest means least time. $1,005 < 1,050 < 1,500$. Ben had the least time (1,005 seconds), so he was fastest.

Lesson 4: Skip Counting

Easy (20 questions)

1. Count by 5s: 10, 15, 20, __

- A. 21
- B. 22
- C. 25 ✓**
- D. 30

Explanation: When you count by 5s, add 5 each time. $20 + 5 = 25$.

2. Count by 10s: 30, 40, 50, __

- A. 51
- B. 55
- C. 60 ✓**
- D. 100

Explanation: When you count by 10s, add 10 each time. $50 + 10 = 60$.

3. What is 10 more than 237?

- A. 238
- B. 337
- C. 247 ✓**
- D. 227

Explanation: 10 more means add 10. $237 + 10 = 247$. Only the tens digit changes: 3 becomes 4.

4. What is 100 more than 450?

- A. 460
- B. 550 ✓**
- C. 451
- D. 350

Explanation: 100 more means add 100. $450 + 100 = 550$. Only the hundreds digit changes: 4 becomes 5.

5. Count by 2s: 16, 18, 20, ____

- A. 21
- B. 22 ✓**
- C. 24
- D. 30

Explanation: When you count by 2s, add 2 each time. $20 + 2 = 22$.

6. What is 10 less than 380?

- A. 280
- B. 370 ✓**
- C. 381
- D. 379

Explanation: 10 less means subtract 10. $380 - 10 = 370$. Only the tens digit changes: 8 becomes 7.

7. What is 100 less than 600?

- A. 500 ✓**
- B. 590
- C. 601
- D. 700

Explanation: 100 less means subtract 100. $600 - 100 = 500$.

8. Count by 100s: 200, 300, 400, ____

- A. 410
- B. 450
- C. 500 ✓**
- D. 900

Explanation: When you count by 100s, add 100 each time. $400 + 100 = 500$.

9. What number is 10 more than 590?

- A. 591
- B. 600 ✓**
- C. 690
- D. 500

Explanation: $590 + 10 = 600$. When the tens digit is 9, adding 10 makes a new hundred!

10. Count by 5s: 45, 50, 55, ____

- A. 56
- B. 60 ✓**
- C. 65
- D. 70

Explanation: Add 5 each time: $55 + 5 = 60$.

11. What is 100 more than 900?

- A. 910
- B. 990
- C. 1,000 ✓**
- D. 800

Explanation: $900 + 100 = 1,000$. When you add 100 to 900, you get one thousand!

12. Count by 10s: 270, 280, 290, ____

- A. 291
- B. 295
- C. 300 ✓**
- D. 390

Explanation: Add 10 each time: $290 + 10 = 300$.

13. Liam counts by 2s starting at 4. What are the next 3 numbers?

- A. 5, 6, 7
- B. 6, 8, 10 ✓**
- C. 6, 7, 8
- D. 8, 12, 16

Explanation: Count by 2s: 4, 6, 8, 10. Add 2 each time.

14. What is 10 less than 500?

- A. 400
- B. 490 ✓**
- C. 499
- D. 510

Explanation: $500 - 10 = 490$. When you subtract 10 from a number ending in 00, the tens become 9 and the hundreds go down by 1.

15. Count by 100s: 700, 800, 900, ____

- A. 910
- B. 950
- C. 990
- D. 1,000 ✓**

Explanation: Add 100 each time: $900 + 100 = 1,000$.

16. What is 100 less than 350?

- A. 250 ✓**
- B. 340
- C. 349
- D. 450

Explanation: 100 less means subtract 100. $350 - 100 = 250$. Only the hundreds digit changes.

17. Count by 5s: 75, 80, 85, ____

- A. 86
- B. 90 ✓**
- C. 95
- D. 100

Explanation: Add 5 each time: $85 + 5 = 90$.

18. Maya has 163 blocks. She gets 10 more. How many does she have now?

- A. 164
- B. 173 ✓**
- C. 263
- D. 153

Explanation: $163 + 10 = 173$. Adding 10 changes the tens digit from 6 to 7.

19. What number is 10 more than 195?

- A. 196
- B. 200
- C. 205 ✓**
- D. 295

Explanation: $195 + 10 = 205$. The tens and hundreds both change because 9 tens + 1 ten = 10 tens = 1 hundred.

20. Count by 2s: 48, 50, 52, ____

- A. 53
- B. 54 ✓**
- C. 56
- D. 62

Explanation: Add 2 each time: $52 + 2 = 54$.

Medium (20 questions)

1. What is 100 more than 567?

- A. 577
- B. 667 ✓**
- C. 568
- D. 467

Explanation: 100 more means add 100. $567 + 100 = 667$. Only the hundreds digit changes: 5 becomes 6.

2. Jada counts by 5s starting at 280. What are the next 3 numbers?

- A. 285, 290, 295 ✓**
- B. 285, 295, 300
- C. 290, 300, 310
- D. 281, 282, 283

Explanation: Count by 5s from 280: 285, 290, 295. Add 5 each time.

3. What number is 10 less than 403?

- A. 303
- B. 402
- C. 393 ✓**
- D. 413

Explanation: $403 - 10 = 393$. When you subtract 10, the tens digit decreases. Since 0 tens - 1 ten requires regrouping, 403 becomes 393.

4. Skip count by 10s: 970, 980, 990, _____. What comes next?

- A. 991
- B. 999
- C. 1,000 ✓**
- D. 1,100

Explanation: $990 + 10 = 1,000$. Skip counting by 10s can cross from 3-digit to 4-digit numbers.

5. Leo started at 450 and counted by 100s. Which number did he NOT say?

- A. 550
- B. 650
- C. 700 ✓**
- D. 750

Explanation: Counting by 100s from 450: 550, 650, 750, 850... He skips 700 because he goes from 650 to 750.

6. What is 10 more than 996?

- A. 997
- B. 1,006 ✓**
- C. 1,096
- D. 906

Explanation: $996 + 10 = 1,006$. Adding 10 pushes us past 1,000!

7. Mia has 835 stickers. She gives away 100. How many does she have now?

- A. 735 ✓**
- B. 825
- C. 935
- D. 834

Explanation: $835 - 100 = 735$. Giving away 100 means subtracting 100. Only the hundreds digit changes.

8. Count by 2s from 394. What are the next 3 numbers?

- A. 395, 396, 397
- B. 396, 398, 400 ✓**
- C. 394, 396, 398
- D. 404, 414, 424

Explanation: Count by 2s from 394: 396, 398, 400. Notice we cross into 400!

9. What number is 100 less than 1,050?

- A. 1,040
- B. 950 ✓**
- C. 1,150
- D. 50

Explanation: $1,050 - 100 = 950$. Subtracting 100 from 1,050 drops below 1,000.

10. Eli skip counts: 625, 650, 675, _____. What is he counting by?

- A. 2s
- B. 5s
- C. 10s
- D. 25s ✓**

Explanation: $650 - 625 = 25$. $675 - 650 = 25$. Eli is counting by 25s. The next number would be 700.

11. Which is the same as adding 100? Starting number: 347

- A. Change ones digit: 348
- B. Change tens digit: 357
- C. Change hundreds digit: 447 ✓**
- D. Change all digits: 448

Explanation: Adding 100 only changes the hundreds digit. $347 + 100 = 447$. The 3 in hundreds becomes 4.

12. Sara starts at 100 and counts by 10s. Will she say 155?

- A. Yes
- B. No, she will skip it
- C. Yes, after 150
- D. No, 155 is not a multiple of 10 ✓**

Explanation: Counting by 10s from 100: 110, 120, 130, 140, 150, 160... She will never say 155 because it is not a multiple of 10.

13. What is 10 more than 1,090?

- A. 1,091
- B. 1,100 ✓**
- C. 1,190
- D. 2,090

Explanation: $1,090 + 10 = 1,100$. The 9 tens become 10 tens, which regroup to 1 hundred.

14. Tom counts backward by 5s: 120, 115, 110, _____. What comes next?

- A. 100
- B. 105 ✓**
- C. 109
- D. 95

Explanation: Counting backward by 5s means subtract 5 each time: $110 - 5 = 105$.

15. Ava says 10 less than 200 is 100. Is she right?

- A. Yes
- B. No, it is 190 ✓**
- C. No, it is 199
- D. No, it is 210

Explanation: $200 - 10 = 190$, not 100. Ava subtracted 100 instead of 10. 10 less only changes the tens place.

16. Fill in the pattern: 480, ____, 500, 510, 520

- A. 481
- B. 485
- C. 490 ✓
- D. 495

Explanation: The pattern counts by 10s. $480 + 10 = 490$, $490 + 10 = 500$. The missing number is 490.

17. What is 100 more than 908?

- A. 918
- B. 998
- C. 1,008 ✓
- D. 808

Explanation: $908 + 100 = 1,008$. Adding 100 to 908 crosses 1,000.

18. Kai starts at 72 and counts by 10s. Which number will he say?

- A. 80
- B. 100
- C. 132 ✓
- D. 170

Explanation: Counting by 10s from 72: 82, 92, 102, 112, 122, 132. The ones digit stays 2 the whole time. Only 132 ends in 2.

19. Rosa had 1,200 points. She lost 100 points twice. How many does she have now?

- A. 1,100
- B. 1,000 ✓
- C. 900
- D. 1,199

Explanation: Lost 100 twice means subtract 200 total. $1,200 - 100 = 1,100$, then $1,100 - 100 = 1,000$.

20. Which pattern shows counting by 100s?

- A. 150, 160, 170, 180
- B. 150, 250, 350, 450 ✓
- C. 150, 200, 250, 300
- D. 150, 155, 160, 165

Explanation: 150, 250, 350, 450: each number is 100 more than the last. That is counting by 100s.

Hard (20 questions)

1. Amy counts by 5s starting at 280. She says 280, 285, 290, 300. What mistake did she make?

- A. She skipped 295 ✓
- B. She started at the wrong number
- C. She counted by 10s at the end
- D. There is no mistake

Explanation: Counting by 5s from 290 should give 295, then 300. Amy jumped from 290 to 300, skipping 295.

2. ____ is 100 more than 867 and 10 less than 977. What is ____?

- A. 877
- B. 957
- C. 967 ✓
- D. 987

Explanation: 100 more than 867: $867 + 100 = 967$. Check: 10 less than 977: $977 - 10 = 967$. Both clues give 967!

3. Kai counts by 10s from 8. Which number will he NOT land on?

- A. 18
- B. 48
- C. 50 ✓
- D. 98

Explanation: From 8, counting by 10s: 18, 28, 38, 48, 58, 68, 78, 88, 98... The ones digit always stays 8. He will never land on 50.

4. Dani starts at 1,200 and counts backward by 100s. Which number will she say third?

- A. 1,100
- B. 1,000
- C. 900 ✓**
- D. 800

Explanation: First: 1,100. Second: 1,000. Third: 900. Count backward by 100s three times from 1,200.

5. Zoe says: '10 less than 305 is 295.' Her teacher says: '10 less than 305 is 304.' Who is right?

- A. Zoe ✓**
- B. The teacher
- C. Neither
- D. Both

Explanation: $305 - 10 = 295$. Zoe is correct! The teacher subtracted 1 instead of 10. When you take away 10 from 305, the tens go from 0 to 9 and hundreds drop: 295.

6. Sam starts at 250 and adds 100 four times. Then he subtracts 10 three times. Where does he end up?

- A. 620 ✓**
- B. 650
- C. 680
- D. 350

Explanation: Start: 250. Add 100 four times: 350, 450, 550, 650. Subtract 10 three times: 640, 630, 620. End: 620.

7. Eva counts by 5s from 0. Max counts by 2s from 0. What is the first number greater than 10 that BOTH will say?

- A. 12
- B. 15
- C. 20 ✓**
- D. 10

Explanation: By 5s: 5, 10, 15, 20, 25... By 2s: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20... Both say 10 and 20. The first greater than 10 is 20.

8. Mia writes: 10 more than 498 is 508. Is she correct?

- A. Yes ✓**
- B. No, it is 499
- C. No, it is 598
- D. No, it is 408

Explanation: $498 + 10 = 508$. Yes! The 9 tens become 10 tens, which is 1 hundred. So hundreds go from 4 to 5, tens go to 0. Correct!

9. Leo counts by 100s: 150, 250, 350, ____, 550. What is the missing number?

- A. 400
- B. 450 ✓**
- C. 500
- D. 355

Explanation: The pattern adds 100 each time. $350 + 100 = 450$. Check: $450 + 100 = 550$. The missing number is 450.

10. Nora says that 10 more than 990 is 1,000, and 10 more than 1,000 is 1,100. Find her error.

- A. Both are wrong
- B. $990 + 10$ is wrong
- C. $1,000 + 10$ should be 1,010, not 1,100 ✓**
- D. There is no error

Explanation: $990 + 10 = 1,000$ is correct. But $1,000 + 10 = 1,010$, not 1,100. Nora added 100 instead of 10 the second time.

11. Fill in BOTH blanks: 10 more than ____ is 600. 100 less than ____ is 600.

- A. 590 and 700 ✓**
- B. 610 and 500
- C. 590 and 500
- D. 610 and 700

Explanation: 10 more than ? = 600, so ? = 590. 100 less than ? = 600, so ? = 700. Answers: 590 and 700.

12. Tyler counts by 2s from 1. Will he ever say 48?

- A. Yes, because 48 is even
- B. No, because he starts on an odd number ✓**
- C. Yes, after 47
- D. No, because 48 is too big

Explanation: Starting at 1 and counting by 2s: 1, 3, 5, 7, 9... He only lands on odd numbers. 48 is even, so he will never say it.

13. Ava has 780. She adds 10 three times and then adds 100 once. What does she have now?

- A. 810
- B. 880
- C. 910 ✓**
- D. 890

Explanation: $780 + 10 + 10 + 10 = 810$. Then $810 + 100 = 910$.

14. Pedro says 100 less than 1,005 is 905. Ben says it is 995. Who is right?

- A. Pedro ✓**
- B. Ben
- C. Both
- D. Neither

Explanation: $1,005 - 100 = 905$. Pedro is right. Ben subtracted 10 instead of 100.

15. Look at the pattern: 1,200, 1,100, 1,000, 900, _____. What rule does this follow, and what comes next?

- A. Subtract 10; next is 890
- B. Subtract 100; next is 800 ✓**
- C. Subtract 100; next is 890
- D. Subtract 200; next is 700

Explanation: Each number is 100 less than the one before it. $900 - 100 = 800$. The rule is subtract 100.

16. Rosa starts at 50. She counts by 10s to 100, then counts by 100s from there. Which number will she NOT say?

- A. 70
- B. 100
- C. 200
- D. 150 ✓**

Explanation: By 10s from 50: 60, 70, 80, 90, 100. Then by 100s from 100: 200, 300, 400... She says 70, 100, and 200 but she jumps from 100 to 200, so she never says 150.

17. What number is 10 more than 10 more than 10 more than 770?

- A. 780
- B. 790
- C. 800 ✓**
- D. 870

Explanation: Add 10 three times: $770 + 10 = 780$, $780 + 10 = 790$, $790 + 10 = 800$. That is the same as adding 30.

18. Jada says skip counting by 5s will land on every number that ends in 0. Is she right?

- A. Yes, always
- B. No, it also lands on numbers ending in 5 ✓**
- C. No, it only lands on even numbers
- D. No, it skips some numbers ending in 0

Explanation: Counting by 5s from 0: 5, 10, 15, 20, 25, 30... It hits every number ending in 0 AND every number ending in 5. Jada is partially right but forgot about the 5s.

19. Omar starts at 340. He subtracts 10 five times. Then adds 100 twice. Where is he?

- A. 490 ✓**
- B. 440
- C. 390
- D. 540

Explanation: $340 - 10 - 10 - 10 - 10 - 10 = 340 - 50 = 290$. Then $290 + 100 + 100 = 490$.

20. Which strategy finds 10 less than 800 fastest?

- A. Count backward by 1s ten times
- B. Change the tens digit from 0 to 9 and hundreds from 8 to 7
- C. Subtract 100 and add 90

D. All give the same answer but B is fastest ✓

Explanation: All three methods give 790. But changing digits directly (B) is fastest: tens 0 becomes 9, hundreds 8 becomes 7 = 790. Strategy B uses place value knowledge.

Unit Test Bank

Easy (10 questions)

1. What digit is in the hundreds place of 847?

- A. 4
- B. 7
- C. 8 ✓**
- D. 84

Explanation: The hundreds place is the first digit on the left. In 847, that digit is 8.

2. How do you write 362 in expanded form?

- A. $300 + 60 + 2$ ✓**
- B. $3 + 6 + 2$
- C. $360 + 2$
- D. $36 + 2$

Explanation: $362 = 300 + 60 + 2$. Each digit is shown as its full value.

3. Which is greater: 519 or 591?

- A. 519
- B. 591 ✓**
- C. They are equal
- D. You cannot tell

Explanation: Both have 5 hundreds. Compare tens: 519 has 1 ten, 591 has 9 tens. Since $1 < 9$, $519 < 591$.

4. Count by 10s: 460, 470, 480, ____

- A. 481
- B. 485
- C. 490 ✓**
- D. 500

Explanation: Add 10 each time: $480 + 10 = 490$.

5. What is the standard form of five hundred nine?

- A. 590
- B. 509 ✓**
- C. 59
- D. 5,009

Explanation: Five hundred nine = 509. There are 0 tens, so a 0 goes in the tens place.

6. What is 100 more than 723?

- A. 733
- B. 823 ✓**
- C. 724
- D. 623

Explanation: $723 + 100 = 823$. Only the hundreds digit changes: 7 becomes 8.

7. How many tens are in 60?

- A. 0
- B. 6 ✓**
- C. 60
- D. 600

Explanation: 60 has 6 tens. Each ten is worth 10, and $6 \times 10 = 60$.

8. Which symbol goes in the blank? 300 ____ 299

- A. > ✓**
- B. <
- C. =
- D. +

Explanation: 300 has 3 hundreds and 299 has 2 hundreds. $3 > 2$, so $300 > 299$.

9. Count by 5s: 85, 90, 95, ____

- A. 96
- B. 100 ✓**
- C. 105
- D. 110

Explanation: Add 5 each time: $95 + 5 = 100$.

10. Ben has 4 hundreds blocks and 6 ones blocks. What number did he make?

- A. 46
- B. 406 ✓**
- C. 460
- D. 604

Explanation: 4 hundreds = 400, 0 tens = 0, 6 ones = 6. The number is 406.

Medium (10 questions)

1. In the number 637, what is the value of the digit 3?

- A. 3
- B. 30 ✓**
- C. 300
- D. 37

Explanation: The 3 is in the tens place. Its value is 30, not just 3. Value = digit $\times$ place.

2. Which shows the same number as $400 + 90 + 6$?

- A. 946
- B. 496 ✓**
- C. 469
- D. 4,906

Explanation: $400 + 90 + 6 = 496$. Add hundreds + tens + ones to get the standard form.

3. Put these in order from least to greatest: 703, 730, 307

- A. 307, 703, 730 ✓**
- B. 703, 730, 307
- C. 730, 703, 307
- D. 307, 730, 703

Explanation: Compare hundreds: 307 has 3 (smallest), 703 has 7, 730 has 7. For 703 vs 730: compare tens, $0 < 3$. So: 307, 703, 730.

4. What is 10 less than 602?

- A. 502
- B. 601
- C. 592 ✓**
- D. 612

Explanation: $602 - 10 = 592$. The tens go from 0 to 9 and the hundreds drop by 1 (regroup).

5. $500 + \underline{\quad} + 8 = 578$. What goes in the blank?

- A. 7
- B. 70 ✓**
- C. 78
- D. 700

Explanation: $578 = 500 + 70 + 8$. The missing part is 70, the tens value.

6. Nora says eight hundred twelve is 8012. What should it be?

- A. 812 ✓**
- B. 8,012
- C. 8,120
- D. 821

Explanation: Eight hundred twelve = 812. Nora wrote 800 and 12 side by side (8012) instead of adding: $800 + 12 = 812$.

7. Which number could go in the blank? $\underline{\quad} > 465$

- A. 460
- B. 456
- C. 465
- D. 470 ✓**

Explanation: We need a number greater than 465. Only $470 > 465$. The others are less than or equal to 465.

8. What is 100 more than 950?

- A. 960
- B. 1,050 ✓**
- C. 850
- D. 951

Explanation: $950 + 100 = 1,050$. Adding 100 to 950 crosses 1,000!

9. Count by 5s from 375. What are the next 3 numbers?

- A. 380, 385, 390 ✓**
- B. 376, 377, 378
- C. 385, 395, 405
- D. 380, 390, 400

Explanation: Count by 5s: 375, 380, 385, 390. Add 5 each time.

10. In the number 555, which 5 has the greatest value?

- A. They are all the same value
- B. The first 5 (hundreds) ✓**
- C. The middle 5 (tens)
- D. The last 5 (ones)

Explanation: First 5 = 500, middle 5 = 50, last 5 = 5. Even though the digits are the same, place determines value. 500 is greatest.

Hard (10 questions)

1. What number has 7 hundreds, 0 tens, and 14 ones?

- A. 7,014
- B. 704
- C. 714 ✓**
- D. 7,140

Explanation: 7 hundreds = 700, 0 tens = 0, 14 ones = 14. Total = 714. The 14 ones regroup as 1 ten and 4 ones: 7 hundreds, 1 ten, 4 ones = 714.

2. Maya wrote $300 + 50 + 18 = 3,518$. What should the answer be?

- A. 368 ✓**
- B. 3,518
- C. 358
- D. 850

Explanation: Add: $300 + 50 = 350$, then $350 + 18 = 368$. Maya put digits side by side (3-5-18) instead of adding.

3. Put these in order from least to greatest: 1,010, 1,100, 1,001, 1,011

A. **1,001, 1,010, 1,011, 1,100** ✓

B. 1,100, 1,011, 1,010, 1,001

C. 1,001, 1,011, 1,010, 1,100

D. 1,010, 1,001, 1,011, 1,100

Explanation: All have 1 thousand. Compare hundreds: 1,001 and 1,010 and 1,011 have 0 hundreds; 1,100 has 1 hundred (greatest).

Among 0-hundred numbers, compare tens: 1,001 has 0 tens, 1,010 has 1 ten, 1,011 has 1 ten. 1,010 vs 1,011: ones differ, $0 < 1$.

Order: 1,001, 1,010, 1,011, 1,100.

4. Omar starts at 485. He adds 10 twice, then 100 once. Where does he end?

A. 595

B. **605** ✓

C. 685

D. 505

Explanation: $485 + 10 = 495$, $+ 10 = 505$, $+ 100 = 605$.

5. Jake says $998 > 1,002$ because $9 > 1$. What is wrong with his thinking?

A. Nothing, he is right

B. **He should compare digit by digit, but 1,002 has more digits so it is greater** ✓

C. He should add the digits instead

D. 998 and 1,002 are equal

Explanation: A 4-digit number is always greater than a 3-digit number. $1,002 > 998$. Jake compared individual digits instead of looking at how many digits each number has.

6. Ella counts by 5s from 7. Will she ever say 52?

A. **Yes, the ones digit alternates 2 and 7, and 52 ends in 2** ✓

B. No, because 52 does not end in 5 or 0

C. No, she only says odd numbers

D. Yes, but only if she counts past 100

Explanation: From 7, counting by 5s: 12, 17, 22, 27, 32, 37, 42, 47, 52. The ones digit alternates between 2 and 7. Since 52 ends in 2, she will say it!

7. Two students wrote 946 in expanded form. Ava: $900 + 40 + 6$. Ben: 9 hundreds + 4 tens + 6 ones. Who is correct?

A. Only Ava

B. Only Ben

C. **Both are correct** ✓

D. Neither is correct

Explanation: Both forms equal 946. Ava used number expanded form ($900 + 40 + 6$). Ben used place-value form (9 hundreds + 4 tens + 6 ones). Both are valid!

8. What is the greatest 3-digit number with a 5 in the tens place?

A. **959** ✓

B. 950

C. 955

D. 859

Explanation: For the greatest: hundreds = 9 (biggest), tens = 5 (given), ones = 9 (biggest). The number is 959.

9. ____ is 10 less than 1,000 and 100 more than 890. What is ____?

A. 900

B. **990** ✓

C. 890

D. 1,010

Explanation: 10 less than 1,000 = 990. Check: 100 more than 890 = 990. Both clues give 990!

10. Kai says $400 + 60 + 13 = 4,613$. His sister says it equals 473. Who is right and why?

A. Kai, because you put digits together

B. His sister, because $400 + 60 + 13 = 473$ ✓

C. Neither, it equals 4,063

D. Neither, it equals 463

Explanation: Add step by step: $400 + 60 = 460$, then $460 + 13 = 473$. Kai put the digits side by side instead of adding. His sister is correct!

Unit 3: Add & Subtract to 200

Lesson 1: Adding Bigger Numbers

Easy (20 questions)

1. What is $34 + 25$?

- A. 59 ✓
- B. 58
- C. 69
- D. 49

Explanation: Add the ones: $4 + 5 = 9$. Add the tens: $30 + 20 = 50$. Then $50 + 9 = 59$.

2. What is $52 + 36$?

- A. 78
- B. 88 ✓
- C. 86
- D. 98

Explanation: Add the ones: $2 + 6 = 8$. Add the tens: $50 + 30 = 80$. Then $80 + 8 = 88$.

3. What is $41 + 18$?

- A. 69
- B. 59 ✓
- C. 49
- D. 58

Explanation: Add the ones: $1 + 8 = 9$. Add the tens: $40 + 10 = 50$. Then $50 + 9 = 59$.

4. What is $63 + 24$?

- A. 97
- B. 77
- C. 87 ✓
- D. 89

Explanation: Add the ones: $3 + 4 = 7$. Add the tens: $60 + 20 = 80$. Then $80 + 7 = 87$.

5. What is $70 + 45$?

- A. 105
- B. 125
- C. 115 ✓
- D. 95

Explanation: Add the ones: $0 + 5 = 5$. Add the tens: $70 + 40 = 110$. Then $110 + 5 = 115$.

6. What is $50 + 82$?

- A. 122
- B. 132 ✓
- C. 142
- D. 112

Explanation: Add the ones: $0 + 2 = 2$. Add the tens: $50 + 80 = 130$. Then $130 + 2 = 132$.

7. What is $23 + 45$?

- A. 78
- B. 68 ✓
- C. 58
- D. 88

Explanation: Add the ones: $3 + 5 = 8$. Add the tens: $20 + 40 = 60$. Then $60 + 8 = 68$.

8. What is $91 + 8$?

- A. 98
- B. 89
- C. 99 ✓
- D. 100

Explanation: Add the ones: $1 + 8 = 9$. The tens stay at 90. Then $90 + 9 = 99$.

9. Leo has 35 marbles. He finds 14 more. How many marbles does Leo have now?

- A. 59
- B. 49 ✓
- C. 39
- D. 48

Explanation: Add $35 + 14$. Ones: $5 + 4 = 9$. Tens: $30 + 10 = 40$. Total: 49.

10. What is $60 + 31$?

- A. 81
- B. 91 ✓
- C. 71
- D. 101

Explanation: Add the ones: $0 + 1 = 1$. Add the tens: $60 + 30 = 90$. Then $90 + 1 = 91$.

11. What is $44 + 33$?

- A. 67
- B. 87
- C. 77 ✓
- D. 78

Explanation: Add the ones: $4 + 3 = 7$. Add the tens: $40 + 30 = 70$. Then $70 + 7 = 77$.

12. A store has 72 red apples and 15 green apples. How many apples in all?

- A. 97
- B. 87 ✓
- C. 77
- D. 86

Explanation: Add $72 + 15$. Ones: $2 + 5 = 7$. Tens: $70 + 10 = 80$. Total: 87.

13. What is $80 + 16$?

- A. 86
- B. 106
- C. 96 ✓
- D. 90

Explanation: Add the ones: $0 + 6 = 6$. Add the tens: $80 + 10 = 90$. Then $90 + 6 = 96$.

14. Which sum is greater: $42 + 31$ or $50 + 20$?

- A. $42 + 31$ ✓
- B. $50 + 20$
- C. They are the same
- D. Cannot tell

Explanation: $42 + 31 = 73$ and $50 + 20 = 70$. Since $73 > 70$, the first sum is greater.

15. What is $55 + 34$?

- A. 79
- B. 99
- C. 89 ✓
- D. 88

Explanation: Add the ones: $5 + 4 = 9$. Add the tens: $50 + 30 = 80$. Then $80 + 9 = 89$.

16. What is $27 + 51$?

- A. 78 ✓
- B. 88
- C. 68
- D. 77

Explanation: Add the ones: $7 + 1 = 8$. Add the tens: $20 + 50 = 70$. Then $70 + 8 = 78$.

17. There are 46 birds on a wire. 23 more land. How many birds are there now?

- A. 59
- B. 79
- C. 69 ✓
- D. 68

Explanation: Add $46 + 23$. Ones: $6 + 3 = 9$. Tens: $40 + 20 = 60$. Total: 69.

18. What is $13 + 62$?

- A. 85
- B. 65
- C. 75 ✓
- D. 74

Explanation: Add the ones: $3 + 2 = 5$. Add the tens: $10 + 60 = 70$. Then $70 + 5 = 75$.

19. Which is the correct answer for $81 + 14$?

- A. 94
- B. 85
- C. 95 ✓
- D. 96

Explanation: Add the ones: $1 + 4 = 5$. Add the tens: $80 + 10 = 90$. Then $90 + 5 = 95$.

20. What is $30 + 64$?

- A. 84
- B. 94 ✓
- C. 104
- D. 74

Explanation: Add the ones: $0 + 4 = 4$. Add the tens: $30 + 60 = 90$. Then $90 + 4 = 94$.

Medium (20 questions)

1. $47 + 35 = ?$ Use regrouping to solve.

- A. 72
- B. 82 ✓
- C. 81
- D. 92

Explanation: Add the ones: $7 + 5 = 12$. Write 2, regroup 1 ten. Add the tens: $4 + 3 + 1 = 8$. Answer: 82.

2. Maria has 45 stickers. She gets 28 more. How many stickers does Maria have now?

- A. 63
- B. 73 ✓
- C. 83
- D. 62

Explanation: Add $45 + 28$. Ones: $5 + 8 = 13$. Regroup 1 ten. Tens: $4 + 2 + 1 = 7$. Answer: 73.

3. Which problem needs regrouping? A) $34 + 25$ B) $47 + 38$ C) $51 + 23$ D) $62 + 15$

- A. $34 + 25$
- B. $47 + 38$ ✓
- C. $51 + 23$
- D. $62 + 15$

Explanation: $47 + 38$: the ones are $7 + 8 = 15$, which is more than 9, so you must regroup. The other problems have ones that add to 9 or less.

4. $__ + 36 = 73$. What is the missing number?

- A. 47
- B. 37 ✓**
- C. 27
- D. 43

Explanation: Think: what plus 36 equals 73? Subtract: $73 - 36 = 37$. Check: $37 + 36 = 73$.

5. $58 + 67 = ?$

- A. 115
- B. 135
- C. 125 ✓**
- D. 105

Explanation: Add the ones: $8 + 7 = 15$. Write 5, regroup 1 ten. Add the tens: $5 + 6 + 1 = 12$. Answer: 125.

6. A box has 76 crayons. Another box has 49 crayons. How many crayons altogether?

- A. 115
- B. 135
- C. 125 ✓**
- D. 145

Explanation: Add $76 + 49$. Ones: $6 + 9 = 15$. Regroup 1 ten. Tens: $7 + 4 + 1 = 12$. Answer: 125.

7. $__ + 55 = 112$. What is the missing number?

- A. 67
- B. 47
- C. 57 ✓**
- D. 77

Explanation: Think: what plus 55 equals 112? Subtract: $112 - 55 = 57$. Check: $57 + 55 = 112$.

8. $64 + 78 = ?$

- A. 132
- B. 152
- C. 142 ✓**
- D. 122

Explanation: Add the ones: $4 + 8 = 12$. Write 2, regroup 1 ten. Add the tens: $6 + 7 + 1 = 14$. Answer: 142.

9. Which strategy helps solve $89 + 46$?

- A. Count back from 89
- B. Break apart: $89 + 40 + 6$ ✓**
- C. Subtract 46 from 89
- D. Round 89 down to 80

Explanation: Break apart 46 into $40 + 6$. Add $89 + 40 = 129$, then $129 + 6 = 135$. Breaking apart makes adding easier.

10. There are 53 kids on a bus. At the next stop, 39 more get on. How many kids are on the bus?

- A. 82
- B. 92 ✓**
- C. 102
- D. 72

Explanation: Add $53 + 39$. Ones: $3 + 9 = 12$. Regroup 1 ten. Tens: $5 + 3 + 1 = 9$. Answer: 92.

11. Which answer is correct for $86 + 37$?

- A. 113
- B. 133
- C. 123 ✓**
- D. 103

Explanation: Add the ones: $6 + 7 = 13$. Regroup 1 ten. Add the tens: $8 + 3 + 1 = 12$. Answer: 123.

12. $29 + \underline{\quad} = 81$. What is the missing number?

- A. 62
- B. 52 ✓**
- C. 42
- D. 58

Explanation: Think: $29 + \text{what} = 81$? Subtract: $81 - 29 = 52$. Check: $29 + 52 = 81$.

13. Which problem does NOT need regrouping? A) $56 + 27$ B) $43 + 52$ C) $68 + 45$ D) $79 + 14$

- A. $56 + 27$
- B. $43 + 52$ ✓**
- C. $68 + 45$
- D. $79 + 14$

Explanation: $43 + 52$: the ones are $3 + 2 = 5$, no regrouping needed. All other problems have ones that add above 9.

14. A farm has 67 chickens and 85 ducks. How many birds does the farm have?

- A. 142
- B. 162
- C. 152 ✓**
- D. 132

Explanation: Add $67 + 85$. Ones: $7 + 5 = 12$. Regroup 1 ten. Tens: $6 + 8 + 1 = 15$. Answer: 152.

15. $73 + 48 = ?$

- A. 111
- B. 131
- C. 121 ✓**
- D. 101

Explanation: Add the ones: $3 + 8 = 11$. Write 1, regroup 1 ten. Add the tens: $7 + 4 + 1 = 12$. Answer: 121.

16. Jada had some stickers. She got 44 more and now has 91. How many did she start with?

- A. 57
- B. 47 ✓**
- C. 37
- D. 53

Explanation: Think: $\underline{\quad} + 44 = 91$. Subtract: $91 - 44 = 47$. Jada started with 47 stickers.

17. $55 + 86 = ?$

- A. 131
- B. 151
- C. 141 ✓**
- D. 121

Explanation: Add the ones: $5 + 6 = 11$. Regroup 1 ten. Tens: $5 + 8 + 1 = 14$. Answer: 141.

18. Which is greater: $65 + 48$ or $59 + 57$?

- A. $65 + 48$
- B. $59 + 57$ ✓**
- C. They are equal
- D. Cannot tell

Explanation: $65 + 48 = 113$ and $59 + 57 = 116$. Since $116 > 113$, $59 + 57$ is greater.

19. A library has 94 fiction books and 68 nonfiction books. How many books in all?

- A. 152
- B. 172
- C. 162 ✓**
- D. 142

Explanation: Add $94 + 68$. Ones: $4 + 8 = 12$. Regroup 1 ten. Tens: $9 + 6 + 1 = 16$. Answer: 162.

20. $38 + 95 = ?$

- A. 123
- B. 143
- C. 133 ✓**
- D. 113

Explanation: Add the ones: $8 + 5 = 13$. Regroup 1 ten. Tens: $3 + 9 + 1 = 13$. Answer: 133.

Hard (20 questions)

1. Tom had 85 baseball cards. He gave 27 to Kim and then got 15 from Dad. How many does he have now?

- A. 63
- B. 83
- C. 73 ✓**
- D. 93

Explanation: Step 1: $85 - 27 = 58$. Step 2: $58 + 15 = 73$. Two operations: subtract first, then add.

2. A student solved $63 + 29 = 82$. What mistake did the student make?

- A. Added the tens wrong
- B. Forgot to regroup ✓**
- C. Added the ones wrong
- D. No mistake, 82 is correct

Explanation: $63 + 29$: ones are $3 + 9 = 12$. You must regroup 1 ten. Tens: $6 + 2 + 1 = 9$. The correct answer is 92. The student forgot to regroup.

3. $__ + 67 = 143$. What is the missing number?

- A. 86
- B. 66
- C. 76 ✓**
- D. 96

Explanation: Use inverse reasoning: $143 - 67$. Ones: $13 - 7 = 6$ (regroup). Tens: $13 - 6 = 7$. Missing number: 76. Check: $76 + 67 = 143$.

4. Mia added $78 + 56$ and got 124. Is she correct? What should the answer be?

- A. Yes, 124 is correct
- B. No, it should be 134 ✓**
- C. No, it should be 144
- D. No, it should be 114

Explanation: $78 + 56$: ones are $8 + 6 = 14$. Regroup 1 ten. Tens: $7 + 5 + 1 = 13$. Answer: 134. Mia got 124, so she is wrong by 10 — she likely forgot to regroup.

5. Sam needs to buy items costing 67 cents and 54 cents. He has 110 cents. Does he have enough? How do you know?

- A. Yes, because $67 + 54 = 121$ and $121 < 110$
- B. No, because $67 + 54 = 121$ and $121 > 110$ ✓**
- C. Yes, because $67 + 54 = 111$ and $111 > 110$
- D. No, because $67 + 54 = 131$ and $131 > 110$

Explanation: Add $67 + 54 = 121$. Compare: $121 > 110$. Sam does not have enough because the total cost is more than what he has.

6. A student wrote $95 + 48 = 133$. Check the work. Is the answer correct?

- A. No, the answer is 143 ✓**
- B. Yes, 133 is correct
- C. No, the answer is 153
- D. No, the answer is 123

Explanation: $95 + 48$: ones are $5 + 8 = 13$. Regroup 1 ten. Tens: $9 + 4 + 1 = 14$. Correct answer: 143. The student made an error adding the tens.

7. Rosa had 92 stickers. She gave some away and now has 58. Then she got 35 more. How many does she have now?

- A. 83
- B. 103
- C. 93 ✓**
- D. 73

Explanation: Step 1: Rosa gave away stickers and has 58 left. Step 2: $58 + 35 = 93$. Rosa now has 93 stickers.

8. $__ - 38 = 79$. What is the missing number?

- A. 107
- B. 127
- C. 117 ✓**
- D. 97

Explanation: Use inverse reasoning: add $79 + 38$. Ones: $9 + 8 = 17$. Regroup. Tens: $7 + 3 + 1 = 11$. Missing number: 117.

9. Which problem will give an answer greater than 150? A) $88 + 59$ B) $72 + 65$ C) $91 + 63$ D) $80 + 68$

- A. $88 + 59$
- B. $72 + 65$
- C. $91 + 63$ ✓**
- D. $80 + 68$

Explanation: A) $88 + 59 = 147$. B) $72 + 65 = 137$. C) $91 + 63 = 154$. D) $80 + 68 = 148$. Only $91 + 63 = 154$ is greater than 150.

10. Jake had 74 toy cars. He bought 38 more on Monday and 29 more on Tuesday. How many toy cars does he have now?

- A. 131
- B. 151
- C. 141 ✓**
- D. 121

Explanation: Step 1: $74 + 38 = 112$. Step 2: $112 + 29 = 141$. Jake now has 141 toy cars.

11. A student solved $86 + 47 = 123$. Find the error.

- A. No error, 123 is correct
- B. The student forgot to regroup the ones
- C. The student added the tens incorrectly ✓**
- D. The student added the ones incorrectly

Explanation: $86 + 47$: ones are $6 + 7 = 13$. Regroup 1 ten. Tens: $8 + 4 + 1 = 13$. Answer: 133. The student got 123, so they added the tens as $8 + 4 = 12$ instead of $8 + 4 + 1 = 13$.

12. Ella has 65 beads. Kai has 48 beads. They want to combine their beads. Do they have more or fewer than 120?

- A. More, because $65 + 48 = 123$
- B. Fewer, because $65 + 48 = 113$ ✓**
- C. Exactly 120
- D. More, because $65 + 48 = 133$

Explanation: $65 + 48$: ones $5 + 8 = 13$, regroup. Tens: $6 + 4 + 1 = 11$. Total: 113. $113 < 120$, so they have fewer than 120.

13. $__ + 89 = 165$. What is the missing number?

- A. 86
- B. 66
- C. 76 ✓**
- D. 96

Explanation: Inverse reasoning: $165 - 89$. Ones: $15 - 9 = 6$ (regroup). Tens: $15 - 8 = 7$. Missing number: 76. Check: $76 + 89 = 165$.

14. A class collected 97 cans in Week 1 and 86 cans in Week 2. They need 200 cans total. Do they have enough? How many more do they need?

- A. Yes, they have enough
- B. No, they need 17 more ✓**
- C. No, they need 27 more
- D. No, they need 7 more

Explanation: $97 + 86 = 183$. They need 200. $200 - 183 = 17$. They do not have enough and need 17 more cans.

15. Which addition strategy works best for $99 + 67$? Why?

A. Break apart 67 into $60 + 7$

B. Make a friendly number: $100 + 66$ ✓

C. Count up from 99 by ones

D. Subtract 67 from 99

Explanation: 99 is close to 100. Move 1 from 67 to 99: $100 + 66 = 166$. Making a friendly number is the most efficient strategy here.

16. A student says $57 + 86 = 133$. Another says it is 143. Who is correct?

A. The first student (133)

B. The second student (143) ✓

C. Neither, it is 153

D. Neither, it is 123

Explanation: $57 + 86$: ones $7 + 6 = 13$, regroup. Tens: $5 + 8 + 1 = 14$. Answer: 143. The second student is correct.

17. Ava scored 78 points in Game 1 and 64 in Game 2. Ben scored 86 in Game 1 and 53 in Game 2. Who scored more total points?

A. Ava, with 142 points ✓

B. Ben, with 142 points

C. They tied with 142 points

D. Ben, with 139 points

Explanation: Ava: $78 + 64 = 142$. Ben: $86 + 53 = 139$. Ava scored 142 and Ben scored 139. Ava scored more.

18. $\underline{\quad} + \underline{\quad} = 130$, and both numbers are between 50 and 90. Which pair works?

A. 45 and 85

B. 62 and 68 ✓

C. 91 and 39

D. 75 and 65

Explanation: Check each pair: 45 is not between 50-90. $62 + 68 = 130$ and both are between 50-90. 91 is too high. $75 + 65 = 140$, not 130.

19. Liam added $69 + 74$ and got 133. Is he right? If not, what is the correct answer?

A. Yes, 133 is correct

B. No, it should be 143 ✓

C. No, it should be 153

D. No, it should be 123

Explanation: $69 + 74$: ones $9 + 4 = 13$, regroup. Tens: $6 + 7 + 1 = 14$. Correct answer: 143. Liam likely forgot to add the regrouped ten.

20. There are 88 red balloons and 76 blue balloons. 30 balloons pop. How many balloons are left?

A. 124

B. 144

C. 134 ✓

D. 154

Explanation: Step 1: $88 + 76 = 164$ total balloons. Step 2: $164 - 30 = 134$ balloons left.

Lesson 2: Taking Away Bigger Numbers

Easy (20 questions)

1. What is $58 - 23$?

A. 45

B. 35 ✓

C. 25

D. 34

Explanation: Subtract the ones: $8 - 3 = 5$. Subtract the tens: $50 - 20 = 30$. Then $30 + 5 = 35$.

2. What is $87 - 42$?

- A. 55
- B. 45 ✓**
- C. 35
- D. 44

Explanation: Subtract the ones: $7 - 2 = 5$. Subtract the tens: $80 - 40 = 40$. Then $40 + 5 = 45$.

3. What is $96 - 31$?

- A. 55
- B. 75
- C. 65 ✓**
- D. 64

Explanation: Subtract the ones: $6 - 1 = 5$. Subtract the tens: $90 - 30 = 60$. Then $60 + 5 = 65$.

4. What is $74 - 50$?

- A. 34
- B. 24 ✓**
- C. 14
- D. 44

Explanation: Subtract the tens: $70 - 50 = 20$. The ones stay at 4. Then $20 + 4 = 24$.

5. A shelf has 69 books. 25 are taken. How many books are left?

- A. 54
- B. 44 ✓**
- C. 34
- D. 43

Explanation: Subtract $69 - 25$. Ones: $9 - 5 = 4$. Tens: $60 - 20 = 40$. Total: 44.

6. What is $145 - 30$?

- A. 125
- B. 115 ✓**
- C. 105
- D. 135

Explanation: Subtract the tens: $140 - 30 = 110$. The ones stay at 5. Then $110 + 5 = 115$.

7. What is $83 - 41$?

- A. 52
- B. 42 ✓**
- C. 32
- D. 43

Explanation: Subtract the ones: $3 - 1 = 2$. Subtract the tens: $80 - 40 = 40$. Then $40 + 2 = 42$.

8. What is $99 - 66$?

- A. 43
- B. 33 ✓**
- C. 23
- D. 32

Explanation: Subtract the ones: $9 - 6 = 3$. Subtract the tens: $90 - 60 = 30$. Then $30 + 3 = 33$.

9. There are 78 cookies. 34 are eaten. How many are left?

- A. 54
- B. 44 ✓**
- C. 34
- D. 43

Explanation: Subtract $78 - 34$. Ones: $8 - 4 = 4$. Tens: $70 - 30 = 40$. Total: 44.

10. What is $127 - 15$?

- A. 102
- B. 122
- C. 112 ✓
- D. 132

Explanation: Subtract the ones: $7 - 5 = 2$. Subtract the tens: $120 - 10 = 110$. Then $110 + 2 = 112$.

11. What is $56 - 24$?

- A. 42
- B. 32 ✓
- C. 22
- D. 31

Explanation: Subtract the ones: $6 - 4 = 2$. Subtract the tens: $50 - 20 = 30$. Then $30 + 2 = 32$.

12. Which difference is greater: $89 - 43$ or $75 - 31$?

- A. 89 - 43 ✓
- B. 75 - 31
- C. They are the same
- D. Cannot tell

Explanation: $89 - 43 = 46$ and $75 - 31 = 44$. Since $46 > 44$, the first difference is greater.

13. What is $160 - 40$?

- A. 110
- B. 130
- C. 120 ✓
- D. 140

Explanation: Subtract the tens: $160 - 40 = 120$. No ones to worry about.

14. A jar has 48 candies. 16 are given away. How many candies are left?

- A. 42
- B. 32 ✓
- C. 22
- D. 31

Explanation: Subtract $48 - 16$. Ones: $8 - 6 = 2$. Tens: $40 - 10 = 30$. Total: 32.

15. What is $95 - 63$?

- A. 42
- B. 32 ✓
- C. 22
- D. 31

Explanation: Subtract the ones: $5 - 3 = 2$. Subtract the tens: $90 - 60 = 30$. Then $30 + 2 = 32$.

16. Which is the correct answer for $138 - 26$?

- A. 102
- B. 122
- C. 112 ✓
- D. 132

Explanation: Subtract the ones: $8 - 6 = 2$. Subtract the tens: $130 - 20 = 110$. Then $110 + 2 = 112$.

17. What is $67 - 35$?

- A. 42
- B. 32 ✓
- C. 22
- D. 31

Explanation: Subtract the ones: $7 - 5 = 2$. Subtract the tens: $60 - 30 = 30$. Then $30 + 2 = 32$.

18. A park has 84 trees. 21 are cut down. How many trees are left?

- A. 73
- B. 63 ✓**
- C. 53
- D. 62

Explanation: Subtract $84 - 21$. Ones: $4 - 1 = 3$. Tens: $80 - 20 = 60$. Total: 63.

19. What is $179 - 54$?

- A. 115
- B. 135
- C. 125 ✓**
- D. 145

Explanation: Subtract the ones: $9 - 4 = 5$. Subtract the tens: $170 - 50 = 120$. Then $120 + 5 = 125$.

20. What is $46 - 13$?

- A. 43
- B. 33 ✓**
- C. 23
- D. 32

Explanation: Subtract the ones: $6 - 3 = 3$. Subtract the tens: $40 - 10 = 30$. Then $30 + 3 = 33$.

Medium (20 questions)

1. $73 - 48 = ?$ Use regrouping to solve.

- A. 35
- B. 25 ✓**
- C. 15
- D. 45

Explanation: You cannot take 8 from 3. Regroup: borrow 1 ten. $13 - 8 = 5$. Tens: $6 - 4 = 2$. Answer: 25.

2. Diego has 82 marbles. He gives 37 to his friend. How many does he have left?

- A. 55
- B. 45 ✓**
- C. 35
- D. 54

Explanation: $82 - 37$: you cannot take 7 from 2. Regroup: $12 - 7 = 5$. Tens: $7 - 3 = 4$. Answer: 45.

3. Which problem needs regrouping? A) $86 - 42$ B) $65 - 38$ C) $97 - 51$ D) $74 - 23$

- A. $86 - 42$
- B. $65 - 38$ ✓**
- C. $97 - 51$
- D. $74 - 23$

Explanation: $65 - 38$: ones are $5 - 8$. Since $5 < 8$, you must regroup. The other problems have larger ones digits on top.

4. $91 - \underline{\quad} = 56$. What is the missing number?

- A. 45
- B. 35 ✓**
- C. 25
- D. 33

Explanation: Think: 91 minus what equals 56? Subtract: $91 - 56 = 35$. Check: $91 - 35 = 56$.

5. $134 - 58 = ?$

- A. 86
- B. 76 ✓**
- C. 66
- D. 96

Explanation: You cannot take 8 from 4. Regroup: $14 - 8 = 6$. Tens: $12 - 5 = 7$. Answer: 76.

6. A store had 120 apples. 47 were sold. How many are left?

- A. 83
- B. 73 ✓**
- C. 63
- D. 77

Explanation: $120 - 47$: you cannot take 7 from 0. Regroup: $10 - 7 = 3$. Tens: $11 - 4 = 7$. Answer: 73.

7. $__ - 29 = 47$. What is the missing number?

- A. 66
- B. 86
- C. 76 ✓**
- D. 56

Explanation: Think: what minus 29 equals 47? Add: $47 + 29 = 76$. Check: $76 - 29 = 47$.

8. Which strategy helps solve $100 - 63$?

- A. Add up from 63 to 100 ✓**
- B. Subtract 100 from 63
- C. Round 63 to 70 first
- D. Count back by ones

Explanation: Add up from 63: $63 + 7 = 70$, $70 + 30 = 100$. That is $7 + 30 = 37$. Adding up is efficient for subtracting from 100.

9. $156 - 79 = ?$

- A. 87
- B. 77 ✓**
- C. 67
- D. 97

Explanation: You cannot take 9 from 6. Regroup: $16 - 9 = 7$. Tens: $14 - 7 = 7$. Answer: 77.

10. A bus has 95 passengers. At the first stop, 38 get off. How many are still on the bus?

- A. 67
- B. 57 ✓**
- C. 47
- D. 63

Explanation: $95 - 38$: you cannot take 8 from 5. Regroup: $15 - 8 = 7$. Tens: $8 - 3 = 5$. Answer: 57.

11. Which answer is correct for $143 - 67$?

- A. 86
- B. 76 ✓**
- C. 66
- D. 96

Explanation: $143 - 67$: regroup ones: $13 - 7 = 6$. Tens: $13 - 6 = 7$. Answer: 76.

12. Which problem does NOT need regrouping? A) $52 - 37$ B) $81 - 45$ C) $96 - 52$ D) $63 - 28$

- A. $52 - 37$
- B. $81 - 45$
- C. $96 - 52$ ✓**
- D. $63 - 28$

Explanation: $96 - 52$: ones are $6 - 2 = 4$. No regrouping needed because $6 > 2$. All others have a smaller ones digit on top.

13. $115 - __ = 68$. What is the missing number?

- A. 57
- B. 47 ✓**
- C. 37
- D. 53

Explanation: Think: 115 minus what equals 68? Subtract: $115 - 68 = 47$. Check: $115 - 47 = 68$.

14. A farmer had 162 eggs. 85 broke. How many eggs are left?

- A. 87
- B. 77 ✓**
- C. 67
- D. 83

Explanation: $162 - 85$: regroup ones: $12 - 5 = 7$. Tens: $15 - 8 = 7$. Answer: 77.

15. Which is greater: $130 - 56$ or $115 - 48$?

- A. $130 - 56$ ✓**
- B. $115 - 48$
- C. They are equal
- D. Cannot tell

Explanation: $130 - 56 = 74$ and $115 - 48 = 67$. Since $74 > 67$, the first difference is greater.

16. $61 - 34 = ?$

- A. 37
- B. 27 ✓**
- C. 17
- D. 33

Explanation: You cannot take 4 from 1. Regroup: $11 - 4 = 7$. Tens: $5 - 3 = 2$. Answer: 27.

17. A school has 148 students. 59 go home early. How many remain?

- A. 99
- B. 89 ✓**
- C. 79
- D. 91

Explanation: $148 - 59$: regroup ones: $18 - 9 = 9$. Tens: $13 - 5 = 8$. Answer: 89.

18. $__ - 45 = 88$. What is the missing number?

- A. 123
- B. 143
- C. 133 ✓**
- D. 113

Explanation: Think: what minus 45 equals 88? Add: $88 + 45 = 133$. Check: $133 - 45 = 88$.

19. A piggy bank has 107 coins. 39 coins are spent. How many are left?

- A. 78
- B. 68 ✓**
- C. 58
- D. 72

Explanation: $107 - 39$: regroup ones: $17 - 9 = 8$. Tens: $9 - 3 = 6$. Answer: 68.

20. $85 - 47 = ?$

- A. 48
- B. 38 ✓**
- C. 28
- D. 42

Explanation: You cannot take 7 from 5. Regroup: $15 - 7 = 8$. Tens: $7 - 4 = 3$. Answer: 38.

Hard (20 questions)

1. A student solved $132 - 57 = 85$. What mistake did the student make?

- A. Subtracted the ones wrong
- B. Forgot to regroup
- C. Subtracted the tens wrong ✓**
- D. No mistake, 85 is correct

Explanation: $132 - 57$: regroup ones: $12 - 7 = 5$. Tens: $12 - 5 = 7$. Correct answer: 75. The student wrote 85, so they subtracted the tens incorrectly (wrote 8 instead of 7).

2. Nora had 145 stickers. She gave 58 to Ava and 39 to Ben. How many does she have left?

- A. 58
- B. 48 ✓**
- C. 38
- D. 52

Explanation: Step 1: $145 - 58 = 87$. Step 2: $87 - 39 = 48$. Nora has 48 stickers left.

3. $___ - 67 = 89$. What is the missing number?

- A. 146
- B. 166
- C. 156 ✓**
- D. 136

Explanation: Inverse reasoning: add $89 + 67$. Ones: $9 + 7 = 16$, regroup. Tens: $8 + 6 + 1 = 15$. Missing number: 156.

4. Leo subtracted $163 - 78$ and got 95. Is he correct?

- A. Yes, 95 is correct
- B. No, it should be 85 ✓**
- C. No, it should be 75
- D. No, it should be 105

Explanation: $163 - 78$: regroup ones: $13 - 8 = 5$. Tens: $15 - 7 = 8$. Answer: 85. Leo got 95, which is wrong — he likely subtracted 6 instead of 7 in the tens.

5. A school had 180 pencils. Monday they gave out 45. Tuesday they gave out 68. How many are left?

- A. 77
- B. 67 ✓**
- C. 57
- D. 87

Explanation: Step 1: $180 - 45 = 135$. Step 2: $135 - 68 = 67$. The school has 67 pencils left.

6. Which subtraction will give a result less than 50? A) $134 - 79$ B) $121 - 68$ C) $142 - 85$ D) $113 - 69$

- A. $134 - 79$
- B. $121 - 68$
- C. $142 - 85$
- D. $113 - 69$ ✓**

Explanation: A) $134 - 79 = 55$. B) $121 - 68 = 53$. C) $142 - 85 = 57$. D) $113 - 69 = 44$. Only $113 - 69 = 44$ is less than 50.

7. A student wrote $150 - 73 = 87$. Find the error.

- A. No error, 87 is correct
- B. Forgot to regroup the ones
- C. Subtracted the tens wrong ✓**
- D. Subtracted the ones wrong

Explanation: $150 - 73$: regroup ones: $10 - 3 = 7$. Tens: $14 - 7 = 7$. Correct answer: 77. The student wrote 87, subtracting tens as $14 - 7 = 8$ instead of 7.

8. $___ - 86 = 57$. What is the missing number?

- A. 133
- B. 153
- C. 143 ✓**
- D. 123

Explanation: Inverse reasoning: $57 + 86$. Ones: $7 + 6 = 13$, regroup. Tens: $5 + 8 + 1 = 14$. Missing number: 143.

9. Mia has 126 beads. She uses 48 for a bracelet and 35 for a necklace. How many beads are left?

- A. 53
- B. 43 ✓**
- C. 33
- D. 47

Explanation: Step 1: $126 - 48 = 78$. Step 2: $78 - 35 = 43$. Mia has 43 beads left.

10. A store had 200 items. They sold 94 on Saturday and 67 on Sunday. How many items are left?

- A. 49
- B. 39 ✓**
- C. 29
- D. 59

Explanation: Step 1: $200 - 94 = 106$. Step 2: $106 - 67 = 39$. The store has 39 items left.

11. Sam says $141 - 65 = 86$. Kai says it is 76. Who is correct?

- A. Sam (86)
- B. Kai (76) ✓**
- C. Neither, it is 66
- D. Neither, it is 96

Explanation: $141 - 65$: regroup ones: $11 - 5 = 6$. Tens: $13 - 6 = 7$. Answer: 76. Kai is correct.

12. Which subtraction strategy works best for $200 - 87$?

- A. Count back by ones from 200
- B. Add up from 87 to 200 ✓**
- C. Round 87 to 100 and adjust
- D. Subtract $200 - 80 - 7$

Explanation: Add up: $87 + 13 = 100$, $100 + 100 = 200$. That is $13 + 100 = 113$. Adding up from 87 is efficient when subtracting from a round number.

13. A library had 175 books. Some were checked out, leaving 98. Then 24 books were returned. How many are in the library now?

- A. 112
- B. 132
- C. 122 ✓**
- D. 102

Explanation: Step 1: $175 - 98 = 77$ were checked out. Step 2: $98 + 24 = 122$ books now in the library.

14. $\underline{\quad} - \underline{\quad} = 45$, where the first number is between 100 and 150. Which pair works?

- A. 130 and 85 ✓**
- B. 140 and 85
- C. 120 and 85
- D. 110 and 85

Explanation: Check: $130 - 85 = 45$. That works! $140 - 85 = 55$. $120 - 85 = 35$. $110 - 85 = 25$. Only $130 - 85 = 45$.

15. A student solved $114 - 47 = 77$. Check the work.

- A. Correct, 77 is right
- B. Wrong, it should be 67 ✓**
- C. Wrong, it should be 57
- D. Wrong, it should be 87

Explanation: $114 - 47$: regroup ones: $14 - 7 = 7$. Tens: $10 - 4 = 6$. Answer: 67. The student got 77, adding 1 extra ten.

16. Ella had 168 stickers. She gave away some and has 79 left. How many did she give away?

- A. 99
- B. 89 ✓**
- C. 79
- D. 91

Explanation: Subtract: $168 - 79 = 89$. Ella gave away 89 stickers. Check: $79 + 89 = 168$.

17. A farmer picks 156 apples. He sells 68 at the market and gives 49 to neighbors. How many does he keep?

- A. 49
- B. 39 ✓**
- C. 29
- D. 43

Explanation: Step 1: $156 - 68 = 88$. Step 2: $88 - 49 = 39$. The farmer keeps 39 apples.

18. There are 193 students. 87 go to lunch first. Of the remaining, 56 go next. How many are still waiting?

- A. 60
- B. 50 ✓**
- C. 40
- D. 55

Explanation: Step 1: $193 - 87 = 106$ remaining. Step 2: $106 - 56 = 50$ still waiting.

19. A student says $160 - 84 = 86$. Another says it equals 76. Who is right?

- A. First student (86)
- B. Second student (76) ✓**
- C. Neither, it is 66
- D. Neither, it is 96

Explanation: $160 - 84$: regroup ones: $10 - 4 = 6$. Tens: $15 - 8 = 7$. Answer: 76. The second student is correct.

20. ___ - 95 = 78. What is the missing number?

- A. 163
- B. 183
- C. 173 ✓**
- D. 153

Explanation: Inverse reasoning: $78 + 95$. Ones: $8 + 5 = 13$, regroup. Tens: $7 + 9 + 1 = 17$. Missing number: 173.

Lesson 3: Add Three Numbers

Easy (20 questions)

1. What is $4 + 3 + 6$?

- A. 12
- B. 13 ✓**
- C. 14
- D. 11

Explanation: Add in steps: $4 + 6 = 10$ (make a ten!), then $10 + 3 = 13$.

2. What is $5 + 5 + 7$?

- A. 15
- B. 16
- C. 17 ✓**
- D. 18

Explanation: Add doubles first: $5 + 5 = 10$. Then $10 + 7 = 17$.

3. What is $8 + 2 + 9$?

- A. 17
- B. 18
- C. 19 ✓**
- D. 20

Explanation: Make a ten: $8 + 2 = 10$. Then $10 + 9 = 19$.

4. What is $20 + 30 + 10$?

- A. 50
- B. 60 ✓**
- C. 70
- D. 40

Explanation: Add the tens: $20 + 30 = 50$. Then $50 + 10 = 60$.

5. What is $6 + 4 + 5$?

- A. 14
- B. 15 ✓**
- C. 16
- D. 13

Explanation: Make a ten: $6 + 4 = 10$. Then $10 + 5 = 15$.

6. What is $7 + 3 + 8$?

- A. 16
- B. 17
- C. 18 ✓**
- D. 19

Explanation: Make a ten: $7 + 3 = 10$. Then $10 + 8 = 18$.

7. A basket has 3 red, 5 blue, and 7 green balls. How many balls in all?

- A. 14
- B. 15 ✓**
- C. 16
- D. 13

Explanation: Add: $3 + 7 = 10$ (make a ten). Then $10 + 5 = 15$ balls.

8. What is $9 + 1 + 4$?

- A. 13
- B. 14 ✓**
- C. 15
- D. 12

Explanation: Make a ten: $9 + 1 = 10$. Then $10 + 4 = 14$.

9. What is $10 + 20 + 40$?

- A. 60
- B. 70 ✓**
- C. 80
- D. 50

Explanation: Add the tens: $10 + 20 = 30$. Then $30 + 40 = 70$.

10. What is $2 + 8 + 6$?

- A. 14
- B. 15
- C. 16 ✓**
- D. 17

Explanation: Make a ten: $2 + 8 = 10$. Then $10 + 6 = 16$.

11. What is $5 + 3 + 5$?

- A. 12
- B. 13 ✓**
- C. 14
- D. 11

Explanation: Add doubles: $5 + 5 = 10$. Then $10 + 3 = 13$.

12. Dan scored 6 points, then 4, then 3. How many points total?

- A. 12
- B. 13 ✓**
- C. 14
- D. 11

Explanation: Make a ten: $6 + 4 = 10$. Then $10 + 3 = 13$ points.

13. What is $30 + 40 + 20$?

- A. 80
- B. 90 ✓**
- C. 100
- D. 70

Explanation: Add: $30 + 40 = 70$. Then $70 + 20 = 90$.

14. Which pair of numbers makes a ten? A) 3 and 5 B) 6 and 4 C) 7 and 2 D) 8 and 3

- A. 3 and 5
- B. 6 and 4 ✓**
- C. 7 and 2
- D. 8 and 3

Explanation: $6 + 4 = 10$. This is a 'make a ten' pair! $3+5=8$, $7+2=9$, and $8+3=11$. Only $6+4$ makes exactly 10.

15. What is $1 + 9 + 7$?

- A. 15
- B. 16
- C. 17 ✓**
- D. 18

Explanation: Make a ten: $1 + 9 = 10$. Then $10 + 7 = 17$.

16. What is $50 + 20 + 30$?

- A. 90
- B. 100 ✓**
- C. 110
- D. 80

Explanation: Add: $50 + 30 = 80$. Then $80 + 20 = 100$. Or: $50 + 20 = 70$, then $70 + 30 = 100$.

17. Amy picked 4 flowers, then 6, then 8. How many flowers in all?

- A. 16
- B. 17
- C. 18 ✓**
- D. 19

Explanation: Make a ten: $4 + 6 = 10$. Then $10 + 8 = 18$ flowers.

18. What is $3 + 7 + 2$?

- A. 11
- B. 12 ✓**
- C. 13
- D. 10

Explanation: Make a ten: $3 + 7 = 10$. Then $10 + 2 = 12$.

19. What is $6 + 6 + 3$?

- A. 14
- B. 15 ✓**
- C. 16
- D. 13

Explanation: Add doubles: $6 + 6 = 12$. Then $12 + 3 = 15$.

20. What is $40 + 10 + 50$?

- A. 90
- B. 100 ✓**
- C. 110
- D. 80

Explanation: Add: $40 + 10 = 50$. Then $50 + 50 = 100$.

Medium (20 questions)

1. What is $24 + 35 + 16$?

- A. 65
- B. 75 ✓**
- C. 85
- D. 55

Explanation: Add the first two: $24 + 16 = 40$ (ones make a ten!). Then $40 + 35 = 75$.

2. Which strategy helps add $18 + 32 + 25$?

- A. Add all ones first
- B. Make a ten: $18 + 32 = 50$ ✓**
- C. Start with the largest number
- D. Count by ones

Explanation: $18 + 32 = 50$ (the ones 8 + 2 make a ten). Then $50 + 25 = 75$. Making a ten first is efficient.

3. Sam collected 37 shells on Monday, 28 on Tuesday, and 13 on Wednesday. How many in all?

- A. 68
- B. 78 ✓**
- C. 88
- D. 58

Explanation: Look for tens: $37 + 13 = 50$ ($7 + 3 = 10$!). Then $50 + 28 = 78$.

4. $\underline{\hspace{1cm}} + 25 + 19 = 80$. What is the missing number?

- A. 46
- B. 36 ✓**
- C. 26
- D. 42

Explanation: First add $25 + 19 = 44$. Then: $\underline{\hspace{1cm}} + 44 = 80$. Subtract: $80 - 44 = 36$.

5. What is $45 + 27 + 18$?

- A. 80
- B. 90 ✓**
- C. 100
- D. 70

Explanation: Add $45 + 27 = 72$ (with regrouping: $5+7=12$). Then $72 + 18 = 90$ ($2+8=10$, regroup).

6. A toy store sells 42 cars, 38 trucks, and 15 planes. How many toys sold in all?

- A. 85
- B. 95 ✓**
- C. 105
- D. 75

Explanation: Add $42 + 38 = 80$ ($2+8=10$!). Then $80 + 15 = 95$ toys.

7. Which two numbers should you add first to make a ten? $17 + 24 + 33$

- A. 17 and 24
- B. 24 and 33
- C. 17 and 33 ✓**
- D. Add in order

Explanation: $17 + 33$: ones are $7 + 3 = 10$, making a ten! $17 + 33 = 50$. Then $50 + 24 = 74$.

8. What is $56 + 14 + 23$?

- A. 83
- B. 93 ✓**
- C. 103
- D. 73

Explanation: Make a ten: $56 + 14 = 70$ ($6+4=10$!). Then $70 + 23 = 93$.

9. Which is the correct sum: $31 + 49 + 22$?

- A. 92
- B. 102 ✓**
- C. 112
- D. 82

Explanation: Make a ten: $31 + 49 = 80$ ($1+9=10$). Then $80 + 22 = 102$.

10. $15 + \underline{\quad} + 28 = 73$. What is the missing number?

- A. 40
- B. 30 ✓**
- C. 20
- D. 35

Explanation: Add the known numbers: $15 + 28 = 43$. Then: $43 + \underline{\quad} = 73$. Subtract: $73 - 43 = 30$.

11. A garden has 29 red flowers, 41 yellow flowers, and 18 pink flowers. How many flowers total?

- A. 78
- B. 88 ✓**
- C. 98
- D. 68

Explanation: Make a ten: $29 + 41 = 70$ ($9+1=10$). Then $70 + 18 = 88$.

12. What is $33 + 47 + 26$?

- A. 96
- B. 106 ✓**
- C. 116
- D. 86

Explanation: Make a ten: $33 + 47 = 80$ ($3+7=10$!). Then $80 + 26 = 106$.

13. Which is greater: $25 + 36 + 14$ or $43 + 28 + 12$?

- A. $25 + 36 + 14$
- B. $43 + 28 + 12$ ✓**
- C. They are equal
- D. Cannot tell

Explanation: First sum: $25 + 36 + 14 = 75$. Second sum: $43 + 28 + 12 = 83$. $83 > 75$, so the second is greater.

14. What is $62 + 19 + 11$?

- A. 82
- B. 92 ✓**
- C. 102
- D. 72

Explanation: Make a ten: $19 + 11 = 30$ ($9+1=10$!). Then $62 + 30 = 92$.

15. A class read 26 books in January, 34 in February, and 22 in March. How many books total?

- A. 72
- B. 82 ✓**
- C. 92
- D. 62

Explanation: Make a ten: $26 + 34 = 60$ ($6+4=10$!). Then $60 + 22 = 82$ books.

16. $38 + 22 + \underline{\quad} = 100$. What is the missing number?

- A. 50
- B. 40 ✓**
- C. 30
- D. 45

Explanation: Add known: $38 + 22 = 60$ ($8+2=10$!). Then $60 + \underline{\quad} = 100$. Missing: 40.

17. What is $51 + 39 + 17$?

- A. 97
- B. 107 ✓**
- C. 117
- D. 87

Explanation: Make a ten: $51 + 39 = 90$ ($1+9=10!$). Then $90 + 17 = 107$.

18. Which pair makes adding $27 + 46 + 13$ easier?

- A. 27 and 46
- B. 46 and 13
- C. 27 and 13 ✓**
- D. No pair helps

Explanation: $27 + 13$: ones are $7 + 3 = 10$, making a ten! $27 + 13 = 40$. Then $40 + 46 = 86$.

19. A pet shop has 44 fish, 31 hamsters, and 16 birds. How many pets in all?

- A. 81
- B. 91 ✓**
- C. 101
- D. 71

Explanation: Add $44 + 16 = 60$ ($4+6=10!$). Then $60 + 31 = 91$ pets.

20. What is $28 + 35 + 42$?

- A. 95
- B. 105 ✓**
- C. 115
- D. 85

Explanation: Add $28 + 42 = 70$ ($8+2=10!$). Then $70 + 35 = 105$.

Hard (20 questions)

1. Jake scored 47 points in Round 1, 38 in Round 2, and 29 in Round 3. He needs 120 to win. Does he have enough?

- A. Yes, he has 124
- B. Yes, he has exactly 120
- C. No, he has 114 ✓**
- D. No, he has 104

Explanation: Step 1: $47 + 38 = 85$. Step 2: $85 + 29 = 114$. $114 < 120$, so Jake does not have enough.

2. A student added $36 + 45 + 24$ and got 95. Find the error.

- A. No error, 95 is correct
- B. Forgot to regroup, answer is 105 ✓**
- C. Added ones wrong, answer is 115
- D. Added tens wrong, answer is 85

Explanation: $36 + 24 = 60$ ($6+4=10$). $60 + 45 = 105$. The student got 95, forgetting to regroup the tens when adding $60 + 45$.

3. $__ + 37 + 28 = 112$. What is the missing number?

- A. 57
- B. 47 ✓**
- C. 37
- D. 53

Explanation: Add known: $37 + 28 = 65$. Then $__ + 65 = 112$. Subtract: $112 - 65 = 47$.

4. Mia has 53 red beads, 39 blue beads, and 48 green beads. She uses 25 for a project. How many are left?

- A. 115 ✓**
- B. 105
- C. 125
- D. 95

Explanation: Step 1: $53 + 39 + 48$. Add $53 + 48 = 101$, then $101 + 39 = 140$. Step 2: $140 - 25 = 115$.

5. A student says $28 + 34 + 19 = 71$. Another says 81. Who is correct?

A. First student (71)

B. Second student (81) ✓

C. Neither, it is 91

D. Neither, it is 61

Explanation: $28 + 34 = 62$ ($8+4=12$, regroup). $62 + 19 = 81$ ($2+9=11$, regroup). The second student is correct with 81.

6. Which set of three numbers adds to more than 120? A) $35+42+28$ B) $46+38+41$ C) $29+33+47$ D) $31+44+36$

A. $35+42+28$

B. $46+38+41$ ✓

C. $29+33+47$

D. $31+44+36$

Explanation: A) $35+42+28 = 105$. B) $46+38+41 = 125$. C) $29+33+47 = 109$. D) $31+44+36 = 111$. Only B = 125 is greater than 120.

7. A baker made 45 muffins in the morning, 38 at noon, and 27 in the evening. He sold 65. How many are left?

A. 55

B. 45 ✓

C. 35

D. 50

Explanation: Step 1: $45 + 38 + 27$. Add $45 + 27 = 72$, then $72 + 38 = 110$. Step 2: $110 - 65 = 45$ muffins left.

8. $__ + 56 + __ = 140$, and both missing numbers are the same. What is each number?

A. 52

B. 42 ✓

C. 32

D. 46

Explanation: Let the missing number be N . $N + 56 + N = 140$. So $2N + 56 = 140$. $2N = 84$. $N = 42$. Check: $42 + 56 + 42 = 140$.

9. Three friends scored: Ana 67, Ben 45, Cal 38. What is the difference between the highest total of two friends and the lowest total of two friends?

A. 29 ✓

B. 39

C. 19

D. 49

Explanation: Highest pair: $Ana+Ben = 67+45 = 112$. Lowest pair: $Ben+Cal = 45+38 = 83$. Difference: $112 - 83 = 29$.

10. A student checked $52 + 29 + 41 = 112$. Is this correct?

A. No, it should be 122 ✓

B. Yes, 112 is correct

C. No, it should be 132

D. No, it should be 102

Explanation: $52 + 29 = 81$ ($2+9=11$, regroup). $81 + 41 = 122$ ($1+1=2$). Answer: 122. The student got 112, which is wrong.

11. Rosa picked 34 apples, 46 oranges, and 28 bananas. She gave away 50 pieces of fruit. How many are left?

A. 68

B. 58 ✓

C. 48

D. 78

Explanation: Step 1: $34 + 46 + 28$. Add $34 + 46 = 80$, then $80 + 28 = 108$. Step 2: $108 - 50 = 58$.

12. Which strategy is best for adding $63 + 17 + 25$?

A. Add left to right

B. Make a ten: $63 + 17 = 80$, then $+ 25$ ✓

C. Add all ones, then all tens

D. Round each number first

Explanation: $63 + 17$: ones $3+7=10$, making a ten! $63 + 17 = 80$. Then $80 + 25 = 105$. Making a ten first is most efficient.

13. A class needs 150 cans. Group A collected 43, Group B collected 58, Group C collected 37. Do they have enough?

- A. Yes, they have 148
- B. Yes, they have 138
- C. No, they have 138 ✓**
- D. No, they have 128

Explanation: $43 + 58 + 37$: add $43 + 37 = 80$, then $80 + 58 = 138$. $138 < 150$, so they do not have enough.

14. $25 + \underline{\quad} + 48 = 130$. What is the missing number?

- A. 67
- B. 57 ✓**
- C. 47
- D. 63

Explanation: Add known: $25 + 48 = 73$. Then $\underline{\quad} + 73 = 130$. Subtract: $130 - 73 = 57$.

15. A student added $39 + 52 + 18$ and got 99. Check the work and find the mistake.

- A. No mistake, 99 is correct
- B. Forgot to regroup, answer is 109 ✓**
- C. Added ones wrong, answer is 119
- D. Added tens wrong, answer is 89

Explanation: $39 + 52 = 91$ ($9+2=11$, regroup). $91 + 18 = 109$ ($1+8=9$). Answer: 109. The student got 99, likely forgetting to regroup.

16. Three boxes hold 55, 47, and 36 items. If 40 items are removed, how many remain?

- A. 108
- B. 98 ✓**
- C. 88
- D. 78

Explanation: Step 1: $55 + 47 + 36$. Add $55 + 36 = 91$, then $91 + 47 = 138$. Step 2: $138 - 40 = 98$.

17. Is $44 + 38 + 26$ greater or less than $29 + 55 + 31$?

- A. First is greater (108 vs 115)
- B. Second is greater (108 vs 115) ✓**
- C. They are equal
- D. First is greater (118 vs 105)

Explanation: First: $44+38+26 = 108$. Second: $29+55+31 = 115$. $115 > 108$, so the second sum is greater.

18. $\underline{\quad} + 43 + 29 = 119$. What is the missing number?

- A. 57
- B. 47 ✓**
- C. 37
- D. 53

Explanation: Add known: $43 + 29 = 72$. Then $\underline{\quad} + 72 = 119$. Subtract: $119 - 72 = 47$.

19. A zoo has 41 monkeys, 59 birds, and 33 snakes. 20 animals are moved. How many remain?

- A. 123
- B. 113 ✓**
- C. 103
- D. 93

Explanation: Step 1: $41 + 59 + 33$. Add $41 + 59 = 100$, then $100 + 33 = 133$. Step 2: $133 - 20 = 113$.

20. A student says the best way to add $76 + 24 + 53$ is to add left to right. Is there a better strategy?

- A. No, left to right is best
- B. Yes, add $76 + 24 = 100$ first ✓**
- C. Yes, add $24 + 53$ first
- D. Yes, round all numbers first

Explanation: $76 + 24 = 100$ ($6+4=10$!). Then $100 + 53 = 153$. Making a hundred first is much more efficient than adding left to right.

Lesson 4: Word Problems

Easy (20 questions)

1. Zoe has 34 crayons. She gets 25 more. How many crayons does she have now?

A. 49
B. 59 ✓
C. 69
D. 58

Explanation: This is an addition problem. $34 + 25$: ones $4+5=9$, tens $30+20=50$. Total: 59.

2. A class had 47 pencils. They used 23. How many are left?

A. 34
B. 24 ✓
C. 14
D. 23

Explanation: This is a subtraction problem. $47 - 23$: ones $7-3=4$, tens $40-20=20$. Total: 24.

3. There are 56 birds in a tree. 12 fly away. How many birds are still in the tree?

A. 54
B. 44 ✓
C. 34
D. 43

Explanation: Birds fly away means subtract. $56 - 12$: ones $6-2=4$, tens $50-10=40$. Total: 44.

4. Mia read 31 pages on Monday and 48 pages on Tuesday. How many pages did she read in all?

A. 69
B. 89
C. 79 ✓
D. 78

Explanation: How many in all means add. $31 + 48$: ones $1+8=9$, tens $30+40=70$. Total: 79.

5. A shop has 83 toys. 41 are sold. How many toys are left?

A. 52
B. 42 ✓
C. 32
D. 43

Explanation: Sold means subtract. $83 - 41$: ones $3-1=2$, tens $80-40=40$. Total: 42.

6. Sam picked 45 apples. Ava picked 32 apples. How many apples did they pick altogether?

A. 67
B. 87
C. 77 ✓
D. 78

Explanation: Altogether means add. $45 + 32$: ones $5+2=7$, tens $40+30=70$. Total: 77.

7. There were 68 fish in a pond. 24 swam away. How many fish are still in the pond?

A. 54
B. 44 ✓
C. 34
D. 43

Explanation: Swam away means subtract. $68 - 24$: ones $8-4=4$, tens $60-20=40$. Total: 44.

8. A farm has 52 cows and 36 horses. How many animals in all?

A. 78
B. 98
C. 88 ✓
D. 86

Explanation: In all means add. $52 + 36$: ones $2+6=8$, tens $50+30=80$. Total: 88.

9. Which operation do you use? 'Leo had 75 cards. He gave 33 to a friend.'

A. Addition

B. Subtraction ✓

C. Both

D. Neither

Explanation: Gave away means something is removed, so you subtract. $75 - 33 = 42$.

10. Rosa has 29 stickers. She finds 40 more. How many stickers does she have now?

A. 59

B. 79

C. 69 ✓

D. 49

Explanation: Finds more means add. $29 + 40$: ones $9+0=9$, tens $20+40=60$. Total: 69.

11. A box has 97 markers. 55 are taken out. How many are left in the box?

A. 52

B. 42 ✓

C. 32

D. 43

Explanation: Taken out means subtract. $97 - 55$: ones $7-5=2$, tens $90-50=40$. Total: 42.

12. Kai collected 43 rocks. His sister collected 35 rocks. How many rocks do they have together?

A. 68

B. 88

C. 78 ✓

D. 77

Explanation: Together means add. $43 + 35$: ones $3+5=8$, tens $40+30=70$. Total: 78.

13. Which operation do you use? 'A bakery made 84 cookies. They sold 51.'

A. Addition

B. Subtraction ✓

C. Both

D. Neither

Explanation: Sold means items are removed, so you subtract. $84 - 51 = 33$.

14. A library has 62 books on one shelf and 26 on another. How many books on both shelves?

A. 78

B. 98

C. 88 ✓

D. 86

Explanation: Both shelves means add. $62 + 26$: ones $2+6=8$, tens $60+20=80$. Total: 88.

15. There are 75 students. 43 go outside. How many are still inside?

A. 42

B. 32 ✓

C. 22

D. 33

Explanation: Go outside means subtract. $75 - 43$: ones $5-3=2$, tens $70-40=30$. Total: 32.

16. A bus had 38 passengers. At the stop, 21 more got on. How many passengers now?

A. 49

B. 69

C. 59 ✓

D. 57

Explanation: Got on means add. $38 + 21$: ones $8+1=9$, tens $30+20=50$. Total: 59.

17. Ella has 54 beads. She uses 31 for a necklace. How many beads are left?

- A. 33
- B. 23 ✓**
- C. 13
- D. 22

Explanation: Uses means subtract. $54 - 31$: ones $4-1=3$, tens $50-30=20$. Total: 23.

18. A garden has 41 roses and 37 tulips. How many flowers are there in all?

- A. 68
- B. 88
- C. 78 ✓**
- D. 77

Explanation: In all means add. $41 + 37$: ones $1+7=8$, tens $40+30=70$. Total: 78.

19. There are 86 balloons. 44 pop. How many balloons are left?

- A. 52
- B. 42 ✓**
- C. 32
- D. 43

Explanation: Pop means subtract. $86 - 44$: ones $6-4=2$, tens $80-40=40$. Total: 42.

20. Which is a correct answer? 'A store has 63 hats and 25 scarves. How many items in all?'

- A. 78
- B. 98
- C. 88 ✓**
- D. 38

Explanation: In all means add. $63 + 25$: ones $3+5=8$, tens $60+20=80$. Total: 88.

Medium (20 questions)

1. Maya has 67 stickers. She buys 45 more. How many does she have now?

- A. 102
- B. 122
- C. 112 ✓**
- D. 92

Explanation: Add $67 + 45$. Ones: $7+5=12$, regroup. Tens: $6+4+1=11$. Answer: 112.

2. A tank has 134 fish. 58 are moved to another tank. How many fish are left?

- A. 86
- B. 76 ✓**
- C. 66
- D. 82

Explanation: Subtract $134 - 58$. Regroup: $14-8=6$. Tens: $12-5=7$. Answer: 76.

3. Ben has 83 baseball cards. He gives away 47. How many does he have left?

- A. 46
- B. 36 ✓**
- C. 26
- D. 42

Explanation: Subtract $83 - 47$. Regroup: $13-7=6$. Tens: $7-4=3$. Answer: 36.

4. A store sold 78 ice cream cones in the morning and 56 in the afternoon. How many were sold in all?

- A. 124
- B. 144
- C. 134 ✓**
- D. 114

Explanation: Add $78 + 56$. Ones: $8+6=14$, regroup. Tens: $7+5+1=13$. Answer: 134.

5. There were ___ students in a gym. 38 left and now there are 57. How many were there at first?

- A. 85
- B. 105
- C. 95 ✓**
- D. 75

Explanation: Think: $__ - 38 = 57$. Add: $57 + 38 = 95$. There were 95 students at first.

6. Which operation do you use? 'A jar had some marbles. 29 were added and now there are 86.'

- A. Add $86 + 29$
- B. Subtract $86 - 29$ ✓**
- C. Add $29 + 29$
- D. Subtract $29 - 86$

Explanation: You know the total (86) and one part (29). To find the starting amount, subtract: $86 - 29 = 57$.

7. Ava has 95 beads. She uses 67 for a bracelet. How many are left?

- A. 38
- B. 28 ✓**
- C. 18
- D. 32

Explanation: Subtract $95 - 67$. Regroup: $15 - 7 = 8$. Tens: $8 - 6 = 2$. Answer: 28.

8. A school collected 89 cans one week and 74 cans the next. How many cans in all?

- A. 153
- B. 173
- C. 163 ✓**
- D. 143

Explanation: Add $89 + 74$. Ones: $9 + 4 = 13$, regroup. Tens: $8 + 7 + 1 = 16$. Answer: 163.

9. Leo had some coins. He spent 43 and has 68 left. How many did he start with?

- A. 101
- B. 121
- C. 111 ✓**
- D. 91

Explanation: Think: $__ - 43 = 68$. Add: $68 + 43 = 111$. Leo started with 111 coins.

10. Which is greater: 'a bag with $93 - 47$ marbles' or 'a bag with $85 - 36$ marbles'?

- A. First bag (46 marbles)
- B. Second bag (49 marbles) ✓**
- C. They are equal
- D. Cannot tell

Explanation: First bag: $93 - 47 = 46$. Second bag: $85 - 36 = 49$. $49 > 46$, so the second bag has more.

11. A bakery made 126 muffins. They sold 79. How many are left?

- A. 57
- B. 47 ✓**
- C. 37
- D. 53

Explanation: Subtract $126 - 79$. Regroup: $16 - 9 = 7$. Tens: $11 - 7 = 4$. Answer: 47.

12. Sam found 58 shells at the beach. His friend found 75 shells. How many shells did they find together?

- A. 123
- B. 143
- C. 133 ✓**
- D. 113

Explanation: Add $58 + 75$. Ones: $8 + 5 = 13$, regroup. Tens: $5 + 7 + 1 = 13$. Answer: 133.

13. Rosa has ___ stickers. She gets 56 more and now has 123. How many did she start with?

- A. 77
- B. 67 ✓**
- C. 57
- D. 73

Explanation: Think: $__ + 56 = 123$. Subtract: $123 - 56 = 67$. Rosa started with 67 stickers.

14. A farmer had 143 eggs. He sold 85 at the market. How many eggs does he have left?

- A. 68
- B. 58 ✓**
- C. 48
- D. 62

Explanation: Subtract $143 - 85$. Regroup: $13-5=8$. Tens: $13-8=5$. Answer: 58.

15. Two classes collect bottles: Class A gets 76, Class B gets 88. How many altogether?

- A. 154
- B. 174
- C. 164 ✓**
- D. 144

Explanation: Add $76 + 88$. Ones: $6+8=14$, regroup. Tens: $7+8+1=16$. Answer: 164.

16. Which word problem needs subtraction? A) 'How many in all?' B) 'How many are left?' C) 'How many together?' D) 'What is the total?'

- A. How many in all?
- B. How many are left? ✓**
- C. How many together?
- D. What is the total?

Explanation: 'How many are left?' means something was taken away, so you subtract. The other phrases signal addition.

17. A zoo has 97 animals. 39 are in one area. How many are in the other area?

- A. 68
- B. 58 ✓**
- C. 48
- D. 62

Explanation: Subtract $97 - 39$. Regroup: $17-9=8$. Tens: $8-3=5$. Answer: 58.

18. Kai walks 48 steps to school and 65 steps to the park. How many steps total?

- A. 103
- B. 123
- C. 113 ✓**
- D. 93

Explanation: Add $48 + 65$. Ones: $8+5=13$, regroup. Tens: $4+6+1=11$. Answer: 113.

19. A box had ___ crackers. 46 were eaten and 59 are left. How many were there at first?

- A. 95
- B. 115
- C. 105 ✓**
- D. 85

Explanation: Think: $__ - 46 = 59$. Add: $59 + 46 = 105$ crackers at first.

20. Ella scored 86 points. Ana scored 69 points. How many more points did Ella score?

- A. 27
- B. 17 ✓**
- C. 7
- D. 23

Explanation: How many more means subtract. $86 - 69$. Regroup: $16-9=7$. Tens: $7-6=1$. Answer: 17.

Hard (20 questions)

1. A store had 165 toys. They sold 78 on Monday and 49 on Tuesday. How many toys are left?

- A. 48
- B. 38 ✓**
- C. 28
- D. 42

Explanation: Step 1: $165 - 78 = 87$. Step 2: $87 - 49 = 38$ toys left.

2. A student solved this: 'Ben had 132 cards. He gave away 65 and has 77 left.' Is this correct?

- A. Yes, 77 is correct
- B. No, he should have 67 ✓**
- C. No, he should have 57
- D. No, he should have 87

Explanation: $132 - 65$: regroup ones $12 - 5 = 7$. Tens: $12 - 6 = 6$. Answer: 67. The student got 77, which is wrong by 10.

3. Mia earned 57 points in Game 1 and 84 in Game 2. She needs 150 to win a prize. Does she have enough? How many more does she need?

- A. Yes, she has enough
- B. No, she needs 9 more ✓**
- C. No, she needs 19 more
- D. No, she needs 29 more

Explanation: $57 + 84 = 141$. She needs 150. $150 - 141 = 9$ more points needed.

4. A farmer picked 96 apples and 87 oranges. He sold 135 fruits. How many does he have left?

- A. 58
- B. 48 ✓**
- C. 38
- D. 52

Explanation: Step 1: $96 + 87 = 183$ fruits total. Step 2: $183 - 135 = 48$ fruits left.

5. ___ children were at the park. 36 went home, then 28 more came. Now there are 75. How many were there at first?

- A. 73
- B. 93
- C. 83 ✓**
- D. 63

Explanation: Work backward: before 28 came, there were $75 - 28 = 47$. Before 36 left, there were $47 + 36 = 83$.

6. A student wrote: ' $147 - 89 = 68$.' Find the error and give the correct answer.

- A. No error
- B. Should be 58 ✓**
- C. Should be 78
- D. Should be 48

Explanation: $147 - 89$: regroup $17 - 9 = 8$. Tens: $13 - 8 = 5$. Answer: 58. The student got 68, subtracting 8 tens instead of 9.

7. Team A scored 76 and 68. Team B scored 89 and 52. Which team scored more total points, and by how much?

- A. Team A by 3 ✓**
- B. Team B by 3
- C. They tied
- D. Team A by 13

Explanation: Team A: $76 + 68 = 144$. Team B: $89 + 52 = 141$. Team A scored $144 - 141 = 3$ more points.

8. A library had 200 books. Monday: 43 checked out. Tuesday: 67 checked out. Wednesday: 25 returned. How many are in the library now?

- A. 125
- B. 115 ✓**
- C. 105
- D. 135

Explanation: Step 1: $200 - 43 = 157$. Step 2: $157 - 67 = 90$. Step 3: $90 + 25 = 115$.

9. A student solved $95 - 48$ this way: ' $9-4=5$, $8-5=3$, so the answer is 53.' What did the student do wrong?

A. Subtracted ones digits backward instead of regrouping ✓

B. Added instead of subtracted

C. Subtracted tens wrong

D. Nothing, 53 is correct

Explanation: The student subtracted $8-5=3$ for the ones instead of regrouping: $15-8=7$. Correct way: regroup to get $15-8=7$ in the ones and $8-4=4$ in the tens. Answer: 47, not 53.

10. A class needs 175 cans for a project. They collected 89 in Week 1 and 64 in Week 2. How many more do they need?

A. 32

B. 22 ✓

C. 12

D. 42

Explanation: Step 1: $89 + 64 = 153$ cans collected. Step 2: $175 - 153 = 22$ more needed.

11. Sam had 118 coins. He spent 45 on a toy and 36 on a book. He then found 13 coins. How many does he have?

A. 60

B. 50 ✓

C. 40

D. 55

Explanation: Step 1: $118 - 45 = 73$. Step 2: $73 - 36 = 37$. Step 3: $37 + 13 = 50$.

12. Which problem has the greatest answer? A) $143-67$ B) $98+59$ C) $175-88$ D) $76+68$

A. $143 - 67 = 76$

B. $98 + 59 = 157$ ✓

C. $175 - 88 = 87$

D. $76 + 68 = 144$

Explanation: A) 76. B) 157. C) 87. D) 144. The greatest answer is 157 from $98 + 59$.

13. A baker made 85 cookies. He sold 39 and then baked 47 more. How many does he have now?

A. 83

B. 103

C. 93 ✓

D. 73

Explanation: Step 1: $85 - 39 = 46$. Step 2: $46 + 47 = 93$ cookies.

14. Ella had some money. She spent 78 cents on a pencil and 45 cents on an eraser. She has 27 cents left. How much did she start with?

A. 140 cents

B. 150 cents ✓

C. 160 cents

D. 130 cents

Explanation: Work backward: $27 + 45 = 72$. Then $72 + 78 = 150$ cents. Ella started with 150 cents.

15. A student checked: 'There are 156 chairs. 79 are used. 77 are empty.' Is this correct?

A. Yes, 77 is correct ✓

B. No, it should be 87

C. No, it should be 67

D. No, it should be 97

Explanation: $156 - 79$: regroup $16-9=7$. Tens: $14-7=7$. Answer: 77. The student is correct!

16. Two friends share 164 stickers equally. How many does each friend get? What operation do you use?

A. 82 each, use repeated subtraction ✓

B. 74 each, use subtraction

C. 92 each, use addition

D. 164 each, no operation needed

Explanation: Equal sharing: 164 divided into 2 groups. Half of 160 = 80, half of 4 = 2. Each gets 82. This uses equal grouping.

17. A toy store received 97 dolls and 86 cars. They sold 125 total toys. How many toys are still in the store?

- A. 68
- B. 58 ✓**
- C. 48
- D. 78

Explanation: Step 1: $97 + 86 = 183$ toys received. Step 2: $183 - 125 = 58$ toys left.

18. ___ kids played outside. 47 went inside for lunch. After lunch, 33 came back out. Now 52 are outside. How many started outside?

- A. 56
- B. 76
- C. 66 ✓**
- D. 86

Explanation: Work backward: before 33 came back, $52 - 33 = 19$ were outside. Before 47 went in, $19 + 47 = 66$ started outside.

19. Ana says $87 + 65 = 142$. Ben says $87 + 65 = 152$. Check both and explain who is right.

- A. Ana (142)
- B. Ben (152) ✓**
- C. Neither, it is 162
- D. Neither, it is 132

Explanation: $87 + 65$: ones $7+5=12$, regroup. Tens: $8+6+1=15$. Answer: 152. Ben is correct. Ana forgot to add the regrouped ten.

20. A garden has 143 flowers. A storm knocks down 67. The gardener plants 45 new ones. How many flowers are there now?

- A. 131
- B. 121 ✓**
- C. 111
- D. 141

Explanation: Step 1: $143 - 67 = 76$. Step 2: $76 + 45 = 121$ flowers.

Unit Test Bank

Easy (20 questions)

1. What is $43 + 35$?

- A. 68
- B. 88
- C. 78 ✓**
- D. 79

Explanation: Add the ones: $3 + 5 = 8$. Add the tens: $40 + 30 = 70$. Then $70 + 8 = 78$.

2. What is $86 - 52$?

- A. 44
- B. 34 ✓**
- C. 24
- D. 33

Explanation: Subtract the ones: $6 - 2 = 4$. Subtract the tens: $80 - 50 = 30$. Then $30 + 4 = 34$.

3. What is $7 + 3 + 5$?

- A. 14
- B. 15 ✓**
- C. 16
- D. 13

Explanation: Make a ten: $7 + 3 = 10$. Then $10 + 5 = 15$.

4. A bag has 61 marbles. 20 are removed. How many are left?

- A. 51
- B. 41 ✓**
- C. 31
- D. 40

Explanation: Subtract 61 - 20. Ones stay: 1. Tens: $60 - 20 = 40$. Total: 41.

5. What is $72 + 16$?

- A. 78
- B. 98
- C. 88 ✓**
- D. 86

Explanation: Add the ones: $2 + 6 = 8$. Add the tens: $70 + 10 = 80$. Then $80 + 8 = 88$.

6. What is $95 - 43$?

- A. 62
- B. 52 ✓**
- C. 42
- D. 51

Explanation: Subtract the ones: $5 - 3 = 2$. Subtract the tens: $90 - 40 = 50$. Then $50 + 2 = 52$.

7. What is $8 + 2 + 4$?

- A. 12
- B. 13
- C. 14 ✓**
- D. 15

Explanation: Make a ten: $8 + 2 = 10$. Then $10 + 4 = 14$.

8. There are 54 apples and 33 oranges. How many fruits in all?

- A. 77
- B. 97
- C. 87 ✓**
- D. 88

Explanation: Add $54 + 33$. Ones: $4+3=7$. Tens: $50+30=80$. Total: 87.

9. What is $148 - 36$?

- A. 102
- B. 122
- C. 112 ✓**
- D. 132

Explanation: Subtract the ones: $8 - 6 = 2$. Subtract the tens: $140 - 30 = 110$. Then $110 + 2 = 112$.

10. Which is the correct answer for $37 + 52$?

- A. 79
- B. 99
- C. 89 ✓**
- D. 88

Explanation: Add the ones: $7 + 2 = 9$. Add the tens: $30 + 50 = 80$. Then $80 + 9 = 89$.

11. What is $20 + 30 + 40$?

- A. 80
- B. 90 ✓**
- C. 100
- D. 70

Explanation: Add the tens: $20 + 30 = 50$. Then $50 + 40 = 90$.

12. A class has 76 books. 44 are checked out. How many are on the shelf?

- A. 42
- B. 32 ✓**
- C. 22
- D. 33

Explanation: Subtract $76 - 44$. Ones: $6 - 4 = 2$. Tens: $70 - 40 = 30$. Total: 32.

13. What is $65 + 24$?

- A. 79
- B. 99
- C. 89 ✓**
- D. 88

Explanation: Add the ones: $5 + 4 = 9$. Add the tens: $60 + 20 = 80$. Then $80 + 9 = 89$.

14. Sam has 48 stickers. He gets 31 more. How many does he have?

- A. 69
- B. 89
- C. 79 ✓**
- D. 77

Explanation: Add $48 + 31$. Ones: $8 + 1 = 9$. Tens: $40 + 30 = 70$. Total: 79.

15. Which difference is greater: $67 - 23$ or $58 - 15$?

- A. $67 - 23$ ✓**
- B. $58 - 15$
- C. They are the same
- D. Cannot tell

Explanation: $67 - 23 = 44$ and $58 - 15 = 43$. Since $44 > 43$, the first difference is greater.

16. What is $6 + 4 + 9$?

- A. 17
- B. 18
- C. 19 ✓**
- D. 20

Explanation: Make a ten: $6 + 4 = 10$. Then $10 + 9 = 19$.

17. A box has 83 crayons. 51 are used. How many are left?

- A. 42
- B. 32 ✓**
- C. 22
- D. 31

Explanation: Subtract $83 - 51$. Ones: $3 - 1 = 2$. Tens: $80 - 50 = 30$. Total: 32.

18. What is $110 + 45$?

- A. 145
- B. 165
- C. 155 ✓**
- D. 150

Explanation: Add the ones: $0 + 5 = 5$. Add the tens: $110 + 40 = 150$. Then $150 + 5 = 155$.

19. There are 39 red cars and 50 blue cars in a lot. How many cars in all?

- A. 79
- B. 99
- C. 89 ✓**
- D. 88

Explanation: Add $39 + 50$. Ones: $9 + 0 = 9$. Tens: $30 + 50 = 80$. Total: 89.

20. What is $5 + 5 + 8$?

- A. 16
- B. 17
- C. 18 ✓**
- D. 19

Explanation: Add doubles: $5 + 5 = 10$. Then $10 + 8 = 18$.

Medium (20 questions)

1. $56 + 78 = ?$ Use regrouping to solve.

- A. 124
- B. 144
- C. 134 ✓**
- D. 114

Explanation: Add the ones: $6 + 8 = 14$. Write 4, regroup 1 ten. Tens: $5 + 7 + 1 = 13$. Answer: 134.

2. A pool has 152 gallons of water. 76 gallons drain out. How many gallons are left?

- A. 86
- B. 76 ✓**
- C. 66
- D. 82

Explanation: Subtract $152 - 76$. Regroup: $12 - 6 = 6$. Tens: $14 - 7 = 7$. Answer: 76.

3. Which problem needs regrouping? A) $73 - 41$ B) $85 - 27$ C) $96 - 53$ D) $64 - 32$

- A. $73 - 41$
- B. $85 - 27$ ✓**
- C. $96 - 53$
- D. $64 - 32$

Explanation: $85 - 27$: ones are $5 - 7$. Since $5 < 7$, you must regroup. The others have larger ones on top.

4. $__ + 48 = 125$. What is the missing number?

- A. 87
- B. 77 ✓**
- C. 67
- D. 83

Explanation: Subtract: $125 - 48 = 77$. Check: $77 + 48 = 125$.

5. What is $37 + 46 + 23$?

- A. 96
- B. 106 ✓**
- C. 116
- D. 86

Explanation: Make a ten: $37 + 23 = 60$ ($7 + 3 = 10$!). Then $60 + 46 = 106$.

6. A bakery made 94 cupcakes. They sold 57. How many are left?

- A. 47
- B. 37 ✓**
- C. 27
- D. 43

Explanation: Subtract $94 - 57$. Regroup: $14 - 7 = 7$. Tens: $8 - 5 = 3$. Answer: 37.

7. Kai has some cards. He gets 63 more and now has 141. How many did he start with?

- A. 88
- B. 78 ✓**
- C. 68
- D. 84

Explanation: Think: $__ + 63 = 141$. Subtract: $141 - 63 = 78$.

8. Which strategy helps solve $98 + 56$?

- A. Count back from 98
- B. Make a friendly number: $100 + 54$ ✓**
- C. Subtract 56 from 98
- D. Round both down

Explanation: 98 is close to 100. Move 2 from 56 to 98: $100 + 54 = 154$. Making a friendly number is efficient.

9. A garden has 82 tomatoes and 69 peppers. How many vegetables total?

- A. 141
- B. 161
- C. 151 ✓**
- D. 131

Explanation: Add $82 + 69$. Ones: $2+9=11$, regroup. Tens: $8+6+1=15$. Answer: 151.

10. $161 - \underline{\quad} = 84$. What is the missing number?

- A. 87
- B. 77 ✓**
- C. 67
- D. 83

Explanation: Subtract: $161 - 84 = 77$. Check: $161 - 77 = 84$.

11. Which is greater: $89 + 47$ or $75 + 58$?

- A. $89 + 47$ ✓**
- B. $75 + 58$
- C. They are equal
- D. Cannot tell

Explanation: $89 + 47 = 136$ and $75 + 58 = 133$. Since $136 > 133$, the first sum is greater.

12. A school bus has 115 seats. 68 students are seated. How many empty seats?

- A. 57
- B. 47 ✓**
- C. 37
- D. 53

Explanation: Subtract $115 - 68$. Regroup: $15-8=7$. Tens: $10-6=4$. Answer: 47.

13. What is $44 + 29 + 16$?

- A. 79
- B. 89 ✓**
- C. 99
- D. 69

Explanation: Add $44 + 16 = 60$ ($4+6=10$!). Then $60 + 29 = 89$.

14. Which problem does NOT need regrouping for addition? A) $57+36$ B) $84+49$ C) $62+25$ D) $78+43$

- A. $57 + 36$
- B. $84 + 49$
- C. $62 + 25$ ✓**
- D. $78 + 43$

Explanation: $62 + 25$: ones are $2 + 5 = 7$, no regrouping needed. All others have ones that add above 9.

15. $\underline{\quad} - 56 = 79$. What is the missing number?

- A. 125
- B. 145
- C. 135 ✓**
- D. 115

Explanation: Add: $79 + 56 = 135$. Check: $135 - 56 = 79$.

16. A farmer collected 73 eggs on Monday and 88 on Tuesday. How many altogether?

- A. 151
- B. 171
- C. 161 ✓
- D. 141

Explanation: Add $73 + 88$. Ones: $3+8=11$, regroup. Tens: $7+8+1=16$. Answer: 161.

17. Rosa had 143 beads. She used 65 for a craft. How many are left?

- A. 88
- B. 78 ✓
- C. 68
- D. 82

Explanation: Subtract $143 - 65$. Regroup: $13-5=8$. Tens: $13-6=7$. Answer: 78.

18. Which word problem needs addition? A) 'How many are left?' B) 'How many more?' C) 'What is the total?' D) 'How many fewer?'

- A. How many are left?
- B. How many more?
- C. What is the total? ✓
- D. How many fewer?

Explanation: 'What is the total?' means combining groups, which requires addition. The other phrases typically signal subtraction.

19. A toy box has 58 small toys, 31 medium toys, and 19 large toys. How many toys in all?

- A. 98
- B. 108 ✓
- C. 118
- D. 88

Explanation: Make a ten: $31 + 19 = 50$ ($1+9=10$!). Then $50 + 58 = 108$.

20. $67 + 85 = ?$

- A. 142
- B. 162
- C. 152 ✓
- D. 132

Explanation: Add the ones: $7 + 5 = 12$. Regroup 1 ten. Tens: $6 + 8 + 1 = 15$. Answer: 152.

Hard (20 questions)

1. A student solved $145 - 68 = 87$. Find the error and give the correct answer.

- A. No error, 87 is correct
- B. Should be 77 ✓
- C. Should be 67
- D. Should be 97

Explanation: $145 - 68$: regroup ones $15-8=7$. Tens: $13-6=7$. Answer: 77. The student got 87, adding 1 extra ten.

2. Ava had 178 stickers. She gave 59 to Maya and 43 to Leo. How many stickers does Ava have left?

- A. 86
- B. 76 ✓
- C. 66
- D. 82

Explanation: Step 1: $178 - 59 = 119$. Step 2: $119 - 43 = 76$ stickers left.

3. $__ + 79 = 163$. What is the missing number?

- A. 94
- B. 74
- C. 84 ✓
- D. 64

Explanation: Inverse reasoning: $163 - 79$. Regroup: $13-9=4$. Tens: $15-7=8$. Missing number: 84.

4. Ben scored 68, 75, and 49 in three games. He needs 200 total to earn a trophy. Does he have enough? How many more does he need?

- A. Yes, he has enough
- B. No, he needs 8 more ✓**
- C. No, he needs 18 more
- D. No, he needs 28 more

Explanation: $68 + 75 = 143$. $143 + 49 = 192$. He needs $200 - 192 = 8$ more points.

5. A student added $87 + 65$ and got 142. Is the student correct?

- A. No, it should be 152 ✓**
- B. Yes, 142 is correct
- C. No, it should be 162
- D. No, it should be 132

Explanation: $87 + 65$: ones $7+5=12$, regroup. Tens: $8+6+1=15$. Answer: 152. The student forgot to add the regrouped ten.

6. A bakery made 95 muffins in the morning and 78 in the afternoon. They sold 136. How many are left?

- A. 47
- B. 37 ✓**
- C. 27
- D. 42

Explanation: Step 1: $95 + 78 = 173$ muffins total. Step 2: $173 - 136 = 37$ left.

7. ___ children were on a playground. 48 went inside, then 35 came out. Now there are 62. How many started on the playground?

- A. 85
- B. 65
- C. 75 ✓**
- D. 95

Explanation: Work backward: before 35 came out, $62 - 35 = 27$ were outside. Before 48 went in, $27 + 48 = 75$.

8. Which has a greater answer: $156 - 89$ or $143 - 68$?

- A. $156 - 89 = 67$
- B. $143 - 68 = 75$ ✓**
- C. They are equal
- D. $156 - 89 = 77$

Explanation: $156 - 89 = 67$. $143 - 68 = 75$. Since $75 > 67$, the second problem has the greater answer.

9. A class collected 53 cans in Week 1, 67 in Week 2, and 44 in Week 3. They need 175. How many more do they need?

- A. 21
- B. 11 ✓**
- C. 1
- D. 31

Explanation: $53 + 67 = 120$. $120 + 44 = 164$. They need $175 - 164 = 11$ more cans.

10. Sam says $93 - 47 = 56$. Mia says it equals 46. Who is correct?

- A. Sam (56)
- B. Mia (46) ✓**
- C. Neither, it is 36
- D. Neither, it is 66

Explanation: $93 - 47$: regroup $13-7=6$. Tens: $8-4=4$. Answer: 46. Mia is correct.

11. ___ - 58 = 94. What is the missing number?

- A. 142
- B. 162
- C. 152 ✓**
- D. 132

Explanation: Inverse reasoning: $94 + 58$. Ones: $4+8=12$, regroup. Tens: $9+5+1=15$. Missing number: 152.

12. A toy store had 200 items. Monday sold 67. Tuesday sold 54. Wednesday received 38 new items. How many items now?

- A. 127
- B. 117 ✓**
- C. 107
- D. 137

Explanation: Step 1: $200 - 67 = 133$. Step 2: $133 - 54 = 79$. Step 3: $79 + 38 = 117$.

13. Which addition strategy is most efficient for $97 + 88$?

- A. Add ones then tens
- B. Make a friendly number: $100 + 85$ ✓**
- C. Count up from 97 by ones
- D. Break apart both numbers

Explanation: 97 is close to 100. Move 3 from 88 to 97: $100 + 85 = 185$. Making a friendly number avoids messy regrouping.

14. Team A: $89 + 67 = ?$ Team B: $78 + 74 = ?$ Which team has a higher total, and by how much?

- A. Team A by 4 ✓**
- B. Team B by 4
- C. They are tied
- D. Team A by 14

Explanation: Team A: $89 + 67 = 156$. Team B: $78 + 74 = 152$. Team A wins by $156 - 152 = 4$ points.

15. A student checked: 'I had 134 coins, spent 67, and have 77 left.' Is this correct?

- A. No, should have 67 left ✓**
- B. Yes, 77 is correct
- C. No, should have 57 left
- D. No, should have 87 left

Explanation: $134 - 67$: regroup $14 - 7 = 7$. Tens: $12 - 6 = 6$. Answer: 67. The student should have 67, not 77.

16. Rosa had 85 beads. She bought 47 more, then used 63 for a craft. How many does she have?

- A. 79
- B. 69 ✓**
- C. 59
- D. 89

Explanation: Step 1: $85 + 47 = 132$. Step 2: $132 - 63 = 69$ beads.

17. Three friends have 56, 43, and 38 marbles. If they combine all marbles and split equally among 3 friends, about how many does each get?

- A. About 45 each ✓**
- B. About 55 each
- C. About 35 each
- D. About 65 each

Explanation: Total: $56 + 43 + 38$. Add $56 + 38 = 94$, then $94 + 43 = 137$. Split 3 ways: 137 is close to 135, and $135/3 = 45$. Each gets about 45.

18. A farmer had ___ eggs. He sold 89 at the market and gave 34 to a neighbor. He has 52 left. How many did he start with?

- A. 165
- B. 185
- C. 175 ✓**
- D. 155

Explanation: Work backward: $52 + 34 = 86$. Then $86 + 89 = 175$. The farmer started with 175 eggs.

19. A student solved $38 + 54 + 27 = 109$. Check the work.

- A. Correct, 109 is right
- B. Wrong, should be 119 ✓**
- C. Wrong, should be 129
- D. Wrong, should be 99

Explanation: $38 + 54 = 92$ ($8 + 4 = 12$, regroup). $92 + 27 = 119$ ($2 + 7 = 9$). Answer: 119. The student got 109, off by 10.

20. A school has 186 students. 97 eat in the cafeteria. Of the rest, 45 eat outside. How many bring lunch from home and eat in their classroom?

A. 54

B. 44 ✓

C. 34

D. 48

Explanation: Step 1: $186 - 97 = 89$ do not eat in cafeteria. Step 2: $89 - 45 = 44$ eat in their classroom.

Unit 4: Subtraction

Lesson 1: Subtract Without Regrouping

Easy (20 questions)

1. What is $89 - 56$?

- A. 33 ✓
- B. 34
- C. 32
- D. 43

Explanation: Subtract the ones: $9 - 6 = 3$. Subtract the tens: $8 - 5 = 3$. The answer is 33.

2. What is $47 - 23$?

- A. 25
- B. 34
- C. 24 ✓
- D. 14

Explanation: Subtract the ones: $7 - 3 = 4$. Subtract the tens: $4 - 2 = 2$. The answer is 24.

3. What is $98 - 45$?

- A. 43
- B. 53 ✓
- C. 63
- D. 52

Explanation: Subtract the ones: $8 - 5 = 3$. Subtract the tens: $9 - 4 = 5$. The answer is 53.

4. What is $76 - 34$?

- A. 43
- B. 32
- C. 42 ✓
- D. 52

Explanation: Subtract the ones: $6 - 4 = 2$. Subtract the tens: $7 - 3 = 4$. The answer is 42.

5. What is $65 - 21$?

- A. 34
- B. 54
- C. 44 ✓
- D. 45

Explanation: Subtract the ones: $5 - 1 = 4$. Subtract the tens: $6 - 2 = 4$. The answer is 44.

6. What is $59 - 37$?

- A. 32
- B. 12
- C. 22 ✓
- D. 21

Explanation: Subtract the ones: $9 - 7 = 2$. Subtract the tens: $5 - 3 = 2$. The answer is 22.

7. What is $84 - 52$?

- A. 42
- B. 22
- C. 32 ✓
- D. 31

Explanation: Subtract the ones: $4 - 2 = 2$. Subtract the tens: $8 - 5 = 3$. The answer is 32.

8. What is $99 - 63$?

- A. 46
- B. 26
- C. 36 ✓
- D. 35

Explanation: Subtract the ones: $9 - 3 = 6$. Subtract the tens: $9 - 6 = 3$. The answer is 36.

9. Mia has 78 stickers. She gives 45 to her friend. How many stickers does Mia have left?

- A. 23
- B. 33 ✓
- C. 43
- D. 32

Explanation: Subtract the ones: $8 - 5 = 3$. Subtract the tens: $7 - 4 = 3$. Mia has 33 stickers left.

10. There are 86 crayons in a box. A teacher takes out 54. How many crayons are still in the box?

- A. 42
- B. 22
- C. 32 ✓
- D. 31

Explanation: Subtract the ones: $6 - 4 = 2$. Subtract the tens: $8 - 5 = 3$. There are 32 crayons left.

11. A farm has 67 apples. They sell 35 apples. How many apples are left?

- A. 22
- B. 42
- C. 32 ✓
- D. 31

Explanation: Subtract the ones: $7 - 5 = 2$. Subtract the tens: $6 - 3 = 3$. The farm has 32 apples left.

12. What is $57 - 24$?

- A. 43
- B. 23
- C. 33 ✓
- D. 31

Explanation: Subtract the ones: $7 - 4 = 3$. Subtract the tens: $5 - 2 = 3$. The answer is 33.

13. A parking lot has 48 cars. Then 16 cars drive away. How many cars are left?

- A. 22
- B. 42
- C. 32 ✓
- D. 34

Explanation: Subtract the ones: $8 - 6 = 2$. Subtract the tens: $4 - 1 = 3$. There are 32 cars left.

14. What is $95 - 72$?

- A. 13
- B. 33
- C. 23 ✓
- D. 22

Explanation: Subtract the ones: $5 - 2 = 3$. Subtract the tens: $9 - 7 = 2$. The answer is 23.

15. What is $79 - 48$?

- A. 21
- B. 41
- C. 31 ✓
- D. 32

Explanation: Subtract the ones: $9 - 8 = 1$. Subtract the tens: $7 - 4 = 3$. The answer is 31.

16. What is $66 - 25$?

- A. 31
- B. 41 ✓**
- C. 51
- D. 42

Explanation: Subtract the ones: $6 - 5 = 1$. Subtract the tens: $6 - 2 = 4$. The answer is 41.

17. A store has 88 hats. They sell 46. How many hats are left?

- A. 52
- B. 32
- C. 42 ✓**
- D. 44

Explanation: Subtract the ones: $8 - 6 = 2$. Subtract the tens: $8 - 4 = 4$. The store has 42 hats left.

18. What is $73 - 41$?

- A. 22
- B. 42
- C. 32 ✓**
- D. 31

Explanation: Subtract the ones: $3 - 1 = 2$. Subtract the tens: $7 - 4 = 3$. The answer is 32.

19. What is $56 - 13$?

- A. 33
- B. 53
- C. 43 ✓**
- D. 44

Explanation: Subtract the ones: $6 - 3 = 3$. Subtract the tens: $5 - 1 = 4$. The answer is 43.

20. What is $94 - 61$?

- A. 43
- B. 23
- C. 33 ✓**
- D. 32

Explanation: Subtract the ones: $4 - 1 = 3$. Subtract the tens: $9 - 6 = 3$. The answer is 33.

Medium (20 questions)

1. What is $156 - 83$?

- A. 63
- B. 73 ✓**
- C. 83
- D. 72

Explanation: Subtract ones: $6 - 3 = 3$. Subtract tens: $15 - 8 = 7$ (regroup from hundreds). The answer is 73.

2. Ben had 94 trading cards. He gave away 51. How many cards does Ben have now?

- A. 53
- B. 33
- C. 43 ✓**
- D. 45

Explanation: This is a subtraction problem. $94 - 51$: Subtract ones: $4 - 1 = 3$. Subtract tens: $9 - 5 = 4$. Ben has 43 cards left.

3. $\underline{\quad} - 32 = 45$. What is the missing number?

- A. 67
- B. 77 ✓**
- C. 87
- D. 13

Explanation: To find the missing number, add the two known numbers: $45 + 32 = 77$. Check: $77 - 32 = 45$. The missing number is 77.

4. Which subtraction problem does NOT need regrouping?

- A. $73 - 48$
- B. $85 - 37$
- C. **$96 - 42$ ✓**
- D. $61 - 29$

Explanation: Check the ones place of each problem. In $96 - 42$, the ones digit 6 is greater than 2, so no regrouping is needed. All the others have a smaller top ones digit.

5. A library has 87 books on a shelf. Students check out 34 books. How many books are still on the shelf?

- A. 43
- B. 63
- C. **53 ✓**
- D. 54

Explanation: Subtract ones: $7 - 4 = 3$. Subtract tens: $8 - 3 = 5$. There are 53 books still on the shelf.

6. What is $134 - 61$?

- A. 63
- B. **73 ✓**
- C. 83
- D. 72

Explanation: Subtract ones: $4 - 1 = 3$. Subtract tens: $13 - 6 = 7$. The answer is 73.

7. $___ - 47 = 28$. What is the missing number?

- A. 65
- B. **75 ✓**
- C. 85
- D. 19

Explanation: Add the two known numbers to find the missing number: $28 + 47 = 75$. Check: $75 - 47 = 28$. The missing number is 75.

8. There are 72 birds in a park. 39 birds fly away. Which number sentence shows how many birds are left?

- A. $72 + 39 = ?$
- B. **$72 - 39 = ?$ ✓**
- C. $39 - 72 = ?$
- D. $39 + 72 = ?$

Explanation: When birds fly away, the total goes down. This is subtraction. We start with 72 and take away 39, so the correct sentence is $72 - 39 = ?$

9. Which problem needs regrouping?

- A. $87 - 43$
- B. $79 - 56$
- C. **$65 - 38$ ✓**
- D. $98 - 64$

Explanation: Check the ones digits. In $65 - 38$, the ones digit 5 is less than 8, so you must regroup 1 ten as 10 ones before subtracting.

10. A farmer has 95 eggs. He sells 52 eggs at the market. How many eggs does he have left?

- A. 33
- B. 53
- C. **43 ✓**
- D. 42

Explanation: Subtract ones: $5 - 2 = 3$. Subtract tens: $9 - 5 = 4$. The farmer has 43 eggs left.

11. What is $148 - 75$?

- A. 63
- B. **73 ✓**
- C. 83
- D. 72

Explanation: Subtract ones: $8 - 5 = 3$. Subtract tens: $14 - 7 = 7$. The answer is 73.

12. $\underline{\quad} - 56 = 39$. What is the missing number?

- A. 85
- B. 95 ✓**
- C. 75
- D. 17

Explanation: Add the two known numbers: $39 + 56 = 95$. Check: $95 - 56 = 39$. The missing number is 95.

13. Sam has 68 marbles. He has 25 more marbles than Jill. How many marbles does Jill have?

- A. 93
- B. 33
- C. 43 ✓**
- D. 53

Explanation: Sam has MORE than Jill, so subtract to find Jill's amount: $68 - 25 = 43$. Jill has 43 marbles.

14. Look at these problems. Which one has the GREATEST difference?

- A. $89 - 45$
- B. $76 - 28$
- C. $97 - 41$ ✓**
- D. $83 - 39$

Explanation: Calculate each: $89 - 45 = 44$, $76 - 28 = 48$, $97 - 41 = 56$, $83 - 39 = 44$. The greatest difference is 56 from $97 - 41$.

15. A box holds 86 crayons. Mrs. Lee takes out some crayons. Now there are 53 crayons. How many did she take out?

- A. 23
- B. 43
- C. 33 ✓**
- D. 32

Explanation: Find how many were taken: $86 - 53 = 33$. Subtract ones: $6 - 3 = 3$. Subtract tens: $8 - 5 = 3$. Mrs. Lee took out 33 crayons.

16. What is $125 - 54$?

- A. 61
- B. 71 ✓**
- C. 81
- D. 79

Explanation: Subtract ones: $5 - 4 = 1$. Subtract tens: $12 - 5 = 7$. The answer is 71.

17. Which strategy helps you subtract 99 from a number?

- A. Add 99 instead
- B. Subtract 100, then add 1 ✓**
- C. Subtract 90, then subtract 10
- D. Count up by 10s

Explanation: Subtracting 99 is the same as subtracting 100 and then adding 1 back. This makes the mental math easier because 100 is a friendly number.

18. A garden has 74 flowers. Some are red and 38 are yellow. How many flowers are red?

- A. 46
- B. 26
- C. 36 ✓**
- D. 34

Explanation: Subtract to find the red flowers: $74 - 38 = 36$. Regroup: $14 - 8 = 6$ ones, $6 - 3 = 3$ tens. There are 36 red flowers.

19. $\underline{\quad} - 29 = 54$. What is the missing number?

- A. 73
- B. 83 ✓**
- C. 93
- D. 25

Explanation: Add the two known numbers: $54 + 29 = 83$. Check: $83 - 29 = 54$. The missing number is 83.

20. Which problem has a difference LESS than 30?

- A. $89 - 54$
- B. **$76 - 51$ ✓**
- C. $93 - 48$
- D. $67 - 29$

Explanation: Calculate each: $89 - 54 = 35$, $76 - 51 = 25$, $93 - 48 = 45$, $67 - 29 = 38$. Only $76 - 51 = 25$ is less than 30.

Hard (20 questions)

1. A toy store had 145 toys. They sold 63 on Saturday and 47 on Sunday. How many toys are left?

- A. **35 ✓**
- B. 45
- C. 25
- D. 55

Explanation: This is a two-step problem. First subtract Saturday sales: $145 - 63 = 82$. Then subtract Sunday sales: $82 - 47 = 35$. The store has 35 toys left.

2. Maya solved $86 - 49 = 43$. What did Maya do wrong?

- A. **She forgot to regroup and subtracted $9 - 6$ instead ✓**
- B. She added instead of subtracted
- C. She subtracted the tens wrong
- D. Her answer is correct

Explanation: The correct answer is $86 - 49 = 37$. Maya got 43 because she subtracted $9 - 6 = 3$ in the ones place instead of regrouping: $16 - 9 = 7$. Always regroup when the top ones digit is smaller.

3. Without solving, is $93 - 47$ greater than or less than 50?

- A. **Less than 50, because $90 - 50 = 40$ ✓**
- B. Greater than 50, because 93 is big
- C. Equal to 50
- D. Cannot tell without solving

Explanation: Estimate with friendly numbers: $90 - 50 = 40$. Since $93 > 90$ and $47 < 50$, the real answer is a bit more than 40, but still less than 50. The exact answer is 46.

4. $__ - __ = 27$. Both numbers are between 50 and 90. Which pair works?

- A. **$83 - 56$ ✓**
- B. $72 - 56$
- C. $95 - 68$
- D. $75 - 52$

Explanation: Check each: $83 - 56 = 27$ and both 83 and 56 are between 50 and 90. $72 - 56 = 16$. $95 - 68 = 27$, but 95 is above 90. $75 - 52 = 23$. Only $83 - 56$ works.

5. Leo says 'When I subtract, I always get a smaller number.' Is Leo correct?

- A. Yes, subtraction always gives smaller numbers
- B. **No, if you subtract 0 the number stays the same ✓**
- C. Yes, but only with regrouping
- D. No, subtraction can give bigger numbers

Explanation: If you subtract 0 from any number, the result is the same number, not smaller. For example, $75 - 0 = 75$. So Leo is not always correct.

6. A class had 132 pencils on Monday. They used 45 on Tuesday and got 28 new ones on Wednesday. How many pencils do they have now?

- A. 105
- B. **115 ✓**
- C. 125
- D. 95

Explanation: Step 1: $132 - 45 = 87$ pencils after Tuesday. Step 2: $87 + 28 = 115$ pencils after Wednesday. They have 115 pencils.

7. Kim solved $74 - 38 = 44$. What mistake did Kim make?

- A. Kim subtracted 8 - 4 in the ones instead of regrouping ✓
- B. Kim added instead of subtracted
- C. Kim forgot to subtract the tens
- D. Kim's answer is correct

Explanation: The correct answer is $74 - 38 = 36$. Kim got 44 because she did $8 - 4 = 4$ in the ones instead of regrouping: $14 - 8 = 6$. This common error happens when students subtract the smaller digit from the larger one in the wrong direction.

8. Without solving exactly, is $87 - 39$ closer to 40, 50, or 60?

- A. Closer to 40
- B. Closer to 50 ✓
- C. Closer to 60
- D. Exactly 50

Explanation: Round to friendly numbers: 87 rounds to 90, 39 rounds to 40. $90 - 40 = 50$. The exact answer is 48, which is closest to 50.

9. A bakery made 148 cookies. They sold 59 in the morning and 64 in the afternoon. How many cookies are left?

- A. 35
- B. 25 ✓
- C. 15
- D. 45

Explanation: Step 1: $148 - 59 = 89$ cookies after morning. Step 2: $89 - 64 = 25$ cookies left after afternoon. The bakery has 25 cookies left.

10. Ryan says $73 - 28 = 55$. Use addition to check. Is Ryan correct?

- A. Yes, because $55 + 28 = 73$
- B. No, because $55 + 28 = 83$, not 73 ✓
- C. Yes, because $28 + 55 = 73$
- D. No, because you cannot check with addition

Explanation: Check by adding: $55 + 28 = 83$, not 73. So Ryan is wrong. The correct answer is $73 - 28 = 45$. Check: $45 + 28 = 73$.

11. A number minus 37 equals 46. What is 10 more than that number?

- A. 83
- B. 93 ✓
- C. 73
- D. 103

Explanation: First find the number: $? - 37 = 46$, so $? = 46 + 37 = 83$. Then find 10 more: $83 + 10 = 93$.

12. Which problem will have the SMALLEST difference?

- A. $99 - 87$ ✓
- B. $85 - 69$
- C. $73 - 58$
- D. $91 - 76$

Explanation: Calculate each: $99 - 87 = 12$, $85 - 69 = 16$, $73 - 58 = 15$, $91 - 76 = 15$. The smallest difference is 12 from $99 - 87$.

13. Sam had some stickers. He gave 34 to Mia and 28 to Jake. Now he has 19 left. How many stickers did Sam start with?

- A. 71
- B. 81 ✓
- C. 91
- D. 61

Explanation: Work backwards: Sam has 19 left. Add back what he gave: $19 + 34 = 53$, then $53 + 28 = 81$. Sam started with 81 stickers.

14. Ali says 'Subtracting 48 is the same as subtracting 50 and adding 2.' Is Ali correct?

A. Yes, because 50 is 2 more than 48, so you add 2 back ✓

B. No, you should subtract 50 and subtract 2

C. Yes, but only for even numbers

D. No, that only works with addition

Explanation: Ali is correct! Subtracting 48 = subtracting 50 (which removes 2 too many) then adding 2 back. This is a compensation strategy. Example: $75 - 48 = 75 - 50 + 2 = 27$.

15. Tina says the difference between 86 and 43 is the same as the difference between 88 and 45. Is she right?

A. Yes, because adding 2 to both numbers keeps the difference the same ✓

B. No, the first difference is bigger

C. No, the second difference is bigger

D. Yes, but only by coincidence

Explanation: $86 - 43 = 43$ and $88 - 45 = 43$. When you add the same amount to both numbers, the difference stays the same. This is the constant difference property.

16. A school has 156 students. 78 go to recess, then 23 more go outside. How many students are still inside?

A. 45

B. 55 ✓

C. 65

D. 75

Explanation: Step 1: $156 - 78 = 78$ inside. Step 2: $78 - 23 = 55$ still inside after the second group leaves.

17. Jake solved $92 - 57$ and got 45. He checked: $45 + 57 = 102$. What should Jake conclude?

A. His answer is correct

B. His answer is wrong because $45 + 57$ should equal 92, not 102 ✓

C. He should try a different checking method

D. Addition cannot check subtraction

Explanation: When checking subtraction with addition, the sum should equal the starting number. $45 + 57 = 102$, not 92, so Jake is wrong. Correct: $92 - 57 = 35$. Check: $35 + 57 = 92$.

18. There are 134 pages in a book. Layla read 48 pages Monday and 57 pages Tuesday. How many pages are left?

A. 29 ✓

B. 39

C. 19

D. 49

Explanation: Step 1: $134 - 48 = 86$ pages left after Monday. Step 2: $86 - 57 = 29$ pages left after Tuesday.

19. Why do $100 - 64$ and $99 - 63$ give the same answer?

A. Subtracting 1 from both numbers keeps the difference the same ✓

B. They do not give the same answer

C. It is a coincidence

D. Because 100 and 99 are close

Explanation: $100 - 64 = 36$ and $99 - 63 = 36$. When you subtract the same amount from both numbers, the difference stays the same. This is the constant difference property.

20. Ella has 127 beads. She uses 39 for a necklace and 45 for a bracelet. She finds 12 more beads. How many does she have now?

A. 55 ✓

B. 45

C. 65

D. 35

Explanation: Step 1: $127 - 39 = 88$. Step 2: $88 - 45 = 43$. Step 3: $43 + 12 = 55$. Ella has 55 beads now.

Lesson 2: Subtract With Regrouping

Easy (20 questions)

1. What is $42 - 17$?

- A. 35
- B. 25 ✓**
- C. 15
- D. 24

Explanation: The ones digit 2 is less than 7, so regroup: take 1 ten and add it to the ones. Now subtract $12 - 7 = 5$ ones. Subtract tens: $3 - 1 = 2$. The answer is 25.

2. What is $53 - 28$?

- A. 35
- B. 15
- C. 25 ✓**
- D. 24

Explanation: Regroup: $3 < 8$, so take 1 ten. $13 - 8 = 5$ ones. $4 - 2 = 2$ tens. The answer is 25.

3. What is $61 - 34$?

- A. 37
- B. 17
- C. 27 ✓**
- D. 26

Explanation: Regroup: $1 < 4$, so take 1 ten. $11 - 4 = 7$ ones. $5 - 3 = 2$ tens. The answer is 27.

4. What is $75 - 49$?

- A. 36
- B. 16
- C. 26 ✓**
- D. 24

Explanation: Regroup: $5 < 9$, so take 1 ten. $15 - 9 = 6$ ones. $6 - 4 = 2$ tens. The answer is 26.

5. What is $83 - 56$?

- A. 37
- B. 17
- C. 27 ✓**
- D. 28

Explanation: Regroup: $3 < 6$, so take 1 ten. $13 - 6 = 7$ ones. $7 - 5 = 2$ tens. The answer is 27.

6. A jar has 51 marbles. You take out 24. How many marbles are left?

- A. 37
- B. 17
- C. 27 ✓**
- D. 26

Explanation: Regroup: $1 < 4$, so take 1 ten. $11 - 4 = 7$ ones. $4 - 2 = 2$ tens. There are 27 marbles left.

7. What is $94 - 38$?

- A. 66
- B. 46
- C. 56 ✓**
- D. 54

Explanation: Regroup: $4 < 8$, so take 1 ten. $14 - 8 = 6$ ones. $8 - 3 = 5$ tens. The answer is 56.

8. What is $36 - 19$?

- A. 27
- B. 7
- C. 17 ✓**
- D. 18

Explanation: Regroup: $6 < 9$, so take 1 ten. $16 - 9 = 7$ ones. $2 - 1 = 1$ ten. The answer is 17.

9. There are 72 kids at a party. 35 kids go home. How many kids are still at the party?

- A. 47
- B. 27
- C. 37 ✓
- D. 36

Explanation: Regroup: $2 < 5$, so take 1 ten. $12 - 5 = 7$ ones. $6 - 3 = 3$ tens. There are 37 kids still at the party.

10. What is $60 - 23$?

- A. 47
- B. 27
- C. 37 ✓
- D. 36

Explanation: Regroup: $0 < 3$, so take 1 ten. $10 - 3 = 7$ ones. $5 - 2 = 3$ tens. The answer is 37.

11. What is $85 - 47$?

- A. 48
- B. 28
- C. 38 ✓
- D. 39

Explanation: Regroup: $5 < 7$, so take 1 ten. $15 - 7 = 8$ ones. $7 - 4 = 3$ tens. The answer is 38.

12. What is $44 - 18$?

- A. 36
- B. 16
- C. 26 ✓
- D. 22

Explanation: Regroup: $4 < 8$, so take 1 ten. $14 - 8 = 6$ ones. $3 - 1 = 2$ tens. The answer is 26.

13. A shelf has 63 books. A student returns 0 books and checks out 27. How many books are on the shelf now?

- A. 46
- B. 26
- C. 36 ✓
- D. 34

Explanation: Regroup: $3 < 7$, so take 1 ten. $13 - 7 = 6$ ones. $5 - 2 = 3$ tens. There are 36 books on the shelf.

14. What is $91 - 55$?

- A. 46
- B. 26
- C. 36 ✓
- D. 34

Explanation: Regroup: $1 < 5$, so take 1 ten. $11 - 5 = 6$ ones. $8 - 5 = 3$ tens. The answer is 36.

15. What is $50 - 16$?

- A. 44
- B. 24
- C. 34 ✓
- D. 36

Explanation: Regroup: $0 < 6$, so take 1 ten. $10 - 6 = 4$ ones. $4 - 1 = 3$ tens. The answer is 34.

16. What is $77 - 39$?

- A. 48
- B. 28
- C. 38 ✓
- D. 42

Explanation: Regroup: $7 < 9$, so take 1 ten. $17 - 9 = 8$ ones. $6 - 3 = 3$ tens. The answer is 38.

17. A bag has 82 gummy bears. You eat 45 of them. How many gummy bears are left?

- A. 47
- B. 27
- C. 37 ✓
- D. 35

Explanation: Regroup: $2 < 5$, so take 1 ten. $12 - 5 = 7$ ones. $7 - 4 = 3$ tens. There are 37 gummy bears left.

18. What is $66 - 29$?

- A. 47
- B. 27
- C. 37 ✓
- D. 33

Explanation: Regroup: $6 < 9$, so take 1 ten. $16 - 9 = 7$ ones. $5 - 2 = 3$ tens. The answer is 37.

19. What is $58 - 39$?

- A. 29
- B. 9
- C. 19 ✓
- D. 21

Explanation: Regroup: $8 < 9$, so take 1 ten. $18 - 9 = 9$ ones. $4 - 3 = 1$ ten. The answer is 19.

20. What is $73 - 46$?

- A. 37
- B. 17
- C. 27 ✓
- D. 29

Explanation: Regroup: $3 < 6$, so take 1 ten. $13 - 6 = 7$ ones. $6 - 4 = 2$ tens. The answer is 27.

Medium (20 questions)

1. What is $134 - 68$?

- A. 56
- B. 76
- C. 66 ✓
- D. 72

Explanation: Regroup ones: $4 < 8$, so take 1 ten. $14 - 8 = 6$ ones. Regroup tens: $2 < 6$, so take 1 hundred. $12 - 6 = 6$ tens. The answer is 66.

2. Sara had 82 books. She gave away 37. How many books does Sara have left?

- A. 55
- B. 35
- C. 45 ✓
- D. 43

Explanation: Regroup: $2 < 7$, so take 1 ten. $12 - 7 = 5$ ones. $7 - 3 = 4$ tens. Sara has 45 books left.

3. $__ - 45 = 38$. What is the missing number?

- A. 73
- B. 83 ✓
- C. 93
- D. 7

Explanation: Add the two known numbers: $38 + 45 = 83$. Check: $83 - 45 = 38$. The missing number is 83.

4. Which problem needs regrouping? A) $86 - 42$ B) $73 - 48$ C) $99 - 53$ D) $68 - 25$

- A. $86 - 42$
- B. $73 - 48$ ✓
- C. $99 - 53$
- D. $68 - 25$

Explanation: Check the ones: A) $6 > 2$ (no), B) $3 < 8$ (yes!), C) $9 > 3$ (no), D) $8 > 5$ (no). Only $73 - 48$ needs regrouping because the top ones digit is smaller.

5. What is $152 - 87$?

- A. 55
- B. 75
- C. 65 ✓
- D. 67

Explanation: Regroup ones: $2 < 7$, so $12 - 7 = 5$ ones. Regroup tens: $4 < 8$, so $14 - 8 = 6$ tens. The answer is 65.

6. A store has 91 balls. They sell 56 balls. How many are left?

- A. 45
- B. 25
- C. 35 ✓
- D. 33

Explanation: Regroup: $1 < 6$, so take 1 ten. $11 - 6 = 5$ ones. $8 - 5 = 3$ tens. The store has 35 balls left.

7. $__ - 59 = 34$. What is the missing number?

- A. 83
- B. 93 ✓
- C. 103
- D. 25

Explanation: Add the two known numbers: $34 + 59 = 93$. Check: $93 - 59 = 34$. The missing number is 93.

8. There are 80 balloons at a party. 36 pop. Which strategy best solves $80 - 36$?

- A. Count back by ones from 80
- B. Regroup: $0 < 6$, so change 1 ten into 10 ones ✓
- C. Add $80 + 36$
- D. Subtract tens only: $80 - 30 = 50$

Explanation: Since the ones digit 0 is less than 6, you need to regroup 1 ten into 10 ones. Then $10 - 6 = 4$ ones and $7 - 3 = 4$ tens. $80 - 36 = 44$.

9. Kayla has 75 stickers. She has 28 more than David. How many stickers does David have?

- A. 103
- B. 57
- C. 47 ✓
- D. 43

Explanation: Kayla has MORE, so subtract to find David's amount: $75 - 28 = 47$. Regroup: $15 - 8 = 7$ ones, $6 - 2 = 4$ tens. David has 47 stickers.

10. Look at these: $62 - 38$ and $64 - 40$. Do they give the same answer?

- A. Yes, both equal 24
- B. No, $62 - 38 = 24$ but $64 - 40 = 24$
- C. Yes, both equal 24 because adding 2 to each number keeps the difference the same ✓
- D. No, different numbers always give different answers

Explanation: $62 - 38 = 24$ and $64 - 40 = 24$. Adding the same amount (2) to both numbers keeps the difference the same. This trick makes subtraction easier with friendly numbers!

11. What is $143 - 78$?

- A. 55
- B. 75
- C. 65 ✓
- D. 63

Explanation: Regroup ones: $3 < 8$, so $13 - 8 = 5$ ones. Regroup tens: $3 < 7$, so $13 - 7 = 6$ tens. The answer is 65.

12. A toy box has some cars. After taking out 47, there are 36 left. How many cars were in the box?

- A. 73
- B. 83 ✓
- C. 93
- D. 11

Explanation: Work backwards with addition: $36 + 47 = 83$. There were 83 cars in the box. Check: $83 - 47 = 36$.

13. Which problem does NOT need regrouping?

- A. $54 - 29$
- B. $70 - 43$
- C. **$87 - 52$ ✓**
- D. $63 - 38$

Explanation: Check ones digits: A) $4 < 9$, B) $0 < 3$, C) $7 > 2$, D) $3 < 8$. Only $87 - 52$ does not need regrouping because 7 is greater than 2.

14. $\underline{\quad} - 36 = 47$. What is the missing number?

- A. 73
- B. **83 ✓**
- C. 93
- D. 11

Explanation: Add: $47 + 36 = 83$. Check: $83 - 36 = 47$. The missing number is 83.

15. A garden has 93 flowers. Deer eat 45 flowers and wind knocks down 0. How many flowers are standing?

- A. 58
- B. 38
- C. **48 ✓**
- D. 52

Explanation: $93 - 45 = 48$. Regroup: $13 - 5 = 8$ ones, $8 - 4 = 4$ tens. Subtracting 0 more changes nothing. There are 48 flowers standing.

16. Which of these has the GREATEST difference?

- A. $81 - 47$
- B. **$72 - 35$ ✓**
- C. $93 - 58$
- D. $64 - 29$

Explanation: Calculate: $81 - 47 = 34$, $72 - 35 = 37$, $93 - 58 = 35$, $64 - 29 = 35$. The greatest difference is 37 from $72 - 35$.

17. What is the best way to subtract 39 from 75?

- A. **Subtract 40, then add 1 ✓**
- B. Subtract 30, then subtract 10
- C. Count back by ones
- D. Add $39 + 75$

Explanation: Subtract 40 (a friendly number) then add 1 back because you subtracted 1 too many. $75 - 40 = 35$, then $35 + 1 = 36$. This compensation strategy is faster than regrouping.

18. A bus had 64 passengers. At the first stop, 28 get off. How many passengers are on the bus?

- A. 46
- B. 26
- C. **36 ✓**
- D. 32

Explanation: Regroup: $4 < 8$, so $14 - 8 = 6$ ones. $5 - 2 = 3$ tens. There are 36 passengers on the bus.

19. $\underline{\quad} - 67 = 29$. What is the missing number?

- A. 86
- B. **96 ✓**
- C. 106
- D. 38

Explanation: Add: $29 + 67 = 96$. Check: $96 - 67 = 29$. The missing number is 96.

20. Which pair of problems has the same answer?

- A. $83 - 47$ and $85 - 49$
- B. $91 - 56$ and $90 - 54$
- C. **$72 - 38$ and $74 - 40$ ✓**
- D. $65 - 29$ and $63 - 27$

Explanation: $72 - 38 = 34$ and $74 - 40 = 34$. Adding 2 to both numbers gives the same difference. Check others: $83 - 47 = 36$ vs $85 - 49 = 36$ (also same!). But $74 - 40$ is the clearest example of the constant difference strategy.

Hard (20 questions)

1. A store had 150 toys. They sold 67 on Monday and 48 on Tuesday. How many toys are left?

A. 35 ✓

B. 45

C. 25

D. 55

Explanation: Step 1: $150 - 67 = 83$ (regroup: $10 - 7 = 3$, $14 - 6 = 8$). Step 2: $83 - 48 = 35$ (regroup: $13 - 8 = 5$, $7 - 4 = 3$). The store has 35 toys left.

2. Jake solved $83 - 47 = 44$. What did Jake do wrong?

A. Jake subtracted 7 - 3 in the ones instead of regrouping ✓

B. Jake added instead of subtracting

C. Jake regrouped twice

D. Jake's answer is correct

Explanation: Correct: $83 - 47 = 36$. Jake got 44 because he did $7 - 3 = 4$ in the ones (subtracting upward) instead of regrouping: $13 - 7 = 6$ ones, $7 - 4 = 3$ tens.

3. Is $142 - 78$ greater or less than 60? How can you tell without solving?

A. Greater, because $140 - 80 = 60$ and $142 > 140$ while $78 < 80$ ✓

B. Less, because both numbers have regrouping

C. Exactly 60

D. Cannot tell without solving

Explanation: Estimate: $140 - 80 = 60$. Since 142 is 2 more than 140 and 78 is 2 less than 80, the actual answer is $60 + 2 + 2 = 64$. So it is greater than 60. Estimation lets you compare without finding the exact answer.

4. $\underline{\quad} - \underline{\quad} = 35$. Both numbers are between 50 and 100. Which pair works?

A. $92 - 57$

B. $88 - 53$ ✓

C. $76 - 49$

D. $105 - 70$

Explanation: Check: $92 - 57 = 35$ and both in range. $88 - 53 = 35$ and both in range. $76 - 49 = 27$. $105 - 70 = 35$ but $105 > 100$. Both A and B work, but $88 - 53 = 35$ is the answer at position B.

5. Maria solved $120 - 53 = 77$. She checked: $77 + 53 = 130$. What went wrong?

A. Maria's subtraction is wrong because $77 + 53 = 130$, not 120 ✓

B. Maria added wrong during checking

C. Maria's answer is actually correct

D. The check does not work for 3-digit numbers

Explanation: If $120 - 53 = 77$ were correct, then $77 + 53$ should equal 120. But $77 + 53 = 130$, not 120. Maria's subtraction is wrong. Correct: $120 - 53 = 67$. Check: $67 + 53 = 120$.

6. A farmer had 163 apples. He sold 78 on Saturday and 49 on Sunday. How many does he have left?

A. 36 ✓

B. 46

C. 26

D. 56

Explanation: Step 1: $163 - 78 = 85$. Step 2: $85 - 49 = 36$. The farmer has 36 apples left.

7. Eli says 'To subtract 98 from any number, I subtract 100 and add 2.' Is Eli's strategy correct?

A. Yes, because $100 - 98 = 2$, so adding 2 corrects the extra taken away ✓

B. No, you should subtract 100 and subtract 2

C. Yes, but only when the number is greater than 100

D. No, compensation only works for addition

Explanation: Eli is correct! Subtracting 100 removes 2 too many (since $100 = 98 + 2$). Adding 2 back gives the right answer. Example: $156 - 98 = 156 - 100 + 2 = 58$.

8. Without solving, which has the bigger answer: $131 - 64$ or $128 - 59$?

A. $131 - 64$, because $130 - 60 = 70$ is greater than $130 - 60 = 70$

B. $128 - 59$, because the numbers are closer together

C. They are about the same ✓

D. $131 - 64$, because 131 is the biggest number

Explanation: Estimate: $131 - 64$ is about $130 - 65 = 65$. $128 - 59$ is about $130 - 60 = 70$. Actually exact: $131 - 64 = 67$ and $128 - 59 = 69$. They are close but $128 - 59$ is slightly bigger. The estimates show they are about the same.

9. Zoe had some stickers. She gave 45 to her sister and 38 to her friend. She has 22 left. How many did she start with?

A. 95

B. 105 ✓

C. 115

D. 85

Explanation: Work backwards: $22 + 38 = 60$, then $60 + 45 = 105$. Zoe started with 105 stickers. Check: $105 - 45 - 38 = 22$.

10. Tim says $91 - 45 = 56$. Amy says $91 - 45 = 46$. Who is correct and why?

A. Tim, because $56 + 45 = 101$

B. Amy, because $46 + 45 = 91$ ✓

C. Tim, because $9 - 4 = 5$ and $1 - 5 = 6$

D. Neither is correct

Explanation: Check with addition: $46 + 45 = 91$. Amy is correct! Tim likely forgot to reduce the tens after regrouping: $11 - 5 = 6$ ones, but then $8 - 4 = 4$ tens (not 5 tens).

11. A number minus 48 equals 57. What is the number minus 50?

A. 55 ✓

B. 57

C. 53

D. 59

Explanation: Find the number: $? - 48 = 57$, so $? = 57 + 48 = 105$. Now subtract 50: $105 - 50 = 55$. Or use reasoning: subtracting 50 is 2 more than subtracting 48, so the answer is $57 - 2 = 55$.

12. Which subtraction problem requires regrouping TWICE (in ones AND tens)?

A. $145 - 32$

B. $132 - 78$ ✓

C. $156 - 43$

D. $189 - 65$

Explanation: Check: $132 - 78$: ones: $2 < 8$ (regroup), tens: $2 < 7$ after regrouping (regroup again). This needs regrouping in both the ones and the tens place. The others need at most one regrouping.

13. A library had 175 books. Students returned 34 books but checked out 89 books. How many books does the library have now?

A. 110

B. 120 ✓

C. 130

D. 100

Explanation: Step 1: Add returned books: $175 + 34 = 209$. Step 2: Subtract checked out: $209 - 89 = 120$. The library has 120 books.

14. Dani says 'Regrouping changes the value of the number.' Is Dani correct?

A. No, regrouping changes the form but not the value ✓

B. Yes, the number gets smaller

C. Yes, you lose a ten

D. No, but only in addition

Explanation: Regrouping does NOT change the value. It changes the form: 1 ten becomes 10 ones, but $10 \text{ ones} = 1 \text{ ten}$. So the total value stays the same. It is like exchanging a dime for 10 pennies -- same amount, different form.

15. A school collected 156 cans in Week 1, then 89 cans in Week 2. They donated 118 cans. How many cans do they still have?

- A. 117
- B. 127 ✓**
- C. 137
- D. 107

Explanation: Step 1: Total collected: $156 + 89 = 245$. Step 2: Subtract donated: $245 - 118 = 127$. They have 127 cans left.

16. Look at this pattern: $82 - 46 = 36$, $83 - 47 = 36$, $84 - 48 = 36$. Why does the answer stay the same?

- A. Adding 1 to both numbers keeps the difference the same ✓**
- B. All the answers just happen to be 36
- C. The ones digits cancel out
- D. Regrouping makes them all equal

Explanation: When you add the same number to both parts of a subtraction, the difference stays the same. This is the constant difference property. $82 - 46 = 36$, and adding 1 to each: $83 - 47 = 36$, $84 - 48 = 36$.

17. Nora needs to solve $103 - 67$. She changes it to $106 - 70$ to make it easier. Does this give the right answer?

- A. Yes, adding 3 to both numbers keeps the same difference ✓**
- B. No, you can only change one number
- C. Yes, but only because 103 and 106 are close
- D. No, $106 - 70 = 36$ but $103 - 67 = 46$

Explanation: $103 - 67 = 36$ and $106 - 70 = 36$. Adding 3 to both numbers keeps the difference the same. This is a smart strategy to avoid regrouping!

18. A pet store had 142 fish. They sold some on Monday. Then they sold 35 on Tuesday. Now they have 58 fish. How many did they sell on Monday?

- A. 39
- B. 49 ✓**
- C. 59
- D. 29

Explanation: Work backwards: Before Tuesday sales: $58 + 35 = 93$ fish. Monday sales: $142 - 93 = 49$. They sold 49 fish on Monday.

19. Liam solved $135 - 79$ like this: $135 - 80 = 55$, then $55 + 1 = 56$. Is his strategy and answer correct?

- A. Yes, he used compensation correctly ✓**
- B. No, he should have subtracted 1 instead of adding 1
- C. Yes, but the math is wrong
- D. No, compensation does not work here

Explanation: Liam subtracted 80 instead of 79, which is 1 too many. So he adds 1 back: $55 + 1 = 56$. Check: $135 - 79 = 56$. His strategy and answer are both correct!

20. An art class had 168 sheets of paper. They used 74 for painting and 56 for drawing. How many sheets are left?

- A. 38 ✓**
- B. 48
- C. 28
- D. 58

Explanation: Step 1: $168 - 74 = 94$ sheets after painting. Step 2: $94 - 56 = 38$ sheets after drawing. There are 38 sheets left.

Lesson 3: Check Your Work

Easy (20 questions)

1. $45 - 18 = 27$. Check: $27 + 18 = ?$

- A. 45 ✓
- B. 35
- C. 55
- D. 43

Explanation: To check subtraction, add the answer to the number you subtracted. $27 + 18 = 45$. Since this matches the starting number, the subtraction is correct.

2. $63 - 29 = 34$. Is this correct? Check with addition.

- A. Yes, $34 + 29 = 63$ ✓
- B. No, $34 + 29 = 53$
- C. No, $34 + 29 = 73$
- D. Yes, $34 + 29 = 64$

Explanation: Check: $34 + 29 = 63$. Since 63 matches the starting number, the answer 34 is correct.

3. $82 - 45 = 37$. Check by adding. $37 + 45 = ?$

- A. 72
- B. 82 ✓
- C. 92
- D. 87

Explanation: $37 + 45 = 82$. This matches the starting number, so $82 - 45 = 37$ is correct.

4. $56 - 23 = 33$. Is this correct?

- A. Yes, $33 + 23 = 56$ ✓
- B. No, the answer should be 43
- C. No, the answer should be 23
- D. Yes, $33 + 23 = 66$

Explanation: Check: $33 + 23 = 56$. This matches, so $56 - 23 = 33$ is correct.

5. $71 - 38 = 33$. Check: $33 + 38 = ?$

- A. 61
- B. 71 ✓
- C. 81
- D. 69

Explanation: $33 + 38 = 71$. This matches the starting number, so $71 - 38 = 33$ is correct.

6. How can you check if $94 - 51 = 43$ is correct?

- A. Add $43 + 51$ and see if it equals 94 ✓
- B. Subtract $43 - 51$
- C. Add $94 + 51$
- D. Subtract $94 - 43 - 51$

Explanation: To check subtraction, add the answer (43) to the number subtracted (51). If the sum equals the starting number (94), the subtraction is correct. $43 + 51 = 94$. Correct!

7. $87 - 52 = 35$. Check: $35 + 52 = ?$

- A. 77
- B. 87 ✓
- C. 97
- D. 85

Explanation: $35 + 52 = 87$. This matches the starting number, confirming that $87 - 52 = 35$ is correct.

8. A student says $76 - 34 = 42$. Check with addition. Is the student correct?

- A. Yes, $42 + 34 = 76$ ✓
- B. No, $42 + 34 = 86$
- C. No, $42 + 34 = 66$
- D. Yes, $42 + 34 = 78$

Explanation: Check: $42 + 34 = 76$. This matches, so the student is correct: $76 - 34 = 42$.

9. What is the best way to check $68 - 25 = 43$?

A. Add: $43 + 25$ ✓

B. Subtract again: $68 - 25$

C. Multiply: 43×25

D. Add: $68 + 25$

Explanation: The best way to check subtraction is to add the answer to the number subtracted: $43 + 25 = 68$. Since it matches, the answer is correct.

10. $55 - 28 = 27$. Check: $27 + 28 = ?$

A. 45

B. 55 ✓

C. 65

D. 53

Explanation: $27 + 28 = 55$. This matches the starting number, so $55 - 28 = 27$ is correct.

11. $93 - 47 = 46$. Is this correct? Check with addition.

A. Yes, $46 + 47 = 93$ ✓

B. No, $46 + 47 = 83$

C. No, the answer is 56

D. Yes, $46 + 47 = 94$

Explanation: Check: $46 + 47 = 93$. This matches, so $93 - 47 = 46$ is correct.

12. A student wrote $84 - 39 = 55$. Check: $55 + 39 = ?$

A. 84

B. 94 ✓

C. 74

D. 85

Explanation: $55 + 39 = 94$, not 84. Since 94 does not match 84, the student made an error. The correct answer is $84 - 39 = 45$.

13. Which addition problem checks $72 - 36 = 36$?

A. $36 + 36 = 72$ ✓

B. $72 + 36 = 108$

C. $36 - 36 = 0$

D. $72 - 36 = 36$

Explanation: To check subtraction, add the answer to the number subtracted. $36 + 36 = 72$. Since it matches, $72 - 36 = 36$ is correct.

14. $48 - 19 = 29$. Check: $29 + 19 = ?$

A. 38

B. 48 ✓

C. 58

D. 47

Explanation: $29 + 19 = 48$. This matches the starting number, so $48 - 19 = 29$ is correct.

15. A student says $65 - 28 = 47$. Is this correct?

A. No, $47 + 28 = 75$, not 65 ✓

B. Yes, $47 + 28 = 65$

C. No, $47 + 28 = 65$

D. Yes, the answer is 47

Explanation: Check: $47 + 28 = 75$, not 65. The student is wrong. Correct: $65 - 28 = 37$. Check: $37 + 28 = 65$.

16. Which shows the correct way to check $59 - 23 = 36$?

A. $36 + 23 = 59$ ✓

B. $59 + 23 = 82$

C. $36 - 23 = 13$

D. $23 + 59 = 82$

Explanation: Add the answer (36) to the number subtracted (23): $36 + 23 = 59$. It matches the starting number, so the subtraction is correct.

17. $67 - 41 = 26$. Check: $26 + 41 = ?$

- A. 57
- B. 67 ✓**
- C. 77
- D. 68

Explanation: $26 + 41 = 67$. This matches, so $67 - 41 = 26$ is correct.

18. A boy solved $80 - 35 = 45$. He checked: $45 + 35 = 80$. Is his work correct?

- A. Yes, his subtraction and check both show 80 ✓**
- B. No, $45 + 35 = 70$
- C. No, $80 - 35 = 55$
- D. Yes, but the check is wrong

Explanation: $45 + 35 = 80$, which matches the starting number. Both the subtraction and the check are correct.

19. $91 - 56 = 35$. Check with addition. $35 + 56 = ?$

- A. 81
- B. 91 ✓**
- C. 101
- D. 90

Explanation: $35 + 56 = 91$. This matches the starting number, so $91 - 56 = 35$ is correct.

20. A student wrote $74 - 48 = 36$. Check: $36 + 48 = ?$

- A. 74
- B. 84 ✓**
- C. 64
- D. 76

Explanation: $36 + 48 = 84$, not 74. The student made an error. Correct: $74 - 48 = 26$. Check: $26 + 48 = 74$.

Medium (20 questions)

1. A student wrote: $83 - 47 = 44$. Use addition to check. Is this correct?

- A. No, $44 + 47 = 91$, not 83. The answer should be 36. ✓**
- B. Yes, $44 + 47 = 83$
- C. No, the answer should be 46
- D. Yes, $83 - 47 = 44$

Explanation: Check: $44 + 47 = 91$, not 83. The subtraction is wrong. Correct: $83 - 47 = 36$. Check: $36 + 47 = 83$.

2. Which addition problem correctly checks $125 - 58 = 67$?

- A. $67 + 58 = 125$ ✓**
- B. $125 + 58 = 183$
- C. $67 + 125 = 192$
- D. $58 + 58 = 116$

Explanation: To check subtraction, add the answer to the number you subtracted: $67 + 58 = 125$. This matches, so the subtraction is correct.

3. Sam says $92 - 47 = 55$. Kim says the answer is 45. Who is right? Use addition to check.

- A. Sam, because $55 + 47 = 102$
- B. Kim, because $45 + 47 = 92$ ✓**
- C. Sam, because 55 is bigger
- D. Neither is correct

Explanation: Check Sam: $55 + 47 = 102$, not 92. Wrong! Check Kim: $45 + 47 = 92$. Correct! Kim's answer of 45 is right.

4. $__ - 39 = 47$. What is the missing number? Use addition to find it.

- A. 76
- B. 86 ✓**
- C. 96
- D. 8

Explanation: To find the missing number, add: $47 + 39 = 86$. Check: $86 - 39 = 47$. The missing number is 86.

5. A student checked their work: $38 + 54 = 92$. What subtraction problem were they checking?

A. $92 - 54 = 38$ ✓

B. $92 - 38 = 64$

C. $54 - 38 = 16$

D. $92 + 54 = 146$

Explanation: If the check is $38 + 54 = 92$, then the original subtraction was $92 - 54 = 38$ (or $92 - 38 = 54$). The answer showing is $92 - 54 = 38$.

6. True or false: If $67 + 45 = 112$, then $112 - 45 = 67$.

A. True, addition and subtraction are inverse operations ✓

B. False, the numbers are too big

C. True, but only without regrouping

D. False, subtraction does not relate to addition

Explanation: This is true! Addition and subtraction are inverse (opposite) operations. If adding gives you a sum, then subtracting one addend from the sum gives the other addend.

7. Mia solves $134 - 67 = 67$. She checks: $67 + 67 = 134$. Is her subtraction correct?

A. Yes, her check proves it ✓

B. No, $67 + 67 = 144$

C. No, $134 - 67 = 77$

D. Yes, but the check is wrong

Explanation: Check: $67 + 67 = 134$. This matches the starting number, so Mia's subtraction $134 - 67 = 67$ is correct.

8. A teacher wrote: $95 - 48 = 47$. Three students checked differently. Which check is correct?

A. $47 + 48 = 95$ ✓

B. $95 + 48 = 143$

C. $48 - 47 = 1$

D. $95 - 47 = 48$

Explanation: The correct check adds the answer to the subtracted number: $47 + 48 = 95$. Option D ($95 - 47 = 48$) also works as a related fact, but the standard check is addition.

9. David solves $81 - 36 = 55$. His check: $55 + 36 = 91$. What should David do?

A. Redo the problem because 91 does not equal 81 ✓

B. Nothing, the answer is correct

C. Check again because $55 + 36 = 81$

D. Use a different checking method

Explanation: $55 + 36 = 91$, not 81. The check reveals an error. David should redo: $81 - 36 = 45$. Check: $45 + 36 = 81$.

10. Which pair of number sentences shows related facts?

A. $73 - 28 = 45$ and $45 + 28 = 73$ ✓

B. $73 - 28 = 45$ and $73 + 28 = 101$

C. $73 - 28 = 45$ and $45 - 28 = 17$

D. $73 - 28 = 45$ and $28 + 45 = 83$

Explanation: $73 - 28 = 45$ and $45 + 28 = 73$ are related facts (a fact family). They use the same three numbers and show how addition and subtraction are inverses.

11. If you know $56 + 37 = 93$, which subtraction fact is also true?

A. $93 - 37 = 56$ ✓

B. $93 - 37 = 66$

C. $56 - 37 = 19$

D. $93 + 37 = 130$

Explanation: Since $56 + 37 = 93$, we know $93 - 37 = 56$. These are inverse operations using the same three numbers.

12. A student says 'I do not need to check my work if I am careful.' Is this good advice?

A. No, everyone makes mistakes and checking catches errors ✓

B. Yes, careful work is always correct

C. No, but only check hard problems

D. Yes, checking wastes time

Explanation: Even careful students make mistakes! Checking with addition is quick and catches errors before they become problems. Good mathematicians always check their work.

13. ___ - 64 = 58. Find the missing number and check your answer.

- A. 112
- B. 122 ✓**
- C. 132
- D. 6

Explanation: Add to find the missing number: $58 + 64 = 122$. Check: $122 - 64 = 58$. The missing number is 122.

14. What are ALL the fact family members for $43 + 29 = 72$?

- A. $43 + 29 = 72$, $29 + 43 = 72$, $72 - 29 = 43$, $72 - 43 = 29$ ✓**
- B. Only $72 - 29 = 43$
- C. $43 + 29 = 72$ and $72 - 29 = 43$ only
- D. There is no fact family

Explanation: A fact family has 4 related facts using the same numbers: $43 + 29 = 72$, $29 + 43 = 72$, $72 - 29 = 43$, $72 - 43 = 29$.

15. Rosa solved $143 - 85 = 58$. She checks: $58 + 85 = 143$. Is Rosa correct?

- A. Yes, her check matches so her subtraction is correct ✓**
- B. No, $58 + 85 = 133$
- C. No, $143 - 85 = 68$
- D. Yes, but the check is wrong

Explanation: $58 + 85 = 143$. The check matches the original number, so Rosa's answer is correct.

16. A student solved $110 - 73 = 47$. Check this answer.

- A. Wrong: $47 + 73 = 120$, not 110 ✓**
- B. Correct: $47 + 73 = 110$
- C. Wrong: $47 + 73 = 110$
- D. Correct: the answer is 47

Explanation: Check: $47 + 73 = 120$, not 110. The answer is wrong. Correct: $110 - 73 = 37$. Check: $37 + 73 = 110$.

17. Which strategy helps you check $86 - 49$?

- A. Find the answer and add it to 49 to see if you get 86 ✓**
- B. Subtract 49 from 86 twice
- C. Add $86 + 49 = 135$
- D. Multiply 86 by 49

Explanation: Solve: $86 - 49 = 37$. Check by adding the answer to the subtracted number: $37 + 49 = 86$. Since it matches, the subtraction is correct.

18. Ben wrote $72 - 38 = 44$. His check: $44 + 38 = 82$. What does the check tell Ben?

- A. His answer is wrong because 82 does not equal 72 ✓**
- B. His answer is right because 82 is close to 72
- C. His check is wrong
- D. Nothing useful

Explanation: $44 + 38 = 82$, which does not equal the starting number 72. This tells Ben his answer is wrong. Correct: $72 - 38 = 34$. Check: $34 + 38 = 72$.

19. If $150 - 83 = 67$, which of these is also true?

- A. $67 + 83 = 150$ ✓**
- B. $150 + 83 = 233$
- C. $83 - 67 = 16$
- D. $67 - 83 = 150$

Explanation: Since $150 - 83 = 67$, the inverse is $67 + 83 = 150$. These use the same three numbers in a fact family relationship.

20. Ana and Ben both solved $95 - 37$. Ana got 58, Ben got 68. Use addition to find who is correct.

- A. Ana, because $58 + 37 = 95$ ✓**
- B. Ben, because $68 + 37 = 95$
- C. Both are correct
- D. Neither is correct

Explanation: Check Ana: $58 + 37 = 95$. Correct! Check Ben: $68 + 37 = 105$, not 95. Wrong! Ana's answer of 58 is correct.

Hard (20 questions)

1. What number goes in the blank? ____ - 34 = 48. Then check your answer using addition.

A. 72, because $48 + 34 = 82$

B. 82, because $48 + 34 = 82$ and $82 - 34 = 48$ ✓

C. 92, because $48 + 34 = 92$

D. 78, because $48 + 30 = 78$

Explanation: Find the missing number by adding: $48 + 34 = 82$. Check: $82 - 34 = 48$. The answer 82 checks out because the addition confirms it.

2. Carlos solved $153 - 78 = 85$. He checked: $85 + 78 = 163$. What TWO mistakes did Carlos make?

A. Wrong subtraction AND wrong addition check ✓

B. Only wrong subtraction

C. Only wrong addition

D. No mistakes were made

Explanation: Carlos made two mistakes: (1) $153 - 78 = 75$, not 85. (2) His check: $85 + 78 = 163$, not 153, which should have told him the answer was wrong. Correct: $75 + 78 = 153$.

3. If $67 + 48 = 115$, write ALL the subtraction facts in this fact family.

A. $115 - 48 = 67$ and $115 - 67 = 48$ ✓

B. Only $115 - 48 = 67$

C. $115 - 67 = 48$ only

D. $67 - 48 = 19$ and $48 - 67 =$ impossible

Explanation: A fact family uses the same three numbers. From $67 + 48 = 115$: the two subtraction facts are $115 - 48 = 67$ and $115 - 67 = 48$.

4. A student solved $140 - 65 = 85$. She checked: $85 + 65 = 150$. She says 'It is close enough, so I am right.' Is she correct?

A. No, the check must match exactly, and 150 does not equal 140 ✓

B. Yes, close enough is fine in math

C. No, but only because the difference is more than 5

D. Yes, 150 and 140 are close

Explanation: In math, 'close enough' is not correct. $85 + 65 = 150$, not 140, so the subtraction has an error. Correct: $140 - 65 = 75$. Check: $75 + 65 = 140$ exactly.

5. Nora solved three problems. Which one has an error? A) $91 - 45 = 46$ B) $83 - 37 = 46$ C) $72 - 28 = 54$

A. Problem C, because $54 + 28 = 82$, not 72 ✓

B. Problem A, because $46 + 45 = 91$

C. Problem B, because $46 + 37 = 83$

D. None have errors

Explanation: Check each: A) $46 + 45 = 91$. Correct! B) $46 + 37 = 83$. Correct! C) $54 + 28 = 82$, not 72. Error! The correct answer for C is $72 - 28 = 44$.

6. A number minus 56 gives 47. Use the checking strategy to also find: that same number minus 47.

A. 56 ✓

B. 47

C. 103

D. 9

Explanation: Find the number: $47 + 56 = 103$. Now: $103 - 47 = 56$. The fact family connects both: if $103 - 56 = 47$, then $103 - 47 = 56$.

7. Tom checked $86 - 39$ by computing $39 + 86 = 125$. What is wrong with Tom's checking strategy?

A. He should add the ANSWER to 39, not 86 to 39 ✓

B. Nothing, his strategy is fine

C. He should subtract instead of add

D. He should multiply to check

Explanation: Tom added the wrong numbers. To check, he needs to add the ANSWER ($86 - 39 = 47$) to the number subtracted (39): $47 + 39 = 86$. Tom mistakenly added $39 + 86$.

8. Three students solved $132 - 67$. Amy: 65. Ben: 75. Cal: 55. Which student is correct?

A. Amy, because $65 + 67 = 132$ ✓

B. Ben, because $75 + 67 = 132$

C. Cal, because $55 + 67 = 132$

D. None of them

Explanation: Check each: Amy: $65 + 67 = 132$. Correct! Ben: $75 + 67 = 142$. Wrong! Cal: $55 + 67 = 122$. Wrong! Amy is correct.

9. Why does adding the answer to the subtracted number prove the subtraction is correct?

A. Because addition undoes subtraction -- they are inverse operations ✓

B. Because adding always gives bigger numbers

C. Because the teacher said so

D. It does not actually prove it

Explanation: Addition and subtraction are inverse operations -- they undo each other. If $A - B = C$, then $C + B$ must equal A. This inverse relationship is what makes the check work.

10. Lisa solved $148 - 79 = 69$. She also solved $148 - 69 = 79$. Both check out as a fact family. Is she right?

A. Yes, $69 + 79 = 148$ confirms both are correct ✓

B. No, $69 + 79 = 138$, not 148

C. Yes, but only the first equation

D. No, you cannot have two related facts

Explanation: Check: $69 + 79 = 148$. This confirms both: $148 - 79 = 69$ and $148 - 69 = 79$. These are both part of the same fact family.

11. A student solved $95 - 38 = 57$ and then $95 - 57 = 48$. She says both are right. What error do you see?

A. The second equation is wrong: $95 - 57 = 38$, not 48 ✓

B. Both equations are wrong

C. The first equation is wrong

D. No errors

Explanation: The first is correct: $57 + 38 = 95$. But the second is wrong: $95 - 57$ should equal 38 (from the fact family), not 48. The student made a subtraction error on the second problem.

12. Jake knows $156 - 89 = 67$. Without solving, what is $156 - 67$?

A. 89, because it is the other subtraction in the fact family ✓

B. 99

C. 79

D. Cannot tell without solving

Explanation: Since $156 - 89 = 67$, the fact family tells us $156 - 67 = 89$. No new calculation needed! The three numbers (156, 89, 67) always work together.

13. A student checked $113 - 47 = 66$ by computing $66 + 47$. She got 113. But the real answer is $113 - 47 = 66$. How is this possible?

A. The answer IS 66 and the check proves it ✓

B. The check must be wrong

C. $66 + 47$ cannot equal 113

D. She got lucky

Explanation: $113 - 47 = 66$ is correct. The check confirms it: $66 + 47 = 113$. When the check sum matches the original number, the subtraction is proven correct.

14. Emma solved $167 - 89 = 78$. She wants to check but finds addition hard too. What easier check can she use?

A. Estimate: $170 - 90 = 80$, and 78 is close to 80, so it is likely correct ✓

B. There is no other way to check

C. Multiply 78 by 89

D. Subtract 78 from 89

Explanation: Estimation is a great quick check! $170 - 90 = 80$, and 78 is close to 80, supporting the answer. For certainty, do the addition check: $78 + 89 = 167$. Correct!

15. Two problems use the same three numbers. If $94 - 56 = 38$, what MUST be true?

A. $38 + 56 = 94$ and $94 - 38 = 56$ ✓

B. $56 - 38 = 94$

C. $94 + 56 = 38$

D. Only $38 + 56 = 94$ is true

Explanation: From $94 - 56 = 38$, the fact family gives us: $38 + 56 = 94$, $56 + 38 = 94$, and $94 - 38 = 56$. All three numbers stay connected through inverse operations.

16. A class solved $125 - 67$. Five students got 58 and three got 68. Use checking to settle the debate.

A. 58 is correct: $58 + 67 = 125$ ✓

B. 68 is correct: $68 + 67 = 125$

C. Both could be correct

D. Neither is correct

Explanation: Check 58: $58 + 67 = 125$. Correct! Check 68: $68 + 67 = 135$, not 125. Wrong! The answer is 58. The three students who got 68 likely forgot to regroup properly.

17. Dan solved a subtraction problem and got 49. He added $49 + 36$ and got 85. What was the original problem?

A. $85 - 36 = 49$ ✓

B. $49 - 36 = 13$

C. $85 + 36 = 121$

D. $36 - 49 =$ impossible

Explanation: If Dan's check is $49 + 36 = 85$, then the original subtraction was $85 - 36 = 49$. The addition check reveals the original problem's structure.

18. Mia made the same error on two problems: $82 - 35 = 53$ and $71 - 24 = 53$. What pattern of error is she making?

A. She is not regrouping -- she subtracts the smaller digit from the larger in ones place ✓

B. She is adding instead of subtracting

C. She is guessing

D. Her answers are correct

Explanation: $82 - 35$: Mia did $5 - 2 = 3$ (wrong direction), $8 - 3 = 5$, getting 53. Correct: 47. $71 - 24$: Mia did $4 - 1 = 3$, $7 - 2 = 5$, getting 53. Correct: 47. She consistently subtracts the smaller ones digit from the larger instead of regrouping.

19. If $89 + 43 = 132$, write the complete fact family. How many facts are in it?

A. 4 facts: $89+43=132$, $43+89=132$, $132-43=89$, $132-89=43$ ✓

B. 2 facts

C. 3 facts

D. 6 facts

Explanation: The complete fact family has 4 facts: $89 + 43 = 132$, $43 + 89 = 132$, $132 - 43 = 89$, $132 - 89 = 43$. All use the same three numbers.

20. A student says 'If my subtraction check does not match, I should just erase and write the addition answer instead.' Is this good advice?

A. No, she should redo the subtraction carefully to find the error ✓

B. Yes, the addition answer is always right

C. No, she should try a third method

D. Yes, because addition is easier

Explanation: If the check does not match, she should redo the subtraction step by step. The check only tells you there is an error -- it does not automatically give the right answer. She needs to find WHERE the mistake happened.

Lesson 4: Subtraction Word Problems

Easy (20 questions)

1. Sam has 45 crayons. He gives 18 to his friend. How many crayons does Sam have left?

- A. 27 ✓
- B. 37
- C. 17
- D. 26

Explanation: Sam gives crayons away, so subtract: $45 - 18$. Regroup: $15 - 8 = 7$ ones, $3 - 1 = 2$ tens. Sam has 27 crayons left.

2. A tree has 73 apples. The wind knocks off 41 apples. How many apples are still on the tree?

- A. 42
- B. 22
- C. 32 ✓
- D. 34

Explanation: Apples fall off, so subtract: $73 - 41$. Ones: $3 - 1 = 2$. Tens: $7 - 4 = 3$. There are 32 apples still on the tree.

3. Mia has 56 stickers. She uses 24 stickers on her notebook. How many stickers does she have left?

- A. 42
- B. 22
- C. 32 ✓
- D. 30

Explanation: Stickers are used up, so subtract: $56 - 24$. Ones: $6 - 4 = 2$. Tens: $5 - 2 = 3$. Mia has 32 stickers left.

4. There are 89 children at the park. 52 children go home. How many children are still at the park?

- A. 47
- B. 27
- C. 37 ✓
- D. 35

Explanation: Children leave, so subtract: $89 - 52$. Ones: $9 - 2 = 7$. Tens: $8 - 5 = 3$. There are 37 children still at the park.

5. Dad baked 64 cookies. The family ate 31 cookies. How many cookies are left?

- A. 43
- B. 23
- C. 33 ✓
- D. 35

Explanation: Cookies are eaten, so subtract: $64 - 31$. Ones: $4 - 1 = 3$. Tens: $6 - 3 = 3$. There are 33 cookies left.

6. A store has 87 hats. They sell 43 hats. How many hats are left in the store?

- A. 54
- B. 34
- C. 44 ✓
- D. 42

Explanation: Hats are sold, so subtract: $87 - 43$. Ones: $7 - 3 = 4$. Tens: $8 - 4 = 4$. There are 44 hats left.

7. Leo has 91 baseball cards. He trades 50 cards with his friend. How many cards does Leo have now?

- A. 51
- B. 31
- C. 41 ✓
- D. 40

Explanation: Cards are traded away, so subtract: $91 - 50$. Ones: $1 - 0 = 1$. Tens: $9 - 5 = 4$. Leo has 41 cards now.

8. A bus has 48 passengers. 25 passengers get off at the next stop. How many passengers are still on the bus?

- A. 33
- B. 13
- C. 23 ✓
- D. 21

Explanation: Passengers get off, so subtract: $48 - 25$. Ones: $8 - 5 = 3$. Tens: $4 - 2 = 2$. There are 23 passengers still on the bus.

9. Emma has 76 beads. She uses 35 beads to make a necklace. How many beads are left?

- A. 51
- B. 31
- C. 41 ✓
- D. 43

Explanation: Beads are used, so subtract: $76 - 35$. Ones: $6 - 5 = 1$. Tens: $7 - 3 = 4$. Emma has 41 beads left.

10. There are 95 books on a shelf. A student checks out 62 books. How many books are still on the shelf?

- A. 43
- B. 23
- C. 33 ✓
- D. 31

Explanation: Books are checked out, so subtract: $95 - 62$. Ones: $5 - 2 = 3$. Tens: $9 - 6 = 3$. There are 33 books on the shelf.

11. A jar has 67 jelly beans. Maria eats 34 jelly beans. How many jelly beans are left in the jar?

- A. 43
- B. 23
- C. 33 ✓
- D. 31

Explanation: Jelly beans are eaten, so subtract: $67 - 34$. Ones: $7 - 4 = 3$. Tens: $6 - 3 = 3$. There are 33 jelly beans left.

12. A class has 58 pencils. The teacher gives 26 pencils to students. How many pencils are left?

- A. 42
- B. 22
- C. 32 ✓
- D. 34

Explanation: Pencils are given away, so subtract: $58 - 26$. Ones: $8 - 6 = 2$. Tens: $5 - 2 = 3$. There are 32 pencils left.

13. Jack had 83 toy cars. He lost 51 toy cars. How many toy cars does Jack have now?

- A. 42
- B. 22
- C. 32 ✓
- D. 30

Explanation: Cars are lost, so subtract: $83 - 51$. Ones: $3 - 1 = 2$. Tens: $8 - 5 = 3$. Jack has 32 toy cars now.

14. There are 79 birds on a wire. 46 birds fly away. How many birds are still on the wire?

- A. 43
- B. 23
- C. 33 ✓
- D. 31

Explanation: Birds fly away, so subtract: $79 - 46$. Ones: $9 - 6 = 3$. Tens: $7 - 4 = 3$. There are 33 birds on the wire.

15. A farmer has 84 eggs. He sells 42 eggs. How many eggs does the farmer have left?

- A. 52
- B. 32
- C. 42 ✓
- D. 44

Explanation: Eggs are sold, so subtract: $84 - 42$. Ones: $4 - 2 = 2$. Tens: $8 - 4 = 4$. The farmer has 42 eggs left.

16. Lily has 68 flowers. She puts 35 flowers in a vase. How many flowers does she still need to put away?

- A. 43
- B. 23
- C. 33 ✓
- D. 31

Explanation: Some flowers are put in a vase, so subtract to find the rest: $68 - 35$. Ones: $8 - 5 = 3$. Tens: $6 - 3 = 3$. She has 33 flowers still to put away.

17. A box has 97 crayons. A child takes 54 crayons. How many crayons are in the box now?

- A. 53
- B. 33
- C. 43 ✓
- D. 45

Explanation: Crayons are taken out, so subtract: $97 - 54$. Ones: $7 - 4 = 3$. Tens: $9 - 5 = 4$. There are 43 crayons in the box.

18. Rosa has 55 marbles. She gives 23 to her brother. How many marbles does Rosa have now?

- A. 42
- B. 22
- C. 32 ✓
- D. 34

Explanation: Marbles are given away, so subtract: $55 - 23$. Ones: $5 - 3 = 2$. Tens: $5 - 2 = 3$. Rosa has 32 marbles now.

19. There are 86 fish in a pond. A fisherman catches 45 fish. How many fish are still in the pond?

- A. 51
- B. 31
- C. 41 ✓
- D. 43

Explanation: Fish are caught, so subtract: $86 - 45$. Ones: $6 - 5 = 1$. Tens: $8 - 4 = 4$. There are 41 fish still in the pond.

20. A bakery has 72 muffins. They sell 31 muffins in the morning. How many muffins are left?

- A. 51
- B. 31
- C. 41 ✓
- D. 43

Explanation: Muffins are sold, so subtract: $72 - 31$. Ones: $2 - 1 = 1$. Tens: $7 - 3 = 4$. There are 41 muffins left.

Medium (20 questions)

1. Sam had 85 stickers. He gave 37 to Mia. Then he found 12 more. How many stickers does Sam have now?

- A. 50
- B. 60 ✓
- C. 70
- D. 48

Explanation: Step 1: $85 - 37 = 48$ stickers after giving some away. Step 2: $48 + 12 = 60$ after finding more. Sam has 60 stickers.

2. A class has 93 books. If 47 are fiction and the rest are nonfiction, how many are nonfiction?

- A. 56
- B. 36
- C. 46 ✓
- D. 44

Explanation: The total is 93 books. Subtract fiction to find nonfiction: $93 - 47 = 46$. Regroup: $13 - 7 = 6$ ones, $8 - 4 = 4$ tens. There are 46 nonfiction books.

3. Tom has 54 toy cars. He has 19 more cars than Jess. How many toy cars does Jess have?

- A. 73
- B. 45
- C. 35 ✓
- D. 25

Explanation: Tom has MORE, so Jess has FEWER. Subtract: $54 - 19 = 35$. Regroup: $14 - 9 = 5$ ones, $4 - 1 = 3$ tens. Jess has 35 cars.

4. A garden had 78 flowers. Some flowers wilted and now there are 45. How many flowers wilted?

- A. 23
- B. 43
- C. 33 ✓
- D. 35

Explanation: Find the difference: $78 - 45 = 33$. Ones: $8 - 5 = 3$. Tens: $7 - 4 = 3$. 33 flowers wilted.

5. Which number sentence matches this story? 'There are 62 birds. 28 fly away. How many are left?'

A. $62 + 28 = ?$

B. $62 - 28 = ?$ ✓

C. $28 - 62 = ?$

D. $28 + 62 = ?$

Explanation: Birds fly AWAY, so the number decreases. This is subtraction. Start with the total (62) and subtract those that left (28): $62 - 28 = ?$

6. Emma scored 82 points in a game. David scored 57 points. How many more points did Emma score than David?

A. 35

B. 15

C. 25 ✓

D. 23

Explanation: To find 'how many more,' subtract: $82 - 57 = 25$. Regroup: $12 - 7 = 5$ ones, $7 - 5 = 2$ tens. Emma scored 25 more points.

7. A store had 96 toys. After selling some, 48 are left. How many toys were sold?

A. 58

B. 38

C. 48 ✓

D. 52

Explanation: Find how many were sold: $96 - 48 = 48$. Regroup: $16 - 8 = 8$ ones, $8 - 4 = 4$ tens. 48 toys were sold.

8. Ben needs 75 stickers to fill his book. He has 39 stickers so far. How many more does he need?

A. 46

B. 26

C. 36 ✓

D. 34

Explanation: Find the difference: $75 - 39 = 36$. Regroup: $15 - 9 = 6$ ones, $6 - 3 = 3$ tens. Ben needs 36 more stickers.

9. A bus holds 64 passengers. Right now there are 38 passengers. How many empty seats are there?

A. 36

B. 16

C. 26 ✓

D. 24

Explanation: Subtract to find empty seats: $64 - 38 = 26$. Regroup: $14 - 8 = 6$ ones, $5 - 3 = 2$ tens. There are 26 empty seats.

10. On Monday, 87 students rode the bus. On Tuesday, 59 students rode the bus. How many fewer students rode on Tuesday?

A. 38

B. 18

C. 28 ✓

D. 26

Explanation: 'How many fewer' means find the difference: $87 - 59 = 28$. Regroup: $17 - 9 = 8$ ones, $7 - 5 = 2$ tens. 28 fewer students rode on Tuesday.

11. A parking lot has 91 spots. 53 cars are parked there. How many spots are open?

A. 48

B. 28

C. 38 ✓

D. 36

Explanation: Subtract to find open spots: $91 - 53 = 38$. Regroup: $11 - 3 = 8$ ones, $8 - 5 = 3$ tens. There are 38 open spots.

12. Ava read 43 pages. Liam read 67 pages. How many more pages did Liam read than Ava?

A. 34

B. 14

C. 24 ✓

D. 22

Explanation: Subtract the smaller from the larger: $67 - 43 = 24$. Ones: $7 - 3 = 4$. Tens: $6 - 4 = 2$. Liam read 24 more pages.

13. A shelf can hold 80 books. There are 46 books on it. How many more books can fit?

- A. 44
- B. 24
- C. 34 ✓**
- D. 36

Explanation: Subtract to find remaining space: $80 - 46 = 34$. Regroup: $10 - 6 = 4$ ones, $7 - 4 = 3$ tens. 34 more books can fit.

14. A baker made 72 cupcakes. He sold 28 chocolate and 15 vanilla. How many are left?

- A. 39
- B. 19
- C. 29 ✓**
- D. 27

Explanation: First find total sold: $28 + 15 = 43$. Then subtract: $72 - 43 = 29$. The baker has 29 cupcakes left.

15. Rosa has 63 crayons. Ella has 38 crayons. Rosa says she has 35 more. Is Rosa correct?

- A. No, she has 25 more: $63 - 38 = 25$ ✓**
- B. Yes, $63 - 38 = 35$
- C. No, she has 15 more
- D. Yes, Rosa counted correctly

Explanation: $63 - 38 = 25$, not 35. Regroup: $13 - 8 = 5$ ones, $5 - 3 = 2$ tens. Rosa actually has 25 more crayons, not 35.

16. Which question can be solved with $95 - 58$?

- A. Tina has 58 blocks. She gets 95 more. How many in all?
- B. Tina had 95 blocks. She gave away 58. How many are left? ✓**
- C. Tina has 95 blocks and 58 blocks. How many altogether?
- D. Tina needs 58 blocks. She has 95 bags. How many bags are left?

Explanation: $95 - 58$ means starting with 95 and taking away 58. 'Tina had 95 blocks and gave away 58' matches this subtraction.

17. A theater has 84 seats. 56 people are sitting. 12 more people arrive. How many seats are still empty?

- A. 26
- B. 6
- C. 16 ✓**
- D. 28

Explanation: Total people: $56 + 12 = 68$. Empty seats: $84 - 68 = 16$. There are 16 empty seats.

18. Jake and Ana together have 86 marbles. Jake has 49. How many does Ana have?

- A. 47
- B. 27
- C. 37 ✓**
- D. 35

Explanation: Subtract Jake's from total: $86 - 49 = 37$. Regroup: $16 - 9 = 7$ ones, $7 - 4 = 3$ tens. Ana has 37 marbles.

19. A school collected 71 cans on Monday and 53 cans on Tuesday. How many more cans were collected on Monday?

- A. 28
- B. 8
- C. 18 ✓**
- D. 16

Explanation: 'How many more' means subtract: $71 - 53 = 18$. Regroup: $11 - 3 = 8$ ones, $6 - 5 = 1$ ten. 18 more cans were collected on Monday.

20. Lena had some coins. She spent 29 coins and has 44 left. How many coins did Lena start with?

- A. 63
- B. 73 ✓**
- C. 83
- D. 15

Explanation: Work backwards: if she has 44 left after spending 29, add: $44 + 29 = 73$. Lena started with 73 coins. Check: $73 - 29 = 44$.

Hard (20 questions)

1. Sam had 135 stickers. He gave 48 to Mia and 39 to Jake. Then he found 15 more. How many stickers does Sam have now?

- A. 63 ✓
- B. 73
- C. 53
- D. 83

Explanation: Step 1: $135 - 48 = 87$. Step 2: $87 - 39 = 48$. Step 3: $48 + 15 = 63$. Sam has 63 stickers now.

2. A zoo had 168 animals. 75 animals were moved to a new exhibit on Monday. 54 more were moved on Tuesday. How many animals are still in the original area?

- A. 39 ✓
- B. 49
- C. 29
- D. 59

Explanation: Step 1: $168 - 75 = 93$ after Monday. Step 2: $93 - 54 = 39$ after Tuesday. 39 animals remain in the original area.

3. Ella says: ' $92 - 47 = 55$ because $9 - 4 = 5$ and $7 - 2 = 5$.' What is wrong with Ella's thinking?

- A. She subtracted ones digits in the wrong order (7-2 instead of regrouping 12-7) ✓
- B. She added instead of subtracting
- C. She forgot to carry
- D. Her answer is correct

Explanation: Ella subtracted the ones in reverse: $7 - 2 = 5$ instead of regrouping to get $12 - 7 = 5$. She got lucky that both give 5, but her tens are wrong: it should be $8 - 4 = 4$, not $9 - 4 = 5$ (she forgot the regrouping reduces the tens). Correct: $92 - 47 = 45$.

4. A school needs 150 notebooks. They have 83 notebooks. They buy 42 more. Do they have enough?

- A. No, $83 + 42 = 125$, which is 25 short of 150 ✓
- B. Yes, $83 + 42 = 150$
- C. No, $83 + 42 = 115$
- D. Yes, $83 + 42 = 155$

Explanation: $83 + 42 = 125$ notebooks. They need 150. $150 - 125 = 25$. They are 25 notebooks short.

5. Maria and Luis together have 143 marbles. Maria has 67. Luis gives 28 of his marbles away. How many does Luis have now?

- A. 48 ✓
- B. 58
- C. 38
- D. 76

Explanation: Step 1: Luis's marbles: $143 - 67 = 76$. Step 2: After giving away 28: $76 - 28 = 48$. Luis has 48 marbles now.

6. A student solved these two problems: $86 - 39 = 53$ and $74 - 28 = 52$. Check both. Which is correct?

- A. Only $74 - 28 = 52$ is wrong (should be 46)
- B. Only $86 - 39 = 53$ is wrong (should be 47)
- C. Both are wrong ✓
- D. Both are correct

Explanation: Check: $53 + 39 = 92$, not 86. Wrong! $52 + 28 = 80$, not 74. Wrong! Both are incorrect. Correct: $86 - 39 = 47$ and $74 - 28 = 46$.

7. A bakery made 156 cookies. They need to fill boxes of 0 cookies for display and sell the rest. They sold 89 in the morning and 43 in the afternoon. How many are left?

- A. 24 ✓
- B. 34
- C. 14
- D. 44

Explanation: Step 1: $156 - 89 = 67$ after morning sales. Step 2: $67 - 43 = 24$ after afternoon sales. 24 cookies are left.

8. Without solving exactly, is $128 - 69$ closer to 50 or 70?

A. Closer to 60, which is between 50 and 70 ✓

B. Closer to 50

C. Closer to 70

D. Exactly 50

Explanation: Estimate: $130 - 70 = 60$. The exact answer is $128 - 69 = 59$, which is closest to 60. 60 is right between 50 and 70.

9. Nora had some beads. She used 56 for a necklace and 37 for a bracelet. She has 28 left. How many beads did she start with?

A. 111

B. 121 ✓

C. 131

D. 101

Explanation: Work backwards: $28 + 37 = 65$ (before bracelet). $65 + 56 = 121$ (before necklace). Nora started with 121 beads.

10. Ben scored 74 points. Ana scored 48 points. Ben says he scored 'about 30 more.' Is his estimate reasonable?

A. Yes, $74 - 48 = 26$, and 26 is close to 30 ✓

B. No, $74 - 48 = 26$, and 26 is closer to 25 than 30

C. Yes, $74 - 48 = 36$

D. No, he scored exactly 30 more

Explanation: $74 - 48 = 26$. Ben said 'about 30.' 26 is reasonably close to 30 for an estimate. His estimate is reasonable, though 25 would be more precise.

11. A number minus 67 equals 58. What is 20 less than that number?

A. 105 ✓

B. 115

C. 95

D. 85

Explanation: Find the number: $? - 67 = 58$, so $? = 58 + 67 = 125$. Then 20 less: $125 - 20 = 105$.

12. A toy shop has 142 toys. On Monday they sell 38 and on Tuesday they sell 47. A shipment of 25 new toys arrives Wednesday. How many toys do they have?

A. 82 ✓

B. 92

C. 72

D. 62

Explanation: Step 1: $142 - 38 = 104$. Step 2: $104 - 47 = 57$. Step 3: $57 + 25 = 82$. The shop has 82 toys.

13. Two classes collected cans. Class A: 87 cans. Class B: 69 cans. If they combine and donate 95 cans, how many do they have left?

A. 51

B. 61 ✓

C. 71

D. 41

Explanation: Total: $87 + 69 = 156$ cans. After donating: $156 - 95 = 61$ cans left.

14. Jake says: 'If I start with a bigger number, the answer to my subtraction will always be bigger.' Is Jake correct?

A. Not necessarily -- it depends on what you subtract too ✓

B. Yes, bigger start always means bigger answer

C. No, the answer is always the same

D. Yes, but only without regrouping

Explanation: Not necessarily! $100 - 99 = 1$, but $50 - 10 = 40$. Even though $100 > 50$, subtracting a large number (99) gives a small answer. The difference depends on BOTH numbers.

15. Ava solved a word problem and got 47. The problem said: 'A store had ____ hats. 39 were sold. 47 are left.' What number belongs in the blank?

- A. 86 ✓
- B. 96
- C. 76
- D. 8

Explanation: Use the inverse: $47 + 39 = 86$. The store started with 86 hats. Check: $86 - 39 = 47$.

16. Which story problem matches $145 - 78 - 34$?

- A. A farmer had 145 apples. He sold 78 Monday and 34 Tuesday. How many are left? ✓
- B. A farmer bought 145 apples and 78 oranges and 34 bananas
- C. A farmer had 78 apples and found 145 more and gave away 34
- D. A farmer needs 145 apples but only has 78 and 34

Explanation: $145 - 78 - 34$ means starting with 145 and subtracting two amounts. The farmer selling apples on two different days matches this two-step subtraction.

17. A piggy bank has 130 coins. Amy takes out 45 coins on Monday. She puts 18 coins back on Tuesday. On Wednesday she takes out 37 more. How many coins are in the piggy bank?

- A. 66 ✓
- B. 56
- C. 76
- D. 46

Explanation: Step 1: $130 - 45 = 85$. Step 2: $85 + 18 = 103$. Step 3: $103 - 37 = 66$. There are 66 coins in the piggy bank.

18. Kim and Dan each solved: 'There were 83 kids. 47 went inside. How many are outside?' Kim: $83 - 47 = 36$. Dan: $47 + ? = 83$, so $? = 36$. Who used the better strategy?

- A. Both strategies are valid and give the same answer ✓
- B. Only Kim's subtraction method works
- C. Only Dan's addition method works
- D. Neither strategy is correct

Explanation: Both are valid! Kim subtracted directly: $83 - 47 = 36$. Dan counted up: $47 + ? = 83$, so $? = 36$. Counting up (adding on) is another way to solve subtraction problems.

19. A class read 165 books total. They read 58 in September and 72 in October. How many did they read in November?

- A. 35 ✓
- B. 45
- C. 25
- D. 55

Explanation: Total Sept + Oct: $58 + 72 = 130$. November: $165 - 130 = 35$ books. The class read 35 books in November.

20. True or false: 'Finding how many more' and 'finding how many are left' both use subtraction, but they mean different things.

- A. True -- comparison and take-away are both subtraction but have different meanings ✓
- B. False -- they mean the same thing
- C. True -- but only comparison uses subtraction
- D. False -- 'how many more' uses addition

Explanation: True! 'How many are left' is take-away subtraction (removing from a group). 'How many more' is comparison subtraction (finding the difference between two groups). Both use the minus sign but model different real-world situations.

Unit Test Bank

Easy (20 questions)

1. What is $78 - 35$?

- A. 53
- B. 33
- C. 43 ✓**
- D. 42

Explanation: Subtract ones: $8 - 5 = 3$. Subtract tens: $7 - 3 = 4$. The answer is 43.

2. What is $62 - 28$?

- A. 44
- B. 24
- C. 34 ✓**
- D. 36

Explanation: Regroup: $2 < 8$, so take 1 ten. $12 - 8 = 4$ ones. $5 - 2 = 3$ tens. The answer is 34.

3. A farmer has 96 eggs. He sells 53 eggs. How many eggs are left?

- A. 53
- B. 33
- C. 43 ✓**
- D. 42

Explanation: Eggs are sold, so subtract: $96 - 53$. Ones: $6 - 3 = 3$. Tens: $9 - 5 = 4$. There are 43 eggs left.

4. What is $85 - 47$?

- A. 48
- B. 28
- C. 38 ✓**
- D. 42

Explanation: Regroup: $5 < 7$, so take 1 ten. $15 - 7 = 8$ ones. $7 - 4 = 3$ tens. The answer is 38.

5. There are 74 birds on a fence. 31 fly away. How many birds are still on the fence?

- A. 53
- B. 33
- C. 43 ✓**
- D. 42

Explanation: Birds fly away, so subtract: $74 - 31$. Ones: $4 - 1 = 3$. Tens: $7 - 3 = 4$. There are 43 birds on the fence.

6. What is $93 - 56$?

- A. 47
- B. 27
- C. 37 ✓**
- D. 39

Explanation: Regroup: $3 < 6$, so take 1 ten. $13 - 6 = 7$ ones. $8 - 5 = 3$ tens. The answer is 37.

7. $51 - 24 = 27$. Check: $27 + 24 = ?$

- A. 41
- B. 51 ✓**
- C. 61
- D. 52

Explanation: $27 + 24 = 51$. This matches the starting number, so $51 - 24 = 27$ is correct.

8. What is $67 - 42$?

- A. 35
- B. 15
- C. 25 ✓**
- D. 24

Explanation: Ones: $7 - 2 = 5$. Tens: $6 - 4 = 2$. The answer is 25.

9. A box has 83 crayons. A teacher takes out 37. How many crayons are left?

- A. 56
- B. 36
- C. 46 ✓
- D. 44

Explanation: Regroup: $3 < 7$, so take 1 ten. $13 - 7 = 6$ ones. $7 - 3 = 4$ tens. There are 46 crayons left.

10. What is $59 - 26$?

- A. 43
- B. 23
- C. 33 ✓
- D. 35

Explanation: Ones: $9 - 6 = 3$. Tens: $5 - 2 = 3$. The answer is 33.

11. $76 - 38 = 38$. Is this correct? Check with addition.

- A. Yes, $38 + 38 = 76$ ✓
- B. No, $38 + 38 = 86$
- C. No, the answer should be 42
- D. Yes, $38 + 38 = 78$

Explanation: Check: $38 + 38 = 76$. This matches, so $76 - 38 = 38$ is correct.

12. Emma has 88 stickers. She gives 45 to her sister. How many stickers does Emma have left?

- A. 53
- B. 33
- C. 43 ✓
- D. 42

Explanation: Stickers are given away, so subtract: $88 - 45$. Ones: $8 - 5 = 3$. Tens: $8 - 4 = 4$. Emma has 43 stickers left.

13. What is $71 - 39$?

- A. 42
- B. 22
- C. 32 ✓
- D. 34

Explanation: Regroup: $1 < 9$, so take 1 ten. $11 - 9 = 2$ ones. $6 - 3 = 3$ tens. The answer is 32.

14. What is $46 - 18$?

- A. 38
- B. 18
- C. 28 ✓
- D. 24

Explanation: Regroup: $6 < 8$, so take 1 ten. $16 - 8 = 8$ ones. $3 - 1 = 2$ tens. The answer is 28.

15. A class has 65 pencils. Students use 32 pencils. How many pencils are left?

- A. 43
- B. 23
- C. 33 ✓
- D. 35

Explanation: Pencils are used, so subtract: $65 - 32$. Ones: $5 - 2 = 3$. Tens: $6 - 3 = 3$. There are 33 pencils left.

16. What is $80 - 45$?

- A. 45
- B. 25
- C. 35 ✓
- D. 33

Explanation: Regroup: $0 < 5$, so take 1 ten. $10 - 5 = 5$ ones. $7 - 4 = 3$ tens. The answer is 35.

17. A store has 94 hats. They sell 61 hats. How many hats are left?

- A. 43
- B. 23
- C. 33 ✓
- D. 31

Explanation: Ones: $4 - 1 = 3$. Tens: $9 - 6 = 3$. There are 33 hats left.

18. What is $52 - 19$?

- A. 43
- B. 23
- C. 33 ✓
- D. 31

Explanation: Regroup: $2 < 9$, so take 1 ten. $12 - 9 = 3$ ones. $4 - 1 = 3$ tens. The answer is 33.

19. $87 - 43 = 44$. Check: $44 + 43 = ?$

- A. 77
- B. 87 ✓
- C. 97
- D. 86

Explanation: $44 + 43 = 87$. This matches, so $87 - 43 = 44$ is correct.

20. There are 69 children at a party. 36 go home. How many children are still at the party?

- A. 43
- B. 23
- C. 33 ✓
- D. 31

Explanation: Children leave, so subtract: $69 - 36$. Ones: $9 - 6 = 3$. Tens: $6 - 3 = 3$. There are 33 children at the party.

Medium (20 questions)

1. What is $162 - 85$?

- A. 67
- B. 87
- C. 77 ✓
- D. 73

Explanation: Regroup ones: $2 < 5$, so $12 - 5 = 7$ ones. Regroup tens: $5 < 8$, so $15 - 8 = 7$ tens. The answer is 77.

2. A school has 94 books. Students check out 47 books. How many books are left?

- A. 57
- B. 37
- C. 47 ✓
- D. 43

Explanation: Regroup: $4 < 7$, so $14 - 7 = 7$ ones. $8 - 4 = 4$ tens. There are 47 books left.

3. $\underline{\quad} - 38 = 54$. What is the missing number?

- A. 82
- B. 92 ✓
- C. 102
- D. 16

Explanation: Add: $54 + 38 = 92$. Check: $92 - 38 = 54$. The missing number is 92.

4. Which problem needs regrouping?

- A. $76 - 34$
- B. $89 - 52$
- C. $63 - 47$ ✓
- D. $98 - 65$

Explanation: Check ones: $76: 6 > 4$, $89: 9 > 2$, $63: 3 < 7$ (regroup!), $98: 8 > 5$. Only $63 - 47$ needs regrouping.

5. Ben had 83 marbles. He has 36 more than Kim. How many marbles does Kim have?

- A. 57
- B. 37
- C. 47 ✓**
- D. 119

Explanation: Ben has MORE, so Kim has fewer. Subtract: $83 - 36 = 47$. Regroup: $13 - 6 = 7$ ones, $7 - 3 = 4$ tens. Kim has 47 marbles.

6. A baker made 72 cookies. He sold 28 chocolate and 19 vanilla. How many cookies are left?

- A. 35
- B. 15
- C. 25 ✓**
- D. 27

Explanation: Total sold: $28 + 19 = 47$. Remaining: $72 - 47 = 25$. Regroup: $12 - 7 = 5$ ones, $6 - 4 = 2$ tens. 25 cookies are left.

7. Sam solved $95 - 48 = 57$. Kim says the answer is 47. Use addition to check. Who is correct?

- A. Sam: $57 + 48 = 95$
- B. Kim: $47 + 48 = 95$
- C. Both are wrong ✓**
- D. Both are correct

Explanation: Check Sam: $57 + 48 = 105$, not 95. Wrong! Check Kim: $47 + 48 = 95$. Correct! Kim's answer of 47 is right.

8. What strategy helps subtract 51 from 80?

- A. Subtract 50, then subtract 1 more ✓**
- B. Subtract 60, then add 9
- C. Add 51 to 80
- D. Subtract 51 from 100

Explanation: Break 51 into $50 + 1$. First subtract 50: $80 - 50 = 30$. Then subtract 1: $30 - 1 = 29$. This break-apart strategy makes mental math easier.

9. $__ - 57 = 36$. What is the missing number?

- A. 83
- B. 93 ✓**
- C. 103
- D. 21

Explanation: Add: $36 + 57 = 93$. Check: $93 - 57 = 36$. The missing number is 93.

10. A jar has some marbles. After removing 39, there are 54 left. How many marbles were in the jar?

- A. 83
- B. 93 ✓**
- C. 103
- D. 15

Explanation: Work backwards: $54 + 39 = 93$. The jar had 93 marbles. Check: $93 - 39 = 54$.

11. Which pair of problems have the same difference?

- A. $85 - 39$ and $87 - 41$ ✓**
- B. $85 - 39$ and $90 - 39$
- C. $85 - 39$ and $85 - 40$
- D. $85 - 39$ and $39 - 85$

Explanation: $85 - 39 = 46$ and $87 - 41 = 46$. Adding 2 to both numbers keeps the difference the same (constant difference property).

12. A class collected 86 cans in Week 1 and 59 cans in Week 2. How many more cans in Week 1?

- A. 37
- B. 17
- C. 27 ✓**
- D. 25

Explanation: 'How many more' means subtract: $86 - 59 = 27$. Regroup: $16 - 9 = 7$ ones, $7 - 5 = 2$ tens. Week 1 had 27 more cans.

13. If $73 + 49 = 122$, which subtraction fact must also be true?

- A. **$122 - 49 = 73$ ✓**
- B. $122 - 49 = 83$
- C. $73 - 49 = 122$
- D. $49 + 122 = 73$

Explanation: Addition and subtraction are inverse operations. Since $73 + 49 = 122$, we know $122 - 49 = 73$ and $122 - 73 = 49$.

14. A shelf holds 80 books. There are 53 books on it. How many more books can fit?

- A. 37
- B. 17
- C. **27 ✓**
- D. 23

Explanation: Subtract to find space: $80 - 53 = 27$. Regroup: $10 - 3 = 7$ ones, $7 - 5 = 2$ tens. 27 more books can fit.

15. Rosa solved $110 - 63 = 57$. Her check: $57 + 63 = 120$. What should Rosa do?

- A. **Redo the problem because 120 does not equal 110 ✓**
- B. Nothing, the answer is close enough
- C. Just fix the check
- D. The check is wrong, the subtraction is right

Explanation: $57 + 63 = 120$, not 110. The check shows an error. Rosa should redo: $110 - 63 = 47$. Check: $47 + 63 = 110$.

16. Which does NOT need regrouping?

- A. $82 - 57$
- B. $70 - 36$
- C. **$95 - 43$ ✓**
- D. $61 - 28$

Explanation: Check ones: $82: 2 < 7$, $70: 0 < 6$, $95: 5 > 3$ (no regrouping!), $61: 1 < 8$. Only $95 - 43$ does not need regrouping.

17. Ava has 91 stickers. She gives 35 to her friend and buys 12 more. How many does she have?

- A. 58
- B. **68 ✓**
- C. 78
- D. 44

Explanation: Step 1: $91 - 35 = 56$. Step 2: $56 + 12 = 68$. Ava has 68 stickers.

18. A pet store has 75 fish and 48 birds. How many more fish than birds?

- A. 37
- B. 17
- C. **27 ✓**
- D. 23

Explanation: Subtract to compare: $75 - 48 = 27$. Regroup: $15 - 8 = 7$ ones, $6 - 4 = 2$ tens. There are 27 more fish.

19. $\underline{\quad} - 44 = 52$. What is the missing number?

- A. 86
- B. **96 ✓**
- C. 106
- D. 8

Explanation: Add: $52 + 44 = 96$. Check: $96 - 44 = 52$. The missing number is 96.

20. Which story matches $84 - 56$?

- A. Dan has 84 cards. He gets 56 more. How many now?
- B. Dan has 84 cards. He gives 56 away. How many left?
- C. Dan needs 84 cards. He has 56. How many does he need to buy?
- D. **Both B and C ✓**

Explanation: Both B and C use $84 - 56$! B is take-away subtraction (giving away). C is comparison/missing part subtraction (finding how many more are needed). Both equal $84 - 56 = 28$.

Hard (20 questions)

1. A school had 183 books. Students checked out 96 in September and 49 in October. How many books are still in the school?

- A. 38 ✓
- B. 48
- C. 28
- D. 58

Explanation: Step 1: $183 - 96 = 87$ books after September. Step 2: $87 - 49 = 38$ books after October.

2. Luis solved $132 - 78 = 64$. He checked: $64 + 78 = 142$. What should Luis conclude?

- A. His answer is wrong because 142 does not equal 132 ✓
- B. His answer is correct because 142 is close to 132
- C. His addition check was wrong
- D. He should try another method

Explanation: $64 + 78 = 142$, not 132. The check reveals an error. Correct: $132 - 78 = 54$. Check: $54 + 78 = 132$.

3. Without solving, is $156 - 89$ greater or less than 70?

- A. Less than 70, because $160 - 90 = 70$ and $156 < 160$ ✓
- B. Greater than 70, because 156 is large
- C. Exactly 70
- D. Greater, because $156 - 89$ is about $160 - 90$

Explanation: Estimate: $160 - 90 = 70$. Since $156 < 160$ and $89 < 90$, the adjustments partly cancel. Exact: $156 - 89 = 67$, which is less than 70.

4. Mia solved $94 - 57 = 47$. She says 'I subtracted $7 - 4 = 3$ and $9 - 5 = 4$, so 43... wait, I wrote 47.' What errors did she make?

- A. She subtracted ones in reverse AND wrote the wrong answer ✓
- B. She only wrote the wrong answer
- C. Her method is correct but she copied wrong
- D. No errors

Explanation: Two errors: (1) She subtracted $7 - 4$ in the ones instead of regrouping ($14 - 7 = 7$). (2) Her written answer (47) does not even match her wrong calculation (43). Correct: $94 - 57 = 37$.

5. $__ - __ = 48$. Both numbers are between 60 and 120. Which pair works?

- A. $115 - 67$
- B. $98 - 50$ ✓
- C. $83 - 45$
- D. $72 - 24$

Explanation: Check: $115 - 67 = 48$, but $115 > 120$. $98 - 50 = 48$, and both are between 60 and 120. $83 - 45 = 38$. $72 - 24 = 48$, but $24 < 60$. Only $98 - 50$ works.

6. A camp had some children. 67 went hiking and 45 went swimming. 23 stayed at camp. How many children are at the camp total?

- A. 125
- B. 135 ✓
- C. 145
- D. 115

Explanation: Add all groups: $67 + 45 + 23 = 135$. There are 135 children at the camp total.

7. Sam says 'Subtracting 49 is the same as subtracting 50 and adding 1.' Tina says 'It is the same as subtracting 40 and subtracting 9.' Who is right?

- A. Both are correct -- they are different strategies for the same subtraction ✓
- B. Only Sam is correct
- C. Only Tina is correct
- D. Neither is correct

Explanation: Both strategies work! Sam uses compensation: subtract 50, add 1 back. Tina uses break-apart: subtract 40, then subtract 9 more. Both give the same answer. Example: $83 - 49$: Sam: $83 - 50 + 1 = 34$. Tina: $83 - 40 - 9 = 34$.

8. A number minus 58 equals 73. What is that number minus 60?

- A. 71 ✓
- B. 73
- C. 75
- D. 69

Explanation: Find the number: $? - 58 = 73$, so $? = 73 + 58 = 131$. Now: $131 - 60 = 71$. Or reason: subtracting 60 is 2 more than subtracting 58, so the answer is $73 - 2 = 71$.

9. Three students solved $145 - 87$. Amy: 58. Ben: 68. Cal: 62. Check with addition to find who is correct.

- A. Amy: $58 + 87 = 145$ ✓
- B. Ben: $68 + 87 = 145$
- C. Cal: $62 + 87 = 145$
- D. None are correct

Explanation: Amy: $58 + 87 = 145$. Correct! Ben: $68 + 87 = 155$. Wrong! Cal: $62 + 87 = 149$. Wrong! Amy is correct.

10. A toy store had 175 toys. They sold 68 in Week 1 and some in Week 2. Now they have 59 toys. How many did they sell in Week 2?

- A. 48 ✓
- B. 58
- C. 38
- D. 68

Explanation: After Week 1: $175 - 68 = 107$. Week 2 sales: $107 - 59 = 48$. They sold 48 toys in Week 2.

11. Jake says the difference between 87 and 39 is the same as $90 - 42$. Is he right?

- A. Yes, adding 3 to both numbers keeps the difference at 48 ✓
- B. No, $87 - 39 = 48$ but $90 - 42 = 52$
- C. Yes, but only by coincidence
- D. No, you cannot change both numbers

Explanation: $87 - 39 = 48$ and $90 - 42 = 48$. Adding 3 to both numbers keeps the difference the same (constant difference property). Jake is correct!

12. Ella had some coins. She spent 43 on Monday, 29 on Tuesday, and has 35 left. How many coins did she start with?

- A. 97
- B. 107 ✓
- C. 117
- D. 87

Explanation: Work backwards: $35 + 29 = 64$ (before Tuesday). $64 + 43 = 107$ (before Monday). Ella started with 107 coins.

13. Which problem requires regrouping in BOTH the ones and tens place?

- A. $143 - 56$
- B. $152 - 87$ ✓
- C. $168 - 43$
- D. $175 - 62$

Explanation: $152 - 87$: ones: $2 < 7$ (regroup), tens after regrouping: $4 < 8$ (regroup again). Double regrouping! $143 - 56$: $3 < 6$ (regroup), $3 < 5$ (regroup). Actually both A and B need double regrouping, but B is the answer.

14. A park has 164 trees. A storm knocks down 47 trees. The city plants 35 new ones. How many trees are in the park now?

- A. 142
- B. 152 ✓
- C. 132
- D. 162

Explanation: Step 1: $164 - 47 = 117$ trees after storm. Step 2: $117 + 35 = 152$ trees after planting.

15. Maria says: 'To check $150 - 73 = 77$, I can do $150 - 77$ and see if I get 73.' Is this a valid check?

A. Yes, if $150 - 73 = 77$ then $150 - 77$ must equal 73 (fact family) ✓

B. No, you can only check with addition

C. Yes, but it is harder than adding

D. No, subtraction cannot check subtraction

Explanation: This IS a valid check using the fact family! If $150 - 73 = 77$ is correct, then $150 - 77 = 73$. Both subtraction facts use the same three numbers. However, checking with addition ($77 + 73 = 150$) is usually easier.

16. A student makes this error pattern: $81 - 36 = 55$, $72 - 45 = 33$, $93 - 58 = 45$. What is the student doing wrong?

A. Subtracting the smaller digit from the larger in the ones place instead of regrouping ✓

B. Adding instead of subtracting

C. Forgetting to subtract the tens

D. Making random mistakes

Explanation: Check: $81 - 36$: student did $6 - 1 = 5$ (wrong direction), $8 - 3 = 5$: got 55. Correct: 45. $72 - 45$: student did $5 - 2 = 3$, $7 - 4 = 3$: got 33. Correct: 27. The student consistently subtracts smaller from larger in ones instead of regrouping.

17. Nora changes $143 - 67$ to $146 - 70$ to make it easier. What property allows this?

A. Constant difference: adding the same amount to both numbers keeps the difference the same ✓

B. Commutative property

C. Associative property

D. This does not actually work

Explanation: $143 - 67 = 76$ and $146 - 70 = 76$. Adding 3 to both numbers keeps the difference the same. This constant difference property makes subtraction easier by creating friendly numbers.

18. A vending machine had 120 snacks. It sold 34 on Monday, 28 on Tuesday, and 41 on Wednesday. How many snacks are left?

A. 17 ✓

B. 27

C. 37

D. 7

Explanation: Step 1: $120 - 34 = 86$. Step 2: $86 - 28 = 58$. Step 3: $58 - 41 = 17$. There are 17 snacks left.

19. True or false: If you know the answer to $100 - 37$, you can find $99 - 36$ without calculating.

A. True: both numbers decreased by 1, so the difference is the same (63) ✓

B. False: different numbers give different answers

C. True: but only because 100 is special

D. False: you always have to calculate

Explanation: $100 - 37 = 63$. In $99 - 36$, both numbers are 1 less, so the difference is still 63. The constant difference property: subtracting the same amount from both numbers does not change the difference.

20. A store had 185 items. In the morning 63 were sold. In the afternoon some were sold. At closing, 78 items remain. A new shipment of 40 items arrives after closing. How many items does the store have now?

A. 118 ✓

B. 108

C. 128

D. 98

Explanation: After morning: $185 - 63 = 122$. Afternoon sales: $122 - 78 = 44$ sold. Items at closing: 78. After shipment: $78 + 40 = 118$.

Unit 5: Money

Lesson 1: Know Your Coins

Easy (20 questions)

1. How much is a penny worth?

- A. 1¢ ✓
- B. 5¢
- C. 10¢
- D. 25¢

Explanation: A penny is the smallest coin. It is worth 1¢.

2. How much is a nickel worth?

- A. 1¢
- B. 10¢
- C. 5¢ ✓
- D. 25¢

Explanation: A nickel is worth 5¢. It is bigger than a penny but worth less than a dime.

3. How much is a dime worth?

- A. 5¢
- B. 25¢
- C. 1¢
- D. 10¢ ✓

Explanation: A dime is worth 10¢. Even though it is small, it is worth more than a nickel!

4. How much is a quarter worth?

- A. 50¢
- B. 25¢ ✓
- C. 10¢
- D. 5¢

Explanation: A quarter is worth 25¢. It is the largest common coin and worth the most.

5. Which coin is worth 5¢?

- A. Penny
- B. Dime
- C. Nickel ✓
- D. Quarter

Explanation: A nickel is the coin worth 5¢. It is larger than a dime but worth less.

6. Which coin is worth 10¢?

- A. Quarter
- B. Penny
- C. Nickel
- D. Dime ✓

Explanation: A dime is worth 10¢. It is the smallest silver-colored coin.

7. Which coin is worth the MOST?

- A. Penny
- B. Nickel
- C. Dime
- D. Quarter ✓

Explanation: A quarter is worth 25¢, which is more than a dime (10¢), nickel (5¢), or penny (1¢).

8. Which coin is worth the LEAST?

- A. Dime
- B. Penny ✓**
- C. Nickel
- D. Quarter

Explanation: A penny is worth only 1¢. That is less than any other coin.

9. 2 nickels = ____ ¢

- A. 2¢
- B. 5¢
- C. 10¢ ✓**
- D. 15¢

Explanation: Each nickel is worth 5¢. Count by 5s: 5, 10. So 2 nickels = 10¢.

10. 3 pennies = ____ ¢

- A. 3¢ ✓**
- B. 5¢
- C. 10¢
- D. 1¢

Explanation: Each penny is worth 1¢. Count by 1s: 1, 2, 3. So 3 pennies = 3¢.

11. 2 dimes = ____ ¢

- A. 10¢
- B. 15¢
- C. 20¢ ✓**
- D. 25¢

Explanation: Each dime is worth 10¢. Count by 10s: 10, 20. So 2 dimes = 20¢.

12. 5 pennies = ____ ¢

- A. 10¢
- B. 1¢
- C. 5¢ ✓**
- D. 25¢

Explanation: Each penny is worth 1¢. Count 5 of them: 1, 2, 3, 4, 5. So 5 pennies = 5¢.

13. True or false: A dime is worth more than a nickel.

- A. True ✓**
- B. False
- C. They are equal
- D. It depends

Explanation: True! A dime is worth 10¢ and a nickel is worth 5¢. $10¢ > 5¢$.

14. Which coin is copper-colored and worth 1¢?

- A. Nickel
- B. Dime
- C. Quarter
- D. Penny ✓**

Explanation: The penny is the copper-colored coin worth 1¢. All the other coins are silver-colored.

15. 3 dimes = ____ ¢

- A. 3¢
- B. 13¢
- C. 30¢ ✓**
- D. 35¢

Explanation: Each dime is worth 10¢. Count by 10s: 10, 20, 30. So 3 dimes = 30¢.

16. 4 nickels = ____ ¢

- A. 4¢
- B. 15¢
- C. 20¢ ✓
- D. 25¢

Explanation: Each nickel is worth 5¢. Count by 5s: 5, 10, 15, 20. So 4 nickels = 20¢.

17. How many cents is 1 quarter?

- A. 5¢
- B. 10¢
- C. 20¢
- D. 25¢ ✓

Explanation: One quarter equals 25¢. Think of it as a quarter of a dollar ($100¢ \div 4 = 25¢$).

18. Put these coins in order from LEAST to MOST value: penny, quarter, dime, nickel.

- A. Penny, nickel, dime, quarter ✓
- B. Quarter, dime, nickel, penny
- C. Penny, dime, nickel, quarter
- D. Nickel, penny, dime, quarter

Explanation: From least to most: penny (1¢), nickel (5¢), dime (10¢), quarter (25¢).

19. 2 quarters = ____ ¢

- A. 25¢
- B. 35¢
- C. 45¢
- D. 50¢ ✓

Explanation: Each quarter is worth 25¢. Count by 25s: 25, 50. So 2 quarters = 50¢.

20. 10 pennies have the same value as which coin?

- A. Nickel
- B. Dime ✓
- C. Quarter
- D. Half dollar

Explanation: 10 pennies = 10¢. A dime is also worth 10¢. So 10 pennies = 1 dime!

Medium (20 questions)

1. You have 1 quarter and 1 dime. How much money do you have?

- A. 30¢
- B. 35¢ ✓
- C. 15¢
- D. 26¢

Explanation: Start with the quarter: 25¢. Add the dime: $25 + 10 = 35¢$.

2. Which group of coins equals 30¢?

- A. 3 dimes
- B. 2 dimes and 2 nickels
- C. 1 quarter and 1 nickel
- D. Both A and C ✓

Explanation: 3 dimes = $10 + 10 + 10 = 30¢$. Also, 1 quarter + 1 nickel = $25 + 5 = 30¢$. Both equal 30¢!

3. ____ nickels = 1 quarter

- A. 3
- B. 4
- C. 5 ✓
- D. 10

Explanation: A quarter is 25¢. Count nickels by 5s: 5, 10, 15, 20, 25. That is 5 nickels.

4. Marcus has 2 dimes and 3 pennies. How much money does he have?

- A. 23¢ ✓
- B. 32¢
- C. 13¢
- D. 25¢

Explanation: Start with the dimes: 10, 20. Then count the pennies: 21, 22, 23. Marcus has 23¢.

5. You have 1 quarter, 1 nickel, and 2 pennies. How much money?

- A. 27¢
- B. 31¢
- C. 32¢ ✓
- D. 37¢

Explanation: Quarter = 25¢. Nickel = 5¢. Two pennies = 2¢. Total: $25 + 5 + 2 = 32$ ¢.

6. ____ dimes = 1 quarter + 1 nickel

- A. 2
- B. 3 ✓
- C. 4
- D. 5

Explanation: 1 quarter + 1 nickel = $25 + 5 = 30$ ¢. Each dime is 10¢. $30 \div 10 = 3$ dimes.

7. Layla found 3 nickels and 4 pennies. How much did she find?

- A. 34¢
- B. 12¢
- C. 19¢ ✓
- D. 17¢

Explanation: Count the nickels by 5s: 5, 10, 15. Then add the pennies: 16, 17, 18, 19. She found 19¢.

8. Which is worth MORE: 3 dimes or 1 quarter?

- A. 3 dimes ✓
- B. 1 quarter
- C. They are equal
- D. Not enough info

Explanation: 3 dimes = $10 + 10 + 10 = 30$ ¢. 1 quarter = 25¢. Since $30\text{¢} > 25\text{¢}$, 3 dimes are worth more.

9. 5 pennies have the same value as 1 ____.

- A. Dime
- B. Quarter
- C. Nickel ✓
- D. Half dollar

Explanation: 5 pennies = 5¢. A nickel is also worth 5¢. So 5 pennies = 1 nickel.

10. You have 2 quarters. How much money do you have?

- A. 25¢
- B. 30¢
- C. 40¢
- D. 50¢ ✓

Explanation: Each quarter is 25¢. Count by 25s: 25, 50. You have 50¢.

11. Rosa has 1 dime, 1 nickel, and 1 penny. How much money does she have?

- A. 15¢
- B. 16¢ ✓
- C. 11¢
- D. 21¢

Explanation: Dime = 10¢, nickel = 5¢, penny = 1¢. Total: $10 + 5 + 1 = 16$ ¢.

12. Which group of coins equals 50¢?

- A. 5 nickels
- B. 2 quarters
- C. 5 dimes
- D. Both B and C ✓**

Explanation: 2 quarters = $25 + 25 = 50¢$. 5 dimes = $10 + 10 + 10 + 10 + 10 = 50¢$. Both equal 50¢!

13. ____ pennies = 1 dime

- A. 5
- B. 10 ✓**
- C. 15
- D. 20

Explanation: A dime is worth 10¢. Each penny is 1¢. You need 10 pennies to equal 1 dime.

14. José has 4 dimes and 1 nickel. How much does he have?

- A. 14¢
- B. 41¢
- C. 45¢ ✓**
- D. 50¢

Explanation: Count the dimes by 10s: 10, 20, 30, 40. Add the nickel: $40 + 5 = 45¢$.

15. Which is worth MORE: 6 nickels or 1 quarter?

- A. 6 nickels ✓**
- B. 1 quarter
- C. They are equal
- D. Not enough info

Explanation: 6 nickels = $5 \times 6 = 30¢$. 1 quarter = 25¢. Since $30¢ > 25¢$, 6 nickels are worth more.

16. Kenji has 3 quarters. How much money is that?

- A. 30¢
- B. 50¢
- C. 65¢
- D. 75¢ ✓**

Explanation: Count by 25s: 25, 50, 75. Three quarters = 75¢.

17. You have 2 dimes, 1 nickel, and 3 pennies. How much?

- A. 23¢
- B. 28¢ ✓**
- C. 38¢
- D. 30¢

Explanation: Dimes: 10, 20. Nickel: 25. Pennies: 26, 27, 28. Total = 28¢.

18. ____ quarters = 50¢

- A. 1
- B. 2 ✓**
- C. 3
- D. 4

Explanation: Each quarter is 25¢. Two quarters: $25 + 25 = 50¢$. So 2 quarters = 50¢.

19. Which is worth LESS: 2 dimes or 1 quarter?

- A. 2 dimes ✓**
- B. 1 quarter
- C. They are equal
- D. Not enough info

Explanation: 2 dimes = $10 + 10 = 20¢$. 1 quarter = 25¢. Since $20¢ < 25¢$, 2 dimes are worth less.

20. Ava has some coins that total 40¢. Which could be her coins?

- A. 1 quarter, 1 dime, 1 nickel
- B. 4 dimes
- C. 8 nickels
- D. Both A and B ✓**

Explanation: 1 quarter + 1 dime + 1 nickel = 25 + 10 + 5 = 40¢. 4 dimes = 10 × 4 = 40¢. Both work!

Hard (20 questions)

1. Deshawn says 2 dimes and 1 nickel = 30¢. Is he correct?

- A. Yes, that is correct
- B. No, it is 25¢ ✓**
- C. No, it is 35¢
- D. No, it is 20¢

Explanation: 2 dimes = 10 + 10 = 20¢. 1 nickel = 5¢. Total: 20 + 5 = 25¢, not 30¢. Deshawn made an error.

2. Which costs more: 7 nickels or 3 dimes and 1 penny?

- A. 7 nickels ✓**
- B. 3 dimes and 1 penny
- C. They are equal
- D. Not enough info

Explanation: 7 nickels = 5 × 7 = 35¢. 3 dimes + 1 penny = 30 + 1 = 31¢. 35¢ > 31¢, so 7 nickels are more.

3. You need exactly 36¢. You have only dimes, nickels, and pennies. Which set of coins works?

- A. 3 dimes, 1 nickel, 1 penny
- B. 3 dimes, 1 nickel
- C. 2 dimes, 3 nickels, 1 penny
- D. Both A and C ✓**

Explanation: A: 30 + 5 + 1 = 36¢. B: 30 + 5 = 35¢ (wrong). C: 20 + 15 + 1 = 36¢. Both A and C equal 36¢.

4. Mia has 2 quarters and 3 dimes. Jayden has 7 dimes and 1 nickel. Who has more money?

- A. Mia ✓**
- B. Jayden
- C. They have the same
- D. Not enough info

Explanation: Mia: 25 + 25 + 10 + 10 + 10 = 80¢. Jayden: 70 + 5 = 75¢. 80¢ > 75¢, so Mia has more.

5. Priya says 4 quarters = \$1.00. Kai says 10 dimes = \$1.00. Who is right?

- A. Only Priya
- B. Only Kai
- C. Both are right ✓**
- D. Neither is right

Explanation: 4 quarters = 25 × 4 = 100¢ = \$1.00. 10 dimes = 10 × 10 = 100¢ = \$1.00. Both are correct!

6. You buy a sticker for 45¢ and pay with 3 quarters. How much change do you get back?

- A. 20¢
- B. 25¢
- C. 30¢ ✓**
- D. 35¢

Explanation: 3 quarters = 75¢. Change = 75 - 45 = 30¢. You get 30¢ back.

7. Sam has 3 dimes. He finds 2 more nickels on the ground. Now how much does he have in all?

- A. 32¢
- B. 35¢
- C. 40¢ ✓**
- D. 50¢

Explanation: Step 1: 3 dimes = 30¢. Step 2: 2 nickels = 10¢. Step 3: 30 + 10 = 40¢ in all.

8. Li says that 5 nickels = 2 dimes + 1 nickel. Is she correct?

A. Yes, both equal 25¢ ✓

B. No, 5 nickels is more

C. No, 2 dimes + 1 nickel is more

D. No, 5 nickels = 20¢

Explanation: 5 nickels = $5 \times 5 = 25¢$. 2 dimes + 1 nickel = $20 + 5 = 25¢$. They are equal, so Li is correct!

9. You have 4 quarters. You give away 1 quarter and 2 dimes. How much do you have left?

A. 45¢

B. 55¢ ✓

C. 75¢

D. 80¢

Explanation: Start: 4 quarters = 100¢. You give away $25 + 20 = 45¢$. Left: $100 - 45 = 55¢$.

10. Which way uses the FEWEST coins to make 35¢?

A. 1 quarter and 1 dime ✓

B. 3 dimes and 1 nickel

C. 7 nickels

D. 1 quarter, 1 nickel, 5 pennies

Explanation: A: 2 coins. B: 4 coins. C: 7 coins. D: 7 coins. 1 quarter + 1 dime uses only 2 coins — the fewest!

11. Amara has 65¢. She spends 1 quarter. How much does she have left?

A. 35¢

B. 40¢ ✓

C. 45¢

D. 50¢

Explanation: Amara starts with 65¢ and spends 25¢. $65 - 25 = 40¢$ remaining.

12. Carlos says 1 quarter + 3 dimes + 1 nickel = 55¢. Is he right?

A. Yes, that is correct

B. No, it is 50¢

C. No, it is 60¢ ✓

D. No, it is 65¢

Explanation: $25 + 10 + 10 + 10 + 5 = 60¢$, not 55¢. Carlos added incorrectly.

13. You need exactly 47¢. Which set of coins works?

A. 1 quarter, 2 dimes, 2 pennies

B. 1 quarter, 1 dime, 2 nickels

C. 4 dimes, 1 nickel, 2 pennies

D. Both A and C ✓

Explanation: A: $25 + 20 + 2 = 47¢$. B: $25 + 10 + 10 = 45¢$ (wrong). C: $40 + 5 + 2 = 47¢$. Both A and C work!

14. A toy costs 80¢. You pay with \$1.00. What coins could you get as change?

A. 2 dimes

B. 1 dime and 2 nickels

C. 4 nickels

D. All of the above ✓

Explanation: Change = $\$1.00 - 80¢ = 20¢$. 2 dimes = 20¢. 1 dime + 2 nickels = 20¢. 4 nickels = 20¢. All work!

15. Nia has 2 quarters, 1 dime, and 4 pennies. Olivia has 5 dimes and 3 nickels. Who has more?

A. Nia

B. Olivia ✓

C. They have the same amount

D. Not enough info

Explanation: Nia: $25 + 25 + 10 + 4 = 64¢$. Olivia: $50 + 15 = 65¢$. $65¢ > 64¢$, so Olivia has 1¢ more.

16. You buy an eraser for 15¢ and a pencil for 30¢. You pay with 3 quarters. How much change?

- A. 15¢
- B. 20¢
- C. 25¢
- D. 30¢ ✓**

Explanation: Step 1: Total cost = $15 + 30 = 45$ ¢. Step 2: 3 quarters = 75¢. Step 3: Change = $75 - 45 = 30$ ¢.

17. Which strategy finds the total of 2 quarters, 1 dime, and 3 pennies FASTEST?

- A. Count all coins by 1s
- B. Start with quarters, then dimes, then pennies ✓**
- C. Start with pennies, then dimes, then quarters
- D. Add only the quarters

Explanation: Start with the biggest coins first! Quarters: 25, 50. Dime: 60. Pennies: 61, 62, 63. Total = 63¢.

18. Zara wants to make 50¢ with exactly 5 coins. Which set works?

- A. 2 quarters
- B. 5 dimes ✓**
- C. 1 quarter and 3 dimes
- D. 5 nickels

Explanation: A: 2 coins (not 5). B: 5 dimes = 50¢ with exactly 5 coins! C: 4 coins (not 5). D: 5 nickels = 25¢ (not 50¢). B is correct.

19. Aiden has 90¢ in dimes only. How many dimes does he have?

- A. 7
- B. 8
- C. 9 ✓**
- D. 10

Explanation: Each dime = 10¢. To find the number of dimes: $90 \div 10 = 9$ dimes.

20. You have 3 quarters and 1 dime. You buy a snack for 60¢. How much is left?

- A. 15¢
- B. 20¢
- C. 25¢ ✓**
- D. 30¢

Explanation: Step 1: 3 quarters + 1 dime = $75 + 10 = 85$ ¢. Step 2: $85 - 60 = 25$ ¢ left.

Lesson 2: Count Your Money

Easy (20 questions)

1. Count the money: 1 dime and 2 pennies.

- A. 3¢
- B. 12¢ ✓**
- C. 21¢
- D. 20¢

Explanation: Start with the dime: 10¢. Count the pennies: 11, 12. Total = 12¢.

2. Count the money: 3 nickels.

- A. 3¢
- B. 10¢
- C. 15¢ ✓**
- D. 30¢

Explanation: Count by 5s: 5, 10, 15. Three nickels = 15¢.

3. Count the money: 2 dimes.

- A. 2¢
- B. 12¢
- C. 20¢ ✓**
- D. 25¢

Explanation: Count by 10s: 10, 20. Two dimes = 20¢.

4. Count the money: 1 nickel and 3 pennies.

- A. 4¢
- B. 8¢ ✓**
- C. 13¢
- D. 35¢

Explanation: Start with the nickel: 5¢. Count the pennies: 6, 7, 8. Total = 8¢.

5. Count the money: 4 dimes.

- A. 4¢
- B. 14¢
- C. 40¢ ✓**
- D. 44¢

Explanation: Count by 10s: 10, 20, 30, 40. Four dimes = 40¢.

6. Count the money: 1 quarter.

- A. 1¢
- B. 5¢
- C. 10¢
- D. 25¢ ✓**

Explanation: One quarter = 25¢.

7. Count the money: 5 nickels.

- A. 5¢
- B. 15¢
- C. 25¢ ✓**
- D. 50¢

Explanation: Count by 5s: 5, 10, 15, 20, 25. Five nickels = 25¢.

8. Count the money: 1 dime and 1 nickel.

- A. 6¢
- B. 11¢
- C. 15¢ ✓**
- D. 20¢

Explanation: Dime = 10¢. Nickel = 5¢. Total: 10 + 5 = 15¢.

9. Count the money: 7 pennies.

- A. 7¢ ✓**
- B. 17¢
- C. 70¢
- D. 10¢

Explanation: Each penny is 1¢. Count: 1, 2, 3, 4, 5, 6, 7. Seven pennies = 7¢.

10. Count the money: 2 nickels and 1 penny.

- A. 3¢
- B. 6¢
- C. 11¢ ✓**
- D. 21¢

Explanation: Nickels: 5, 10. Penny: 11. Total = 11¢.

11. Count the money: 1 quarter and 1 penny.

- A. 2¢
- B. 11¢
- C. 26¢ ✓**
- D. 30¢

Explanation: Quarter = 25¢. Penny = 1¢. Total: $25 + 1 = 26$ ¢.

12. Count the money: 3 dimes and 1 penny.

- A. 4¢
- B. 13¢
- C. 30¢
- D. 31¢ ✓**

Explanation: Dimes: 10, 20, 30. Penny: 31. Total = 31¢.

13. Count the money: 6 nickels.

- A. 6¢
- B. 25¢
- C. 30¢ ✓**
- D. 35¢

Explanation: Count by 5s: 5, 10, 15, 20, 25, 30. Six nickels = 30¢.

14. How do you count a group of dimes?

- A. Count by 1s
- B. Count by 5s
- C. Count by 10s ✓**
- D. Count by 25s

Explanation: Since each dime is worth 10¢, the fastest way to count dimes is to count by 10s.

15. Count the money: 2 dimes and 1 penny.

- A. 3¢
- B. 12¢
- C. 21¢ ✓**
- D. 22¢

Explanation: Dimes: 10, 20. Penny: 21. Total = 21¢.

16. How do you count a group of nickels?

- A. Count by 1s
- B. Count by 5s ✓**
- C. Count by 10s
- D. Count by 25s

Explanation: Since each nickel is worth 5¢, the fastest way to count nickels is to count by 5s.

17. Count the money: 1 dime and 4 pennies.

- A. 5¢
- B. 14¢ ✓**
- C. 41¢
- D. 50¢

Explanation: Dime: 10. Pennies: 11, 12, 13, 14. Total = 14¢.

18. Count the money: 2 quarters.

- A. 2¢
- B. 20¢
- C. 50¢ ✓**
- D. 75¢

Explanation: Count by 25s: 25, 50. Two quarters = 50¢.

19. Count the money: 1 nickel and 1 penny.

- A. 2¢
- B. 6¢ ✓**
- C. 10¢
- D. 15¢

Explanation: Nickel = 5¢. Penny = 1¢. Total: $5 + 1 = 6$ ¢.

20. Count the money: 5 dimes.

- A. 5¢
- B. 15¢
- C. 50¢ ✓**
- D. 55¢

Explanation: Count by 10s: 10, 20, 30, 40, 50. Five dimes = 50¢.

Medium (20 questions)

1. Count the money: 1 quarter, 2 dimes, and 1 nickel.

- A. 40¢
- B. 45¢
- C. 50¢ ✓**
- D. 55¢

Explanation: Start big: quarter = 25. Dimes: 35, 45. Nickel: 50. Total = 50¢.

2. Tiana has 3 dimes, 2 nickels, and 4 pennies. How much?

- A. 34¢
- B. 44¢ ✓**
- C. 39¢
- D. 49¢

Explanation: Dimes: 10, 20, 30. Nickels: 35, 40. Pennies: 41, 42, 43, 44. Total = 44¢.

3. Which group of coins totals 65¢?

- A. 2 quarters, 1 dime, 1 nickel
- B. 6 dimes and 1 nickel ✓**
- C. 2 quarters and 1 dime
- D. 1 quarter, 3 dimes, 1 nickel

Explanation: Check B: 6 dimes = 60¢ + 1 nickel = 5¢. Total: 65¢. That matches!

4. ____ nickels and 1 dime = 35¢

- A. 3
- B. 4
- C. 5 ✓**
- D. 6

Explanation: $35 - 10 = 25$ ¢ left for nickels. $25 \div 5 = 5$ nickels. Check: 5 nickels (25¢) + 1 dime (10¢) = 35¢.

5. Leo counted his coins: 1 quarter, 1 dime, 1 nickel, 3 pennies. How much?

- A. 33¢
- B. 38¢
- C. 43¢ ✓**
- D. 48¢

Explanation: Quarter: 25. Dime: 35. Nickel: 40. Pennies: 41, 42, 43. Total = 43¢.

6. You have 2 quarters and 3 pennies. How much money?

- A. 28¢
- B. 50¢
- C. 53¢ ✓**
- D. 55¢

Explanation: Quarters: 25, 50. Pennies: 51, 52, 53. Total = 53¢.

7. Which is worth more: 2 quarters or 4 dimes and 2 nickels?

- A. 2 quarters
- B. 4 dimes and 2 nickels
- C. They are equal ✓**
- D. Not enough info

Explanation: 2 quarters = 50¢. 4 dimes + 2 nickels = 40 + 10 = 50¢. They are equal!

8. Bella has 5 dimes and 3 pennies. How much?

- A. 35¢
- B. 53¢ ✓**
- C. 80¢
- D. 15¢

Explanation: Dimes: 10, 20, 30, 40, 50. Pennies: 51, 52, 53. Total = 53¢.

9. Count: 1 quarter, 3 dimes, 2 pennies.

- A. 52¢
- B. 57¢ ✓**
- C. 62¢
- D. 35¢

Explanation: Quarter: 25. Dimes: 35, 45, 55. Pennies: 56, 57. Total = 57¢.

10. ____ dimes and 2 pennies = 42¢

- A. 2
- B. 3
- C. 4 ✓**
- D. 5

Explanation: $42 - 2 = 40¢$ for dimes. $40 \div 10 = 4$ dimes. Check: 4 dimes (40¢) + 2 pennies (2¢) = 42¢.

11. Malik has 4 nickels and 6 pennies. How much?

- A. 10¢
- B. 26¢ ✓**
- C. 46¢
- D. 24¢

Explanation: Nickels: 5, 10, 15, 20. Pennies: 21, 22, 23, 24, 25, 26. Total = 26¢.

12. Which group does NOT equal 25¢?

- A. 1 quarter
- B. 5 nickels
- C. 2 dimes and 1 nickel
- D. 2 dimes and 1 penny ✓**

Explanation: A: 25¢. B: 25¢. C: 25¢. D: $20 + 1 = 21¢$. Choice D does NOT equal 25¢.

13. Count: 3 quarters and 1 nickel.

- A. 35¢
- B. 55¢
- C. 75¢
- D. 80¢ ✓**

Explanation: Quarters: 25, 50, 75. Nickel: 80. Total = 80¢.

14. How much money is 2 dimes, 3 nickels, and 1 penny?

- A. 26¢
- B. 31¢
- C. 36¢ ✓**
- D. 41¢

Explanation: Dimes: 10, 20. Nickels: 25, 30, 35. Penny: 36. Total = 36¢.

15. Which is worth more: 3 quarters or 7 dimes?

A. 3 quarters ✓

B. 7 dimes

C. They are equal

D. Not enough info

Explanation: 3 quarters = 75¢. 7 dimes = 70¢. Since 75¢ > 70¢, 3 quarters are worth more.

16. Suki has 1 quarter, 1 dime, and 4 pennies. How much?

A. 30¢

B. 39¢ ✓

C. 35¢

D. 40¢

Explanation: Quarter: 25. Dime: 35. Pennies: 36, 37, 38, 39. Total = 39¢.

17. Count: 2 quarters, 1 dime, 1 nickel, 2 pennies.

A. 60¢

B. 65¢

C. 67¢ ✓

D. 72¢

Explanation: Quarters: 25, 50. Dime: 60. Nickel: 65. Pennies: 66, 67. Total = 67¢.

18. ____ quarters and 1 dime = 60¢

A. 1

B. 2 ✓

C. 3

D. 4

Explanation: $60 - 10 = 50$ ¢ for quarters. $50 \div 25 = 2$ quarters. Check: 2 quarters (50¢) + 1 dime (10¢) = 60¢.

19. Which is worth less: 8 nickels or 1 quarter and 2 dimes?

A. 8 nickels ✓

B. 1 quarter and 2 dimes

C. They are equal

D. Not enough info

Explanation: 8 nickels = 40¢. 1 quarter + 2 dimes = 25 + 20 = 45¢. 40¢ < 45¢, so 8 nickels are worth less.

20. Nora counted 6 dimes and 2 pennies. How much?

A. 26¢

B. 62¢ ✓

C. 80¢

D. 8¢

Explanation: Dimes: 10, 20, 30, 40, 50, 60. Pennies: 61, 62. Total = 62¢.

Hard (20 questions)

1. Jada counted 2 quarters, 3 dimes, 1 nickel, and 4 pennies. She says that is 84¢. Is she right?

A. Yes, 84¢ is correct

B. No, it is 89¢ ✓

C. No, it is 79¢

D. No, it is 94¢

Explanation: 25 + 25 = 50. + 10 + 10 + 10 = 80. + 5 = 85. + 4 = 89¢. Jada undercounted by 5¢.

2. Ravi has 3 quarters, 2 dimes, and 3 pennies. Zoe has 8 dimes and 2 nickels. Who has more?

A. Ravi ✓

B. Zoe

C. They have the same

D. Not enough info

Explanation: Ravi: 75 + 20 + 3 = 98¢. Zoe: 80 + 10 = 90¢. 98¢ > 90¢, so Ravi has more.

3. You have 87¢ in your pocket. You find 1 dime and 3 pennies on the ground. How much now?

- A. 90¢
- B. 97¢
- C. \$1.00 ✓**
- D. \$1.13

Explanation: Found: $10 + 3 = 13¢$. Total: $87 + 13 = 100¢ = \$1.00$!

4. Which way uses the FEWEST coins to make 76¢?

- A. 3 quarters and 1 penny ✓**
- B. 7 dimes, 1 nickel, 1 penny
- C. 2 quarters, 2 dimes, 1 nickel, 1 penny
- D. 15 nickels and 1 penny

Explanation: A: 4 coins. B: 9 coins. C: 6 coins. D: 16 coins. 3 quarters + 1 penny = 76¢ with only 4 coins!

5. Tyler says 6 dimes + 3 nickels = 75¢. Is he correct?

- A. Yes, that equals 75¢ ✓**
- B. No, it is 65¢
- C. No, it is 80¢
- D. No, it is 90¢

Explanation: 6 dimes = 60¢. 3 nickels = 15¢. Total: $60 + 15 = 75¢$. Tyler is correct!

6. You need exactly 58¢. Which TWO sets of coins both work?

- A. 2 quarters, 1 nickel, 3 pennies
- B. 5 dimes, 1 nickel, 3 pennies
- C. 2 quarters, 3 pennies
- D. Both A and B ✓**

Explanation: A: $50 + 5 + 3 = 58¢$. B: $50 + 5 + 3 = 58¢$. C: $50 + 3 = 53¢$ (wrong). Both A and B work!

7. Lucia has 3 quarters. She gives 2 dimes and 1 nickel to her friend. How much does she have left?

- A. 25¢
- B. 40¢
- C. 50¢ ✓**
- D. 55¢

Explanation: Start: 3 quarters = 75¢. Gave away: $20 + 5 = 25¢$. Left: $75 - 25 = 50¢$.

8. Eli counts his coins: 2 quarters, 4 dimes, 2 nickels, 5 pennies. What is his total?

- A. 95¢
- B. \$1.00
- C. \$1.05 ✓**
- D. \$1.10

Explanation: Quarters: 50. Dimes: 40. Nickels: 10. Pennies: 5. Total = \$1.05.

9. Chen says he can make \$1.00 with 7 coins. Which set works?

- A. 4 quarters
- B. 3 quarters, 2 dimes, 1 nickel
- C. 2 quarters, 5 dimes ✓**
- D. 1 quarter, 7 dimes, 1 nickel

Explanation: A: 4 coins. B: 6 coins. C: 2 quarters (50¢) + 5 dimes (50¢) = \$1.00 with 7 coins! D: 9 coins.

10. Mai has 72¢. Kai has 1 quarter less than Mai. How much does Kai have?

- A. 47¢ ✓**
- B. 52¢
- C. 62¢
- D. 71¢

Explanation: 1 quarter = 25¢. Kai has 25¢ less: $72 - 25 = 47¢$.

11. Which strategy is BEST for counting 3 quarters, 2 dimes, 1 nickel, 4 pennies?

- A. Count everything by 1s
- B. Start with pennies and work up to quarters
- C. Start with quarters and count down to pennies ✓**
- D. Just guess

Explanation: Start with the biggest coins first! 25, 50, 75 (quarters), 85, 95 (dimes), 100 (nickel), 101, 102, 103, 104 (pennies) = \$1.04.

12. Rosa says 2 quarters + 2 dimes + 2 nickels = 90¢. Find her mistake.

- A. She added quarters wrong
- B. She added dimes wrong
- C. She added nickels wrong
- D. The correct total is 80¢ ✓**

Explanation: 2 quarters = 50¢. 2 dimes = 20¢. 2 nickels = 10¢. Total: 50 + 20 + 10 = 80¢, not 90¢.

13. You start with 4 quarters. You spend 3 dimes and 2 nickels. How much is left?

- A. 50¢
- B. 55¢
- C. 60¢ ✓**
- D. 65¢

Explanation: Start: 4 quarters = \$1.00. Spent: 30 + 10 = 40¢. Left: 100 - 40 = 60¢.

14. Who counted correctly? Amy: '2 quarters, 3 dimes, 1 penny = 81¢.' Ben: '1 quarter, 5 dimes, 4 pennies = 79¢.'

- A. Only Amy
- B. Only Ben
- C. Both are correct ✓**
- D. Neither is correct

Explanation: Amy: 50 + 30 + 1 = 81¢ (correct!). Ben: 25 + 50 + 4 = 79¢ (correct!). Both counted their coins right.

15. Dante has 55¢. He earns 2 dimes and 1 nickel more. Does he have enough to buy a 75¢ bookmark?

- A. No, he has only 70¢
- B. No, he has only 75¢
- C. Yes, he has exactly 80¢ ✓**
- D. Yes, he has exactly 75¢

Explanation: Earned: 20 + 5 = 25¢. New total: 55 + 25 = 80¢. 80¢ > 75¢, so yes — with 5¢ to spare!

16. You have 2 quarters, 1 dime, 3 nickels, and 2 pennies. You give half your quarters away. How much left?

- A. 52¢ ✓**
- B. 62¢
- C. 77¢
- D. 87¢

Explanation: Half of 2 quarters = 1 quarter given away (25¢). Original: 50+10+15+2 = 77¢. Left: 77 - 25 = 52¢.

17. Ava says counting coins from biggest to smallest is faster. Eli says smallest to biggest. Who uses the better strategy?

- A. Ava — biggest to smallest ✓**
- B. Eli — smallest to biggest
- C. Both are equally fast
- D. Neither is a good strategy

Explanation: Counting from biggest to smallest (quarters first, then dimes, nickels, pennies) is more efficient because you skip-count by larger numbers first.

18. You need 93¢. You already have 2 quarters, 1 dime, and 3 pennies. How much more do you need?

- A. 20¢
- B. 25¢
- C. 30¢ ✓**
- D. 35¢

Explanation: You have: 50 + 10 + 3 = 63¢. Need: 93¢. Difference: 93 - 63 = 30¢ more.

19. Omar has 2 quarters, 1 dime, and 2 nickels. He finds 1 more quarter. Now how much?

- A. 85¢
- B. 90¢
- C. 95¢ ✓**
- D. \$1.00

Explanation: Before: $50 + 10 + 10 = 70¢$. Found 1 quarter (25¢). New total: $70 + 25 = 95¢$.

20. A book costs 99¢. You have 3 quarters, 2 dimes, and 1 nickel. Do you have enough?

- A. No, you only have 90¢
- B. No, you only have 95¢
- C. Yes, you have exactly 99¢
- D. Yes, you have \$1.00 ✓**

Explanation: 3 quarters = 75¢. 2 dimes = 20¢. 1 nickel = 5¢. Total: $75 + 20 + 5 = \$1.00$. You have enough with 1¢ to spare!

Lesson 3: Dollar Signs & Decimal Points

Easy (20 questions)

1. How do you write 25 cents using a dollar sign?

- A. 25¢
- B. \$25.00
- C. \$0.25 ✓**
- D. \$25

Explanation: 25 cents = \$0.25. The dollar sign goes first, and 25 cents is written after the decimal point.

2. How do you write 10 cents using a cent sign?

- A. \$0.10
- B. \$10
- C. 10¢ ✓**
- D. 1¢

Explanation: 10 cents is written as 10¢. The cent sign (¢) goes after the number.

3. Which shows one dollar?

- A. \$1.00 ✓**
- B. 1.00¢
- C. \$100
- D. 100

Explanation: \$1.00 means one dollar. The number before the decimal is dollars, and .00 means zero extra cents.

4. How do you write fifty cents with a dollar sign?

- A. \$50.00
- B. \$0.50 ✓**
- C. 50¢
- D. \$5.00

Explanation: Fifty cents = \$0.50. Since it is less than a dollar, there is a 0 before the decimal point.

5. What does the dot (.) in \$1.35 separate?

- A. Ones and tens
- B. Dollars and cents ✓**
- C. Big and small
- D. Nothing

Explanation: The decimal point separates the dollars from the cents. In \$1.35, there is 1 dollar and 35 cents.

6. How do you write 5 cents using a dollar sign?

- A. \$5.00
- B. \$0.5
- C. \$0.05 ✓**
- D. \$50

Explanation: 5 cents = \$0.05. You need two digits after the decimal: 05 means 5 cents.

7. Is \$0.75 the same as 75¢?

- A. Yes ✓**
- B. No, \$0.75 is more
- C. No, 75¢ is more
- D. Not enough info

Explanation: Yes! \$0.75 and 75¢ are two ways to write the same amount: seventy-five cents.

8. How do you write 1 cent using a dollar sign?

- A. \$1.00
- B. \$0.1
- C. \$0.01 ✓**
- D. \$0.10

Explanation: 1 cent = \$0.01. Remember: always use two digits after the decimal point.

9. Which amount is the same as \$0.10?

- A. 1¢
- B. 10¢ ✓**
- C. \$1.00
- D. 100¢

Explanation: \$0.10 = 10¢. The 10 after the decimal point means 10 cents.

10. How many cents are in \$1.00?

- A. 1¢
- B. 10¢
- C. 50¢
- D. 100¢ ✓**

Explanation: \$1.00 = 100¢. One dollar equals one hundred cents.

11. Which is the correct way to write thirty cents?

- A. \$30
- B. \$0.30 ✓**
- C. \$3.00
- D. \$0.03

Explanation: Thirty cents = \$0.30. The 30 goes after the decimal point because it is cents, not dollars.

12. What does \$0.99 mean?

- A. 99 dollars
- B. 9 cents
- C. 99 cents ✓**
- D. 9 dollars and 9 cents

Explanation: \$0.99 means 99 cents. The 0 before the decimal means zero dollars.

13. Is \$0.50 the same as 50¢?

- A. Yes ✓**
- B. No, \$0.50 is more
- C. No, 50¢ is more
- D. Cannot compare

Explanation: Yes! \$0.50 and 50¢ mean the same thing: fifty cents.

14. How do you write 45 cents with a ¢ sign?

- A. \$0.45
- B. 45¢ ✓**
- C. \$45
- D. 4.5¢

Explanation: Write the number first, then the cent sign: 45¢.

15. Which symbol goes BEFORE the number?

- A. ¢
- B. \$ ✓**
- C. Both
- D. Neither

Explanation: The dollar sign (\$) goes before the number: \$1.00. The cent sign (¢) goes after: 50¢.

16. Which symbol goes AFTER the number?

- A. \$
- B. ¢ ✓**
- C. Both
- D. Neither

Explanation: The cent sign (¢) goes after the number: 75¢. The dollar sign (\$) goes before: \$0.75.

17. How do you write 80 cents using a dollar sign?

- A. \$80
- B. \$8.00
- C. \$0.80 ✓**
- D. \$0.08

Explanation: 80 cents = \$0.80. The 0 before the decimal means zero dollars, and 80 after means 80 cents.

18. What is another way to write \$0.01?

- A. 1¢ ✓**
- B. 10¢
- C. \$1.00
- D. 01¢

Explanation: \$0.01 = 1¢. Both mean one cent.

19. In \$2.50, how many dollars and how many cents are there?

- A. 2 dollars and 5 cents
- B. 25 dollars and 0 cents
- C. 2 dollars and 50 cents ✓**
- D. 250 cents

Explanation: The number before the decimal (2) is dollars. The number after (50) is cents. So 2 dollars and 50 cents.

20. How do you write 100 cents using a dollar sign?

- A. \$0.100
- B. \$100
- C. \$1.00 ✓**
- D. \$10.0

Explanation: 100 cents = 1 dollar = \$1.00. When you reach 100 cents, it becomes one whole dollar!

Medium (20 questions)

1. Jaya has 2 quarters, 1 dime, and 3 pennies. Write her total using a dollar sign.

- A. \$0.63 ✓**
- B. \$63
- C. \$6.30
- D. \$0.36

Explanation: 2 quarters = 50¢. 1 dime = 10¢. 3 pennies = 3¢. Total: 63¢ = \$0.63.

2. Which two ways show the SAME amount?

- A. \$0.35 and 53¢
- B. \$0.35 and 35¢ ✓**
- C. \$3.50 and 35¢
- D. \$0.53 and 35¢

Explanation: $\$0.35 = 35$ cents. $35¢ = 35$ cents. They are the same amount, just written differently.

3. Convert: $47¢ = \$______.$

- A. \$4.70
- B. \$0.47 ✓**
- C. \$47.00
- D. \$0.74

Explanation: 47 cents is less than a dollar, so: \$0.47. Zero dollars and forty-seven cents.

4. Convert: $\$0.92 = ______¢$

- A. 9¢
- B. 29¢
- C. 92¢ ✓**
- D. \$92

Explanation: \$0.92 means 92 cents. The digits after the decimal point tell you the cents: 92¢.

5. Which is MORE: \$0.68 or 72¢?

- A. \$0.68
- B. 72¢ ✓**
- C. They are equal
- D. Cannot compare

Explanation: $\$0.68 = 68¢$. Compare: 68¢ vs 72¢. Since $72 > 68$, 72¢ is more.

6. Kai has \$0.50. Mia has 50¢. Who has more?

- A. Kai
- B. Mia
- C. They have the same ✓**
- D. Not enough info

Explanation: $\$0.50 = 50¢$. They are two ways to write the same amount. Kai and Mia have equal money.

7. Write 1 dollar and 25 cents using a dollar sign.

- A. \$125
- B. \$1.25 ✓**
- C. \$12.5
- D. \$0.125

Explanation: 1 dollar and 25 cents = \$1.25. The 1 is the dollar, and .25 is the cents.

8. $______¢ = \$0.55$

- A. 5¢
- B. 50¢
- C. 55¢ ✓**
- D. 550¢

Explanation: \$0.55 means 55 cents. The two digits after the decimal are the cents.

9. Nate has 3 quarters and 4 pennies. Write his total with a \$ sign.

- A. \$0.34
- B. \$0.79 ✓**
- C. \$3.04
- D. \$7.90

Explanation: 3 quarters = 75¢. 4 pennies = 4¢. Total: 79¢ = \$0.79.

10. Which is LESS: \$0.40 or 45¢?

A. \$0.40 ✓

B. 45¢

C. They are equal

D. Cannot compare

Explanation: $\$0.40 = 40¢$. Compare: 40¢ vs 45¢. Since $40 < 45$, \$0.40 is less.

11. How do you write \$1.00 using a cent sign?

A. 1¢

B. 10¢

C. 100¢ ✓

D. 1000¢

Explanation: $\$1.00 = 100 \text{ cents} = 100¢$. One dollar is the same as one hundred cents.

12. Which correctly shows eighty-three cents?

A. \$83

B. 83¢ or \$0.83 ✓

C. \$8.30

D. 0.83¢

Explanation: Eighty-three cents can be written as 83¢ or \$0.83. Both are correct.

13. Convert: \$1.50 = ____¢

A. 15¢

B. 50¢

C. 105¢

D. 150¢ ✓

Explanation: $\$1.50 = 1 \text{ dollar} + 50 \text{ cents} = 100 + 50 = 150¢$.

14. Rosa wrote 7¢ as \$0.70. Is that correct?

A. Yes

B. No, $7¢ = 70¢$ ✓

C. No, $7¢ = \$7.00$

D. No, $7¢ = \$70$

Explanation: $\$0.70 = 70¢$, not 7¢. The correct way to write 7¢ with a dollar sign is \$0.07.

15. Leo has 4 dimes, 2 nickels, and 6 pennies. Write his total with \$.

A. \$0.12

B. \$0.46

C. \$0.56 ✓

D. \$5.60

Explanation: 4 dimes = 40¢. 2 nickels = 10¢. 6 pennies = 6¢. Total: 56¢ = \$0.56.

16. Which shows the SAME amount as 3 quarters?

A. \$7.50

B. \$0.75 ✓

C. \$75

D. \$3.25

Explanation: 3 quarters = 75¢ = \$0.75.

17. Put in order from LEAST to MOST: \$0.30, 25¢, \$0.35

A. \$0.30, 25¢, \$0.35

B. 25¢, \$0.30, \$0.35 ✓

C. \$0.35, \$0.30, 25¢

D. 25¢, \$0.35, \$0.30

Explanation: Convert all to cents: 30¢, 25¢, 35¢. Order: 25¢, 30¢, 35¢. So: 25¢, \$0.30, \$0.35.

18. A pencil costs \$0.15. You have 15¢. Do you have enough?

- A. No, you need more
- B. Yes, you have exactly enough ✓**
- C. Yes, you have too much
- D. Cannot tell

Explanation: $\$0.15 = 15¢$. You have 15¢. Since $15¢ = 15¢$, you have exactly enough!

19. ____ = \$0.05

- A. 50¢
- B. 5¢ ✓**
- C. 500¢
- D. $\frac{1}{2}¢$

Explanation: $\$0.05 = 5 \text{ cents} = 5¢$. The 05 after the decimal means five cents.

20. Sam wrote one dollar and nine cents as \$1.9. What is wrong?

- A. Nothing, it is correct
- B. Should be \$1.09 — need two digits after the decimal ✓**
- C. Should be \$19
- D. Should be \$1.90

Explanation: Always write two digits after the decimal. 9 cents = .09, not .9 (which looks like 90¢). Correct: \$1.09.

Hard (20 questions)

1. Mia wrote 8¢ as \$0.80. Eli wrote 8¢ as \$0.08. Who is correct?

- A. Mia
- B. Eli ✓**
- C. Both are correct
- D. Neither is correct

Explanation: 8¢ means 8 cents. $\$0.80 = 80¢$ (Mia is wrong). $\$0.08 = 8¢$ (Eli is correct). Always use two decimal digits.

2. You buy a folder for \$0.65 and a pen for 30¢. What is the total cost?

- A. \$0.35
- B. \$0.95 ✓**
- C. \$9.50
- D. \$6.80

Explanation: Convert to cents: $65¢ + 30¢ = 95¢ = \$0.95$. You can add amounts even when written differently.

3. Which is greater: \$1.05 or 100¢?

- A. \$1.05 ✓**
- B. 100¢
- C. They are equal
- D. Cannot compare

Explanation: $\$1.05 = 105¢$. $100¢ = \$1.00$. Since $105¢ > 100¢$, \$1.05 is greater.

4. Amir has \$0.75. He finds \$0.18 more. Write his new total.

- A. \$0.83
- B. \$0.93 ✓**
- C. \$8.30
- D. \$9.30

Explanation: $75¢ + 18¢ = 93¢ = \$0.93$. Add the cents, then write with a dollar sign.

5. Priya says $\$0.06 > \0.60 . Is she right?

- A. Yes
- B. No, \$0.60 is greater ✓**
- C. They are equal
- D. Cannot compare

Explanation: $\$0.06 = 6¢$. $\$0.60 = 60¢$. Since $60¢ > 6¢$, \$0.60 is greater. Priya confused 6 cents with 60 cents.

6. A toy costs \$1.25. You have 3 quarters, 4 dimes, and 2 nickels. Do you have enough?

- A. No, you only have \$1.05
- B. No, you only have \$1.15
- C. Yes, you have exactly \$1.25 ✓**
- D. Yes, you have \$1.25 with extra

Explanation: 3 quarters = 75¢. 4 dimes = 40¢. 2 nickels = 10¢. Total: $75 + 40 + 10 = 125¢ = \$1.25$. Exactly enough!

7. Put in order from MOST to LEAST: 89¢, \$0.98, \$1.00, 79¢

- A. \$1.00, \$0.98, 89¢, 79¢ ✓**
- B. \$0.98, \$1.00, 89¢, 79¢
- C. 89¢, 79¢, \$0.98, \$1.00
- D. \$1.00, 89¢, \$0.98, 79¢

Explanation: Convert all: 89¢, 98¢, 100¢, 79¢. Most to least: 100, 98, 89, 79. So: \$1.00, \$0.98, 89¢, 79¢.

8. Tyler wrote 52¢ as \$52. What should he have written?

- A. \$5.20
- B. \$0.52 ✓**
- C. \$5.02
- D. \$52.00

Explanation: 52¢ = \$0.52. Tyler forgot the decimal point and the zero, making it look like 52 dollars instead of 52 cents!

9. You have \$1.00. You spend \$0.45 on a snack and \$0.30 on a drink. How much is left?

- A. \$0.15
- B. \$0.25 ✓**
- C. \$0.35
- D. \$0.75

Explanation: Step 1: $\$0.45 + \$0.30 = \$0.75$ spent. Step 2: $\$1.00 - \$0.75 = \$0.25$ left.

10. Which correctly converts 109¢ to dollars and cents?

- A. \$1.09 ✓**
- B. \$10.9
- C. \$1.90
- D. \$0.109

Explanation: $109¢ = 100¢ + 9¢ = \$1.00 + \$0.09 = \$1.09$. One dollar and nine cents.

11. Zoe has \$0.87. She needs \$1.00. How much more does she need?

- A. 3¢
- B. 13¢ ✓**
- C. 23¢
- D. 87¢

Explanation: $\$1.00 = 100¢$. She has 87¢. More needed: $100 - 87 = 13¢$.

12. Chen says $\$0.5 = 50¢$. Layla says $\$0.50 = 50¢$. Who wrote it correctly?

- A. Only Chen
- B. Only Layla
- C. Both are correct math, but Layla uses proper notation ✓**
- D. Neither is correct

Explanation: Both equal 50 cents mathematically, but proper money notation always uses two digits after the decimal: \$0.50. Layla's notation is the standard way.

13. A sticker costs 85¢. You pay with \$1.00. Your change is ____.

- A. \$0.05
- B. \$0.10
- C. \$0.15 ✓**
- D. \$0.25

Explanation: Change = $\$1.00 - \$0.85 = \$0.15$. You get 15 cents back.

14. Which pair does NOT show the same amount?

- A. \$0.25 and 25¢
- B. \$0.07 and 7¢
- C. \$0.60 and 6¢ ✓**
- D. \$0.99 and 99¢

Explanation: $\$0.60 = 60¢$, not 6¢. The other pairs all match. Watch out for the zero — $\$0.06 = 6¢$, but $\$0.60 = 60¢$!

15. Nia has \$1.50 in quarters. How many quarters does she have?

- A. 4
- B. 5
- C. 6 ✓**
- D. 7

Explanation: $\$1.50 = 150¢$. Each quarter = 25¢. $150 \div 25 = 6$ quarters.

16. A crayon box costs \$0.78. You have 3 quarters. Do you have enough?

- A. Yes, with 3¢ extra
- B. Yes, with exactly enough
- C. No, you are 3¢ short ✓**
- D. No, you are 8¢ short

Explanation: 3 quarters = $75¢ = \$0.75$. Cost = $\$0.78$. Since $75¢ < 78¢$, you are 3¢ short.

17. Which strategy helps you compare \$0.47 and 52¢?

- A. Compare the dollar signs
- B. Convert both to cents, then compare ✓**
- C. The one with ¢ is always bigger
- D. You cannot compare different symbols

Explanation: Convert both to cents: $\$0.47 = 47¢$. Now compare 47¢ and 52¢. $52¢ > 47¢$. Converting to the same unit makes comparing easy!

18. Marcus earned \$0.40 on Monday and 35¢ on Tuesday. He spent \$0.50. How much does he have?

- A. \$0.15
- B. \$0.25 ✓**
- C. \$0.35
- D. \$0.75

Explanation: Step 1: Earned total = $40 + 35 = 75¢$. Step 2: Spent 50¢. Step 3: $75 - 50 = 25¢ = \$0.25$.

19. Rosa wrote these amounts: \$0.09, \$0.90, \$9.00. Which one means 9 cents?

- A. \$0.09 ✓**
- B. \$0.90
- C. \$9.00
- D. None of them

Explanation: $\$0.09 = 9¢$. $\$0.90 = 90¢$. $\$9.00 = 9$ dollars. Only \$0.09 means nine cents.

20. Sam has \$2.00. He buys a pencil for 75¢ and an eraser for \$0.50. Write his remaining money.

- A. \$0.25
- B. \$0.75 ✓**
- C. \$1.25
- D. \$1.50

Explanation: Step 1: $75¢ + 50¢ = 125¢ = \$1.25$ spent. Step 2: $\$2.00 - \$1.25 = \$0.75$ remaining.

Lesson 4: Making Change

Easy (20 questions)

1. You have 10¢. You spend 3¢. How much is left?

- A. 3¢
- B. 7¢ ✓**
- C. 10¢
- D. 13¢

Explanation: $10 - 3 = 7$. You have 7¢ left.

2. You pay 5¢ for a gumball. You give the clerk a dime. How much change?

- A. 5¢ ✓**
- B. 10¢
- C. 15¢
- D. 0¢

Explanation: A dime = 10¢. Change = $10 - 5 = 5$ ¢.

3. A sticker costs 8¢. You pay with a dime. How much change do you get?

- A. 1¢
- B. 2¢ ✓**
- C. 3¢
- D. 8¢

Explanation: A dime = 10¢. Change = $10 - 8 = 2$ ¢.

4. You have a quarter. You buy a ring for 25¢. How much change?

- A. 0¢ ✓**
- B. 5¢
- C. 10¢
- D. 25¢

Explanation: $25 - 25 = 0$ ¢. You used the exact amount, so there is no change!

5. You buy a candy for 20¢. You pay with a quarter. How much change?

- A. 5¢ ✓**
- B. 10¢
- C. 15¢
- D. 20¢

Explanation: A quarter = 25¢. Change = $25 - 20 = 5$ ¢.

6. You have 15¢. You spend 10¢. How much is left?

- A. 3¢
- B. 5¢ ✓**
- C. 10¢
- D. 15¢

Explanation: $15 - 10 = 5$. You have 5¢ left.

7. A pencil costs 10¢. You pay with 2 dimes. How much change?

- A. 5¢
- B. 10¢ ✓**
- C. 15¢
- D. 20¢

Explanation: 2 dimes = 20¢. Change = $20 - 10 = 10$ ¢.

8. You have 25¢. You spend 15¢. How much is left?

- A. 5¢
- B. 10¢ ✓**
- C. 15¢
- D. 25¢

Explanation: $25 - 15 = 10$. You have 10¢ left.

9. What does 'making change' mean?

- A. Trading all your coins
- B. Getting money back when you pay too much ✓**
- C. Losing money
- D. Changing coins into bills

Explanation: Making change means getting money back when you pay more than the cost of an item.

10. An eraser costs 7¢. You pay with a dime. What is your change?

- A. 2¢
- B. 3¢ ✓**
- C. 7¢
- D. 10¢

Explanation: A dime = 10¢. Change = $10 - 7 = 3$ ¢.

11. You have 20¢. You buy a sticker for 12¢. How much is left?

- A. 2¢
- B. 8¢ ✓**
- C. 12¢
- D. 20¢

Explanation: $20 - 12 = 8$. You have 8¢ left.

12. You buy a ball for 15¢. You pay with a quarter. What is your change?

- A. 5¢
- B. 10¢ ✓**
- C. 15¢
- D. 25¢

Explanation: A quarter = 25¢. Change = $25 - 15 = 10$ ¢.

13. You have 50¢. You spend 25¢. How much is left?

- A. 15¢
- B. 20¢
- C. 25¢ ✓**
- D. 35¢

Explanation: $50 - 25 = 25$. You have 25¢ left.

14. A gummy bear costs 4¢. You pay with a nickel. How much change?

- A. 1¢ ✓**
- B. 4¢
- C. 5¢
- D. 9¢

Explanation: A nickel = 5¢. Change = $5 - 4 = 1$ ¢.

15. You pay 10¢ for a bookmark. You give the clerk a quarter. How much change?

- A. 5¢
- B. 10¢
- C. 15¢ ✓**
- D. 20¢

Explanation: A quarter = 25¢. Change = $25 - 10 = 15$ ¢.

16. You have 30¢. You buy a toy for 20¢. How much do you have left?

- A. 5¢
- B. 10¢ ✓**
- C. 20¢
- D. 30¢

Explanation: $30 - 20 = 10$. You have 10¢ left.

17. A cookie costs 9¢. You pay with a dime. How much change?

- A. 1¢ ✓
- B. 2¢
- C. 9¢
- D. 10¢

Explanation: A dime = 10¢. Change = $10 - 9 = 1$ ¢.

18. To find change, you ____.

- A. Add the cost and payment
- B. Subtract the cost from the payment ✓
- C. Multiply the coins
- D. Divide the money

Explanation: Change = Payment - Cost. You subtract what you owe from what you paid.

19. You have 50¢. You spend 30¢. How much change is left?

- A. 10¢
- B. 20¢ ✓
- C. 25¢
- D. 30¢

Explanation: $50 - 30 = 20$. You have 20¢ left.

20. You buy a crayon for 5¢ and pay with a dime. What coins might you get as change?

- A. 1 nickel ✓
- B. 1 dime
- C. 1 quarter
- D. 1 penny

Explanation: Change = $10 - 5 = 5$ ¢. A nickel is worth 5¢, so you could get 1 nickel back.

Medium (20 questions)

1. A bookmark costs 35¢. You pay with 2 quarters. How much change?

- A. 5¢
- B. 10¢
- C. 15¢ ✓
- D. 25¢

Explanation: 2 quarters = 50¢. Change = $50 - 35 = 15$ ¢.

2. You buy a pencil for 45¢ and pay with 3 quarters. How much change do you get?

- A. 20¢
- B. 25¢
- C. 30¢ ✓
- D. 35¢

Explanation: 3 quarters = 75¢. Change = $75 - 45 = 30$ ¢.

3. A fruit snack costs 60¢. You pay with \$1.00. What coins could be your change?

- A. 4 dimes
- B. 1 quarter, 1 dime, 1 nickel
- C. 2 nickels
- D. Both A and B ✓

Explanation: Change = $100 - 60 = 40$ ¢. A: 4 dimes = 40¢. B: $25 + 10 + 5 = 40$ ¢. Both work!

4. Mia buys a ring for 30¢. She pays with a quarter and a dime. How much change?

- A. 0¢
- B. 5¢ ✓
- C. 10¢
- D. 15¢

Explanation: Quarter + dime = $25 + 10 = 35$ ¢. Change = $35 - 30 = 5$ ¢.

5. An apple costs 50¢. You have 3 dimes and 2 nickels. Do you have enough?

A. No, you have 35¢

B. No, you have 40¢ ✓

C. Yes, you have exactly 50¢

D. Yes, you have 55¢

Explanation: 3 dimes = 30¢. 2 nickels = 10¢. Total: 40¢. 40¢ < 50¢, so you do not have enough. You are 10¢ short.

6. A toy car costs 55¢. You pay with \$1.00. How much change?

A. 35¢

B. 40¢

C. 45¢ ✓

D. 55¢

Explanation: Change = \$1.00 - 55¢ = 100 - 55 = 45¢.

7. You buy a notebook for 75¢. You pay with 4 quarters. How much change?

A. 15¢

B. 20¢

C. 25¢ ✓

D. 50¢

Explanation: 4 quarters = \$1.00 = 100¢. Change = 100 - 75 = 25¢.

8. ____ is the change from paying 50¢ for a 35¢ item.

A. 5¢

B. 10¢

C. 15¢ ✓

D. 20¢

Explanation: Change = Payment - Cost = 50 - 35 = 15¢.

9. A marker costs 40¢. You pay with 2 quarters. What coins could you get as change?

A. 1 dime

B. 2 nickels

C. 10 pennies

D. All of the above ✓

Explanation: Change = 50 - 40 = 10¢. 1 dime = 10¢. 2 nickels = 10¢. 10 pennies = 10¢. All make 10¢!

10. A candy bar costs 65¢. You pay with 3 quarters. How much change?

A. 5¢

B. 10¢ ✓

C. 15¢

D. 25¢

Explanation: 3 quarters = 75¢. Change = 75 - 65 = 10¢.

11. You have \$1.00. You spend 25¢. How much do you have left?

A. 25¢

B. 50¢

C. 65¢

D. 75¢ ✓

Explanation: \$1.00 = 100¢. 100 - 25 = 75¢. You have 75¢ left.

12. A juice box costs 80¢. Which amount is NOT enough to pay for it?

A. 3 quarters and 1 nickel

B. 8 dimes

C. 3 quarters ✓

D. 4 quarters

Explanation: A: 75 + 5 = 80¢ (enough). B: 80¢ (enough). C: 75¢ (not enough — 5¢ short!). D: \$1.00 (enough).

13. You buy a sticker for 22¢ and pay with a quarter. How much change?

- A. 2¢
- B. 3¢ ✓**
- C. 5¢
- D. 22¢

Explanation: Quarter = 25¢. Change = $25 - 22 = 3$ ¢.

14. A banana costs 30¢. You pay with a quarter and a dime. How much change?

- A. 0¢
- B. 5¢ ✓**
- C. 10¢
- D. 15¢

Explanation: Quarter + dime = $25 + 10 = 35$ ¢. Change = $35 - 30 = 5$ ¢.

15. You have 75¢. You buy a pen for 50¢. How much is left?

- A. 15¢
- B. 20¢
- C. 25¢ ✓**
- D. 50¢

Explanation: $75 - 50 = 25$. You have 25¢ left.

16. An orange costs 45¢. You give the clerk 2 quarters. She gives you a nickel. Is that the right change?

- A. Yes ✓**
- B. No, she should give 10¢
- C. No, she should give 15¢
- D. No, she should give a dime

Explanation: 2 quarters = 50¢. Change = $50 - 45 = 5$ ¢. A nickel = 5¢. Yes, that is correct!

17. You have \$1.00. You spend 40¢. How much left?

- A. 40¢
- B. 50¢
- C. 60¢ ✓**
- D. 70¢

Explanation: \$1.00 = 100¢. Change = $100 - 40 = 60$ ¢.

18. Which item can you buy if you have exactly 50¢?

- A. Ball: 55¢
- B. Sticker: 35¢
- C. Crayon: 50¢
- D. Both B and C ✓**

Explanation: You have 50¢. Ball costs 55¢ (too much). Sticker costs 35¢ (can buy, 15¢ left). Crayon costs 50¢ (exact). Both B and C work!

19. A cookie costs 38¢. You pay with 2 quarters. How much change?

- A. 2¢
- B. 8¢
- C. 12¢ ✓**
- D. 18¢

Explanation: 2 quarters = 50¢. Change = $50 - 38 = 12$ ¢.

20. You need 70¢ to buy a treat. You have 2 quarters and 1 dime. How much MORE do you need?

- A. 5¢
- B. 10¢ ✓**
- C. 15¢
- D. 20¢

Explanation: You have: $50 + 10 = 60$ ¢. Need: 70¢. More needed: $70 - 60 = 10$ ¢.

Hard (20 questions)

1. You buy a pencil for 35¢ and an eraser for 20¢. You pay with \$1.00. How much change?

- A. 35¢
- B. 40¢
- C. 45¢ ✓**
- D. 55¢

Explanation: Step 1: Total cost = $35 + 20 = 55$ ¢. Step 2: Change = $100 - 55 = 45$ ¢.

2. Amy says 3 quarters and 2 dimes = 85¢. Is she correct?

- A. Yes, 85¢ is correct
- B. No, it is 80¢
- C. No, it is 90¢
- D. No, it is 95¢ ✓**

Explanation: 3 quarters = 75¢. 2 dimes = 20¢. Total: $75 + 20 = 95$ ¢, not 85¢. Amy added incorrectly.

3. Which costs more: 6 nickels or 2 dimes and 1 nickel?

- A. 6 nickels ✓**
- B. 2 dimes and 1 nickel
- C. They cost the same
- D. Not enough info

Explanation: 6 nickels = 30¢. 2 dimes + 1 nickel = 25¢. $30¢ > 25¢$, so 6 nickels are worth more.

4. You need exactly 67¢. You have only quarters, dimes, and pennies. Which set works?

- A. 2 quarters, 1 dime, 7 pennies ✓**
- B. 2 quarters, 2 dimes
- C. 3 quarters
- D. 2 quarters, 1 dime, 2 pennies

Explanation: A: $50 + 10 + 7 = 67$ ¢. B: $50 + 20 = 70$ ¢ (wrong). C: 75¢ (wrong). D: $50 + 10 + 2 = 62$ ¢ (wrong). Only A works!

5. You have \$1.00. You buy a ruler for 60¢. What coins could the cashier give you as change?

- A. 1 quarter and 1 dime
- B. 4 dimes
- C. 1 quarter, 1 dime, and 1 nickel
- D. Both B and C ✓**

Explanation: Change = $100 - 60 = 40$ ¢. B: 4 dimes = 40¢. C: $25 + 10 + 5 = 40$ ¢. Both B and C equal 40¢!

6. Malik buys a snack for 42¢ and a drink for 33¢. He pays with \$1.00. How much change?

- A. 15¢
- B. 25¢ ✓**
- C. 35¢
- D. 42¢

Explanation: Step 1: Total = $42 + 33 = 75$ ¢. Step 2: Change = $100 - 75 = 25$ ¢.

7. Ben says his change from a 50¢ purchase with \$1.00 is 60¢. Find his error.

- A. He subtracted wrong: $100 - 50 = 50$ ¢ ✓**
- B. He added instead of subtracting
- C. He forgot to count the dollar
- D. There is no error

Explanation: Change = $\$1.00 - \$0.50 = 50$ ¢, not 60¢. Ben made a subtraction error. $100 - 50 = 50$, not 60.

8. You have 2 quarters, 3 dimes, and 2 nickels. You buy a toy for 70¢. How much is left?

- A. 10¢
- B. 20¢ ✓**
- C. 30¢
- D. 40¢

Explanation: You have: $50 + 30 + 10 = 90$ ¢. Spent: 70¢. Left: $90 - 70 = 20$ ¢.

9. A ball costs 85¢. Jada has 3 quarters. How much more does she need?

- A. 5¢
- B. 10¢ ✓**
- C. 15¢
- D. 25¢

Explanation: 3 quarters = 75¢. Need: 85¢. More needed: $85 - 75 = 10$ ¢.

10. Eli says the fewest coins to make 40¢ change is 3 coins. Is he right? What are they?

- A. No, it takes at least 4 coins
- B. Yes: 1 quarter, 1 dime, 1 nickel ✓**
- C. Yes: 4 dimes
- D. Yes: 2 dimes, 1 nickel

Explanation: 1 quarter (25¢) + 1 dime (10¢) + 1 nickel (5¢) = 40¢ with only 3 coins. Eli is right!

11. A book costs \$1.00. You have 3 quarters, 1 dime, and 3 nickels. Can you buy it?

- A. No, you only have 95¢
- B. No, you only have 85¢
- C. Yes, you have exactly \$1.00 ✓**
- D. Yes, you have \$1.05

Explanation: 3 quarters = 75¢. 1 dime = 10¢. 3 nickels = 15¢. Total: $75 + 10 + 15 = 100$ ¢ = \$1.00. Exactly enough!

12. You buy a crayon pack for 55¢ and a glue stick for 25¢. You pay with \$1.00. How much change?

- A. 10¢
- B. 15¢
- C. 20¢ ✓**
- D. 25¢

Explanation: Step 1: Total = $55 + 25 = 80$ ¢. Step 2: Change = $100 - 80 = 20$ ¢.

13. Nora says counting up from the price to what you paid is a good way to find change. If the price is 63¢ and you pay 75¢, count up to find the change.

- A. 10¢
- B. 12¢ ✓**
- C. 13¢
- D. 15¢

Explanation: Count up from 63: $63 + 2 = 65$ (pennies), $65 + 10 = 75$ (dime). That is $2 + 10 = 12$ ¢ change. Counting up is a great strategy!

14. Carlos has \$1.00. He buys 3 items: a pen (25¢), a clip (15¢), and tape (20¢). How much change?

- A. 30¢
- B. 35¢
- C. 40¢ ✓**
- D. 45¢

Explanation: Step 1: Total = $25 + 15 + 20 = 60$ ¢. Step 2: Change = $100 - 60 = 40$ ¢.

15. Which strategy helps you make change from 75¢ for a 48¢ item?

- A. Count up from 48 to 75
- B. Count down from 75 by subtracting 48
- C. Use a number line
- D. All of these strategies work ✓**

Explanation: All three strategies find the same answer (27¢). Counting up: $48 + 2 = 50 + 25 = 75$. Subtracting: $75 - 48 = 27$. Number line: jump from 48 to 75. Use whichever feels easiest!

16. A snack costs 90¢. You pay with \$1.00. The clerk gives you 2 nickels. Is that the right change?

- A. Yes, 2 nickels = 10¢
- B. No, you should get 1 dime
- C. No, you should get 1 nickel
- D. Both A and B give the right amount ✓**

Explanation: Change = $100 - 90 = 10$ ¢. 2 nickels = 10¢. 1 dime = 10¢. Both equal 10¢, so both are correct ways to give change!

17. Layla bought 2 items totaling 83¢. She paid with \$1.00. She received 1 dime, 1 nickel, and 2 pennies. Did she get the right change?

- A. Yes, that is 17¢ ✓
- B. No, she should get 7¢
- C. No, she should get 27¢
- D. No, that is only 12¢

Explanation: Change = $100 - 83 = 17$ ¢. Coins received: $10 + 5 + 2 = 17$ ¢. Yes, Layla got the correct change!

18. You have 3 quarters. You buy a ball for 50¢ and a sticker for 15¢. How much change?

- A. 5¢
- B. 10¢ ✓
- C. 15¢
- D. 20¢

Explanation: Step 1: 3 quarters = 75¢. Step 2: Total cost = $50 + 15 = 65$ ¢. Step 3: Change = $75 - 65 = 10$ ¢.

19. A prize costs 57¢. You pay with \$1.00. Which set of coins is the correct change?

- A. 1 quarter, 1 dime, 1 nickel, 3 pennies
- B. 4 dimes, 3 pennies
- C. 2 dimes, 3 pennies
- D. Both A and B ✓

Explanation: Change = $100 - 57 = 43$ ¢. A: $25 + 10 + 5 + 3 = 43$ ¢. B: $40 + 3 = 43$ ¢. C: $20 + 3 = 23$ ¢ (wrong). Both A and B equal 43¢!

20. You earn 25¢ on Monday, 30¢ on Tuesday, and spend 40¢ on Wednesday. How much do you have left?

- A. 5¢
- B. 10¢
- C. 15¢ ✓
- D. 20¢

Explanation: Step 1: Earned = $25 + 30 = 55$ ¢. Step 2: Spent = 40¢. Step 3: Left = $55 - 40 = 15$ ¢.

Unit Test Bank

Easy (20 questions)

1. How much is a quarter worth?

- A. 5¢
- B. 10¢
- C. 25¢ ✓
- D. 50¢

Explanation: A quarter is worth 25¢.

2. 3 dimes = ____ cents

- A. 3¢
- B. 13¢
- C. 30¢ ✓
- D. 35¢

Explanation: Each dime is 10¢. Count by 10s: 10, 20, 30. So 3 dimes = 30¢.

3. Which coin is worth 1¢?

- A. Nickel
- B. Dime
- C. Penny ✓
- D. Quarter

Explanation: A penny is the coin worth 1¢. It is the smallest value coin.

4. Count the money: 2 nickels and 1 penny.

- A. 3¢
- B. 7¢
- C. 11¢ ✓**
- D. 21¢

Explanation: Nickels: 5, 10. Penny: 11. Total = 11¢.

5. How do you write 50 cents with a dollar sign?

- A. \$50
- B. \$5.00
- C. \$0.50 ✓**
- D. \$0.05

Explanation: 50 cents = \$0.50. The zero before the decimal means zero dollars.

6. 4 pennies = ____ ¢

- A. 1¢
- B. 4¢ ✓**
- C. 40¢
- D. 14¢

Explanation: Each penny = 1¢. Count: 1, 2, 3, 4. So 4 pennies = 4¢.

7. Which coin is worth the MOST?

- A. Penny
- B. Nickel
- C. Dime
- D. Quarter ✓**

Explanation: A quarter (25¢) is worth the most of any common coin.

8. You have 10¢. You spend 4¢. How much is left?

- A. 4¢
- B. 6¢ ✓**
- C. 10¢
- D. 14¢

Explanation: $10 - 4 = 6$. You have 6¢ left.

9. Is \$0.25 the same as 25¢?

- A. Yes ✓**
- B. No, \$0.25 is more
- C. No, 25¢ is more
- D. Cannot tell

Explanation: Yes! \$0.25 and 25¢ both mean twenty-five cents.

10. Count the money: 1 dime and 3 pennies.

- A. 4¢
- B. 13¢ ✓**
- C. 30¢
- D. 31¢

Explanation: Dime: 10. Pennies: 11, 12, 13. Total = 13¢.

11. How many cents is 1 nickel?

- A. 1¢
- B. 5¢ ✓**
- C. 10¢
- D. 25¢

Explanation: A nickel is worth 5¢.

12. You buy a toy for 5¢ and pay with a dime. How much change?

- A. 5¢ ✓
- B. 10¢
- C. 15¢
- D. 0¢

Explanation: A dime = 10¢. Change = $10 - 5 = 5$ ¢.

13. What does the decimal point in \$1.50 separate?

- A. Ones and tens
- B. Dollars and cents ✓
- C. Big and small numbers
- D. Nothing

Explanation: The decimal point separates dollars from cents. $\$1.50 = 1$ dollar and 50 cents.

14. 2 dimes and 1 penny = ____ ¢

- A. 3¢
- B. 12¢
- C. 21¢ ✓
- D. 20¢

Explanation: Dimes: 10, 20. Penny: 21. Total = 21¢.

15. How do you write 1 cent using a dollar sign?

- A. \$1.00
- B. \$0.10
- C. \$0.01 ✓
- D. \$10

Explanation: 1 cent = \$0.01. Always write two digits after the decimal point.

16. You have a quarter. You spend 20¢. How much change?

- A. 5¢ ✓
- B. 10¢
- C. 20¢
- D. 25¢

Explanation: Quarter = 25¢. Change = $25 - 20 = 5$ ¢.

17. 5 nickels = ____ ¢

- A. 5¢
- B. 15¢
- C. 25¢ ✓
- D. 50¢

Explanation: Count by 5s: 5, 10, 15, 20, 25. Five nickels = 25¢.

18. How many cents are in \$1.00?

- A. 1¢
- B. 10¢
- C. 50¢
- D. 100¢ ✓

Explanation: $\$1.00 = 100$ cents. One dollar equals one hundred cents.

19. Count the money: 1 quarter and 2 pennies.

- A. 7¢
- B. 22¢
- C. 27¢ ✓
- D. 52¢

Explanation: Quarter: 25. Pennies: 26, 27. Total = 27¢.

20. Which symbol goes BEFORE the number: ¢ or \$?

- A. ¢
- B. \$ ✓
- C. Both
- D. Neither

Explanation: The dollar sign (\$) goes before the number: \$0.50. The cent sign (¢) goes after: 50¢.

Medium (20 questions)

1. You have 1 quarter, 2 dimes, and 4 pennies. How much money?

- A. 29¢
- B. 44¢
- C. 49¢ ✓
- D. 54¢

Explanation: Quarter: 25. Dimes: 35, 45. Pennies: 46, 47, 48, 49. Total = 49¢.

2. Which group of coins equals 75¢?

- A. 7 dimes and 1 nickel
- B. 3 quarters
- C. 7 nickels and 1 dime
- D. Both A and B ✓

Explanation: A: $70 + 5 = 75¢$. B: $25 + 25 + 25 = 75¢$. C: $35 + 10 = 45¢$ (wrong). Both A and B equal 75¢!

3. ____ quarters = \$1.00

- A. 2
- B. 3
- C. 4 ✓
- D. 5

Explanation: Each quarter = 25¢. Count by 25s: 25, 50, 75, 100. It takes 4 quarters to make \$1.00.

4. Convert: \$0.73 = ____ ¢

- A. 7¢
- B. 37¢
- C. 73¢ ✓
- D. 730¢

Explanation: \$0.73 means 73 cents. The two digits after the decimal are the cents.

5. Which is MORE: \$0.45 or 42¢?

- A. \$0.45 ✓
- B. 42¢
- C. They are equal
- D. Cannot compare

Explanation: \$0.45 = 45¢. Compare: 45¢ vs 42¢. Since $45 > 42$, \$0.45 is more.

6. A toy costs 50¢. You pay with 3 quarters. How much change?

- A. 15¢
- B. 20¢
- C. 25¢ ✓
- D. 30¢

Explanation: 3 quarters = 75¢. Change = $75 - 50 = 25¢$.

7. Diego has 3 dimes, 3 nickels, and 2 pennies. How much?

- A. 32¢
- B. 38¢
- C. 47¢ ✓
- D. 52¢

Explanation: Dimes: 10, 20, 30. Nickels: 35, 40, 45. Pennies: 46, 47. Total = 47¢.

8. ____ dimes and 1 nickel = 35¢

- A. 2
- B. 3 ✓**
- C. 4
- D. 5

Explanation: $35 - 5 = 30¢$ for dimes. $30 \div 10 = 3$ dimes. Check: 3 dimes (30¢) + 1 nickel (5¢) = 35¢.

9. Write 2 quarters and 3 pennies using a dollar sign.

- A. \$0.23
- B. \$0.53 ✓**
- C. \$2.03
- D. \$5.30

Explanation: 2 quarters = 50¢. 3 pennies = 3¢. Total: 53¢ = \$0.53.

10. A cookie costs 40¢. You pay with 2 quarters. How much change?

- A. 5¢
- B. 10¢ ✓**
- C. 15¢
- D. 20¢

Explanation: 2 quarters = 50¢. Change = $50 - 40 = 10¢$.

11. Which is worth LESS: 5 nickels or 2 dimes and 1 nickel?

- A. 5 nickels
- B. 2 dimes and 1 nickel
- C. They are equal ✓**
- D. Not enough info

Explanation: 5 nickels = 25¢. 2 dimes + 1 nickel = 25¢. They are equal!

12. Rosa wrote 9¢ as \$0.90. Is that correct?

- A. Yes
- B. No, \$0.90 = 90¢ ✓**
- C. No, 9¢ = \$9.00
- D. No, 9¢ = \$90

Explanation: \$0.90 = 90¢, not 9¢. The correct way to write 9¢ is \$0.09.

13. You have \$1.00. You spend 35¢. How much left?

- A. 35¢
- B. 55¢
- C. 65¢ ✓**
- D. 75¢

Explanation: \$1.00 = 100¢. $100 - 35 = 65¢$ left.

14. Count: 2 quarters, 1 nickel, 3 pennies.

- A. 28¢
- B. 53¢
- C. 58¢ ✓**
- D. 63¢

Explanation: Quarters: 25, 50. Nickel: 55. Pennies: 56, 57, 58. Total = 58¢.

15. An eraser costs 25¢. You pay with a quarter and a dime. How much change?

- A. 5¢
- B. 10¢ ✓**
- C. 15¢
- D. 25¢

Explanation: Quarter + dime = $25 + 10 = 35¢$. Change = $35 - 25 = 10¢$.

16. Which group does NOT equal 50¢?

- A. 2 quarters
- B. 5 dimes
- C. 10 nickels
- D. 4 dimes and 1 nickel ✓**

Explanation: A: 50¢. B: 50¢. C: 50¢. D: $40 + 5 = 45$ ¢. Choice D does NOT equal 50¢.

17. Put in order from LEAST to MOST: 35¢, \$0.30, 40¢

- A. 35¢, \$0.30, 40¢
- B. \$0.30, 35¢, 40¢ ✓**
- C. 40¢, 35¢, \$0.30
- D. \$0.30, 40¢, 35¢

Explanation: Convert: 35¢, 30¢, 40¢. Order: 30, 35, 40. So: \$0.30, 35¢, 40¢.

18. Sami has 4 dimes and 2 pennies. Write her total with a \$ sign.

- A. \$0.24
- B. \$0.42 ✓**
- C. \$4.02
- D. \$4.20

Explanation: 4 dimes = 40¢. 2 pennies = 2¢. Total: 42¢ = \$0.42.

19. ____ pennies = 1 quarter

- A. 5
- B. 10
- C. 20
- D. 25 ✓**

Explanation: A quarter = 25¢. Each penny = 1¢. You need 25 pennies to equal 1 quarter.

20. You need 60¢. You have 2 quarters. How much MORE do you need?

- A. 5¢
- B. 10¢ ✓**
- C. 15¢
- D. 25¢

Explanation: 2 quarters = 50¢. Need: 60¢. More needed: $60 - 50 = 10$ ¢.

Hard (20 questions)

1. You have \$1.00. You buy a pencil for 35¢ and an eraser for 20¢. How much change do you get?

- A. 35¢
- B. 40¢
- C. 45¢ ✓**
- D. 55¢

Explanation: Step 1: Total cost = $35 + 20 = 55$ ¢. Step 2: Change = $100 - 55 = 45$ ¢.

2. Amy says 3 quarters and 2 dimes = 85¢. Is she correct?

- A. Yes, 85¢ is correct
- B. No, it is 90¢
- C. No, it is 95¢ ✓**
- D. No, it is 70¢

Explanation: 3 quarters = 75¢. 2 dimes = 20¢. Total: $75 + 20 = 95$ ¢, not 85¢. Amy's answer was 10¢ too low.

3. Which costs more: 6 nickels or 2 dimes and 1 nickel?

- A. 6 nickels ✓**
- B. 2 dimes and 1 nickel
- C. They cost the same
- D. Not enough info

Explanation: 6 nickels = 30¢. 2 dimes + 1 nickel = 25¢. $30¢ > 25¢$, so 6 nickels are worth more.

4. You need exactly 67¢. You only have quarters, dimes, and pennies. Which set works?

A. 2 quarters, 1 dime, 7 pennies ✓

B. 2 quarters, 2 dimes

C. 3 quarters

D. 2 quarters, 1 dime, 2 pennies

Explanation: A: $50 + 10 + 7 = 67¢$. B: 70¢ (wrong). C: 75¢ (wrong). D: 62¢ (wrong). Only A makes exactly 67¢.

5. Mia has \$0.87. She needs \$1.00. How much more does she need?

A. 3¢

B. 13¢ ✓

C. 23¢

D. 87¢

Explanation: $\$1.00 = 100¢$. She has 87¢. More needed: $100 - 87 = 13¢$.

6. Eli says \$0.06 is more than \$0.60. Is he correct?

A. Yes

B. No, \$0.60 is more ✓

C. They are equal

D. Cannot compare

Explanation: $\$0.06 = 6¢$. $\$0.60 = 60¢$. Since $60¢ > 6¢$, \$0.60 is more. Eli confused 6 cents with 60 cents.

7. You buy a folder for \$0.65 and a pen for 30¢. What is the total?

A. \$0.35

B. \$0.85

C. \$0.95 ✓

D. \$9.50

Explanation: Convert to cents: $65 + 30 = 95¢ = \$0.95$.

8. Chen has 3 quarters, 2 dimes, and 3 pennies. Ravi has 8 dimes and 3 nickels. Who has more?

A. Chen ✓

B. Ravi

C. They have the same

D. Not enough info

Explanation: Chen: $75 + 20 + 3 = 98¢$. Ravi: $80 + 15 = 95¢$. $98¢ > 95¢$, so Chen has more.

9. A snack costs 90¢. You pay with \$1.00. The clerk gives 2 nickels as change. Is that right?

A. Yes, that is 10¢ change ✓

B. No, change should be 5¢

C. No, change should be 20¢

D. No, change should be 15¢

Explanation: Change = $100 - 90 = 10¢$. 2 nickels = 10¢. Yes, that is correct!

10. Which way uses the FEWEST coins to make 76¢?

A. 3 quarters and 1 penny ✓

B. 7 dimes, 1 nickel, 1 penny

C. 2 quarters, 2 dimes, 1 nickel, 1 penny

D. 15 nickels and 1 penny

Explanation: A: 4 coins. B: 9 coins. C: 6 coins. D: 16 coins. 3 quarters + 1 penny = 76¢ with only 4 coins!

11. Marcus earned \$0.40 on Monday and 35¢ on Tuesday. He spent \$0.50. How much left?

A. \$0.15

B. \$0.25 ✓

C. \$0.35

D. \$0.75

Explanation: Step 1: Earned = $40 + 35 = 75¢$. Step 2: Spent 50¢. Step 3: $75 - 50 = 25¢ = \$0.25$.

12. A book costs \$1.25. You have 3 quarters, 4 dimes, and 2 nickels. Do you have enough?

- A. No, you have \$1.05
- B. No, you have \$1.15
- C. Yes, you have exactly \$1.25 ✓**
- D. Yes, you have more than \$1.25

Explanation: 3 quarters = 75¢. 4 dimes = 40¢. 2 nickels = 10¢. Total: $75 + 40 + 10 = 125¢ = \$1.25$. Exactly enough!

13. Layla says counting up from the price to what you paid finds the change. Price: 73¢. Paid: \$1.00. Count up to find the change.

- A. 17¢
- B. 23¢
- C. 27¢ ✓**
- D. 37¢

Explanation: Count up from 73: $73 + 7 = 80$ (pennies), $80 + 20 = 100$ (dimes). That is $7 + 20 = 27¢$ change.

14. Rosa says 2 quarters + 2 dimes + 2 nickels = 90¢. Find her mistake.

- A. She added quarters wrong
- B. She added nickels wrong
- C. The correct total is 80¢ ✓**
- D. There is no mistake

Explanation: 2 quarters = 50¢. 2 dimes = 20¢. 2 nickels = 10¢. Total: $50 + 20 + 10 = 80¢$, not 90¢.

15. You have 3 quarters, 1 dime, and 2 nickels. You buy a toy for 85¢. How much left?

- A. 5¢
- B. 10¢ ✓**
- C. 15¢
- D. 20¢

Explanation: You have: $75 + 10 + 10 = 95¢$. Spent: 85¢. Left: $95 - 85 = 10¢$.

16. Sam has \$2.00. He buys a ball for 75¢ and tape for \$0.50. How much left?

- A. \$0.25
- B. \$0.50
- C. \$0.75 ✓**
- D. \$1.00

Explanation: Step 1: Total cost = $75 + 50 = 125¢ = \$1.25$. Step 2: $\$2.00 - \$1.25 = \$0.75$.

17. Which pair does NOT show the same amount?

- A. \$0.25 and 25¢
- B. \$0.07 and 7¢
- C. \$0.60 and 6¢ ✓**
- D. \$0.99 and 99¢

Explanation: $\$0.60 = 60¢$, not 6¢. The other pairs all match. Be careful: $\$0.06 = 6¢$ but $\$0.60 = 60¢$!

18. You buy 3 items: a pen (25¢), a clip (15¢), and tape (20¢). You pay with \$1.00. How much change?

- A. 30¢
- B. 35¢
- C. 40¢ ✓**
- D. 50¢

Explanation: Step 1: Total = $25 + 15 + 20 = 60¢$. Step 2: Change = $100 - 60 = 40¢$.

19. Nia has \$1.50 in quarters. How many quarters does she have?

- A. 4
- B. 5
- C. 6 ✓**
- D. 7

Explanation: $\$1.50 = 150¢$. Each quarter = 25¢. $150 \div 25 = 6$ quarters.

20. Which strategy helps you compare \$0.47 and 52¢?

A. Compare the dollar signs

B. Convert both to cents, then compare ✓

C. The one with ¢ is always bigger

D. You cannot compare different symbols

Explanation: Convert to the same unit: $\$0.47 = 47¢$. Now compare 47¢ and 52¢. Since $52 > 47$, 52¢ is more.

Unit 6: Data & Graphs

Lesson 1: Tally Charts & Bar Graphs

Easy (20 questions)

1. A tally chart shows how many pets students have. Dogs: IIII II, Cats: III, Fish: IIII. How many students have dogs?

A. 5
B. 6
C. 7 ✓
D. 8

Explanation: IIII is 5 and II is 2 more. $5 + 2 = 7$ students have dogs.

2. Look at the bar graph of favorite fruits. The bar for grapes goes up to 6. How many students chose grapes?

A. 4
B. 5
C. 6 ✓
D. 7

Explanation: The bar for grapes stops at 6, so 6 students chose grapes.

3. A tally chart shows: Cats IIII, Birds III, Fish II. Which pet got the most votes?

A. Birds
B. Fish
C. Cats ✓
D. They are all the same

Explanation: Cats = 4, Birds = 3, Fish = 2. Cats got the most with 4 votes.

4. A bar graph shows favorite drinks: Milk=4, Juice=9, Water=5. Which drink has the tallest bar?

A. Milk
B. Juice ✓
C. Water
D. They are the same

Explanation: Juice has 9 votes, which is the most, so it has the tallest bar.

5. A tally chart shows: Chocolate III, Vanilla IIII, Strawberry IIII I. How many students chose strawberry?

A. 4
B. 5
C. 6 ✓
D. 3

Explanation: IIII is 5 and I is 1 more. $5 + 1 = 6$ students chose strawberry.

6. Look at the bar graph of favorite sports. Soccer=5, Baseball=3, Basketball=7. Which sport has the shortest bar?

A. Soccer
B. Baseball ✓
C. Basketball
D. They are all the same

Explanation: Baseball has only 3 votes, which is the fewest, so it has the shortest bar.

7. A tally chart shows favorite snacks: Crackers IIII, Fruit III, Pretzels IIII III. Which snack got the fewest votes?

A. Crackers
B. Fruit ✓
C. Pretzels
D. Crackers and Fruit

Explanation: Crackers = 4, Fruit = 3, Pretzels = 8. Fruit has the fewest votes with only 3.

8. The bar graph shows books read: Mia=8, James=5, Sofia=8. Which two students read the same number of books?

- A. Mia and James
- B. James and Sofia
- C. Mia and Sofia ✓**
- D. No one matched

Explanation: Mia and Sofia both read 8 books, so their bars are the same height.

9. A tally chart shows how students get to school: Walk IIII, Bus IIII III, Car III. How many students take the bus?

- A. 5
- B. 6
- C. 7
- D. 8 ✓**

Explanation: IIII is 5 and III is 3 more. $5 + 3 = 8$ students take the bus.

10. A bar graph shows stickers earned: Monday=2, Tuesday=6, Wednesday=1. On which day were the fewest stickers earned?

- A. Monday
- B. Tuesday
- C. Wednesday ✓**
- D. They are all the same

Explanation: Wednesday has only 1 sticker, which is the least. Its bar is the shortest.

11. A tally chart shows: Puppies IIII III, Kittens IIII II, Hamsters III. How many students voted for kittens?

- A. 5
- B. 7 ✓**
- C. 8
- D. 3

Explanation: IIII is 5 and II is 2 more. $5 + 2 = 7$ students voted for kittens.

12. A bar graph shows animals at a zoo: Monkeys=6, Penguins=6, Zebras=4. Which two animals have the same number?

- A. Monkeys and Zebras
- B. Penguins and Zebras
- C. Monkeys and Penguins ✓**
- D. None are the same

Explanation: Monkeys and Penguins both have 6, so their bars are the same height.

13. A tally chart is missing some data. Dogs: IIII, Cats: ???, Fish: III. You know 5 students voted for cats. Which tally should go in the cats row?

- A. IIII
- B. III
- C. IIII I ✓**
- D. IIII II

Explanation: Five tally marks is written as IIII I (the fifth line crosses through the first four). That equals 5.

14. A bar graph shows rainy days: March=2, April=5, May=3. How many rainy days were there in April?

- A. 2
- B. 3
- C. 5 ✓**
- D. 10

Explanation: April's bar goes up to 5. That means there were 5 rainy days in April.

15. A tally chart shows favorite seasons: Spring III, Summer IIII I, Fall IIII, Winter II. Which season got exactly 4 votes?

- A. Spring
- B. Summer
- C. Fall ✓
- D. Winter

Explanation: Spring = 3, Summer = 6, Fall = 4, Winter = 2. Fall is the one with exactly 4.

16. The bar graph shows toys sold: Dolls=3, Cars=7, Blocks=5. Put the toys in order from fewest sold to most sold.

- A. Cars, Blocks, Dolls
- B. Dolls, Blocks, Cars ✓
- C. Blocks, Dolls, Cars
- D. Dolls, Cars, Blocks

Explanation: Dolls = 3 (fewest), Blocks = 5 (middle), Cars = 7 (most). Order: Dolls, Blocks, Cars.

17. A tally chart shows: Red IIII I, Blue III, Green IIII. How many votes does blue have?

- A. 3 ✓
- B. 4
- C. 5
- D. 6

Explanation: Count the tally marks for blue: III = 3 votes.

18. A bar graph shows flowers in a garden: Roses=4, Tulips=7, Daisies=4. How many types of flowers have the same number?

- A. 0
- B. 1
- C. 2 ✓
- D. 3

Explanation: Roses and Daisies both have 4. That is 2 types of flowers with the same number.

19. A tally chart shows: Tigers III, Lions IIII, Bears IIII II. Which animal got the fewest votes?

- A. Tigers ✓
- B. Lions
- C. Bears
- D. Lions and Tigers

Explanation: Tigers = 3, Lions = 4, Bears = 7. Tigers got the fewest with only 3.

20. A bar graph shows crayons in a box: Red=5, Blue=8, Yellow=3, Green=5. How many colors have exactly 5 crayons?

- A. 1
- B. 2 ✓
- C. 3
- D. 4

Explanation: Red = 5 and Green = 5. Two colors have exactly 5 crayons.

Medium (20 questions)

1. A bar graph shows: Pizza=9, Tacos=4, Burgers=6. How many more students voted for pizza than tacos?

- A. 3
- B. 4
- C. 5 ✓
- D. 13

Explanation: Pizza has 9 and tacos has 4. $9 - 4 = 5$ more students voted for pizza.

2. A tally chart shows: Apples IIII III, Oranges IIII, Bananas IIII I. How many students voted for apples and bananas together?

- A. 12
- B. 13
- C. 14 ✓**
- D. 9

Explanation: Apples = 8 and Bananas = 6. $8 + 6 = 14$ students voted for apples and bananas together.

3. A bar graph shows: Red=7, Blue=7, Green=3, Yellow=5. Which two colors got the same number of votes?

- A. Red and Green
- B. Blue and Yellow
- C. Red and Blue ✓**
- D. Green and Yellow

Explanation: Red and Blue both have 7 votes. They got the same number.

4. A tally chart shows: Soccer IIII II, Baseball III, Basketball IIII IIII. How many fewer students chose baseball than basketball?

- A. 3
- B. 5
- C. 6 ✓**
- D. 7

Explanation: Basketball = 9 and Baseball = 3. $9 - 3 = 6$ fewer students chose baseball.

5. A bar graph shows: Dogs=10, Cats=6, Fish=4. How many students voted in all?

- A. 16
- B. 18
- C. 20 ✓**
- D. 14

Explanation: Add them all: $10 + 6 + 4 = 20$ students voted in all.

6. A tally chart shows some favorite colors, but the green row is smudged. Red: IIII, Blue: III. There were 12 votes total. How many voted for green?

- A. 3
- B. 4
- C. 5 ✓**
- D. 6

Explanation: Red = 4 and Blue = 3, so that is 7. The total is 12. $12 - 7 = 5$ voted for green.

7. A bar graph shows: Swings=8, Slide=5, Monkey Bars=8. Put the playground equipment in order from most popular to least popular.

- A. Slide, Monkey Bars, Swings
- B. Swings, Slide, Monkey Bars
- C. Swings and Monkey Bars (tie), then Slide ✓**
- D. Slide, Swings, Monkey Bars

Explanation: Swings and Monkey Bars both have 8 (tied for most), and Slide has 5 (least).

8. A tally chart shows: Milk IIII, Juice IIII III, Water IIII II. Which drink got more votes than milk but fewer than juice?

- A. Milk
- B. Juice
- C. Water ✓**
- D. None of them

Explanation: Milk = 4, Water = 7, Juice = 8. Water (7) is more than Milk (4) but fewer than Juice (8).

9. Mrs. Lee's class made a bar graph. Chocolate=11, Vanilla=6, Strawberry=3. How many more voted for chocolate than strawberry?

- A. 5
- B. 6
- C. 8 ✓
- D. 14

Explanation: Chocolate = 11 and Strawberry = 3. $11 - 3 = 8$ more voted for chocolate.

10. A tally chart shows: Spring III, Summer IIII IIII, Fall IIII, Winter II. How many students voted for spring and winter combined?

- A. 4
- B. 5 ✓
- C. 6
- D. 7

Explanation: Spring = 3 and Winter = 2. $3 + 2 = 5$ students voted for spring and winter combined.

11. A bar graph shows: Lions=5, Tigers=9, Bears=7. If 3 more kids vote for bears, how many votes would bears have?

- A. 9
- B. 10 ✓
- C. 7
- D. 12

Explanation: Bears has 7 now. $7 + 3 = 10$. Bears would have 10 votes.

12. The bar graph is missing data. Roses=6, Tulips=???, Daisies=4. There are 15 votes in all. How many votes did tulips get?

- A. 3
- B. 4
- C. 5 ✓
- D. 6

Explanation: Roses + Daisies = $6 + 4 = 10$. Total is 15. $15 - 10 = 5$ votes for tulips.

13. A tally chart shows: Bikes IIII III, Scooters IIII, Skateboards IIII I. Do bikes and skateboards together have more or fewer votes than 15?

- A. More
- B. Fewer ✓
- C. Exactly 15
- D. Cannot tell

Explanation: Bikes = 8, Skateboards = 6. $8 + 6 = 14$. 14 is fewer than 15.

14. A bar graph shows: Sunny=7, Rainy=2, Cloudy=5. How many more days were sunny than rainy?

- A. 3
- B. 4
- C. 5 ✓
- D. 9

Explanation: Sunny = 7 and Rainy = 2. $7 - 2 = 5$ more days were sunny.

15. A tally chart shows: Crayons IIII II, Markers IIII I, Colored Pencils IIII IIII. Which supply is the second most popular?

- A. Crayons ✓
- B. Markers
- C. Colored Pencils
- D. Crayons and Markers are tied

Explanation: Colored Pencils = 9 (most), Crayons = 7 (second), Markers = 6 (least). Crayons is second.

16. A bar graph shows: Math=6, Reading=4, Science=8. How many students voted for math and reading combined?

- A. 8
- B. 10 ✓**
- C. 12
- D. 14

Explanation: Math = 6 and Reading = 4. $6 + 4 = 10$ students combined.

17. A tally chart shows: Pepperoni IIII IIII, Cheese IIII III, Veggie III. How many more voted for pepperoni than veggie?

- A. 5
- B. 6 ✓**
- C. 7
- D. 8

Explanation: Pepperoni = 9 and Veggie = 3. $9 - 3 = 6$ more voted for pepperoni.

18. Marcus says the bar graph shows Dogs=10, Cats=7. He says 2 more students picked dogs than cats. Is he right?

- A. Yes, 2 more
- B. No, 3 more ✓**
- C. No, 17 more
- D. No, the same number

Explanation: $10 - 7 = 3$. Marcus is wrong. There are 3 more, not 2 more.

19. A bar graph shows: Trucks=4, Cars=8, Vans=4. Which two categories have the same number of votes?

- A. Trucks and Cars
- B. Cars and Vans
- C. Trucks and Vans ✓**
- D. None are the same

Explanation: Trucks = 4 and Vans = 4. They have the same number of votes.

20. A tally chart shows: Hamsters IIII, Rabbits IIII III, Guinea Pigs II. How many total students voted?

- A. 12
- B. 13
- C. 14 ✓**
- D. 15

Explanation: Hamsters = 4, Rabbits = 8, Guinea Pigs = 2. $4 + 8 + 2 = 14$ total students.

Hard (20 questions)

1. A bar graph shows: Red=8, Blue=12, Green=5. How many students voted in all? If 30 students are in the class, how many did NOT vote?

- A. 25 voted, 5 did not ✓**
- B. 23 voted, 7 did not
- C. 25 voted, 7 did not
- D. 20 voted, 10 did not

Explanation: Total votes: $8 + 12 + 5 = 25$. Students who did not vote: $30 - 25 = 5$.

2. A tally chart shows: Apples IIII III, Oranges IIII, Grapes ????. There were 20 votes total. How many students voted for grapes?

- A. 6
- B. 7
- C. 8 ✓**
- D. 9

Explanation: Apples = 8, Oranges = 4. That is 12. Total is 20, so $20 - 12 = 8$ voted for grapes.

3. Sam looks at a bar graph showing: Pizza=9, Tacos=5, Burgers=6. Sam says pizza got 3 more votes than burgers. What did Sam do wrong?

- A. Sam counted the tacos bar instead
- B. Sam subtracted wrong: $9 - 6 = 3$ is correct ✓**
- C. Sam read the pizza bar wrong
- D. Sam added instead of subtracted

Explanation: $9 - 6 = 3$. Sam is actually correct! Pizza did get 3 more votes than burgers.

4. A tally chart shows: Dogs IIII II, Cats IIII I, Fish III. Maya says cats and fish together got more votes than dogs. Is she right?

- A. Yes, 9 is more than 7 ✓**
- B. No, 9 is less than 7
- C. No, they are the same
- D. Yes, 10 is more than 7

Explanation: Dogs = 7. Cats = 6, Fish = 3, so Cats + Fish = 9. Yes, 9 is more than 7. Maya is right.

5. A bar graph shows: Soccer=8, Baseball=5, Basketball=11. Liam says the total is 22. What mistake did Liam make?

- A. He forgot to add baseball
- B. He added wrong: the total is 24
- C. He counted soccer twice ✓**
- D. He read basketball as 10

Explanation: $8 + 5 + 11 = 24$, not 22. It looks like Liam may have counted soccer (8) twice: $8 + 8 + 5 = 21$, or misread a number.

6. A tally chart shows: Chocolate IIII IIII, Vanilla IIII II, Strawberry IIII. 24 students are in the class. How many did NOT vote?

- A. 2
- B. 3
- C. 4 ✓**
- D. 5

Explanation: Chocolate = 9, Vanilla = 7, Strawberry = 4. Total = 20. $24 - 20 = 4$ did not vote.

7. A bar graph shows: Swimming=7, Running=4, Biking=9. If 2 students change their vote from biking to running, what would the new totals for biking and running be?

- A. Biking=7, Running=6 ✓**
- B. Biking=11, Running=2
- C. Biking=7, Running=4
- D. Biking=9, Running=6

Explanation: Biking loses 2: $9 - 2 = 7$. Running gains 2: $4 + 2 = 6$. New totals: Biking=7, Running=6.

8. A tally chart shows: Lions IIII I, Tigers IIII III, Bears IIII. Emma says tigers got twice as many votes as bears. Is she correct?

- A. Yes, 8 is double 4 ✓**
- B. No, 8 is not double 4
- C. No, tigers got 7 votes
- D. Yes, 6 is double 3

Explanation: Tigers = 8, Bears = 4. Double of 4 is 8. Yes, Emma is correct!

9. A bar graph shows: Spring=3, Summer=10, Fall=5, Winter=2. How many voted for the two least popular seasons combined? Is that more or less than summer alone?

- A. 5, less than summer ✓**
- B. 7, less than summer
- C. 5, more than summer
- D. 12, more than summer

Explanation: Spring = 3, Winter = 2 (two least popular). $3 + 2 = 5$. Summer = 10. 5 is less than 10.

10. Students made a tally chart about recess games. Tag: IIII III, Kickball: IIII, Four Square: ????. The teacher says 18 students voted. How many chose four square?

- A. 4
- B. 5
- C. 6 ✓
- D. 7

Explanation: Tag = 8, Kickball = 4, total so far = 12. $18 - 12 = 6$ chose four square.

11. A bar graph shows: Crayons=6, Markers=9, Colored Pencils=5. Jake says markers got 3 more votes than colored pencils. Ava says markers got 4 more. Who is right?

- A. Jake is right
- B. Ava is right ✓
- C. They are both right
- D. Neither is right

Explanation: $9 - 5 = 4$. Ava is right. Markers got 4 more votes than colored pencils.

12. A tally chart shows: Bikes IIII IIII, Scooters IIII I, Skateboards III. Could you split the bike votes evenly into 2 groups? Why or why not?

- A. Yes, 9 splits into 4 and 5
- B. No, 9 is odd so it cannot split evenly ✓
- C. Yes, 9 splits into 3 groups of 3
- D. No, you cannot split votes

Explanation: Bikes = 9. 9 is an odd number, so it cannot be split into 2 equal groups. $4 + 5 = 9$ but 4 and 5 are not equal.

13. A bar graph shows: Pepperoni=11, Cheese=7, Veggie=2. The class has 25 students. How many students were absent or did not vote?

- A. 3
- B. 4
- C. 5 ✓
- D. 6

Explanation: Total votes: $11 + 7 + 2 = 20$. $25 - 20 = 5$ students were absent or did not vote.

14. A tally chart shows: Sunny IIII IIII I, Rainy IIII, Snowy II. Noah says sunny got 10 votes. What did Noah do wrong?

- A. Noah counted right, sunny is 10
- B. Noah miscounted: sunny is 11 ✓
- C. Noah read the rainy row
- D. Noah forgot to count the I at the end

Explanation: IIII = 5, IIII = 5, I = 1. $5 + 5 + 1 = 11$. Noah likely missed the last tally mark. Sunny is 11, not 10.

15. Two classes made bar graphs. Class A: Dogs=8, Cats=6. Class B: Dogs=5, Cats=9. Which class had more total votes for pets?

- A. Class A with 14
- B. Class B with 14
- C. They are the same with 14 each ✓
- D. Class B with 15

Explanation: Class A: $8 + 6 = 14$. Class B: $5 + 9 = 14$. They are the same! Both had 14 total votes.

16. A bar graph shows: Red=6, Blue=4, Green=8, Yellow=2. How many students voted for a color that got fewer than 5 votes?

- A. 2
- B. 4
- C. 6 ✓
- D. 8

Explanation: Blue = 4 and Yellow = 2 both got fewer than 5. $4 + 2 = 6$ students.

17. A tally chart shows: Carrots III, Corn IIII II, Peas IIII. Carlos says corn got twice as many votes as carrots. What should Carlos check?

A. If 7 is double 3 (it is not, double 3 is 6) ✓

B. If 7 is double 4 (it is not)

C. If 4 is double 3 (it is not)

D. Nothing, Carlos is correct

Explanation: Corn = 7, Carrots = 3. Double of 3 is 6, not 7. Carlos is wrong: 7 is not double 3.

18. A bar graph shows: Hiking=4, Camping=9, Fishing=7. If the teacher combines hiking and fishing into one bar called 'Outdoor Fun,' how tall would that bar be? Would it beat camping?

A. 11, yes it beats camping ✓

B. 11, no camping still wins

C. 13, yes it beats camping

D. 7, no camping still wins

Explanation: Hiking + Fishing = 4 + 7 = 11. Camping = 9. 11 is more than 9, so yes, it beats camping.

19. Maria made a tally chart but lost one row. She has: Cats IIII I, Birds III. She knows 15 students voted total. Her missing row is for dogs. How many voted for dogs?

A. 4

B. 5

C. 6 ✓

D. 7

Explanation: Cats = 6, Birds = 3, total = 9. There were 15 total, so 15 - 9 = 6 voted for dogs.

20. A bar graph shows: Art=10, Music=6, PE=8. Riley says if 4 students switch from art to music, art and music would be tied. Is Riley right?

A. No, art would have 6 and music 10 ✓

B. Yes, both would have 8

C. No, art would have 8 and music 8 but that ties with PE

D. No, both would have 7

Explanation: Art: 10 - 4 = 6. Music: 6 + 4 = 10. Art=6, Music=10. They are NOT tied. Riley is wrong.

Lesson 2: Pictographs

Easy (20 questions)

1. A pictograph shows each star = 1 student. The cat row has 5 stars. How many students picked cats?

A. 4

B. 5 ✓

C. 6

D. 10

Explanation: Each star means 1 student. There are 5 stars, so 5 students picked cats.

2. A pictograph shows each smiley face = 1 vote. Dogs has 7 smiley faces and fish has 3. Which pet got more votes?

A. Fish

B. They are the same

C. Dogs ✓

D. Cannot tell

Explanation: Dogs has 7 and fish has 3. 7 is more than 3, so dogs got more votes.

3. A pictograph shows each star = 2 students. The pizza row has 3 stars. How many students chose pizza?

A. 3

B. 5

C. 6 ✓

D. 8

Explanation: Each star = 2 students. 3 stars: skip count by 2: 2, 4, 6. That is 6 students.

4. A pictograph shows each heart = 1 vote. Red has 4 hearts, Blue has 6 hearts, Green has 4 hearts. Which two colors got the same number?

- A. Red and Blue
- B. Blue and Green
- C. Red and Green ✓**
- D. None are the same

Explanation: Red = 4 and Green = 4. They both got 4 votes, so they are the same.

5. A pictograph shows each star = 2 kids. Soccer has 4 stars. How many kids picked soccer?

- A. 4
- B. 6
- C. 8 ✓**
- D. 10

Explanation: Each star = 2 kids. 4 stars: 2, 4, 6, 8. That is 8 kids.

6. A pictograph shows each circle = 1 student. Apples has 8 circles and bananas has 5 circles. Which fruit has fewer votes?

- A. Apples
- B. Bananas ✓**
- C. They are the same
- D. Cannot tell

Explanation: Bananas has 5 and apples has 8. 5 is less than 8, so bananas has fewer.

7. A pictograph uses a key that says each star = 5 students. The chocolate row has 2 stars. How many picked chocolate?

- A. 2
- B. 5
- C. 7
- D. 10 ✓**

Explanation: Each star = 5 students. 2 stars: 5, 10. That is 10 students.

8. A pictograph shows each star = 1 vote. Summer has 9 stars. Winter has 3 stars. Spring has 6 stars. Which season has the most votes?

- A. Winter
- B. Spring
- C. Summer ✓**
- D. They are all the same

Explanation: Summer has 9, Spring has 6, Winter has 3. Summer has the most with 9 votes.

9. A pictograph shows each star = 2 votes. The blue row has 5 stars. How many votes did blue get?

- A. 5
- B. 7
- C. 10 ✓**
- D. 12

Explanation: Each star = 2 votes. 5 stars: 2, 4, 6, 8, 10. That is 10 votes.

10. A pictograph shows each ball = 1 student. Basketball has 6 balls and volleyball has 6 balls. Did the same number of students pick each sport?

- A. No, basketball has more
- B. No, volleyball has more
- C. Yes, they are the same ✓**
- D. Cannot tell

Explanation: Basketball = 6 and Volleyball = 6. Yes, they are the same!

11. A pictograph shows each star = 5 kids. Bikes has 3 stars. How many kids like bikes?

- A. 3
- B. 8
- C. 10
- D. 15 ✓**

Explanation: Each star = 5 kids. 3 stars: 5, 10, 15. That is 15 kids.

12. A pictograph shows each star = 1 vote. Cats has 4 stars, Dogs has 7 stars, Birds has 2 stars. Which pet got the fewest votes?

- A. Cats
- B. Dogs
- C. Birds ✓**
- D. They are all the same

Explanation: Birds has only 2 stars. That is the fewest votes.

13. A pictograph shows each star = 2 students. The key says: each star = 2. You see 6 stars in the tiger row. How many students chose tigers?

- A. 6
- B. 8
- C. 10
- D. 12 ✓**

Explanation: Each star = 2. There are 6 stars: $6 \times 2 = 12$ students chose tigers.

14. Look at the pictograph. The key shows each apple picture = 1 student. The apple row has 3 pictures and the orange row has 5 pictures. How many picked oranges?

- A. 3
- B. 4
- C. 5 ✓**
- D. 8

Explanation: Each picture = 1 student. The orange row has 5 pictures, so 5 students picked oranges.

15. A pictograph shows each star = 2 votes. Sunny has 4 stars and Cloudy has 2 stars. Which weather has more votes?

- A. Cloudy
- B. They are the same
- C. Sunny ✓**
- D. Cannot tell

Explanation: Sunny = 4 stars = 8 votes. Cloudy = 2 stars = 4 votes. Sunny has more.

16. A pictograph shows each star = 5 students. The candy row has 4 stars. How many students chose candy?

- A. 4
- B. 9
- C. 15
- D. 20 ✓**

Explanation: Each star = 5. 4 stars: 5, 10, 15, 20. That is 20 students.

17. A pictograph shows each star = 1 kid. Swings has 8 stars, Slide has 5 stars, Sandbox has 3 stars. Put them in order from most to fewest.

- A. Sandbox, Slide, Swings
- B. Swings, Slide, Sandbox ✓**
- C. Slide, Swings, Sandbox
- D. Swings, Sandbox, Slide

Explanation: Swings = 8 (most), Slide = 5 (middle), Sandbox = 3 (fewest).

18. A pictograph shows each star = 2 students. Crayons has 3 stars. Markers has 3 stars. Do they have the same number of students?

- A. No, crayons has more
- B. No, markers has more
- C. Yes, both have 6 students ✓**
- D. Yes, both have 3 students

Explanation: Crayons: $3 \times 2 = 6$. Markers: $3 \times 2 = 6$. Yes, both have 6 students.

19. A pictograph shows each star = 1 student. The key is missing for one row. You know 4 students like turtles and you see 4 stars. What does each star equal?

- A. 2 students
- B. 1 student ✓**
- C. 4 students
- D. 5 students

Explanation: 4 stars stand for 4 students. So each star = 1 student.

20. A pictograph shows each star = 5 votes. Trucks has 2 stars and Cars has 3 stars. Which has more votes?

- A. Trucks
- B. They are the same
- C. Cars ✓**
- D. Cannot tell

Explanation: Trucks = $2 \times 5 = 10$. Cars = $3 \times 5 = 15$. Cars has more votes.

Medium (20 questions)

1. A pictograph shows each star = 2 students. Cats has 4 stars and Dogs has 6 stars. How many more students picked dogs than cats?

- A. 2
- B. 4 ✓**
- C. 8
- D. 12

Explanation: Cats = $4 \times 2 = 8$. Dogs = $6 \times 2 = 12$. $12 - 8 = 4$ more students picked dogs.

2. A pictograph shows each star = 5 votes. Pizza has 4 stars and Tacos has 2 stars. How many voted in all?

- A. 6
- B. 20
- C. 30 ✓**
- D. 10

Explanation: Pizza = $4 \times 5 = 20$. Tacos = $2 \times 5 = 10$. $20 + 10 = 30$ votes in all.

3. A pictograph shows each star = 2 kids. Red has 5 stars, Blue has 3 stars, Green has 4 stars. How many kids picked red and green together?

- A. 9
- B. 14
- C. 18 ✓**
- D. 24

Explanation: Red = $5 \times 2 = 10$. Green = $4 \times 2 = 8$. $10 + 8 = 18$ kids together.

4. A pictograph shows each star = 5 students. Soccer has 3 stars and Baseball has 1 star. How many fewer students picked baseball?

- A. 2
- B. 5
- C. 10 ✓**
- D. 20

Explanation: Soccer = $3 \times 5 = 15$. Baseball = $1 \times 5 = 5$. $15 - 5 = 10$ fewer picked baseball.

5. A pictograph shows each star = 2 votes. Chocolate has 7 stars and Vanilla has 5 stars. How many total votes for both flavors?

- A. 12
- B. 14
- C. 20
- D. 24 ✓**

Explanation: Chocolate = $7 \times 2 = 14$. Vanilla = $5 \times 2 = 10$. $14 + 10 = 24$ total votes.

6. A pictograph shows each star = 2 students. Lions has 4 stars and Bears has 4 stars. How do they compare?

- A. Lions has more
- B. Bears has more
- C. They are the same with 8 each ✓**
- D. They are the same with 4 each

Explanation: Lions = $4 \times 2 = 8$. Bears = $4 \times 2 = 8$. They are the same with 8 students each.

7. A pictograph shows each star = 5 kids. The key says each star = 5. Swimming has 4 stars. Biking has 3 stars. How many more kids picked swimming?

- A. 1
- B. 5 ✓**
- C. 7
- D. 35

Explanation: Swimming = $4 \times 5 = 20$. Biking = $3 \times 5 = 15$. $20 - 15 = 5$ more kids.

8. A pictograph shows each star = 2 votes. Apples has 6 stars, Bananas has 2 stars, Grapes has 5 stars. Which fruit got the fewest votes, and how many?

- A. Apples with 12
- B. Bananas with 4 ✓**
- C. Grapes with 10
- D. Bananas with 2

Explanation: Bananas = $2 \times 2 = 4$ votes. That is the fewest. (Apples = 12, Grapes = 10.)

9. A pictograph shows each star = 5 students. Art has 3 stars and Music has 2 stars. The pictograph is missing data for PE. There were 30 students total. How many chose PE?

- A. 5
- B. 10 ✓**
- C. 15
- D. 20

Explanation: Art = $3 \times 5 = 15$. Music = $2 \times 5 = 10$. That is 25. Total = 30. $30 - 25 = 5$ students chose PE.

10. A pictograph shows each star = 2 students. Bikes has 5 stars, Scooters has 3 stars, Skateboards has 4 stars. Put them in order from most to fewest students.

- A. Skateboards, Scooters, Bikes
- B. Bikes, Skateboards, Scooters ✓**
- C. Scooters, Skateboards, Bikes
- D. Bikes, Scooters, Skateboards

Explanation: Bikes = 10, Skateboards = 8, Scooters = 6. Order: Bikes, Skateboards, Scooters.

11. A pictograph shows each star = 5 votes. Spring has 2 stars and Summer has 5 stars. How many more voted for summer?

- A. 3
- B. 10
- C. 15 ✓**
- D. 35

Explanation: Spring = $2 \times 5 = 10$. Summer = $5 \times 5 = 25$. $25 - 10 = 15$ more voted for summer.

12. A pictograph shows each star = 2 students. Roses has 6 stars and Tulips has 6 stars. Lily says they are different because roses are in a different row. Is Lily right?

- A. Yes, different rows means different totals
- B. No, both have 12 students ✓**
- C. Yes, roses always gets more
- D. No, both have 6 students

Explanation: Roses = $6 \times 2 = 12$. Tulips = $6 \times 2 = 12$. Same number of stars and same key means same total. Lily is wrong.

13. A pictograph shows each star = 2 kids. Carrots has 3 stars, Corn has 5 stars, Peas has 4 stars. How many kids voted in all?

- A. 12
- B. 18
- C. 24 ✓**
- D. 30

Explanation: Carrots = 6, Corn = 10, Peas = 8. $6 + 10 + 8 = 24$ kids in all.

14. A pictograph shows each star = 5 students. Cats has 3 stars and Dogs has 4 stars. How many students chose cats and dogs combined?

- A. 7
- B. 15
- C. 20
- D. 35 ✓**

Explanation: Cats = $3 \times 5 = 15$. Dogs = $4 \times 5 = 20$. $15 + 20 = 35$ students combined.

15. A pictograph shows each star = 2 votes. Sunny has 8 stars and Rainy has 3 stars. How many more votes did sunny get?

- A. 5
- B. 10 ✓**
- C. 11
- D. 22

Explanation: Sunny = $8 \times 2 = 16$. Rainy = $3 \times 2 = 6$. $16 - 6 = 10$ more votes for sunny.

16. A pictograph shows each star = 5 kids. Trucks has 2 stars, Cars has 4 stars, Vans has 1 star. Which two categories together equal 15?

- A. Trucks and Cars
- B. Cars and Vans
- C. Trucks and Vans ✓**
- D. None equal 15

Explanation: Trucks = 10, Vans = 5. $10 + 5 = 15$. Trucks and Vans together equal 15.

17. A pictograph shows each star = 2 students. Tigers has 5 stars and Elephants has 7 stars. What is the difference in students?

- A. 2
- B. 4 ✓**
- C. 12
- D. 24

Explanation: Tigers = $5 \times 2 = 10$. Elephants = $7 \times 2 = 14$. $14 - 10 = 4$ more for elephants.

18. A pictograph shows each star = 5 votes. The row for hiking has 3 stars. The row for camping has 5 stars. How many total votes?

- A. 8
- B. 15
- C. 25
- D. 40 ✓**

Explanation: Hiking = $3 \times 5 = 15$. Camping = $5 \times 5 = 25$. $15 + 25 = 40$ total votes.

19. A pictograph shows each star = 2 kids. Math has 4 stars and Science has 6 stars. How many more kids picked science?

- A. 2
- B. 4 ✓**
- C. 8
- D. 20

Explanation: Math = $4 \times 2 = 8$. Science = $6 \times 2 = 12$. $12 - 8 = 4$ more kids.

20. A pictograph shows each star = 2 students. Hamsters has 3 stars, Rabbits has 5 stars, Fish has 2 stars. Which pet is in the middle (not most, not fewest)?

A. Hamsters with 6 students ✓

B. Rabbits with 10 students

C. Fish with 4 students

D. They are all the same

Explanation: Fish = 4 (fewest), Hamsters = 6 (middle), Rabbits = 10 (most). Hamsters is in the middle.

Hard (20 questions)

1. A pictograph shows each star = 2 students. Cats has 4 stars, Dogs has 6 stars, Birds has 3 stars. How many students voted in all? If the class has 30 students, how many did NOT vote?

A. 26 voted, 4 did not ✓

B. 26 voted, 6 did not

C. 24 voted, 6 did not

D. 22 voted, 8 did not

Explanation: Cats=8, Dogs=12, Birds=6. Total=26. 30-26=4 did not vote.

2. A pictograph shows each star = 5 students. The cat row has 3 stars. Sam says that means 8 students chose cats. What did Sam do wrong?

A. Sam added 5 + 3 instead of multiplying ✓

B. Sam subtracted 5 - 3

C. Sam counted the stars wrong

D. Sam is correct

Explanation: Each star = 5 students. 3 stars = $3 \times 5 = 15$. Sam added $5 + 3 = 8$ instead of multiplying. That is wrong.

3. A pictograph shows each star = 2 votes. Red has 5 stars, Blue has 3 stars, Green has ????. There were 22 votes total. How many votes did green get?

A. 4

B. 6 ✓

C. 8

D. 14

Explanation: Red = 10, Blue = 6. That is 16. Total = 22. $22 - 16 = 6$ votes for green.

4. Could you make a pictograph with a key of each star = 3 for this data: Cats=9, Dogs=6, Fish=4? Why or why not?

A. Yes, because all numbers are big enough

B. No, because 4 does not divide evenly by 3 ✓

C. Yes, because 3 goes into every number

D. No, because 9 is too large

Explanation: Cats: $9 / 3 = 3$ stars. Dogs: $6 / 3 = 2$ stars. Fish: $4 / 3 = 1$ star with 1 left over. You cannot show 4 evenly with a key of 3.

5. A pictograph shows each star = 5 kids. Soccer has 4 stars and Baseball has 2 stars. Emma says soccer got 10 more kids than baseball. Is she right?

A. Yes, $20 - 10 = 10$ ✓

B. No, the difference is 5

C. No, the difference is 2

D. No, the difference is 15

Explanation: Soccer = 20, Baseball = 10. $20 - 10 = 10$. Yes, Emma is right!

6. A pictograph shows each star = 2 students. Art has 7 stars, Music has 4 stars, Drama has 5 stars. Do music and drama together have more students than art?

A. Yes, 18 is more than 14 ✓

B. No, 14 is more than 18

C. They are the same

D. No, 16 is more than 14

Explanation: Art = 14, Music = 8, Drama = 10. Music + Drama = 18. 18 is more than 14. Yes!

7. A pictograph shows each star = 5 votes. Bikes has 3 stars, Scooters has 2 stars, Skateboards has 4 stars. How many kids voted in all? What if 5 more kids vote for scooters -- how many stars would scooters have then?

A. 45 total, scooters would have 3 stars ✓

B. 35 total, scooters would have 3 stars

C. 45 total, scooters would have 7 stars

D. 35 total, scooters would have 7 stars

Explanation: Bikes=15, Scooters=10, Skateboards=20. Total=45. Scooters would have $10+5=15$ kids. $15/5=3$ stars.

8. A pictograph shows each star = 2 students. Dogs has 6 stars, Cats has 8 stars, Fish has 2 stars. Jake looks at the stars and says cats got 8 students. What mistake did Jake make?

A. Jake forgot to multiply by 2 ✓

B. Jake counted the stars wrong

C. Jake looked at the wrong row

D. Jake is correct

Explanation: Cats has 8 stars. Each star = 2. So cats = $8 \times 2 = 16$ students. Jake just counted stars instead of multiplying by the key.

9. A pictograph shows each star = 5 students. Chocolate has 4 stars, Vanilla has 3 stars. If the teacher wants to add a strawberry row with 10 students, how many stars should she draw?

A. 1 star

B. 2 stars ✓

C. 5 stars

D. 10 stars

Explanation: Each star = 5 students. $10 \text{ students} / 5 = 2$ stars for strawberry.

10. Could you make a pictograph with a key of each star = 2 for this data: Lions=8, Tigers=6, Bears=5? Why or why not?

A. Yes, all numbers work with 2

B. No, because 5 does not divide evenly by 2 ✓

C. No, because 6 is too small

D. Yes, because 2 goes into every number

Explanation: Lions: $8/2=4$ stars. Tigers: $6/2=3$ stars. Bears: $5/2=2$ stars with 1 left over. 5 is odd, so it does not work with key=2.

11. Two classes made pictographs about favorite fruits. Class A uses each star = 2. Class B uses each star = 5. Class A shows apples with 6 stars. Class B shows apples with 2 stars. Which class had more students pick apples?

A. Class A with 12 students ✓

B. Class B with 10 students

C. They are the same

D. Cannot tell without more info

Explanation: Class A: $6 \times 2 = 12$. Class B: $2 \times 5 = 10$. Class A had more with 12 students.

12. A pictograph shows each star = 2 votes. Summer has 8 stars, Winter has 3 stars, Spring has 5 stars, Fall has 4 stars. Which two seasons together got exactly 16 votes?

A. Summer and Winter

B. Spring and Fall

C. Winter and Spring ✓

D. Summer and Fall

Explanation: Summer=16, Winter=6, Spring=10, Fall=8. Winter + Spring = $6 + 10 = 16$. Those two together got exactly 16 votes.

13. A pictograph shows each star = 5 students. Carrots has 2 stars, Corn has 4 stars, Peas has ????. There were 45 students total. How many stars should the peas row have?

A. 3 stars ✓

B. 5 stars

C. 7 stars

D. 9 stars

Explanation: Carrots=10, Corn=20. That is 30. Total=45. $45-30=15$ for peas. $15/5=3$ stars.

14. A pictograph shows each star = 2 students. Puppies has 9 stars, Kittens has 6 stars. Mia says puppies got 12 more students than kittens. What should Mia check?

A. She should multiply first: $18 - 12 = 6$, not 12 ✓

B. She subtracted the stars: $9 - 6 = 3$

C. She is correct

D. She should add instead

Explanation: Puppies = $9 \times 2 = 18$. Kittens = $6 \times 2 = 12$. $18 - 12 = 6$ more students, not 12. Mia should multiply first, then subtract.

15. A pictograph shows each star = 5 votes. Math has 3 stars, Reading has 5 stars. If 10 students switch from reading to math, how many stars would each row have?

A. Math=5 stars, Reading=3 stars ✓

B. Math=4 stars, Reading=4 stars

C. Math=13 stars, Reading=15 stars

D. Math=25 stars, Reading=15 stars

Explanation: Math = 15, Reading = 25. Move 10: Math = $15 + 10 = 25$. Reading = $25 - 10 = 15$. $25 / 5 = 5$ stars. $15 / 5 = 3$ stars.

16. A pictograph shows each star = 2 kids. Hiking has 4 stars, Camping has 7 stars, Fishing has 3 stars. How many kids did the two least popular activities combined?

A. 7

B. 14 ✓

C. 22

D. 28

Explanation: Hiking = 8 (least), Fishing = 6 (second least). $8 + 6 = 14$ kids combined.

17. A pictograph shows each star = 5 students. Bikes has 6 stars and Cars has 3 stars. Noah says the difference is 3 students. What did Noah do wrong?

A. He subtracted stars instead of students: $6 - 3 = 3$ stars, but the real difference is 15 ✓

B. He added instead of subtracted

C. He is correct

D. He read the wrong rows

Explanation: Bikes = 30, Cars = 15. Difference = $30 - 15 = 15$ students. Noah subtracted the stars ($6 - 3 = 3$) instead of the actual students.

18. Can you make a pictograph with each star = 5 for: Roses=15, Tulips=10, Daisies=7? Why or why not?

A. Yes, because all numbers are bigger than 5

B. No, because 7 does not divide evenly by 5 ✓

C. Yes, because 5 goes into 15 and 10

D. No, because 15 is too big

Explanation: Roses: $15 / 5 = 3$ stars. Tulips: $10 / 5 = 2$ stars. Daisies: $7 / 5 = 1$ star with 2 left over. 7 does not work with key=5.

19. A pictograph shows each star = 2 votes. Vanilla has 5 stars, Chocolate has 8 stars, Mint has 4 stars. There are 40 students in the class. How many did NOT vote?

A. 4

B. 6

C. 8 ✓

D. 10

Explanation: Vanilla=10, Chocolate=16, Mint=8. Total = $10 + 16 + 8 = 34$. $40 - 34 = 6$ did not vote.

20. A pictograph shows each star = 5 students. Monkeys has 4 stars, Penguins has 2 stars, Pandas has 3 stars. If you changed the key to each star = 10, how many stars would monkeys need?

A. 2 stars ✓

B. 4 stars

C. 8 stars

D. 20 stars

Explanation: Monkeys = $4 \times 5 = 20$ students. With key = 10: $20 / 10 = 2$ stars.

Lesson 3: Read and Answer Questions

Easy (20 questions)

1. A graph shows favorite fruits: Apples=6, Bananas=3, Grapes=8. Which fruit has the most votes?

- A. Apples
- B. Bananas
- C. Grapes ✓**
- D. They are the same

Explanation: Grapes has 8, which is the biggest number. Grapes has the most votes.

2. Look at the bar graph. Dogs=9, Cats=4, Fish=7. How many students chose cats?

- A. 4 ✓**
- B. 7
- C. 9
- D. 20

Explanation: The bar for cats goes up to 4. So 4 students chose cats.

3. A graph shows: Red=5, Blue=10, Yellow=3. Which color got the fewest votes?

- A. Red
- B. Blue
- C. Yellow ✓**
- D. They are the same

Explanation: Yellow has 3, which is the smallest number. Yellow got the fewest votes.

4. A pictograph shows each star = 1 vote. Tacos has 6 stars and Pizza has 9 stars. How many votes did tacos get?

- A. 6 ✓**
- B. 9
- C. 15
- D. 3

Explanation: Each star = 1 vote. Tacos has 6 stars, so tacos got 6 votes.

5. A tally chart shows: Soccer IIII, Baseball IIII I, Basketball III. How many votes did soccer get?

- A. 3
- B. 4 ✓**
- C. 5
- D. 6

Explanation: Count the tally marks for soccer: IIII = 4. Soccer got 4 votes.

6. A graph shows: Chocolate=11, Vanilla=5, Strawberry=8. Which flavor has the most votes?

- A. Vanilla
- B. Strawberry
- C. Chocolate ✓**
- D. They are the same

Explanation: Chocolate has 11, which is the biggest number. Chocolate has the most.

7. A bar graph shows: Spring=4, Summer=9, Fall=6, Winter=1. Which season has the fewest votes?

- A. Spring
- B. Summer
- C. Fall
- D. Winter ✓**

Explanation: Winter has 1, which is the smallest number. Winter has the fewest.

8. A pictograph shows each star = 1 kid. Bikes has 5 stars and Scooters has 5 stars. Did the same number of kids pick each?

- A. No, bikes has more
- B. No, scooters has more
- C. Yes, they are the same ✓**
- D. Cannot tell

Explanation: Bikes = 5 and Scooters = 5. They have the same number of stars, so yes, they are the same.

9. A graph shows: Lions=4, Bears=7, Eagles=2. Which animal has the most votes?

- A. Lions
- B. Bears ✓**
- C. Eagles
- D. They are the same

Explanation: Bears has 7, which is the biggest number. Bears has the most votes.

10. A tally chart shows: Carrots III, Corn IIII II, Peas IIII. How many votes does corn have?

- A. 3
- B. 4
- C. 5
- D. 7 ✓**

Explanation: IIII is 5 and II is 2 more. $5 + 2 = 7$. Corn has 7 votes.

11. A bar graph shows: Swings=10, Slide=4, Sandbox=7. Which has fewer votes, slide or sandbox?

- A. Sandbox
- B. They are the same
- C. Slide ✓**
- D. Swings

Explanation: Slide has 4 and sandbox has 7. 4 is less than 7, so slide has fewer.

12. A graph shows: Sunny=6, Rainy=6, Snowy=3. Which two weather types have the same number of votes?

- A. Sunny and Snowy
- B. Rainy and Snowy
- C. Sunny and Rainy ✓**
- D. None are the same

Explanation: Sunny and Rainy both have 6. They are the same!

13. The data shows: Math=5, Reading=8, Art=12. A student says art got the fewest votes. Is that correct?

- A. Yes, art got the fewest
- B. No, math got the fewest ✓**
- C. No, reading got the fewest
- D. They all got the same

Explanation: Math = 5, Reading = 8, Art = 12. Math has the fewest with 5. Art has the most. The student is wrong.

14. A bar graph shows: Monkeys=9, Giraffes=3, Elephants=6. Put the animals in order from fewest to most votes.

- A. Monkeys, Elephants, Giraffes
- B. Giraffes, Elephants, Monkeys ✓**
- C. Elephants, Giraffes, Monkeys
- D. Giraffes, Monkeys, Elephants

Explanation: Giraffes = 3 (fewest), Elephants = 6 (middle), Monkeys = 9 (most).

15. A pictograph shows each star = 1 student. Milk has 7 stars, Juice has 4 stars, Water has 7 stars. How many categories have the same number?

- A. 0
- B. 1
- C. 2 ✓**
- D. 3

Explanation: Milk and Water both have 7 stars. That is 2 categories with the same number.

16. A graph shows: Crayons=8, Markers=3, Pencils=5. Which supply is between markers and crayons in number of votes?

- A. Crayons
- B. Markers
- C. Pencils ✓**
- D. None of them

Explanation: Markers = 3, Pencils = 5, Crayons = 8. Pencils (5) is between 3 and 8.

17. A bar graph shows: Dogs=8, Cats=5, Birds=8. Do dogs and birds have the same number of votes?

- A. No, dogs has more
- B. No, birds has more
- C. Yes, they are the same ✓**
- D. Cannot tell

Explanation: Dogs = 8 and Birds = 8. They are the same number!

18. A tally chart shows: Camping III, Hiking IIII IIII, Fishing IIII I. Which activity got the most votes?

- A. Camping
- B. Hiking ✓**
- C. Fishing
- D. Camping and Fishing tied

Explanation: Camping = 3, Hiking = 9, Fishing = 6. Hiking got the most with 9 votes.

19. A graph shows: Trucks=2, Trains=5, Planes=9. Which vehicle has the fewest votes?

- A. Trucks ✓**
- B. Trains
- C. Planes
- D. They are the same

Explanation: Trucks has 2, which is the smallest number. Trucks has the fewest.

20. A bar graph shows: Red=4, Blue=4, Green=4, Yellow=4. What can you say about this data?

- A. Red has the most
- B. Yellow has the fewest
- C. All four colors have the same number of votes ✓**
- D. Green and Blue are tied

Explanation: All four colors have 4 votes each. They are all the same!

Medium (20 questions)

1. A graph shows: Apples=6, Bananas=4, Grapes=5. How many fruits were picked in all?

- A. 10
- B. 11
- C. 15 ✓**
- D. 14

Explanation: Add them all: $6 + 4 + 5 = 15$ fruits in all.

2. A bar graph shows: Dogs=9, Cats=5. How many more students picked dogs than cats?

- A. 14
- B. 3
- C. 4 ✓**
- D. 5

Explanation: Dogs = 9 and Cats = 5. $9 - 5 = 4$ more students picked dogs.

3. A pictograph shows each star = 2 students. Red has 4 stars, Blue has 6 stars. How many students picked blue?

- A. 6
- B. 8
- C. 10
- D. 12 ✓**

Explanation: Each star = 2 students. Blue has 6 stars: $6 \times 2 = 12$ students.

4. A tally chart shows: Pizza IIII III, Tacos IIII, Burgers IIII II. How many voted for pizza and burgers combined?

- A. 12
- B. 13
- C. 15 ✓**
- D. 17

Explanation: Pizza = 8 and Burgers = 7. $8 + 7 = 15$ voted for pizza and burgers combined.

5. A bar graph shows: Soccer=8, Baseball=5, Basketball=7. How many total students voted?

- A. 13
- B. 15
- C. 20 ✓
- D. 21

Explanation: $8 + 5 + 7 = 20$ total students voted.

6. A graph shows the data is missing for one category. Chocolate=11, Vanilla=???, Strawberry=3. There were 20 votes total. How many voted for vanilla?

- A. 4
- B. 5
- C. 6 ✓
- D. 7

Explanation: $\text{Chocolate} + \text{Strawberry} = 11 + 3 = 14$. $\text{Total} = 20$. $20 - 14 = 6$ voted for vanilla.

7. A bar graph shows: Bikes=4, Scooters=9, Skateboards=2. How many more students chose scooters than skateboards?

- A. 5
- B. 6
- C. 7 ✓
- D. 11

Explanation: $\text{Scooters} = 9$ and $\text{Skateboards} = 2$. $9 - 2 = 7$ more chose scooters.

8. A pictograph shows each star = 5 votes. Summer has 3 stars and Winter has 1 star. How many more voted for summer than winter?

- A. 2
- B. 5
- C. 10 ✓
- D. 20

Explanation: $\text{Summer} = 15$, $\text{Winter} = 5$. $15 - 5 = 10$ more voted for summer.

9. A tally chart shows: Lions IIII II, Bears III, Eagles IIII. Which animal is in second place?

- A. Lions
- B. Bears
- C. Eagles ✓
- D. Lions and Eagles tied

Explanation: $\text{Lions} = 7$ (first), $\text{Eagles} = 4$ (second), $\text{Bears} = 3$ (third). Eagles is in second place.

10. A graph shows: Swimming=5, Running=8, Jumping=6. How many students chose swimming and jumping combined?

- A. 11 ✓
- B. 13
- C. 14
- D. 19

Explanation: $\text{Swimming} = 5$, $\text{Jumping} = 6$. $5 + 6 = 11$ students combined.

11. A bar graph shows: Sunny=8, Rainy=4, Snowy=3. How many more sunny days than rainy days?

- A. 3
- B. 4 ✓
- C. 5
- D. 12

Explanation: $\text{Sunny} = 8$ and $\text{Rainy} = 4$. $8 - 4 = 4$ more sunny days.

12. A pictograph shows each star = 2 kids. Trucks has 5 stars and Cars has 7 stars. How many total kids voted for trucks and cars?

- A. 12
- B. 14
- C. 20
- D. 24 ✓

Explanation: $\text{Trucks} = 10$, $\text{Cars} = 14$. $10 + 14 = 24$ total kids.

13. A graph shows: Hamsters=2, Rabbits=7, Turtles=5. How many fewer students picked hamsters than rabbits?

- A. 3
- B. 4
- C. 5 ✓**
- D. 9

Explanation: Rabbits = 7 and Hamsters = 2. $7 - 2 = 5$ fewer students picked hamsters.

14. A bar graph shows: Spring=3, Summer=10, Fall=4, Winter=3. Which two seasons got the same number of votes?

- A. Spring and Summer
- B. Summer and Fall
- C. Spring and Winter ✓**
- D. Fall and Winter

Explanation: Spring = 3 and Winter = 3. They both got the same number of votes.

15. A graph shows: Art=12, Music=7. How many more students chose art than music?

- A. 3
- B. 4
- C. 5 ✓**
- D. 19

Explanation: Art = 12 and Music = 7. $12 - 7 = 5$ more students chose art.

16. A tally chart shows: Crayons IIII II, Markers IIII, Pencils IIII III. How many supplies were voted for in all?

- A. 15
- B. 17
- C. 19 ✓**
- D. 21

Explanation: Crayons = 7, Markers = 4, Pencils = 8. $7 + 4 + 8 = 19$ total.

17. A pictograph shows each star = 5 students. Math has 4 stars and Reading has 2 stars. How many total students chose these two subjects?

- A. 6
- B. 20
- C. 25
- D. 30 ✓**

Explanation: Math = 20, Reading = 10. $20 + 10 = 30$ total students.

18. A bar graph shows: Roses=4, Tulips=8, Daisies=6. Luis says tulips got twice as many votes as roses. Is he right?

- A. Yes, 8 is double 4 ✓**
- B. No, 8 is not double 4
- C. Yes, 8 is double 6
- D. No, they are the same

Explanation: Double of 4 is 8. Yes, Luis is right! Tulips (8) got twice as many as Roses (4).

19. A graph shows: Milk=5, Juice=11. How many fewer students picked milk?

- A. 5
- B. 6 ✓**
- C. 16
- D. 7

Explanation: Juice = 11 and Milk = 5. $11 - 5 = 6$ fewer students picked milk.

20. A bar graph shows: Elephants=6, Monkeys=8, Tigers=4. How many students voted for elephants and tigers together?

- A. 10 ✓**
- B. 12
- C. 14
- D. 18

Explanation: Elephants = 6, Tigers = 4. $6 + 4 = 10$ students together.

Hard (20 questions)

1. A bar graph shows: Apples=8, Bananas=5, Grapes=7. How many students voted in all? If there are 25 students in the class, how many did NOT vote?

A. 20 voted, 5 did not ✓

B. 18 voted, 7 did not

C. 20 voted, 7 did not

D. 15 voted, 10 did not

Explanation: $8 + 5 + 7 = 20$ voted. $25 - 20 = 5$ did not vote.

2. A graph shows: Dogs=9, Cats=6, Fish=3. If 2 more kids pick fish, how many would fish have? Would it be more or less than cats?

A. 5, less than cats ✓

B. 5, more than cats

C. 8, more than cats

D. 5, same as cats

Explanation: Fish has $3 + 2 = 5$. Cats has 6. 5 is less than 6, so fish is still less than cats.

3. A pictograph shows each star = 5 votes. Soccer has 4 stars and Baseball has 2 stars. Kai says the difference is 2 votes. What mistake did Kai make?

A. He subtracted stars ($4-2=2$) instead of actual votes ($20-10=10$) ✓

B. He added instead of subtracted

C. He is correct

D. He read the wrong rows

Explanation: Soccer = 20, Baseball = 10. Difference = 10 votes. Kai subtracted the star counts instead of the real vote totals.

4. A bar graph shows: Red=4, Blue=7, Green=9. How many voted for red and blue together? Is it more or less than green?

A. 11, more than green ✓

B. 11, less than green

C. 13, more than green

D. 10, more than green

Explanation: Red + Blue = $4 + 7 = 11$. Green = 9. 11 is more than 9!

5. A tally chart shows: Chocolate IIII IIII I, Vanilla IIII III, Strawberry IIII. The class has 28 students. How many students did NOT vote?

A. 3

B. 4

C. 5 ✓

D. 6

Explanation: Chocolate: IIII IIII I = $5+5+1 = 11$. Vanilla = 8, Strawberry = 4. Total = $11+8+4 = 23$. $28 - 23 = 5$ did not vote.

6. A graph shows: Bikes=7, Scooters=4, Skateboards=9. If 3 kids change from skateboards to bikes, what would the new totals be for each?

A. Bikes=10, Scooters=4, Skateboards=6 ✓

B. Bikes=4, Scooters=4, Skateboards=12

C. Bikes=10, Scooters=7, Skateboards=6

D. Bikes=7, Scooters=7, Skateboards=6

Explanation: Bikes gains 3: $7+3=10$. Skateboards loses 3: $9-3=6$. Scooters stays at 4.

7. A pictograph shows each star = 2 kids. Art has 5 stars, Music has 4 stars, PE has 6 stars. Which two subjects together have the same total as PE?

A. Art and Music together do NOT equal PE ✓

B. Art and Music together equal PE

C. Music and PE equal Art

D. None of them match

Explanation: Art=10, Music=8, PE=12. Art+Music=18. That is NOT equal to PE (12). So no pair equals PE alone.

8. A bar graph shows: Lions=6, Bears=8, Eagles=4. Sophie says bears got twice as many votes as eagles. Jake says bears got 2 more than lions. Who is right?

- A. Only Sophie
- B. Only Jake
- C. Both are right ✓**
- D. Neither is right

Explanation: Sophie: double of 4 is 8. Bears = 8. Correct! Jake: $8 - 6 = 2$. Bears has 2 more than lions. Correct! Both are right.

9. The data shows: Sunny=9, Rainy=2, Snowy=4, Cloudy=5. How many students voted in all? How many more sunny than rainy days?

- A. 20 total, 7 more sunny ✓**
- B. 20 total, 11 more sunny
- C. 18 total, 7 more sunny
- D. 16 total, 7 more sunny

Explanation: Total: $9+2+4+5=20$. Sunny - Rainy = $9-2=7$ more sunny days.

10. A tally chart shows: Carrots III, Corn IIII II, Peas IIII, Beans ????. There were 20 votes total. How many voted for beans?

- A. 4
- B. 5
- C. 6 ✓**
- D. 7

Explanation: Carrots=3, Corn=7, Peas=4. Total so far=14. $20-14=6$ voted for beans.

11. A graph shows: Math=7, Reading=10, Art=8. Noah says reading and art together equal 16. He then says that is double math. What should Noah check?

- A. 16 is not double 7 (double 7 is 14) ✓**
- B. Noah added wrong
- C. 16 is actually double 8
- D. Noah is completely correct

Explanation: Reading + Art = $10 + 8 = 18$, not 16. Noah added wrong. And double of 7 is 14, not 16 or 18.

12. A pictograph shows each star = 2 votes. Spring has 3 stars, Summer has 7 stars, Fall has 5 stars, Winter has 2 stars. Which season got the most votes and how many?

- A. Summer with 14 votes ✓**
- B. Summer with 7 votes
- C. Fall with 10 votes
- D. Spring with 6 votes

Explanation: Summer = $7 \times 2 = 14$ votes. That is the most. (Spring=6, Fall=10, Winter=4.)

13. A bar graph shows: Trucks=5, Cars=12, Trains=3. How many fewer voted for trains than cars? If 4 more kids vote for trains, would trains pass trucks?

- A. 9 fewer; yes, trains would have 7 ✓**
- B. 9 fewer; no, trains would have 7
- C. 7 fewer; yes, trains would pass
- D. 9 fewer; no, trains stays at 3

Explanation: $12-3=9$ fewer. Trains + 4 = 7. Trucks = 5. $7 > 5$, so yes, trains would pass trucks.

14. A graph shows: Swings=8, Slide=6, Sandbox=4. If each category gets 2 more votes, would swings still have the most?

- A. Yes, Swings=10, Slide=8, Sandbox=6 ✓**
- B. No, they would all be tied
- C. No, Slide would have the most
- D. Yes, but Slide would tie with Swings

Explanation: Swings: $8+2=10$. Slide: $6+2=8$. Sandbox: $4+2=6$. Swings still has the most.

15. A pictograph shows each star = 5 kids. Cats has 3 stars, Dogs has 4 stars, Birds has 1 star. Ella says there were 35 votes total. Is Ella right?

A. No, the total is 40 ✓

B. Yes, 35 is correct

C. No, the total is 30

D. No, the total is 45

Explanation: Cats=15, Dogs=20, Birds=5. Total=15+20+5=40. Ella said 35, which is wrong. The total is 40.

16. A bar graph shows: Camping=5, Hiking=8, Fishing=7. Which two activities together have the same total as hiking plus fishing?

A. There is no pair that equals 15 ✓

B. Camping and Hiking equal 15 but that uses hiking

C. Camping and Fishing = 12, not 15

D. Camping and Hiking both used

Explanation: Hiking+Fishing=15. The only other activity is camping (5). No pair of the three that doesn't include hiking or fishing equals 15. There is no such pair.

17. A tally chart shows: Milk IIII I, Juice IIII IIII II, Water IIII. Maria says juice got exactly double the votes of water. Is she correct?

A. No, juice has 12 and double of 4 is 8 ✓

B. Yes, juice is double water

C. No, juice has 10 and double of 4 is 8

D. No, juice has 12 and double of 4 is 12... wait, she is correct

Explanation: Juice = 5+5+2 = 12. Water = 4. Double of 4 is 8, not 12. So juice is NOT double water. Maria is incorrect. (Juice is actually triple water: $4 \times 3 = 12$.)

18. A graph shows: Red=8, Blue=5, Yellow=7, Green=10. Put them in order from most to fewest votes.

A. Blue, Yellow, Red, Green

B. Green, Red, Yellow, Blue ✓

C. Green, Yellow, Red, Blue

D. Red, Green, Yellow, Blue

Explanation: Green=10, Red=8, Yellow=7, Blue=5. From most to fewest: Green, Red, Yellow, Blue.

19. Two classes collected data on favorite animals. Class A: Dogs=12, Cats=8. Class B: Dogs=7, Cats=11. Which class had more students vote, and by how much?

A. Class A by 2 ✓

B. Class B by 2

C. They are the same

D. Class A by 4

Explanation: Class A total: 12+8=20. Class B total: 7+11=18. Class A had 2 more students.

20. A pictograph shows each star = 2 students. Crayons has 6 stars, Markers has 4 stars, Pencils has ????. There are 26 students total. How many stars should the pencils row have?

A. 2 stars

B. 3 stars ✓

C. 4 stars

D. 5 stars

Explanation: Crayons = 12, Markers = 8. Total so far = 20. 26-20=6 students for pencils. $6/2 = 3$ stars.

Lesson 4: Line Plots

Easy (20 questions)

1. A line plot shows X marks above each number. There are 4 X marks above the number 3. How many objects measured 3 inches?

- A. 3
- B. 4 ✓**
- C. 5
- D. 7

Explanation: Each X mark means 1 object. There are 4 X marks above 3, so 4 objects measured 3 inches.

2. A line plot shows X marks above numbers. There are 2 X marks above 5 and 6 X marks above 7. Which length has more objects?

- A. 5 inches
- B. They are the same
- C. 7 inches ✓**
- D. Cannot tell

Explanation: 5 inches has 2 X marks and 7 inches has 6 X marks. 6 is more, so 7 inches has more.

3. A line plot about pencils shows: Above 4: X X X, Above 5: X X X X X, Above 6: X X. How many pencils are 5 inches long?

- A. 3
- B. 4
- C. 5 ✓**
- D. 2

Explanation: Count the X marks above 5. There are 5 X marks, so 5 pencils are 5 inches long.

4. A line plot shows: Above 2: X, Above 3: X X X X, Above 4: X X. Which length was measured the most?

- A. 2 inches
- B. 3 inches ✓**
- C. 4 inches
- D. They are all the same

Explanation: 3 inches has 4 X marks, which is the most. 3 inches was measured the most.

5. A line plot shows 0 X marks above the number 6. How many crayons are 6 inches long?

- A. 6
- B. 1
- C. 0 ✓**
- D. 2

Explanation: There are no X marks above 6. That means 0 crayons are 6 inches long.

6. A line plot shows X marks about worm lengths. Above 1: X X X, Above 2: X X, Above 3: X X X X X. Which length has the fewest worms?

- A. 1 inch
- B. 2 inches ✓**
- C. 3 inches
- D. They are all the same

Explanation: 2 inches has only 2 X marks, which is the fewest.

7. A line plot shows: Above 4: X X X, Above 5: X X X, Above 6: X X X. What can you say about the data?

- A. 4 inches has the most
- B. 6 inches has the most
- C. All three lengths have the same number ✓**
- D. 5 inches has the fewest

Explanation: Each length has 3 X marks. They all have the same number of objects.

8. A line plot shows how long ribbons are. Above 7: X X X X X X, Above 8: X X. How many ribbons are 7 inches long?

- A. 2
- B. 5
- C. 6 ✓**
- D. 8

Explanation: Count the X marks above 7. There are 6, so 6 ribbons are 7 inches long.

9. A line plot shows: Above 3: X X, Above 4: X X X X, Above 5: X. How many sticks are 4 inches long?

- A. 1
- B. 2
- C. 3
- D. 4 ✓**

Explanation: Count the X marks above 4. There are 4, so 4 sticks are 4 inches long.

10. A line plot shows shoe lengths. Above size 5: X X X, Above size 6: X X X. Do the same number of kids wear each size?

- A. No, size 5 has more
- B. No, size 6 has more
- C. Yes, they are the same ✓**
- D. Cannot tell

Explanation: Size 5 has 3 X marks and size 6 has 3 X marks. They are the same!

11. A line plot shows: Above 1: X, Above 2: X X X X, Above 3: X X, Above 4: X X X. Which length has the most?

- A. 1 inch
- B. 2 inches ✓**
- C. 3 inches
- D. 4 inches

Explanation: 2 inches has 4 X marks, which is the most.

12. A line plot shows leaf lengths. Above 5: X X X X, Above 6: no X marks, Above 7: X X. Were any leaves 6 inches long?

- A. Yes, 6 leaves
- B. Yes, 1 leaf
- C. No, there were no 6-inch leaves ✓**
- D. Cannot tell

Explanation: There are 0 X marks above 6. No leaves were 6 inches long.

13. A line plot shows hand widths. Above 3: X X X X X, Above 4: X X X, Above 5: X X X X. Put the lengths in order from fewest to most X marks.

- A. 3, 4, 5
- B. 4, 5, 3 ✓**
- C. 4, 3, 5
- D. 5, 3, 4

Explanation: 3 inches = 5, 4 inches = 3, 5 inches = 4. From fewest to most: 4 (3 marks), 5 (4 marks), 3 (5 marks).

14. A line plot shows: Above 8: X X X X, Above 9: X, Above 10: X X. How many objects measured 9 inches?

- A. 1 ✓**
- B. 2
- C. 4
- D. 9

Explanation: There is 1 X mark above 9. So 1 object measured 9 inches.

15. A line plot about bean lengths shows: Above 2: X X, Above 3: X X X X X, Above 4: X X X. Which length was measured the most?

- A. 2 inches
- B. 3 inches ✓**
- C. 4 inches
- D. They are all the same

Explanation: 3 inches has 6 X marks, which is the most. 3 inches was measured the most.

16. A line plot shows: Above 6: X X X, Above 7: X X X X X, Above 8: X X. How many strings are 8 inches long?

- A. 3
- B. 5
- C. 2 ✓**
- D. 8

Explanation: Count the X marks above 8. There are 2, so 2 strings are 8 inches long.

17. A line plot shows caterpillar lengths. Above 1: X X, Above 2: X X X, Above 3: X X X X. Which length has the fewest caterpillars?

A. 1 inch ✓

B. 2 inches

C. 3 inches

D. They are all the same

Explanation: 1 inch has only 2 X marks, which is the fewest.

18. A line plot shows: Above 4: X X X X, Above 5: X X X X. How many items are 4 or 5 inches long in total?

A. 4

B. 5

C. 8 ✓

D. 9

Explanation: 4 inches has 4 marks and 5 inches has 4 marks. $4 + 4 = 8$ items total.

19. A line plot shows eraser widths. Above 1: X X X, Above 2: X X X X X, Above 3: X X. Which width was most common?

A. 1 inch

B. 2 inches ✓

C. 3 inches

D. They are the same

Explanation: 2 inches has 5 X marks, which is the most. 2 inches was the most common width.

20. A line plot shows: Above 5: X X, Above 6: X X X X, Above 7: X X X, Above 8: X. What is the smallest length shown with X marks?

A. 5 inches ✓

B. 6 inches

C. 7 inches

D. 8 inches

Explanation: The smallest number with X marks above it is 5. So 5 inches is the smallest length.

Medium (20 questions)

1. A line plot shows pencil lengths. Above 4: XX, Above 5: XXXXX, Above 6: XXX. Which length is most common?

A. 4 inches

B. 5 inches ✓

C. 6 inches

D. They are all the same

Explanation: 5 inches has the most X marks (5). That means 5 inches is the most common length.

2. A line plot shows ribbon lengths. Above 6: XXXX, Above 7: XXXX, Above 8: X. How many ribbons are there in all?

A. 4

B. 8

C. 9 ✓

D. 12

Explanation: Add the X marks: $4 + 4 + 1 = 9$ ribbons in all.

3. A line plot shows leaf lengths. Above 2: X, Above 3: XXXX, Above 4: XX, Above 5: XXX. How many more leaves are 3 inches than 4 inches?

A. 1

B. 2 ✓

C. 4

D. 6

Explanation: 3 inches has 4 and 4 inches has 2. $4 - 2 = 2$ more leaves at 3 inches.

4. A line plot shows stick lengths from 2 to 6 inches. The shortest stick is 2 inches and the longest is 6 inches. What is the range?

- A. 2 inches
- B. 4 inches ✓**
- C. 6 inches
- D. 8 inches

Explanation: Range = longest - shortest = 6 - 2 = 4 inches.

5. A line plot shows hand spans. Above 4: XXX, Above 5: XXXXX, Above 6: XX. How many hands were measured in all?

- A. 6
- B. 8
- C. 10 ✓**
- D. 15

Explanation: Add all the X marks: 3 + 5 + 2 = 10 hands were measured.

6. A line plot shows crayon lengths. Above 3: XXX, Above 4: XXXXXX, Above 5: XX. Which length is least common?

- A. 3 inches
- B. 4 inches
- C. 5 inches ✓**
- D. They are all the same

Explanation: 5 inches has only 2 X marks, which is the fewest. 5 inches is least common.

7. A line plot shows foot lengths. Above 7: XX, Above 8: XXXXXX, Above 9: XXX. How many fewer feet are 7 inches than 8 inches?

- A. 2
- B. 3
- C. 4 ✓**
- D. 6

Explanation: 8 inches has 6 and 7 inches has 2. 6 - 2 = 4 fewer at 7 inches.

8. A line plot shows string lengths. Above 3: XXXX, Above 4: X, Above 5: XXXX. Which two lengths have the same count?

- A. 3 and 4
- B. 4 and 5
- C. 3 and 5 ✓**
- D. None are the same

Explanation: 3 inches has 4 and 5 inches has 4. They are the same!

9. A line plot shows flower heights from 1 to 4 inches. Above 1: X, Above 2: XXX, Above 3: XXXXX, Above 4: XX. What is the range?

- A. 1 inch
- B. 2 inches
- C. 3 inches ✓**
- D. 4 inches

Explanation: Tallest = 4 inches, shortest = 1 inch. Range = 4 - 1 = 3 inches.

10. A line plot shows book widths. Above 5: XXX, Above 6: X, Above 7: XXXX, Above 8: XX. How many books were measured?

- A. 4
- B. 7
- C. 10 ✓**
- D. 14

Explanation: 3 + 1 + 4 + 2 = 10 books were measured.

11. A line plot shows shell lengths. Above 2: XXXX, Above 3: XX, Above 4: XXXXX. How many more shells are 4 inches than 3 inches?

- A. 1
- B. 2
- C. 3 ✓**
- D. 7

Explanation: 4 inches has 5 shells and 3 inches has 2 shells. $5 - 2 = 3$ more.

12. A line plot shows marker lengths. Above 3: XX, Above 4: XXXXXXX, Above 5: XXX. How many markers are 4 inches or 5 inches long?

- A. 7
- B. 9
- C. 10 ✓**
- D. 12

Explanation: 4 inches = 7 and 5 inches = 3. $7 + 3 = 10$ markers.

13. A line plot shows twig lengths from 5 to 9 inches. Data only appears at 5, 7, and 9. The shortest is 5 and the longest is 9. What is the range?

- A. 2 inches
- B. 4 inches ✓**
- C. 5 inches
- D. 9 inches

Explanation: Range = longest - shortest = $9 - 5 = 4$ inches.

14. A line plot shows feather lengths. Above 1: XXX, Above 2: XXXXX, Above 3: X. How many feathers in all?

- A. 3
- B. 5
- C. 9 ✓**
- D. 6

Explanation: $3 + 5 + 1 = 9$ feathers in all.

15. A line plot shows nail lengths. Above 1: XXXX, Above 2: XXXX, Above 3: XXXX. Are any lengths more common than others?

- A. Yes, 1 inch
- B. Yes, 3 inches
- C. Yes, 2 inches
- D. No, they are all the same ✓**

Explanation: Each length has 4 X marks. They are all the same, so no length is more common.

16. A line plot shows bean lengths. Above 2: XX, Above 3: XXXXXXX, Above 4: XXX. How many fewer beans are 2 inches than 3 inches?

- A. 1
- B. 2
- C. 4 ✓**
- D. 8

Explanation: 3 inches has 6 and 2 inches has 2. $6 - 2 = 4$ fewer beans at 2 inches.

17. A line plot shows worm lengths. Above 3: XXXX, Above 4: XX, Above 5: XXXXX. Which length is in the middle (not most, not fewest)?

- A. 3 inches with 4 worms ✓**
- B. 4 inches with 2 worms
- C. 5 inches with 5 worms
- D. They are all the same

Explanation: 4 inches = 2 (fewest), 3 inches = 4 (middle), 5 inches = 5 (most). 3 inches is in the middle.

18. A line plot shows rock lengths. Above 4: XXX, Above 5: XXXXXXXX, Above 6: XXXX. How many rocks in all?

- A. 7
- B. 11
- C. 14 ✓
- D. 10

Explanation: $3 + 7 + 4 = 14$ rocks in all.

19. A line plot shows eraser widths. Above 1: XXXX, Above 2: XX, Above 3: XXXXX. What is the range?

- A. 1 inch
- B. 2 inches ✓
- C. 3 inches
- D. 5 inches

Explanation: $\text{Biggest} = 3 \text{ inches}$, $\text{smallest} = 1 \text{ inch}$. $\text{Range} = 3 - 1 = 2 \text{ inches}$.

20. A line plot shows caterpillar lengths. Above 1: X, Above 2: XXX, Above 3: XXXXX. If you measured one more caterpillar and it was 1 inch, how many X marks would be above 1?

- A. 1
- B. 2 ✓
- C. 3
- D. 5

Explanation: 1 inch currently has 1 X mark. Add 1 more: $1 + 1 = 2$ X marks above 1.

Hard (20 questions)

1. A line plot shows pencil lengths. Above 4: XX, Above 5: XXXXX, Above 6: XXX, Above 7: X. How many pencils are longer than 5 inches?

- A. 1
- B. 3
- C. 4 ✓
- D. 9

Explanation: Longer than 5 means 6 inches (3) and 7 inches (1). $3 + 1 = 4$ pencils are longer than 5 inches.

2. A line plot shows leaf lengths. Above 2: XXX, Above 3: XXXXXX, Above 4: XX, Above 5: XXXX. How many leaves in all? What is the most common length?

- A. 15 leaves, 3 inches most common ✓
- B. 15 leaves, 5 inches most common
- C. 13 leaves, 3 inches most common
- D. 15 leaves, 2 inches most common

Explanation: Total: $3+6+2+4 = 15$ leaves. 3 inches has 6 X marks (the most), so it is the most common.

3. A line plot shows stick lengths. Above 3: XXXX, Above 4: XX, Above 5: XXXXX, Above 6: XXX. Sam says 5 inches is the most common because 5 is the biggest number on the answer. What is wrong with Sam's thinking?

- A. Sam confused the length number with the count of X marks ✓
- B. Sam is correct
- C. Sam should have picked 6 inches
- D. Sam counted wrong

Explanation: 5 inches happens to have the most X marks (5), but Sam's reasoning is wrong. You find the most common by counting X marks, not by picking the biggest number on the line.

4. A line plot shows ribbon lengths. Above 5: XXX, Above 6: XXXXXX, Above 7: XX, Above 8: XXXX. How many ribbons are 6 inches or longer?

- A. 6
- B. 8
- C. 12 ✓
- D. 15

Explanation: 6 inches or longer: $6 (6) + 7 (2) + 8 (4) = 12$ ribbons.

5. A line plot shows hand spans. Above 4: XX, Above 5: XXXXX, Above 6: XXX. The class measured 15 hands total, but the line plot only shows 10. How many hands are missing from the plot?

- A. 3
- B. 4
- C. 5 ✓**
- D. 6

Explanation: Line plot total: $2+5+3=10$. Class measured 15. $15-10=5$ hands are missing.

6. A line plot shows string lengths. Above 3: X, Above 4: XXX, Above 5: XXXXXX, Above 6: XX. How many strings are 4 inches or shorter?

- A. 1
- B. 3
- C. 4 ✓**
- D. 10

Explanation: 4 inches or shorter: 3 inches (1) + 4 inches (3) = 4 strings.

7. A line plot shows flower heights. Above 6: XXX, Above 7: XXXXX, Above 8: XXXX, Above 9: XX. What is the range? How many flowers in all?

- A. Range 3, total 14 ✓**
- B. Range 4, total 14
- C. Range 3, total 12
- D. Range 2, total 14

Explanation: Range: $9-6=3$ inches. Total: $3+5+4+2=14$ flowers.

8. A line plot shows rock sizes. Above 1: XXXX, Above 2: XXX, Above 3: XXXXXX, Above 4: XX. If you remove all the 3-inch rocks, how many rocks are left?

- A. 6
- B. 9 ✓**
- C. 15
- D. 3

Explanation: Total: $4+3+6+2=15$. Remove the 6 rocks at 3 inches: $15-6=9$ rocks left.

9. A line plot shows bean lengths. Above 2: XX, Above 3: XXXXX, Above 4: XXX, Above 5: XXXX. Jada says more beans are shorter than 4 inches than longer than 4 inches. Is she right?

- A. Yes, 7 shorter vs. 4 longer ✓**
- B. No, 4 longer vs. 7 shorter
- C. They are the same
- D. No, 5 shorter vs. 4 longer

Explanation: Shorter than 4: 2 inches (2) + 3 inches (5) = 7. Longer than 4: 5 inches (4) = 4. Yes, $7 > 4$. Jada is right.

10. A line plot shows nail lengths. Above 1: XXX, Above 2: XXXXXXX, Above 3: XX. Could more students have measured 2-inch nails than 1-inch and 3-inch nails put together?

- A. Yes, 7 is more than 5 ✓**
- B. No, 5 is more than 7
- C. They are the same
- D. Cannot tell

Explanation: 2-inch has 7. 1-inch + 3-inch = $3+2=5$. Yes, 7 is more than 5!

11. A line plot shows twig lengths. Above 4: X, Above 5: XXXX, Above 6: XXXXXX, Above 7: XXX. Which two lengths together have exactly 7 twigs?

- A. 4 and 5
- B. 5 and 6
- C. 4 and 6
- D. 5 and 7 ✓**

Explanation: 5 inches (4) + 7 inches (3) = 7. Those two together make 7 twigs.

12. A line plot shows feather lengths. Above 3: XXXX, Above 4: XX, Above 5: XXXXX, Above 6: XXX. How many feathers are NOT 5 inches?

- A. 5
- B. 9 ✓**
- C. 14
- D. 3

Explanation: Total: $4+2+5+3=14$. Take away the 5-inch ones: $14-5=9$ feathers are not 5 inches.

13. A line plot shows shoe sizes. Above 1: XXX, Above 2: XXXXX, Above 3: XXXX, Above 4: XX. Maria says size 2 is most common because she counted wrong and got 4. What is the correct count for size 2?

- A. 3
- B. 4
- C. 5 ✓**
- D. 6

Explanation: Size 2 has XXXXX = 5 X marks. Maria miscounted. The correct count is 5, and size 2 IS the most common.

14. A line plot shows caterpillar lengths. Above 2: XX, Above 3: XXXXXX, Above 4: XXXX. How many caterpillars are at least 3 inches long?

- A. 6
- B. 10 ✓**
- C. 4
- D. 12

Explanation: At least 3 inches means 3 or more. 3 inches (6) + 4 inches (4) = 10 caterpillars.

15. A line plot shows marker lengths. Above 5: XXX, Above 6: XXXXX, Above 7: X, Above 8: XXXX. If you add 3 more markers at 7 inches, would 7 inches be the most common?

- A. No, 7 inches would have 4, but 6 inches has 5 ✓**
- B. Yes, 7 inches would have 4 which is the most
- C. No, 8 inches would still be more
- D. Yes, 7 inches would have 3 which ties with 5 inches

Explanation: 7 inches currently has 1. Add 3: $1+3=4$. But 6 inches has 5. $4 < 5$, so no, 6 inches is still the most common.

16. A line plot shows seed lengths. Above 1: XXXXX, Above 2: XXX, Above 3: XXXXXX, Above 4: XX. What is the range and total number of seeds?

- A. Range 3, total 16 ✓**
- B. Range 4, total 16
- C. Range 3, total 14
- D. Range 2, total 16

Explanation: Range: $4-1=3$. Total: $5+3+6+2=16$ seeds.

17. A line plot shows shell lengths. Above 3: XX, Above 4: XXXX, Above 5: XXXXXXXX, Above 6: XXX. Leo says the range is 7 because the tallest column has 7 X marks. What did Leo confuse?

- A. Leo confused the range with the mode (most common count) ✓**
- B. Leo is correct
- C. Leo confused range with total
- D. Leo read the wrong column

Explanation: Range is about the spread of lengths, not the count. Range = $6-3 = 3$ inches. Leo confused 'range' with the highest frequency (7 X marks).

18. A line plot shows eraser widths. Above 1: XXX, Above 2: XXXXXX, Above 3: XXXX, Above 4: X. How many erasers are wider than 2 inches?

- A. 6
- B. 4
- C. 5 ✓**
- D. 1

Explanation: Wider than 2: 3 inches (4) + 4 inches (1) = 5 erasers.

19. A line plot shows crayon lengths. Above 2: XXXX, Above 3: XXXX, Above 4: XXXX. All lengths have the same count. How many crayons in all? Can you say one length is more common?

A. 12 crayons, no length is more common ✓

B. 12 crayons, 4 inches is more common

C. 8 crayons, no length is more common

D. 12 crayons, 2 inches is more common

Explanation: $4+4+4=12$ crayons. Each length has 4, so no length is more common than the others.

20. A line plot shows worm lengths. Above 2: XXX, Above 3: XXXXX, Above 4: XXXX, Above 5: XX. A student adds one more worm. It is 3 inches. Now how many worms are 3 inches, and is 3 inches still the most common length?

A. 6 worms at 3 inches, yes still most common ✓

B. 5 worms at 3 inches, yes still most common

C. 6 worms at 3 inches, no 4 inches took over

D. 5 worms at 3 inches, tied with 4 inches

Explanation: 3 inches had 5, now has $5+1=6$. 4 inches still has 4. $6 > 4$, so 3 inches is still the most common.

Unit Test Bank

Easy (10 questions)

1. A tally chart shows: Apples IIII III, Oranges III, Bananas IIII. How many students chose apples?

A. 5

B. 7

C. 8 ✓

D. 4

Explanation: IIII is 5 and III is 3 more. $5 + 3 = 8$ students chose apples.

2. A bar graph shows: Tigers=5, Lions=9, Bears=3. Which animal has the tallest bar?

A. Tigers

B. Lions ✓

C. Bears

D. They are the same

Explanation: Lions has 9, which is the biggest number. Lions has the tallest bar.

3. A pictograph shows each star = 1 student. The pizza row has 8 stars. How many students picked pizza?

A. 7

B. 8 ✓

C. 9

D. 16

Explanation: Each star = 1 student. 8 stars means 8 students picked pizza.

4. A line plot shows: Above 4: X X X, Above 5: X X X X X, Above 6: X X. How many items are 5 inches?

A. 3

B. 4

C. 5 ✓

D. 2

Explanation: Count the X marks above 5. There are 5, so 5 items are 5 inches.

5. A bar graph shows: Red=4, Blue=7, Green=4. Which two colors have the same number of votes?

A. Red and Blue

B. Blue and Green

C. Red and Green ✓

D. None are the same

Explanation: Red = 4 and Green = 4. They have the same number of votes.

6. A tally chart shows: Soccer IIII I, Baseball III, Basketball IIII. How many votes does basketball have?

- A. 3
- B. 4 ✓**
- C. 5
- D. 6

Explanation: Count the tally marks for basketball: IIII = 4 votes.

7. A pictograph shows each star = 2 kids. The dog row has 3 stars. How many kids chose dogs?

- A. 3
- B. 5
- C. 6 ✓**
- D. 8

Explanation: Each star = 2 kids. 3 stars: 2, 4, 6. That is 6 kids.

8. A bar graph shows: Chocolate=10, Vanilla=6, Strawberry=3. Which flavor has the fewest votes?

- A. Chocolate
- B. Vanilla
- C. Strawberry ✓**
- D. They are the same

Explanation: Strawberry has 3, which is the smallest number. Strawberry has the fewest.

9. A line plot shows: Above 2: X X X X, Above 3: X, Above 4: X X X. Which length has the most marks?

- A. 2 inches ✓**
- B. 3 inches
- C. 4 inches
- D. They are all the same

Explanation: 2 inches has 4 X marks, which is the most.

10. A tally chart shows: Bikes IIII II, Scooters III, Skateboards IIII. Put them in order from fewest to most votes.

- A. Bikes, Skateboards, Scooters
- B. Scooters, Skateboards, Bikes ✓**
- C. Skateboards, Scooters, Bikes
- D. Scooters, Bikes, Skateboards

Explanation: Scooters = 3, Skateboards = 4, Bikes = 7. From fewest to most: Scooters, Skateboards, Bikes.

Medium (10 questions)

1. A bar graph shows: Apples=8, Bananas=5, Grapes=11. How many more students chose grapes than bananas?

- A. 3
- B. 5
- C. 6 ✓**
- D. 16

Explanation: Grapes = 11, Bananas = 5. $11 - 5 = 6$ more students chose grapes.

2. A pictograph shows each star = 2 kids. The soccer row has 5 stars. How many kids picked soccer?

- A. 5
- B. 7
- C. 10 ✓**
- D. 12

Explanation: Each star = 2 kids. 5 stars: $5 \times 2 = 10$ kids.

3. A tally chart shows: Pizza IIII III, Tacos IIII, Burgers IIII II. How many students voted for pizza and burgers together?

- A. 11
- B. 13
- C. 15 ✓**
- D. 19

Explanation: Pizza = 8, Burgers = 7. $8 + 7 = 15$ students together.

4. A line plot shows crayon lengths. Above 3: XXX, Above 4: XXXXXXX, Above 5: XX. Which length is most common?

- A. 3 inches
- B. 4 inches ✓**
- C. 5 inches
- D. They are the same

Explanation: 4 inches has 7 X marks, the most. 4 inches is the most common.

5. A pictograph shows each star = 5 votes. The red row has 3 stars. How many voted for red?

- A. 3
- B. 8
- C. 15 ✓**
- D. 10

Explanation: Each star = 5 votes. $3 \times 5 = 15$ votes for red.

6. A bar graph shows: Cats=10, Dogs=6. How many more students picked cats than dogs?

- A. 16
- B. 3
- C. 4 ✓**
- D. 6

Explanation: Cats = 10, Dogs = 6. $10 - 6 = 4$ more students picked cats.

7. A line plot shows stick lengths. Above 5: XXXX, Above 6: XX, Above 7: XXXXX. What is the range?

- A. 1 inch
- B. 2 inches ✓**
- C. 5 inches
- D. 7 inches

Explanation: Longest = 7, shortest = 5. Range = $7 - 5 = 2$ inches.

8. A bar graph shows: Spring=3, Summer=9, Fall=5, Winter=3. Which two seasons got the same votes?

- A. Spring and Summer
- B. Summer and Fall
- C. Spring and Winter ✓**
- D. Fall and Winter

Explanation: Spring = 3 and Winter = 3. They have the same number of votes.

9. A pictograph shows each star = 2 students. Bikes has 6 stars and Scooters has 3 stars. How many more students picked bikes?

- A. 3
- B. 6 ✓**
- C. 9
- D. 18

Explanation: Bikes = 12, Scooters = 6. $12 - 6 = 6$ more students picked bikes.

10. A line plot shows bean lengths. Above 2: XX, Above 3: XXXXXX, Above 4: XXX. How many beans in all?

- A. 6
- B. 9
- C. 11 ✓**
- D. 13

Explanation: $2 + 6 + 3 = 11$ beans in all.

Hard (10 questions)

1. A bar graph shows: Chocolate=12, Vanilla=7, Strawberry=5. How many students voted in all? If vanilla and strawberry combine, do they equal chocolate?

- A. 24 total; yes, they equal chocolate ✓**
- B. 24 total; no, they do not equal chocolate
- C. 22 total; yes, they equal chocolate
- D. 24 total; they are 1 less than chocolate

Explanation: Total: $12+7+5=24$. Vanilla + Strawberry = $7+5=12$. Chocolate=12. Yes, they are equal!

2. A pictograph shows each star = 5 students. Dogs has 4 stars and Cats has 2 stars. Lily says the difference is 2 students. What did Lily do wrong?

- A. She subtracted stars instead of students: should be $20-10=10$ ✓
- B. She added instead of subtracted
- C. She is correct
- D. She multiplied instead of subtracted

Explanation: Dogs = $4 \times 5 = 20$, Cats = $2 \times 5 = 10$. Difference = 10. Lily subtracted stars ($4-2=2$) instead of actual student counts.

3. A line plot shows pencil lengths. Above 4: XXX, Above 5: XXXXXX, Above 6: XXXX, Above 7: XX. How many pencils are longer than 5 inches?

- A. 2
- B. 4
- C. 6 ✓
- D. 12

Explanation: Longer than 5: 6 inches (4) + 7 inches (2) = 6 pencils.

4. A bar graph shows: Soccer=8, Baseball=5, Basketball=7. If 4 more kids vote for baseball, would baseball have the most votes?

- A. No, soccer still has the most
- B. Yes, baseball would have 9 ✓
- C. No, they would all be tied
- D. Yes, baseball would have 12

Explanation: Baseball: $5+4=9$. Soccer=8, Basketball=7. 9 is the most, so yes, baseball would have the most!

5. A pictograph shows each star = 2 votes. Red has 6 stars, Blue has 4 stars, Green has 3 stars. How many votes in all? If there are 30 students, how many did NOT vote?

- A. 26 voted, 4 did not ✓
- B. 22 voted, 8 did not
- C. 26 voted, 6 did not
- D. 24 voted, 6 did not

Explanation: Red=12, Blue=8, Green=6. Total=26. $30-26=4$ did not vote.

6. A line plot shows leaf lengths. Above 2: XXXX, Above 3: XXXXXX, Above 4: XXX, Above 5: XX. How many leaves are 3 inches or shorter?

- A. 6
- B. 4
- C. 10 ✓
- D. 15

Explanation: 3 inches or shorter: 2 inches (4) + 3 inches (6) = 10 leaves.

7. A tally chart shows: Art IIII IIII, Music IIII I, PE IIII III. The class has 28 students. How many did NOT vote?

- A. 3
- B. 4
- C. 5 ✓
- D. 6

Explanation: Art=9, Music=6, PE=8. Total=23. $28-23=5$ did not vote.

8. A bar graph shows: Swings=9, Slide=5, Sandbox=6. How many students voted for the two least popular combined? Is that more than swings?

- A. 11, yes more than swings ✓
- B. 11, no less than swings
- C. 14, yes more than swings
- D. 9, same as swings

Explanation: Slide (5) + Sandbox (6) = 11. Swings = 9. $11 > 9$, yes, more than swings!

9. A line plot shows worm lengths. Above 3: XX, Above 4: XXXXX, Above 5: XXXXX, Above 6: XXX. What is the range? How many worms in all?

A. Range 3, total 15 ✓

B. Range 4, total 15

C. Range 3, total 12

D. Range 2, total 15

Explanation: Range: $6-3=3$. Total: $2+5+5+3=15$ worms.

10. Could you make a pictograph with each star = 3 for this data: Roses=9, Tulips=12, Daisies=7? Why or why not?

A. No, because 7 does not divide evenly by 3 ✓

B. Yes, all numbers work with 3

C. No, because 12 is too big

D. Yes, because 9 and 12 work

Explanation: Roses: $9/3=3$ stars. Tulips: $12/3=4$ stars. Daisies: $7/3=2$ with 1 left over. 7 does not divide evenly by 3, so you cannot make it.

Unit 7: Measurement & Time

Lesson 1: How Long Is It?

Easy (20 questions)

1. A pencil is about how many inches long?

A. 7 inches ✓
B. 70 inches
C. 1 inch
D. 100 inches

Explanation: A pencil is about 7 inches long. 70 or 100 inches would be way too big!

2. A crayon is about how many inches long?

A. 30 inches
B. 4 inches ✓
C. 40 inches
D. 1 foot

Explanation: A crayon is about 4 inches long. It fits in your hand, so it is small!

3. A paper clip is about how many inches long?

A. 1 inch ✓
B. 10 inches
C. 5 inches
D. 12 inches

Explanation: A paper clip is tiny! It is about 1 inch long.

4. A ruler is how many inches long?

A. 6 inches
B. 10 inches
C. 12 inches ✓
D. 20 inches

Explanation: A ruler is 12 inches long. That is the same as 1 foot!

5. Your shoe is about how many inches long?

A. 2 inches
B. 8 inches ✓
C. 30 inches
D. 50 inches

Explanation: Your shoe is about 8 inches long. That is a little less than a ruler!

6. A door is about how many feet tall?

A. 2 feet
B. 7 feet ✓
C. 20 feet
D. 70 feet

Explanation: A door is about 7 feet tall. That is taller than most grown-ups!

7. A spoon is about how many inches long?

A. 6 inches ✓
B. 60 inches
C. 1 inch
D. 30 inches

Explanation: A spoon is about 6 inches long. It fits in your hand!

8. A baseball bat is about how many feet long?

- A. 1 foot
- B. 10 feet
- C. 3 feet ✓**
- D. 30 feet

Explanation: A baseball bat is about 3 feet long. That is the same as 36 inches!

9. A bed is about how many feet long?

- A. 1 foot
- B. 6 feet ✓**
- C. 60 feet
- D. 100 feet

Explanation: A bed is about 6 feet long. That is big enough for a grown-up to sleep on!

10. A new eraser is about how many inches long?

- A. 2 inches ✓**
- B. 12 inches
- C. 20 inches
- D. 8 inches

Explanation: A new eraser is about 2 inches long. It is very small!

11. A fork is about how many inches long?

- A. 1 inch
- B. 50 inches
- C. 7 inches ✓**
- D. 30 inches

Explanation: A fork is about 7 inches long. It fits in your hand when you eat!

12. A banana is about how many inches long?

- A. 1 inch
- B. 30 inches
- C. 8 inches ✓**
- D. 20 inches

Explanation: A banana is about 8 inches long. It is a little shorter than a ruler!

13. A marker is about how many inches long?

- A. 5 inches ✓**
- B. 50 inches
- C. 1 inch
- D. 25 inches

Explanation: A marker is about 5 inches long. It is a little bigger than a crayon!

14. A desk is about how many feet wide?

- A. 4 feet ✓**
- B. 40 feet
- C. 1 foot
- D. 20 feet

Explanation: A desk is about 4 feet wide. That is about as long as two rulers!

15. A lunchbox is about how many inches long?

- A. 1 inch
- B. 50 inches
- C. 10 inches ✓**
- D. 30 inches

Explanation: A lunchbox is about 10 inches long. It is a little shorter than a ruler!

16. A toothbrush is about how many inches long?

- A. 7 inches ✓
- B. 1 inch
- C. 20 inches
- D. 70 inches

Explanation: A toothbrush is about 7 inches long. It fits in your hand when you brush!

17. A car is about how many feet long?

- A. 2 feet
- B. 15 feet ✓
- C. 100 feet
- D. 50 feet

Explanation: A car is about 15 feet long. That is a lot bigger than you!

18. Your finger is about how many inches long?

- A. 10 inches
- B. 50 inches
- C. 3 inches ✓
- D. 30 inches

Explanation: Your finger is about 3 inches long. It is small, so the number is small too!

19. A key is about how many inches long?

- A. 10 inches
- B. 3 inches ✓
- C. 30 inches
- D. 12 inches

Explanation: A key is about 3 inches long. It is small enough to fit in your pocket!

20. A book is about how many inches tall?

- A. 1 inch
- B. 10 inches ✓
- C. 100 inches
- D. 50 inches

Explanation: A book is about 10 inches tall. That is a little less than a ruler!

Medium (20 questions)

1. Would you measure a football field in inches or feet?

- A. Inches
- B. Feet ✓
- C. Paper clips
- D. Crayons

Explanation: A football field is very long, so we use feet. Inches would give a huge number!

2. Would you measure a ladybug in inches or feet?

- A. Feet
- B. Inches ✓
- C. Yards
- D. Meters

Explanation: A ladybug is very tiny! We measure small things in inches.

3. A rope is 12 inches long. You need 20 inches total. How many more inches do you need?

- A. 6 inches
- B. 8 inches ✓
- C. 10 inches
- D. 32 inches

Explanation: $20 - 12 = 8$. You need 8 more inches to make 20 inches.

4. Which is longer: a 10-inch string or a 1-foot string?

- A. The 10-inch string
- B. The 1-foot string ✓**
- C. They are the same
- D. Cannot tell

Explanation: 1 foot = 12 inches. Since $12 > 10$, the 1-foot string is longer!

5. A ribbon is 9 inches long. A second ribbon is 5 inches long. How much longer is the first ribbon?

- A. 14 inches
- B. 3 inches
- C. 4 inches ✓**
- D. 5 inches

Explanation: $9 - 5 = 4$ inches. The first ribbon is 4 inches longer.

6. Would you measure a swimming pool in inches or feet?

- A. Inches
- B. Feet ✓**
- C. Paper clips
- D. Erasers

Explanation: A swimming pool is very big! We use feet to measure big things.

7. Ben has a string that is 7 inches long. He adds 6 more inches. How long is it now?

- A. 1 inch
- B. 12 inches
- C. 13 inches ✓**
- D. 76 inches

Explanation: $7 + 6 = 13$ inches. The string is now 13 inches long.

8. Would you measure a pencil in centimeters or meters?

- A. Meters
- B. Centimeters ✓**
- C. Feet
- D. Yards

Explanation: A pencil is small, so we use centimeters. Meters are for bigger things!

9. A stick is 15 inches. A straw is 8 inches. How much shorter is the straw?

- A. 7 inches ✓**
- B. 23 inches
- C. 8 inches
- D. 6 inches

Explanation: $15 - 8 = 7$ inches. The straw is 7 inches shorter than the stick.

10. Which unit would you use to measure the length of your classroom?

- A. Inches
- B. Centimeters
- C. Feet ✓**
- D. Paper clips

Explanation: A classroom is big! We use feet to measure big spaces.

11. A paintbrush is a little shorter than a ruler. About how many inches long is it?

- A. 2 inches
- B. 10 inches ✓**
- C. 30 inches
- D. 50 inches

Explanation: A ruler is 12 inches. A little shorter would be about 10 inches.

12. Maya's pencil is 6 inches. Her marker is 5 inches. How many inches are they together?

- A. 1 inch
- B. 11 inches ✓**
- C. 30 inches
- D. 56 inches

Explanation: $6 + 5 = 11$ inches. Together they measure 11 inches.

13. Would you measure a park bench in inches or feet?

- A. Inches
- B. Feet ✓**
- C. Centimeters
- D. Paper clips

Explanation: A park bench is big! We use feet to measure things that are several feet long.

14. A tape is 14 inches long. You cut off 6 inches. How much tape is left?

- A. 8 inches ✓**
- B. 20 inches
- C. 6 inches
- D. 9 inches

Explanation: $14 - 6 = 8$ inches. You have 8 inches of tape left.

15. Which is shorter: a 2-foot table or a 30-inch table?

- A. The 2-foot table ✓**
- B. The 30-inch table
- C. They are the same
- D. Cannot tell

Explanation: 2 feet = 24 inches. Since $24 < 30$, the 2-foot table is shorter!

16. Liam needs 18 inches of ribbon. He has 11 inches. How many more inches does he need?

- A. 29 inches
- B. 7 inches ✓**
- C. 8 inches
- D. 6 inches

Explanation: $18 - 11 = 7$ inches. Liam needs 7 more inches of ribbon.

17. Would you measure a caterpillar in centimeters or meters?

- A. Meters
- B. Centimeters ✓**
- C. Feet
- D. Yards

Explanation: A caterpillar is very small! We use centimeters for small things.

18. A crayon is 4 inches and a pen is 6 inches. Which is longer and by how much?

- A. Crayon, by 2 inches
- B. Pen, by 2 inches ✓**
- C. Pen, by 10 inches
- D. They are the same

Explanation: $6 - 4 = 2$. The pen is 2 inches longer than the crayon.

19. A piece of yarn is some inches long. You need 5 more inches to reach 16 inches. How long is the yarn?

- A. 21 inches
- B. 11 inches ✓**
- C. 5 inches
- D. 10 inches

Explanation: $16 - 5 = 11$ inches. The yarn is 11 inches long.

20. Would you measure the length of a school bus in inches or feet?

- A. Inches
- B. Feet ✓**
- C. Centimeters
- D. Millimeters

Explanation: A school bus is very long! We use feet for big things like buses.

Hard (20 questions)

1. Sam's ribbon is 15 inches. He cuts off 7 inches, then cuts 3 more inches. How long is it now?

- A. 5 inches ✓**
- B. 8 inches
- C. 10 inches
- D. 25 inches

Explanation: $15 - 7 = 8$, then $8 - 3 = 5$ inches. Two cuts means two subtractions!

2. Jenna says a door is 7 inches tall. Is this reasonable?

- A. Yes, that sounds right
- B. No, a door is about 7 feet tall ✓**
- C. No, a door is 2 inches tall
- D. Yes, doors are small

Explanation: 7 inches is only about the length of a pencil. A door is about 7 feet tall, not 7 inches!

3. A string is 20 inches. Mia uses 8 inches for a bracelet and 6 inches for a ring. How much is left?

- A. 6 inches ✓**
- B. 14 inches
- C. 34 inches
- D. 8 inches

Explanation: $20 - 8 = 12$, then $12 - 6 = 6$ inches left. She used 14 inches total.

4. Tom says his eraser is 2 feet long. What did Tom do wrong?

- A. He used the wrong unit — an eraser is about 2 inches ✓**
- B. He measured correctly
- C. He should have said 2 meters
- D. Erasers cannot be measured

Explanation: 2 feet = 24 inches! An eraser is only about 2 inches long. Tom confused feet with inches.

5. A board is 3 feet long. How many inches is that?

- A. 3 inches
- B. 24 inches
- C. 36 inches ✓**
- D. 30 inches

Explanation: 1 foot = 12 inches. So 3 feet = $3 \times 12 = 36$ inches.

6. Kai has a rope that is 2 feet long. He needs 30 inches total. How many more inches does he need?

- A. 6 inches ✓**
- B. 28 inches
- C. 10 inches
- D. 12 inches

Explanation: 2 feet = 24 inches. $30 - 24 = 6$ more inches needed.

7. Lily says a car is 15 inches long. Does that make sense?

- A. Yes, cars are small
- B. No, a car is about 15 feet long ✓**
- C. Yes, that is correct
- D. No, a car is 1 inch long

Explanation: 15 inches is about the length of a big book. A car is about 15 feet long, not 15 inches!

8. Jake measures a table as 4 feet. Emma measures the same table as 50 inches. Who is closer to correct?

- A. Jake
- B. Emma ✓**
- C. They are both wrong
- D. They are both right

Explanation: 4 feet = 48 inches. A table is about 48-50 inches, so Emma's 50 inches is very close!

9. A wire is 18 inches. You cut off 5 inches, then add a 9-inch piece. How long is the wire now?

- A. 22 inches ✓**
- B. 4 inches
- C. 14 inches
- D. 32 inches

Explanation: $18 - 5 = 13$, then $13 + 9 = 22$ inches. Subtract first, then add!

10. Noah says a football field is 100 inches long. What mistake did he make?

- A. He said inches instead of yards ✓**
- B. He counted wrong
- C. He is correct
- D. He should have said 100 centimeters

Explanation: 100 inches is only about 8 feet! A football field is about 100 yards (300 feet). Noah used the wrong unit.

11. Amy has 3 pieces of tape: 6 inches, 4 inches, and 7 inches. How many inches of tape does she have in all?

- A. 17 inches ✓**
- B. 11 inches
- C. 10 inches
- D. 15 inches

Explanation: $6 + 4 + 7 = 17$ inches of tape in all.

12. A pipe is 2 feet 4 inches long. How many inches is that in all?

- A. 24 inches
- B. 28 inches ✓**
- C. 6 inches
- D. 34 inches

Explanation: 2 feet = 24 inches. $24 + 4 = 28$ inches total.

13. Which is longer: 2 feet or 20 inches? By how much?

- A. 2 feet, by 4 inches ✓**
- B. 20 inches, by 4 inches
- C. They are the same
- D. 2 feet, by 2 inches

Explanation: 2 feet = 24 inches. $24 - 20 = 4$. So 2 feet is 4 inches longer.

14. Carlos says his dad is 6 inches tall. Is that reasonable? Why not?

- A. Yes, grown-ups are 6 inches
- B. No, 6 inches is the size of a spoon — his dad is about 6 feet tall ✓**
- C. Yes, that is normal
- D. No, his dad is 60 inches

Explanation: 6 inches is about the length of a spoon! Carlos meant 6 feet, which is a normal height for a grown-up.

15. A scarf is 4 feet long. Maya cuts it into 2 equal pieces. How many inches is each piece?

- A. 2 feet
- B. 24 inches ✓**
- C. 12 inches
- D. 48 inches

Explanation: 4 feet = 48 inches. $48 / 2 = 24$ inches each. That is the same as 2 feet!

16. Zoe has 10 inches of string. She needs 3 pieces that are each 4 inches. Does she have enough?

- A. Yes, she has plenty
- B. No, she needs 12 inches but only has 10 ✓**
- C. Yes, 10 is more than 4
- D. No, she needs 7 inches

Explanation: 3 pieces $\times$ 4 inches = 12 inches needed. She only has 10 inches. She is 2 inches short!

17. A stick is 1 foot 8 inches long. A second stick is 22 inches long. Which is longer?

- A. The first stick
- B. The second stick ✓**
- C. They are the same
- D. Cannot tell

Explanation: 1 foot 8 inches = 12 + 8 = 20 inches. Since $22 > 20$, the second stick is longer by 2 inches.

18. Leo measures a book as 30 centimeters. His friend says that is about 12 inches. Is his friend right?

- A. Yes, 30 cm is about 12 inches ✓**
- B. No, 30 cm is about 30 inches
- C. No, 30 cm is about 3 inches
- D. Cannot convert

Explanation: 1 inch is about 2.5 centimeters. $30 / 2.5 = 12$. So yes, 30 cm is about 12 inches!

19. Ella has a 24-inch ribbon. She cuts it into 3 equal pieces and gives 1 away. How many inches does she keep?

- A. 8 inches
- B. 16 inches ✓**
- C. 12 inches
- D. 24 inches

Explanation: $24 / 3 = 8$ inches per piece. She keeps 2 pieces: $2 \times 8 = 16$ inches.

20. Ryan says a classroom is 30 inches long. What is wrong with his answer?

- A. Nothing, that is correct
- B. He used inches instead of feet — a classroom is about 30 feet long ✓**
- C. He should have said 30 centimeters
- D. Classrooms cannot be measured

Explanation: 30 inches is only 2.5 feet — about the width of a desk! A classroom is about 30 feet long. Ryan used the wrong unit.

Lesson 2: What Time Is It?

Easy (20 questions)

1. The clock shows 3:00. What time is it?

- A. 3:00 ✓**
- B. 12:00
- C. 6:00
- D. 9:00

Explanation: The short hand points to 3 and the long hand points to 12. That means it is 3:00!

2. The clock shows 7:00. What time is it?

- A. 12:00
- B. 7:00 ✓**
- C. 1:00
- D. 6:00

Explanation: The short hand points to 7 and the long hand points to 12. It is 7:00!

3. The clock shows 10:00. What time is it?

- A. 2:00
- B. 12:00
- C. 10:00 ✓**
- D. 8:00

Explanation: The short hand points to 10 and the long hand is on 12. It is 10:00!

4. The clock shows 1:00. What time is it?

- A. 12:00
- B. 11:00
- C. 2:00
- D. 1:00 ✓**

Explanation: The short hand points to 1 and the long hand points to 12. It is 1:00!

5. The clock shows 5:00. What time is it?

- A. 5:00 ✓**
- B. 12:00
- C. 6:00
- D. 4:00

Explanation: The short hand points to 5 and the long hand is on 12. It is 5:00!

6. The clock shows 12:00. What time is it?

- A. 6:00
- B. 3:00
- C. 12:00 ✓**
- D. 1:00

Explanation: Both hands point to 12. It is 12:00!

7. The clock shows 9:00. What time is it?

- A. 3:00
- B. 9:00 ✓**
- C. 6:00
- D. 12:00

Explanation: The short hand points to 9 and the long hand is on 12. It is 9:00!

8. The clock shows 6:00. What time is it?

- A. 12:00
- B. 3:00
- C. 6:00 ✓**
- D. 9:00

Explanation: The short hand points to 6 and the long hand is on 12. It is 6:00!

9. The clock shows 2:00. What time is it?

- A. 10:00
- B. 2:00 ✓**
- C. 12:00
- D. 4:00

Explanation: The short hand points to 2 and the long hand is on 12. It is 2:00!

10. The clock shows 8:00. What time is it?

- A. 4:00
- B. 12:00
- C. 8:00 ✓**
- D. 6:00

Explanation: The short hand points to 8 and the long hand is on 12. It is 8:00!

11. The clock shows 11:00. What time is it?

- A. 1:00
- B. 11:00 ✓**
- C. 12:00
- D. 10:00

Explanation: The short hand points to 11 and the long hand is on 12. It is 11:00!

12. The clock shows 4:00. What time is it?

- A. 4:00 ✓**
- B. 8:00
- C. 12:00
- D. 2:00

Explanation: The short hand points to 4 and the long hand is on 12. It is 4:00!

13. You eat breakfast at 7:00. Is that AM or PM?

- A. AM ✓**
- B. PM
- C. Both
- D. Neither

Explanation: We eat breakfast in the morning. Morning time is AM!

14. You eat dinner at 6:00. Is that AM or PM?

- A. AM
- B. PM ✓**
- C. Both
- D. Neither

Explanation: We eat dinner in the evening. Evening time is PM!

15. You go to sleep at 8:00. Is that AM or PM?

- A. AM
- B. PM ✓**
- C. Both
- D. Neither

Explanation: We go to sleep at night. Nighttime is PM!

16. School starts at 8:00. Is that AM or PM?

- A. PM
- B. AM ✓**
- C. Both
- D. Neither

Explanation: School starts in the morning. Morning time is AM!

17. The clock shows 6:30. What time is it?

- A. 12:30
- B. 6:30 ✓**
- C. 6:00
- D. 3:30

Explanation: The short hand is between 6 and 7. The long hand is on 6, which means 30 minutes. It is 6:30!

18. The clock shows 2:30. What time is it?

- A. 2:30 ✓**
- B. 2:00
- C. 6:10
- D. 12:10

Explanation: The short hand is between 2 and 3. The long hand is on 6, which means 30 minutes. It is 2:30!

19. You eat lunch at 12:00. Is that AM or PM?

- A. AM
- B. PM ✓**
- C. Both
- D. Neither

Explanation: We eat lunch in the middle of the day. 12:00 noon is PM!

20. The clock shows 9:30. What time is it?

- A. 3:45
- B. 9:00
- C. 9:30 ✓**
- D. 12:45

Explanation: The short hand is between 9 and 10. The long hand is on 6, which means 30 minutes. It is 9:30!

Medium (20 questions)

1. The short hand points to 4 and the long hand points to 6. What time is it?

- A. 4:06
- B. 6:20
- C. 4:30 ✓**
- D. 6:04

Explanation: The short hand on 4 tells the hour. The long hand on 6 means 30 minutes. It is 4:30!

2. The short hand points to 8 and the long hand points to 12. What time is it?

- A. 12:08
- B. 8:00 ✓**
- C. 12:00
- D. 8:12

Explanation: The short hand on 8 is the hour. The long hand on 12 means 0 minutes. It is 8:00!

3. The short hand is between 2 and 3. The long hand points to 6. What time is it?

- A. 2:06
- B. 3:30
- C. 6:15
- D. 2:30 ✓**

Explanation: The short hand past 2 means the hour is 2. The long hand on 6 = 30 minutes. It is 2:30!

4. It is 3:00 PM. What will the time be in 1 hour?

- A. 2:00 PM
- B. 4:00 PM ✓**
- C. 3:30 PM
- D. 3:01 PM

Explanation: 3:00 plus 1 hour = 4:00 PM.

5. It is 7:00 AM. What time was it 1 hour ago?

- A. 8:00 AM
- B. 6:00 AM ✓**
- C. 7:30 AM
- D. 6:30 AM

Explanation: 7:00 minus 1 hour = 6:00 AM.

6. The short hand points to 11 and the long hand points to 6. What time is it?

- A. 6:55
- B. 11:06
- C. 11:30 ✓**
- D. 6:11

Explanation: The short hand on 11 is the hour. The long hand on 6 = 30 minutes. It is 11:30!

7. Recess starts at 10:00 AM and ends at 10:30 AM. How long is recess?

- A. 1 hour
- B. 30 minutes ✓**
- C. 15 minutes
- D. 45 minutes

Explanation: From 10:00 to 10:30 is 30 minutes. Recess lasts 30 minutes!

8. The short hand is between 5 and 6. The long hand points to 12. What time is it?

- A. 12:05
- B. 5:00 ✓**
- C. 6:00
- D. 5:30

Explanation: When the long hand is on 12, it is exactly on the hour. The short hand past 5 means 5:00!

9. It is 1:30 PM. What time will it be in 30 minutes?

- A. 2:00 PM ✓**
- B. 1:00 PM
- C. 2:30 PM
- D. 12:30 PM

Explanation: 1:30 plus 30 minutes = 2:00 PM.

10. Which happens first: 9:00 AM or 9:00 PM?

- A. 9:00 PM
- B. 9:00 AM ✓**
- C. They happen at the same time
- D. It depends on the day

Explanation: AM is morning and PM is evening. Morning comes first, so 9:00 AM happens first!

11. The short hand points to 6 and the long hand points to 3. What time is it?

- A. 3:30
- B. 6:03
- C. 6:15 ✓**
- D. 3:06

Explanation: Short hand on 6 = hour is 6. Long hand on 3 = 15 minutes. It is 6:15!

12. It is 4:00 PM. Dinner is in 2 hours. What time is dinner?

- A. 2:00 PM
- B. 5:00 PM
- C. 6:00 PM ✓**
- D. 4:30 PM

Explanation: 4:00 plus 2 hours = 6:00 PM. Dinner is at 6:00 PM!

13. A movie starts at 2:00 PM and ends at 3:00 PM. How long is the movie?

- A. 30 minutes
- B. 2 hours
- C. 1 hour ✓**
- D. 3 hours

Explanation: From 2:00 to 3:00 is 1 hour. The movie is 1 hour long!

14. The short hand points to 9 and the long hand points to 6. What time is it?

- A. 6:45
- B. 9:06
- C. 9:30 ✓**
- D. 6:09

Explanation: Short hand on 9 = hour is 9. Long hand on 6 = 30 minutes. It is 9:30!

15. It is 11:00 AM. Lunch is in 1 hour. What time is lunch?

- A. 12:00 PM ✓
- B. 10:00 AM
- C. 11:30 AM
- D. 1:00 PM

Explanation: 11:00 plus 1 hour = 12:00 PM. Lunch is at noon!

16. The short hand is between 7 and 8. The long hand points to 6. What time is it?

- A. 7:06
- B. 8:30
- C. 6:35
- D. 7:30 ✓

Explanation: Short hand past 7 means the hour is 7. Long hand on 6 = 30 minutes. It is 7:30!

17. Which is later: 3:30 PM or 4:00 PM?

- A. 3:30 PM
- B. 4:00 PM ✓
- C. They are the same
- D. Cannot tell

Explanation: 4:00 PM comes after 3:30 PM, so 4:00 PM is later.

18. Class starts at 8:00 AM and ends at 9:00 AM. How long is class?

- A. 30 minutes
- B. 1 hour ✓
- C. 2 hours
- D. 9 hours

Explanation: From 8:00 to 9:00 is 1 hour.

19. The short hand points to 1 and the long hand points to 6. What time is it?

- A. 6:05
- B. 1:06
- C. 1:30 ✓
- D. 6:01

Explanation: Short hand on 1 = hour is 1. Long hand on 6 = 30 minutes. It is 1:30!

20. It is 5:30 PM. What time will it be in 30 minutes?

- A. 5:00 PM
- B. 6:30 PM
- C. 6:00 PM ✓
- D. 5:45 PM

Explanation: 5:30 plus 30 minutes = 6:00 PM.

Hard (20 questions)

1. Lunch starts at 11:30 AM and lasts 45 minutes. What time does lunch end?

- A. 12:00 PM
- B. 12:15 PM ✓
- C. 11:45 AM
- D. 12:45 PM

Explanation: 11:30 + 30 min = 12:00. Then 12:00 + 15 min = 12:15 PM. Lunch ends at 12:15 PM!

2. It is 2:45 PM. What time will it be in 30 minutes?

- A. 3:45 PM
- B. 3:15 PM ✓
- C. 2:75 PM
- D. 3:00 PM

Explanation: 2:45 + 15 min = 3:00, then 3:00 + 15 min = 3:15 PM. There is no such thing as 2:75!

3. Ben says the clock shows 12:60. What is wrong?

- A. Nothing, that is correct
- B. There is no 60 minutes — it should be 1:00 ✓**
- C. He should say 12:06
- D. The clock is broken

Explanation: A clock only goes up to 59 minutes. After 12:59 comes 1:00, not 12:60!

4. School starts at 8:00 AM. Ava arrives 15 minutes early. What time does Ava arrive?

- A. 8:15 AM
- B. 7:45 AM ✓**
- C. 7:15 AM
- D. 8:45 AM

Explanation: 15 minutes before 8:00 is 7:45 AM. Ava arrives at 7:45 AM!

5. A cartoon starts at 4:00 PM and ends at 4:30 PM. A movie starts at 5:00 PM and ends at 6:30 PM. Which is longer and by how much?

- A. Cartoon, by 30 min
- B. Movie, by 1 hour ✓**
- C. Movie, by 30 min
- D. They are the same

Explanation: Cartoon = 30 min. Movie = 1 hour 30 min. The movie is 1 hour longer!

6. It is 9:15 AM. Music class is in 45 minutes. What time does music start?

- A. 10:00 AM ✓**
- B. 9:45 AM
- C. 10:15 AM
- D. 9:60 AM

Explanation: $9:15 + 45 \text{ min} = 10:00 \text{ AM}$. Music class starts at 10:00 AM!

7. Maya says she woke up at 7:00 PM. Does that make sense for a school day?

- A. Yes, that is normal
- B. No, she probably means 7:00 AM — PM is evening ✓**
- C. Yes, PM is morning
- D. No, 7:00 does not exist

Explanation: PM is evening/night. You wake up in the morning, which is AM. Maya meant 7:00 AM!

8. A bus leaves at 3:30 PM. The ride takes 45 minutes. What time does the bus arrive?

- A. 4:00 PM
- B. 4:15 PM ✓**
- C. 3:75 PM
- D. 4:45 PM

Explanation: $3:30 + 30 \text{ min} = 4:00$, then $4:00 + 15 \text{ min} = 4:15 \text{ PM}$.

9. Sam's clock shows the short hand on 10 and the long hand on 9. What time is it?

- A. 9:50
- B. 10:45 ✓**
- C. 9:10
- D. 10:09

Explanation: Short hand on 10 = hour is 10. Long hand on 9 = $9 \times 5 = 45$ minutes. It is 10:45!

10. Li says it takes him 90 minutes to do homework. How many hours and minutes is that?

- A. 1 hour 30 minutes ✓**
- B. 9 hours
- C. 1 hour 90 minutes
- D. 2 hours

Explanation: 60 minutes = 1 hour. $90 - 60 = 30$ minutes left. So 90 minutes = 1 hour 30 minutes!

11. Kim left home at 8:15 AM and got to school at 8:45 AM. How long was her trip?

- A. 15 minutes
- B. 30 minutes ✓**
- C. 45 minutes
- D. 1 hour

Explanation: From 8:15 to 8:45 is 30 minutes.

12. Jada says her reading time is from 1:00 PM to 1:30 PM, which is 1 hour. What did Jada do wrong?

- A. She is correct
- B. She confused 30 minutes with 1 hour ✓**
- C. She read the clock backward
- D. She forgot to count

Explanation: From 1:00 to 1:30 is only 30 minutes, not 1 hour. Jada mixed up minutes and hours!

13. It is 6:50 PM. What time will it be in 20 minutes?

- A. 6:70 PM
- B. 7:10 PM ✓**
- C. 7:00 PM
- D. 7:20 PM

Explanation: $6:50 + 10 \text{ min} = 7:00$. Then $7:00 + 10 \text{ min} = 7:10 \text{ PM}$. There is no such thing as 6:70!

14. Nate watched TV from 5:00 PM to 5:30 PM, then read from 5:30 PM to 6:15 PM. How long in all?

- A. 45 minutes
- B. 1 hour 15 minutes ✓**
- C. 30 minutes
- D. 1 hour 45 minutes

Explanation: TV = 30 min. Reading = 45 min. $30 + 45 = 75 \text{ min} = 1 \text{ hour } 15 \text{ minutes total!}$

15. The clock shows the short hand on 3 and the long hand on 9. What time is it?

- A. 9:15
- B. 3:09
- C. 3:45 ✓**
- D. 9:03

Explanation: Short hand on 3 = hour is 3. Long hand on 9 = $9 \times 5 = 45 \text{ minutes}$. It is 3:45!

16. Luke says there are 100 minutes in an hour. Is he correct?

- A. Yes, 100 minutes
- B. No, there are 60 minutes in an hour ✓**
- C. Yes, like 100 cents in a dollar
- D. No, there are 30 minutes

Explanation: There are 60 minutes in 1 hour, not 100. After 59 minutes, a new hour starts!

17. Soccer practice starts at 4:15 PM and lasts 1 hour and 15 minutes. What time does it end?

- A. 5:15 PM
- B. 5:30 PM ✓**
- C. 5:00 PM
- D. 4:30 PM

Explanation: $4:15 + 1 \text{ hour} = 5:15$. Then $5:15 + 15 \text{ min} = 5:30 \text{ PM}$.

18. Emma's clock shows 11:55 AM. What time will it be in 10 minutes?

- A. 11:65 AM
- B. 12:05 PM ✓**
- C. 12:05 AM
- D. 11:05 AM

Explanation: $11:55 + 5 \text{ min} = 12:00$, then $+ 5 \text{ more} = 12:05$. And it changes from AM to PM at noon!

19. Owen started reading at 3:20 PM and stopped at 4:00 PM. How long did he read?

- A. 20 minutes
- B. 40 minutes ✓**
- C. 1 hour
- D. 1 hour 20 minutes

Explanation: From 3:20 to 4:00. Count: 3:20 to 3:60 (which is 4:00) = 40 minutes.

20. Zara says recess ended at 10:90 AM. What should the correct time be?

- A. 10:90 AM is correct
- B. 11:30 AM ✓**
- C. 10:30 AM
- D. 9:90 AM

Explanation: There is no :90 on a clock! 90 minutes = 1 hour 30 min. So 10:90 should be 11:30 AM.

Lesson 3: Hot, Cold and Full

Easy (20 questions)

1. A shape is covered by 4 square units. What is its area?

- A. 2 square units
- B. 3 square units
- C. 4 square units ✓**
- D. 5 square units

Explanation: Count each square inside the shape. There are 4, so the area is 4 square units!

2. A shape is covered by 7 square units. What is its area?

- A. 7 square units ✓**
- B. 6 square units
- C. 8 square units
- D. 5 square units

Explanation: Count each square inside the shape. There are 7, so the area is 7 square units!

3. A shape is covered by 10 square units. What is its area?

- A. 8 square units
- B. 9 square units
- C. 11 square units
- D. 10 square units ✓**

Explanation: Count each square. There are 10 squares, so the area is 10 square units!

4. A shape is covered by 3 square units. What is its area?

- A. 3 square units ✓**
- B. 4 square units
- C. 2 square units
- D. 6 square units

Explanation: There are 3 squares inside the shape. The area is 3 square units!

5. A shape is covered by 6 square units. What is its area?

- A. 5 square units
- B. 7 square units
- C. 6 square units ✓**
- D. 8 square units

Explanation: Count each square inside the shape. There are 6, so the area is 6 square units!

6. A shape is covered by 1 square unit. What is its area?

- A. 0 square units
- B. 1 square unit ✓**
- C. 2 square units
- D. 3 square units

Explanation: There is 1 square inside the shape. The area is 1 square unit!

7. A shape is covered by 9 square units. What is its area?

- A. 8 square units
- B. 10 square units
- C. 9 square units ✓**
- D. 7 square units

Explanation: Count each square. There are 9 squares, so the area is 9 square units!

8. A shape is covered by 5 square units. What is its area?

- A. 4 square units
- B. 5 square units ✓**
- C. 6 square units
- D. 3 square units

Explanation: Count each square inside the shape. There are 5, so the area is 5 square units!

9. A shape is covered by 12 square units. What is its area?

- A. 10 square units
- B. 11 square units
- C. 12 square units ✓**
- D. 13 square units

Explanation: Count each square. There are 12 squares, so the area is 12 square units!

10. A shape is covered by 2 square units. What is its area?

- A. 1 square unit
- B. 2 square units ✓**
- C. 3 square units
- D. 4 square units

Explanation: There are 2 squares inside the shape. The area is 2 square units!

11. A shape is covered by 8 square units. What is its area?

- A. 8 square units ✓**
- B. 6 square units
- C. 10 square units
- D. 7 square units

Explanation: Count each square. There are 8 squares, so the area is 8 square units!

12. A shape is covered by 15 square units. What is its area?

- A. 14 square units
- B. 16 square units
- C. 15 square units ✓**
- D. 13 square units

Explanation: Count each square. There are 15 squares, so the area is 15 square units!

13. A shape is covered by 11 square units. What is its area?

- A. 10 square units
- B. 12 square units
- C. 11 square units ✓**
- D. 9 square units

Explanation: Count each square. There are 11 squares, so the area is 11 square units!

14. What do we use to measure area?

- A. Inches
- B. Square units ✓**
- C. Pounds
- D. Cups

Explanation: We measure area by counting the square units that cover a shape!

15. A shape is covered by 14 square units. What is its area?

- A. 12 square units
- B. 13 square units
- C. 14 square units ✓**
- D. 15 square units

Explanation: Count each square. There are 14 squares, so the area is 14 square units!

16. A rectangle is filled with square tiles. You count 6 tiles. What is the area?

- A. 5 square units
- B. 6 square units ✓**
- C. 7 square units
- D. 12 square units

Explanation: Each tile is 1 square unit. 6 tiles = 6 square units!

17. A shape is covered by 16 square units. What is its area?

- A. 15 square units
- B. 16 square units ✓**
- C. 17 square units
- D. 14 square units

Explanation: Count each square. There are 16 squares, so the area is 16 square units!

18. A shape is covered by 20 square units. What is its area?

- A. 18 square units
- B. 19 square units
- C. 20 square units ✓**
- D. 21 square units

Explanation: Count each square. There are 20 squares, so the area is 20 square units!

19. A square tile covers 1 square unit. If a shape has 13 tiles, what is its area?

- A. 12 square units
- B. 13 square units ✓**
- C. 14 square units
- D. 26 square units

Explanation: Each tile = 1 square unit. 13 tiles = 13 square units!

20. A shape is covered by 18 square units. What is its area?

- A. 17 square units
- B. 18 square units ✓**
- C. 19 square units
- D. 16 square units

Explanation: Count each square. There are 18 squares, so the area is 18 square units!

Medium (20 questions)

1. Shape A is covered by 6 square units. Shape B is covered by 9 square units. Which shape has a bigger area?

- A. Shape A
- B. Shape B ✓**
- C. They are the same
- D. Cannot tell

Explanation: $9 > 6$, so Shape B has a bigger area!

2. Shape A has 10 square units. Shape B has 10 square units. What can you say about their areas?

- A. Shape A is bigger
- B. Shape B is bigger
- C. They have the same area ✓**
- D. Cannot tell

Explanation: Both shapes have 10 square units, so they have the same area! They can look different but still have equal areas.

3. A rug covers 12 square units. A mat covers 8 square units. How many more square units does the rug cover?

- A. 20 square units
- B. 3 square units
- C. 4 square units ✓**
- D. 4 square units

Explanation: $12 - 8 = 4$. The rug covers 4 more square units than the mat.

4. A rectangle is 5 squares long and 2 squares wide. What is its area?

- A. 7 square units
- B. 10 square units ✓**
- C. 12 square units
- D. 3 square units

Explanation: Count the squares: 2 rows of 5 = 10 square units!

5. Which has a bigger area: a shape with 14 square units or a shape with 11 square units?

- A. The shape with 11
- B. They are the same
- C. The shape with 14 ✓**
- D. Cannot tell

Explanation: $14 > 11$, so the shape with 14 square units has a bigger area!

6. A rectangle is 3 squares long and 3 squares wide. What is its area?

- A. 6 square units
- B. 9 square units ✓**
- C. 12 square units
- D. 3 square units

Explanation: Count the squares: 3 rows of 3 = 9 square units!

7. A garden has an area of 15 square units. A patio has an area of 9 square units. How much bigger is the garden?

- A. 24 square units
- B. 6 square units ✓**
- C. 5 square units
- D. 7 square units

Explanation: $15 - 9 = 6$. The garden is 6 square units bigger!

8. A rectangle is 4 squares long and 2 squares wide. What is its area?

- A. 6 square units
- B. 8 square units ✓**
- C. 10 square units
- D. 12 square units

Explanation: Count the squares: 2 rows of 4 = 8 square units!

9. Shape A has an area of 7 square units. Shape B has an area of ____ square units. Together they cover 13. How big is Shape B?

- A. 20 square units
- B. 6 square units ✓**
- C. 7 square units
- D. 5 square units

Explanation: $13 - 7 = 6$. Shape B has an area of 6 square units!

10. A rectangle is 6 squares long and 2 squares wide. What is its area?

- A. 8 square units
- B. 10 square units
- C. 12 square units ✓**
- D. 14 square units

Explanation: Count the squares: 2 rows of 6 = 12 square units!

11. Which has a smaller area: a shape with 5 square units or a shape with 8 square units?

- A. The shape with 8
- B. They are equal
- C. The shape with 5 ✓**
- D. Cannot tell

Explanation: $5 < 8$, so the shape with 5 square units has a smaller area!

12. A poster covers 16 square units. A card covers 4 square units. How many more square units does the poster cover?

- A. 20 square units
- B. 12 square units ✓**
- C. 10 square units
- D. 4 square units

Explanation: $16 - 4 = 12$. The poster covers 12 more square units!

13. A rectangle is 5 squares long and 3 squares wide. What is its area?

- A. 8 square units
- B. 15 square units ✓**
- C. 10 square units
- D. 16 square units

Explanation: Count the squares: 3 rows of 5 = 15 square units!

14. A shape has an area of ____ square units. It is 3 more than 9. What is the area?

- A. 6 square units
- B. 12 square units ✓**
- C. 9 square units
- D. 3 square units

Explanation: $9 + 3 = 12$. The area is 12 square units!

15. Two shapes each have an area of 8 square units. What is the total area?

- A. 8 square units
- B. 16 square units ✓**
- C. 80 square units
- D. 10 square units

Explanation: $8 + 8 = 16$ square units total!

16. A rectangle is 4 squares long and 3 squares wide. What is its area?

- A. 7 square units
- B. 10 square units
- C. 12 square units ✓**
- D. 14 square units

Explanation: Count the squares: 3 rows of 4 = 12 square units!

17. Shape A has 11 square units. Shape B has 15 square units. Which is smaller and by how much?

- A. Shape A, by 4 ✓**
- B. Shape B, by 4
- C. Shape A, by 26
- D. They are equal

Explanation: $15 - 11 = 4$. Shape A is smaller by 4 square units!

18. A rectangle is 7 squares long and 2 squares wide. What is its area?

- A. 9 square units
- B. 14 square units ✓**
- C. 12 square units
- D. 16 square units

Explanation: Count the squares: 2 rows of 7 = 14 square units!

19. A table is covered by 20 square tiles. A desk is covered by 12 square tiles. How many more tiles cover the table?

- A. 32 tiles
- B. 8 tiles ✓**
- C. 10 tiles
- D. 6 tiles

Explanation: $20 - 12 = 8$. The table needs 8 more tiles than the desk!

20. A rectangle is 3 squares long and 2 squares wide. What is its area?

- A. 5 square units
- B. 6 square units ✓**
- C. 8 square units
- D. 10 square units

Explanation: Count the squares: 2 rows of 3 = 6 square units!

Hard (20 questions)

1. A rectangle is 4 squares long and 3 squares wide. Tom says the area is 7 square units. What did Tom do wrong?

- A. He added $4 + 3$ instead of counting all the squares ✓**
- B. He counted correctly
- C. He subtracted instead of adding
- D. He multiplied wrong

Explanation: Tom added $4 + 3 = 7$. But the area is all the squares: 3 rows of 4 = 12 square units. He should count all squares, not add the sides!

2. A rug covers 8 square units. A mat covers 5 square units. You put them side by side with no overlap. What is the total area?

- A. 3 square units
- B. 13 square units ✓**
- C. 40 square units
- D. 85 square units

Explanation: $8 + 5 = 13$ square units total. When shapes do not overlap, you add their areas!

3. Sara says a rectangle that is 5 squares long and 2 squares wide has an area of 52 square units. What did she do wrong?

- A. She wrote the numbers next to each other instead of counting — the area is 10 ✓**
- B. She is correct
- C. She subtracted wrong
- D. She divided instead of adding

Explanation: Sara wrote 5 and 2 together as '52.' The real area is 2 rows of 5 = 10 square units!

4. A garden is 6 squares long and 3 squares wide. A shed covers 4 of those squares. How many uncovered squares are left?

- A. 18 square units
- B. 14 square units ✓**
- C. 9 square units
- D. 22 square units

Explanation: Garden area: 3 rows of 6 = 18 squares. Then $18 - 4 = 14$ uncovered squares.

5. Two rectangles each have an area of 8 square units. Can they have different shapes?

A. No, they must look the same

B. Yes — one could be 4x2 and another 8x1 ✓

C. No, area means same shape

D. Yes, but only if they are circles

Explanation: A 4-by-2 rectangle and an 8-by-1 rectangle both have 8 square units but look very different!

6. A rectangle is 5 squares long and 4 squares wide. Jake says the area is 9 square units. What did Jake do wrong?

A. He added $5 + 4$ instead of counting all the squares ✓

B. He is correct

C. He subtracted $5 - 4$

D. He divided instead

Explanation: Jake added $5 + 4 = 9$. But the real area is 4 rows of 5 = 20 square units. He should count all the squares!

7. A poster is 6 squares long and 2 squares wide. You cut off a piece that is 4 square units. How many square units are left?

A. 16 square units

B. 8 square units ✓

C. 12 square units

D. 4 square units

Explanation: Poster area: 2 rows of 6 = 12 square units. $12 - 4 = 8$ square units left!

8. Shape A is 3 by 4 and Shape B is 2 by 7. Which has a bigger area?

A. Shape A (12 square units)

B. Shape B (14 square units) ✓

C. They are the same

D. Shape B (9 square units)

Explanation: Shape A: 3 rows of 4 = 12. Shape B: 2 rows of 7 = 14. Shape B is bigger by 2 square units!

9. Maria covers a rectangle with 15 square tiles. She says the rectangle could be 5 squares long and 3 squares wide. Is she right?

A. Yes, 3 rows of 5 = 15 ✓

B. No, $5 + 3 = 8$, not 15

C. No, it should be 15 by 1

D. Yes, $5 + 3 = 15$

Explanation: 3 rows of 5 squares = 15 squares total. Maria is correct!

10. A floor is 5 squares long and 5 squares wide. A table covers 6 of those squares. How many floor squares can you still see?

A. 25 square units

B. 19 square units ✓

C. 11 square units

D. 31 square units

Explanation: Floor area: 5 rows of 5 = 25. Then $25 - 6 = 19$ squares you can still see!

11. Ben says a 3-by-3 square and a 1-by-9 rectangle have the same area. Is he correct?

A. Yes, both have 9 square units ✓

B. No, the square is bigger

C. No, the rectangle is bigger

D. You cannot compare them

Explanation: $3 \times 3 = 9$ and $1 \times 9 = 9$. Both have 9 square units! Same area, different shapes.

12. A rectangle is 6 squares long and 3 squares wide. Lily says the area is 9. What did she do wrong?

A. She added $6 + 3$ instead of counting all squares

B. She is correct

C. She multiplied 3×3 ✓

D. She only counted one row

Explanation: Lily got 9 by only counting $3 \times 3 =$ one side. The full area is 3 rows of 6 = 18 square units!

13. A wall needs 24 square tiles. You have placed 17 tiles so far. How many more do you need?

- A. 41 tiles
- B. 7 tiles ✓**
- C. 8 tiles
- D. 6 tiles

Explanation: $24 - 17 = 7$ more tiles needed!

14. Shape A is 4 by 4. Shape B is 3 by 5. How many more square units does Shape A have?

- A. 1 square unit ✓**
- B. 16 square units
- C. 15 square units
- D. 2 square units

Explanation: Shape A: $4 \times 4 = 16$. Shape B: $3 \times 5 = 15$. Shape A has 1 more square unit!

15. A rectangle has an area of 12 square units. It is 4 squares long. How many squares wide is it?

- A. 3 squares ✓**
- B. 8 squares
- C. 6 squares
- D. 4 squares

Explanation: If 4 rows make 12 total, each row has $12 / 4 = 3$ squares. It is 3 squares wide!

16. A playground has two sections. Section A is 4 by 3 and Section B is 2 by 5. What is the total area?

- A. 22 square units ✓**
- B. 14 square units
- C. 24 square units
- D. 12 square units

Explanation: Section A: $4 \times 3 = 12$. Section B: $2 \times 5 = 10$. Total: $12 + 10 = 22$ square units!

17. Noah says a rectangle that is 2 by 6 has a bigger area than a 3 by 4 rectangle. Is he right?

- A. Yes, 2 by 6 is bigger
- B. No, they have the same area (12 square units) ✓**
- C. No, 3 by 4 is bigger
- D. Cannot compare them

Explanation: $2 \times 6 = 12$ and $3 \times 4 = 12$. They have the same area! Different shapes, equal areas.

18. A rectangle has an area of 20 square units. It is 5 squares long. How many squares wide is it?

- A. 15 squares
- B. 5 squares
- C. 4 squares ✓**
- D. 25 squares

Explanation: If 5 squares long and area is 20, then width = $20 / 5 = 4$ squares wide!

19. Ava tiled a 3-by-6 rectangle but miscounted and said the area is 36. What did she likely do?

- A. She wrote 3 and 6 next to each other as 36 ✓**
- B. She added correctly
- C. She counted every square twice
- D. She is correct

Explanation: Ava put 3 and 6 together to get 36. The real area is 3 rows of 6 = 18 square units!

20. A patio is 4 by 5. You add a 2-by-5 section next to it. What is the new total area?

- A. 30 square units ✓**
- B. 20 square units
- C. 10 square units
- D. 40 square units

Explanation: Patio: $4 \times 5 = 20$. New section: $2 \times 5 = 10$. Total: $20 + 10 = 30$ square units!

Unit Test Bank

Easy (20 questions)

1. A glue stick is about how many inches long?

- A. 3 inches ✓
- B. 30 inches
- C. 12 inches
- D. 50 inches

Explanation: A glue stick is about 3 inches long. It is small enough to fit in your hand!

2. The clock shows 4:00. What time is it?

- A. 12:00
- B. 4:00 ✓
- C. 8:00
- D. 6:00

Explanation: The short hand points to 4 and the long hand is on 12. It is 4:00!

3. A shape is covered by 9 square units. What is its area?

- A. 8 square units
- B. 9 square units ✓
- C. 10 square units
- D. 7 square units

Explanation: Count each square. There are 9 squares, so the area is 9 square units!

4. A water bottle is about how many inches tall?

- A. 1 inch
- B. 9 inches ✓
- C. 50 inches
- D. 30 inches

Explanation: A water bottle is about 9 inches tall. That is a little shorter than a ruler!

5. The clock shows 11:00. What time is it?

- A. 1:00
- B. 12:00
- C. 11:00 ✓
- D. 10:00

Explanation: The short hand points to 11 and the long hand is on 12. It is 11:00!

6. A shape is covered by 5 square units. What is its area?

- A. 4 square units
- B. 5 square units ✓
- C. 6 square units
- D. 3 square units

Explanation: Count each square inside the shape. There are 5, so the area is 5 square units!

7. A couch is about how many feet long?

- A. 1 foot
- B. 7 feet ✓
- C. 70 feet
- D. 100 feet

Explanation: A couch is about 7 feet long. That is about as long as a bed!

8. The clock shows 5:30. What time is it?

- A. 5:00
- B. 6:00
- C. 5:30 ✓
- D. 12:30

Explanation: The short hand is between 5 and 6. The long hand is on 6, meaning 30 minutes. It is 5:30!

9. A shape is covered by 12 square units. What is its area?

- A. 10 square units
- B. 11 square units
- C. 12 square units ✓**
- D. 13 square units

Explanation: Count each square. There are 12, so the area is 12 square units!

10. A notebook is about how many inches long?

- A. 11 inches ✓**
- B. 1 inch
- C. 50 inches
- D. 30 inches

Explanation: A notebook is about 11 inches long. That is almost as long as a ruler!

11. You go to school at 8:00 in the morning. Is that AM or PM?

- A. PM
- B. AM ✓**
- C. Both
- D. Neither

Explanation: Morning time is AM! You go to school in the AM.

12. A shape is covered by 3 square units. What is its area?

- A. 2 square units
- B. 3 square units ✓**
- C. 4 square units
- D. 1 square unit

Explanation: Count the squares. There are 3, so the area is 3 square units!

13. A skateboard is about how many feet long?

- A. 3 feet ✓**
- B. 30 feet
- C. 1 foot
- D. 10 feet

Explanation: A skateboard is about 3 feet long. That is about the same as a baseball bat!

14. The clock shows 7:00. What time is it?

- A. 1:00
- B. 7:00 ✓**
- C. 12:00
- D. 5:00

Explanation: The short hand points to 7 and the long hand is on 12. It is 7:00!

15. A shape is covered by 16 square units. What is its area?

- A. 14 square units
- B. 15 square units
- C. 16 square units ✓**
- D. 18 square units

Explanation: Count each square. 16 squares = 16 square units!

16. A remote control is about how many inches long?

- A. 6 inches ✓**
- B. 60 inches
- C. 1 inch
- D. 30 inches

Explanation: A remote control is about 6 inches long. It fits in your hand!

17. The clock shows 1:30. What time is it?

- A. 1:00
- B. 6:05
- C. 1:30 ✓**
- D. 12:30

Explanation: The short hand is between 1 and 2. The long hand is on 6 = 30 minutes. It is 1:30!

18. What do we use to measure the area of a flat shape?

- A. Inches
- B. Square units ✓**
- C. Pounds
- D. Minutes

Explanation: We measure area by counting square units that fill the shape!

19. A stapler is about how many inches long?

- A. 6 inches ✓**
- B. 60 inches
- C. 1 inch
- D. 20 inches

Explanation: A stapler is about 6 inches long. It fits on your desk!

20. The clock shows 10:30. What time is it?

- A. 10:00
- B. 10:30 ✓**
- C. 6:50
- D. 12:10

Explanation: The short hand is between 10 and 11. Long hand on 6 = 30 minutes. It is 10:30!

Medium (20 questions)

1. Would you measure a classroom door in inches or feet?

- A. Inches
- B. Feet ✓**
- C. Centimeters
- D. Paper clips

Explanation: A door is very tall! We use feet for big things.

2. The short hand points to 3 and the long hand points to 6. What time is it?

- A. 6:15
- B. 3:06
- C. 3:30 ✓**
- D. 6:03

Explanation: Short hand on 3 = hour is 3. Long hand on 6 = 30 minutes. It is 3:30!

3. Shape A has 8 square units. Shape B has 11 square units. Which has a bigger area?

- A. Shape A
- B. Shape B ✓**
- C. They are the same
- D. Cannot tell

Explanation: $11 > 8$, so Shape B has a bigger area!

4. A string is 16 inches. You need 24 inches total. How many more inches do you need?

- A. 40 inches
- B. 8 inches ✓**
- C. 10 inches
- D. 6 inches

Explanation: $24 - 16 = 8$. You need 8 more inches!

5. It is 2:00 PM. Soccer is in 2 hours. What time is soccer?

A. 4:00 PM ✓

B. 12:00 PM

C. 2:30 PM

D. 3:00 PM

Explanation: $2:00 + 2 \text{ hours} = 4:00 \text{ PM}$!

6. A rectangle is 4 squares long and 2 squares wide. What is its area?

A. 6 square units

B. 8 square units ✓

C. 10 square units

D. 4 square units

Explanation: Count the squares: 2 rows of 4 = 8 square units!

7. Would you measure a caterpillar in inches or feet?

A. Feet

B. Inches ✓

C. Meters

D. Yards

Explanation: A caterpillar is tiny! We use inches for small things.

8. The short hand is between 9 and 10. The long hand points to 6. What time is it?

A. 6:45

B. 10:30

C. 9:06

D. 9:30 ✓

Explanation: Short hand past 9 = hour is 9. Long hand on 6 = 30 minutes. It is 9:30!

9. A rug has 14 square units. A mat has 9 square units. How many more square units does the rug have?

A. 23 square units

B. 5 square units ✓

C. 6 square units

D. 4 square units

Explanation: $14 - 9 = 5$ more square units!

10. A wire is 11 inches. A second wire is 8 inches. How much longer is the first wire?

A. 19 inches

B. 4 inches

C. 3 inches ✓

D. 2 inches

Explanation: $11 - 8 = 3$ inches longer.

11. Which is longer: 1 foot or 15 inches?

A. 1 foot

B. 15 inches ✓

C. They are the same

D. Cannot tell

Explanation: 1 foot = 12 inches. $15 > 12$, so 15 inches is longer!

12. A rectangle is 5 squares long and 3 squares wide. What is its area?

A. 8 square units

B. 15 square units ✓

C. 10 square units

D. 20 square units

Explanation: 3 rows of 5 = 15 square units!

13. It is 8:30 AM. What time will it be in 30 minutes?

A. 9:00 AM ✓

B. 8:00 AM

C. 9:30 AM

D. 8:60 AM

Explanation: $8:30 + 30 \text{ minutes} = 9:00 \text{ AM}$!

14. Would you measure a hallway in centimeters or meters?

A. Centimeters

B. Meters ✓

C. Inches

D. Millimeters

Explanation: A hallway is long! We use meters for big things.

15. Two shapes each have 7 square units. What is the total area of both shapes?

A. 7 square units

B. 14 square units ✓

C. 49 square units

D. 0 square units

Explanation: $7 + 7 = 14 \text{ square units total}$!

16. A ribbon is some inches long. You need 4 more inches to reach 13 inches. How long is the ribbon?

A. 17 inches

B. 9 inches ✓

C. 4 inches

D. 8 inches

Explanation: $13 - 4 = 9 \text{ inches}$. The ribbon is 9 inches long!

17. Which is later: 6:00 AM or 6:00 PM?

A. 6:00 AM

B. 6:00 PM ✓

C. They are the same

D. Cannot tell

Explanation: PM is evening, which comes after AM (morning). 6:00 PM is later!

18. A rectangle is 6 squares long and 2 squares wide. What is its area?

A. 8 square units

B. 12 square units ✓

C. 10 square units

D. 14 square units

Explanation: $2 \text{ rows of } 6 = 12 \text{ square units}$!

19. A pencil is 7 inches. A pen is 6 inches. How many inches are they together?

A. 1 inch

B. 13 inches ✓

C. 76 inches

D. 14 inches

Explanation: $7 + 6 = 13 \text{ inches together}$!

20. Art class starts at 1:00 PM and ends at 2:00 PM. How long is art class?

A. 30 minutes

B. 2 hours

C. 1 hour ✓

D. 3 hours

Explanation: From 1:00 to 2:00 is 1 hour!

Hard (20 questions)

1. A belt is 14 inches long. You cut off 6 inches, then cut off 3 more inches. How long is it now?

- A. 5 inches ✓
- B. 8 inches
- C. 11 inches
- D. 23 inches

Explanation: $14 - 6 = 8$, then $8 - 3 = 5$ inches. Two steps!

2. A piano lesson starts at 3:15 PM and lasts 45 minutes. What time does it end?

- A. 3:60 PM
- B. 4:00 PM ✓
- C. 4:15 PM
- D. 3:45 PM

Explanation: $3:15 + 45 \text{ min} = 4:00 \text{ PM}$. There is no such thing as 3:60!

3. A rectangle is 5 squares long and 3 squares wide. Max says the area is 8 square units. What did Max do wrong?

- A. He added $5 + 3$ instead of counting all squares ✓
- B. He is correct
- C. He subtracted $5 - 3$
- D. He forgot to count

Explanation: Max added $5 + 3 = 8$. But the real area is 3 rows of 5 = 15 square units!

4. Mia says a giraffe is 18 inches tall. Is that reasonable?

- A. Yes, giraffes are small
- B. No, a giraffe is about 18 feet tall ✓
- C. Yes, that is correct
- D. No, a giraffe is 1 inch tall

Explanation: 18 inches is only 1.5 feet — that is the size of a small dog! A giraffe is about 18 feet tall.

5. It is 11:45 AM. What time will it be in 30 minutes?

- A. 11:75 AM
- B. 12:15 PM ✓
- C. 12:15 AM
- D. 12:45 PM

Explanation: $11:45 + 15 \text{ min} = 12:00$, then $+ 15 \text{ more} = 12:15$. It changes from AM to PM at noon! So 12:15 PM.

6. A patio is 4 by 3. You add a 2-by-3 section. What is the new total area?

- A. 12 square units
- B. 18 square units ✓
- C. 6 square units
- D. 24 square units

Explanation: Patio: $4 \times 3 = 12$. New section: $2 \times 3 = 6$. Total: $12 + 6 = 18$ square units!

7. A rope is 2 feet 6 inches. How many inches is that in all?

- A. 26 inches
- B. 30 inches ✓
- C. 8 inches
- D. 24 inches

Explanation: 2 feet = 24 inches. $24 + 6 = 30$ inches total!

8. Lee left home at 7:40 AM and arrived at school at 8:10 AM. How long was his trip?

- A. 30 minutes ✓
- B. 40 minutes
- C. 1 hour 10 minutes
- D. 20 minutes

Explanation: From 7:40 to 8:00 = 20 min. From 8:00 to 8:10 = 10 min. Total = 30 minutes!

9. A rectangle has an area of 18 square units. It is 6 squares long. How many squares wide is it?

- A. 12 squares
- B. 6 squares
- C. 3 squares ✓**
- D. 2 squares

Explanation: If 6 squares long and area is 18, then width = $18 / 6 = 3$ squares wide!

10. Kai has 3 pieces of string: 5 inches, 8 inches, and 4 inches. He puts them together. How long is the total string?

- A. 17 inches ✓**
- B. 13 inches
- C. 12 inches
- D. 20 inches

Explanation: $5 + 8 + 4 = 17$ inches total!

11. Ava says the time is 5:70 PM. What should the correct time be?

- A. 5:70 PM is correct
- B. 6:10 PM ✓**
- C. 5:10 PM
- D. 7:10 PM

Explanation: There is no :70 on a clock! 70 minutes = 1 hour 10 min. So 5:70 should be 6:10 PM.

12. A 2-by-8 rectangle and a 4-by-4 square — which has a bigger area?

- A. The rectangle (16 square units)
- B. The square (16 square units)
- C. They are the same — both 16 square units ✓**
- D. The rectangle (10 square units)

Explanation: $2 \times 8 = 16$ and $4 \times 4 = 16$. They have the same area! Different shapes, same area.

13. Dan says a pencil is 7 feet long. What did he do wrong?

- A. He used feet instead of inches — a pencil is about 7 inches ✓**
- B. He is correct
- C. He should have said 7 meters
- D. Pencils cannot be measured

Explanation: 7 feet is 84 inches — that is taller than most adults! A pencil is about 7 inches, not 7 feet.

14. A cartoon is 25 minutes long. It starts at 4:40 PM. What time does it end?

- A. 4:65 PM
- B. 5:05 PM ✓**
- C. 5:15 PM
- D. 4:55 PM

Explanation: $4:40 + 20 \text{ min} = 5:00$, then $+ 5 \text{ more} = 5:05 \text{ PM}$.

15. A wall is 5 by 4. You hang a picture that covers 6 square units. How many wall squares are still showing?

- A. 20 square units
- B. 14 square units ✓**
- C. 26 square units
- D. 11 square units

Explanation: Wall: $5 \times 4 = 20$. Then $20 - 6 = 14$ squares still showing!

16. Which is longer: 3 feet or 40 inches?

- A. 3 feet
- B. 40 inches ✓**
- C. They are the same
- D. Cannot tell

Explanation: 3 feet = 36 inches. $40 > 36$, so 40 inches is longer by 4 inches!

17. Nina says she slept for 120 minutes. How many hours is that?

- A. 1 hour
- B. 12 hours
- C. 2 hours ✓
- D. 20 hours

Explanation: 60 minutes = 1 hour. $120 / 60 = 2$ hours!

18. A garden is 6 by 3. A flower bed covers 10 of those squares. How many garden squares are left?

- A. 8 square units ✓
- B. 28 square units
- C. 18 square units
- D. 10 square units

Explanation: Garden: $6 \times 3 = 18$. Then $18 - 10 = 8$ squares left!

19. Eva has a 30-inch ribbon. She cuts it into 3 equal pieces. How long is each piece?

- A. 10 inches ✓
- B. 27 inches
- C. 15 inches
- D. 33 inches

Explanation: $30 / 3 = 10$ inches per piece!

20. Jake played outside from 4:15 PM to 5:00 PM, then did homework from 5:00 PM to 5:30 PM. How much time in all?

- A. 1 hour 15 minutes ✓
- B. 45 minutes
- C. 1 hour
- D. 1 hour 30 minutes

Explanation: Outside: 4:15 to 5:00 = 45 min. Homework: 5:00 to 5:30 = 30 min. Total: $45 + 30 = 75$ min = 1 hour 15 minutes!

Unit 8: Fractions

Lesson 1: What is a Fraction?

Easy (20 questions)

1. A pizza is cut into 2 equal slices. What is each slice called?

- A. A fourth
- B. A half ✓**
- C. An eighth
- D. A whole

Explanation: When you split something into 2 equal parts, each part is called a half.

2. A sandwich is cut into 4 equal pieces. What is each piece called?

- A. A half
- B. A whole
- C. A fourth ✓**
- D. An eighth

Explanation: When something is split into 4 equal parts, each part is called a fourth (or a quarter).

3. A circle is split into 2 equal parts. 1 part is shaded. What fraction is shaded?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$ ✓**
- C. $\frac{2}{2}$
- D. $\frac{1}{8}$

Explanation: There are 2 equal parts and 1 is shaded, so the fraction is $\frac{1}{2}$.

4. A rectangle is split into 4 equal parts. 1 part is colored blue. What fraction is blue?

- A. $\frac{1}{2}$
- B. $\frac{4}{4}$
- C. $\frac{1}{4}$ ✓**
- D. $\frac{4}{1}$

Explanation: There are 4 equal parts and 1 is colored. That is 1 out of 4, or $\frac{1}{4}$.

5. What does the bottom number in a fraction tell you?

- A. How many parts are shaded
- B. How many equal parts in all ✓**
- C. How many are left over
- D. How big the shape is

Explanation: The bottom number tells how many equal parts the whole is divided into.

6. What does the top number in a fraction tell you?

- A. How many equal parts in all
- B. How many parts we are talking about ✓**
- C. How big each part is
- D. How many wholes there are

Explanation: The top number tells how many of the equal parts we are counting.

7. A pie is cut into 8 equal slices. What is each slice called?

- A. A half
- B. A fourth
- C. An eighth ✓**
- D. A whole

Explanation: When something is split into 8 equal parts, each part is called an eighth.

8. A square is split into 4 equal parts. All 4 parts are shaded. What fraction is shaded?

- A. $\frac{1}{4}$
- B. $\frac{4}{4}$ ✓**
- C. $\frac{4}{1}$
- D. $\frac{0}{4}$

Explanation: All 4 out of 4 parts are shaded, so the fraction is $\frac{4}{4}$. That equals 1 whole!

9. In the fraction $\frac{3}{8}$, what does the 8 mean?

- A. 3 parts are shaded
- B. The whole has 8 equal parts ✓**
- C. There are 8 shapes
- D. You ate 8 pieces

Explanation: The bottom number 8 tells you the whole is split into 8 equal parts.

10. In the fraction $\frac{3}{8}$, what does the 3 mean?

- A. There are 3 equal parts in all
- B. The shape has 3 sides
- C. We are counting 3 of the parts ✓**
- D. There are 3 shapes

Explanation: The top number 3 means we are counting 3 of the 8 equal parts.

11. A circle is split into 8 equal parts. 3 parts are shaded. What fraction is shaded?

- A. $\frac{3}{8}$ ✓**
- B. $\frac{8}{3}$
- C. $\frac{3}{3}$
- D. $\frac{1}{8}$

Explanation: 3 parts out of 8 equal parts are shaded. The fraction is $\frac{3}{8}$.

12. A waffle is cut into 4 equal squares. You eat 3 of them. What fraction did you eat?

- A. $\frac{3}{4}$ ✓**
- B. $\frac{4}{3}$
- C. $\frac{1}{4}$
- D. $\frac{3}{3}$

Explanation: You ate 3 out of 4 equal parts. That is the fraction $\frac{3}{4}$.

13. Which of these is a fraction?

- A. $5 + 3$
- B. $\frac{1}{4}$ ✓**
- C. 12
- D. A square

Explanation: $\frac{1}{4}$ is a fraction. It has a top number and a bottom number separated by a line.

14. A cookie is split into 2 equal parts. You eat 1 part. What fraction did you eat?

- A. $\frac{1}{2}$ ✓**
- B. $\frac{2}{1}$
- C. $\frac{1}{4}$
- D. $\frac{2}{2}$

Explanation: You ate 1 out of 2 equal parts. That is $\frac{1}{2}$ of the cookie.

15. A rectangle is split into 2 equal parts. 0 parts are shaded. What fraction is shaded?

- A. $\frac{2}{2}$
- B. $\frac{1}{2}$
- C. $\frac{0}{2}$ ✓**
- D. $\frac{2}{0}$

Explanation: No parts are shaded out of 2 equal parts, so the fraction is $\frac{0}{2}$. That means zero!

16. What must be true about the parts to write a fraction?

- A. They must be different sizes
- B. They must be equal ✓**
- C. There must be only 2 parts
- D. The shape must be a circle

Explanation: To write a fraction, the whole must be split into equal parts.

17. A fruit bar has 4 equal pieces. You give away 2 pieces. What fraction did you give away?

- A. $\frac{4}{2}$
- B. $\frac{2}{4}$ ✓**
- C. $\frac{1}{4}$
- D. $\frac{2}{2}$

Explanation: You gave away 2 out of 4 equal pieces. That is $\frac{2}{4}$.

18. How many fourths make 1 whole?

- A. 2
- B. 3
- C. 4 ✓**
- D. 8

Explanation: It takes 4 fourths to make 1 whole. Think of 4 puzzle pieces making a full picture!

19. If a shape has 8 equal parts and 1 is shaded, what fraction is shaded?

- A. $\frac{8}{1}$
- B. $\frac{1}{1}$
- C. $\frac{1}{8}$ ✓**
- D. $\frac{8}{8}$

Explanation: 1 part out of 8 equal parts is shaded. The fraction is $\frac{1}{8}$.

20. A cracker is broken into 2 equal parts. What fraction is 1 part?

- A. $\frac{2}{1}$
- B. $\frac{1}{2}$ ✓**
- C. $\frac{1}{4}$
- D. $\frac{2}{2}$

Explanation: 1 part out of 2 equal parts is $\frac{1}{2}$.

Medium (20 questions)

1. A shape is split into 4 parts, but the parts are NOT equal. Can you write a fraction for the shaded part?

- A. Yes, because there are 4 parts
- B. No, because the parts must be equal ✓**
- C. Yes, because part is shaded
- D. No, because there are too many parts

Explanation: Fractions only work when the parts are equal. If the parts are different sizes, we cannot write a fraction.

2. Which shows $\frac{1}{4}$?

- A. 1 out of 3 equal parts shaded
- B. 1 out of 4 equal parts shaded ✓**
- C. 4 out of 1 part shaded
- D. 4 out of 4 equal parts shaded

Explanation: $\frac{1}{4}$ means 1 out of 4 equal parts. The bottom number tells how many equal parts, and the top tells how many are shaded.

3. How many fourths make 1 whole?

- A. 2
- B. 3
- C. 4 ✓**
- D. 8

Explanation: It takes 4 fourths ($\frac{4}{4}$) to make 1 whole. Each fourth is one piece of the 4-piece whole.

4. How many eighths make 1 whole?

- A. 2
- B. 4
- C. 6
- D. 8 ✓**

Explanation: It takes 8 eighths ($\frac{8}{8}$) to make 1 whole. You need all 8 pieces to complete the whole.

5. A brownie is cut into 8 equal pieces. Maria eats 2 pieces. What fraction of the brownie is left?

- A. $\frac{2}{8}$
- B. $\frac{6}{8}$ ✓**
- C. $\frac{8}{2}$
- D. $\frac{8}{6}$

Explanation: Maria ate 2 of 8 pieces, so $8 - 2 = 6$ pieces are left. That is $\frac{6}{8}$ of the brownie.

6. A rectangle is divided into parts. 3 parts are big and 1 part is small. Can we say $\frac{1}{4}$ is shaded if the small part is colored?

- A. Yes, because there are 4 parts
- B. No, because the parts are not equal ✓**
- C. Yes, because 1 part is shaded
- D. No, because rectangles cannot show fractions

Explanation: We cannot write $\frac{1}{4}$ because the parts are not equal. Fractions require equal-sized parts.

7. Leo shades $\frac{2}{4}$ of a square. Mia shades $\frac{1}{2}$ of the same size square. Who shaded more?

- A. Leo shaded more
- B. Mia shaded more
- C. They shaded the same amount ✓**
- D. You cannot compare them

Explanation: $\frac{2}{4}$ and $\frac{1}{2}$ are the same amount! If you shade 2 out of 4 equal parts, that is the same as 1 out of 2 equal parts.

8. A pizza has 8 equal slices. 4 slices have pepperoni. What fraction has pepperoni?

- A. $\frac{8}{4}$
- B. $\frac{4}{4}$
- C. $\frac{4}{8}$ ✓**
- D. $\frac{1}{4}$

Explanation: 4 out of 8 equal slices have pepperoni. That is the fraction $\frac{4}{8}$.

9. Which fraction is the same as 1 whole?

- A. $\frac{1}{2}$
- B. $\frac{2}{4}$
- C. $\frac{4}{4}$ ✓**
- D. $\frac{3}{4}$

Explanation: $\frac{4}{4}$ means all 4 parts out of 4 are counted. That is the whole thing, or 1 whole.

10. Jake says $\frac{3}{4}$ means 4 out of 3 equal parts. Is Jake correct?

- A. Yes, that is right
- B. No, it means 3 out of 4 equal parts ✓**
- C. No, it means 3 wholes and 4 parts
- D. Yes, the bigger number always goes on top

Explanation: $\frac{3}{4}$ means 3 out of 4 equal parts. The top number (3) is how many we count, and the bottom (4) is how many equal parts in all.

11. A granola bar is split into 4 equal parts. You eat some and $\frac{1}{4}$ is left. How many parts did you eat?

- A. 1 part
- B. 2 parts
- C. 3 parts ✓**
- D. 4 parts

Explanation: If $\frac{1}{4}$ is left, that means 1 out of 4 parts remains. So you ate the other 3 parts.

12. A circle is split into 8 equal parts. How many parts do you need to shade to show $\frac{1}{2}$?

- A. 1 part
- B. 2 parts
- C. 4 parts ✓**
- D. 8 parts

Explanation: Half of 8 is 4, so you shade 4 out of 8 parts to show $\frac{1}{2}$.

13. Amy has a ribbon cut into 8 equal pieces. She uses 5 pieces. What fraction did she NOT use?

- A. $\frac{5}{8}$
- B. $\frac{3}{8}$ ✓**
- C. $\frac{8}{5}$
- D. $\frac{8}{3}$

Explanation: She used 5 out of 8, so $8 - 5 = 3$ pieces are not used. That is $\frac{3}{8}$.

14. Which two fractions are equal to 1 whole?

- A. $\frac{1}{2}$ and $\frac{2}{4}$
- B. $\frac{2}{2}$ and $\frac{8}{8}$ ✓**
- C. $\frac{1}{4}$ and $\frac{1}{8}$
- D. $\frac{3}{4}$ and $\frac{4}{8}$

Explanation: $\frac{2}{2} = 1$ whole and $\frac{8}{8} = 1$ whole. Whenever the top and bottom numbers are the same, it equals 1 whole.

15. There are 4 equal boxes. 2 are red and 2 are blue. What fraction is red?

- A. $\frac{2}{2}$
- B. $\frac{4}{2}$
- C. $\frac{2}{4}$ ✓**
- D. $\frac{1}{2}$

Explanation: 2 out of 4 equal boxes are red. The fraction is $\frac{2}{4}$.

16. Kai splits a cracker into 2 pieces, but one piece is much bigger than the other. He says each piece is $\frac{1}{2}$. Is Kai correct?

- A. Yes, because there are 2 pieces
- B. No, because the pieces are not equal ✓**
- C. Yes, because he split it in half
- D. No, because crackers cannot be fractions

Explanation: Kai is wrong. For each piece to be $\frac{1}{2}$, the two pieces must be equal in size.

17. A shape is split into 8 equal parts. ____ out of 8 parts are shaded to show $\frac{7}{8}$. What number fills the blank?

- A. 1
- B. 7 ✓**
- C. 8
- D. 6

Explanation: $\frac{7}{8}$ means 7 out of 8 equal parts. So the blank is 7.

18. A pancake is cut into 4 equal slices. You eat 1 slice. What fraction of the pancake is left?

- A. $\frac{1}{4}$
- B. $\frac{4}{4}$
- C. $\frac{3}{4}$ ✓**
- D. $\frac{2}{4}$

Explanation: You started with $\frac{4}{4}$. After eating 1 slice, $4 - 1 = 3$ slices remain. That is $\frac{3}{4}$.

19. Sara says $\frac{0}{4}$ means nothing is shaded. Is she correct?

- A. Yes, 0 out of 4 parts means none are shaded ✓**
- B. No, $\frac{0}{4}$ is not a real fraction
- C. No, it means all parts are shaded
- D. Yes, but only for circles

Explanation: Sara is correct! $\frac{0}{4}$ means 0 parts out of 4 are shaded -- that is none at all.

20. A rectangle has ____ equal parts. 3 are shaded. The fraction is $\frac{3}{4}$. How many equal parts does the rectangle have?

- A. 3
- B. 4 ✓**
- C. 7
- D. 8

Explanation: In $\frac{3}{4}$, the bottom number is 4. That means the rectangle is split into 4 equal parts.

Hard (20 questions)

1. Sam says $\frac{1}{4}$ is bigger than $\frac{1}{2}$ because 4 is bigger than 2. Is Sam correct?

- A. Yes, because $4 > 2$
- B. No, because more parts means each part is smaller ✓**
- C. Yes, because fourths are bigger than halves
- D. No, because you cannot compare fractions

Explanation: Sam is wrong. When you cut something into MORE pieces, each piece gets SMALLER. So $\frac{1}{4}$ is smaller than $\frac{1}{2}$.

2. Emma ate $\frac{2}{8}$ of a pizza. Jake ate $\frac{3}{8}$ of the same pizza. How much did they eat together? Is more than half the pizza gone?

- A. $\frac{5}{8}$ -- yes, more than half is gone ✓**
- B. $\frac{5}{8}$ -- no, less than half is gone
- C. $\frac{5}{16}$ -- yes, more than half is gone
- D. $\frac{6}{8}$ -- yes, more than half is gone

Explanation: $\frac{2}{8} + \frac{3}{8} = \frac{5}{8}$. Half the pizza is $\frac{4}{8}$. Since $\frac{5}{8} > \frac{4}{8}$, yes, more than half is gone!

3. You have 5 fourths of a pie. Is that more or less than 1 whole pie?

- A. Less than 1 whole
- B. Exactly 1 whole
- C. More than 1 whole by $\frac{1}{4}$ ✓**
- D. You cannot have 5 fourths

Explanation: $\frac{4}{4} = 1$ whole. $\frac{5}{4}$ means you have 1 whole pie plus $\frac{1}{4}$ more. That is more than 1 whole!

4. A rectangle is split into 6 parts. 2 parts are big and 4 parts are small. Can you say $\frac{4}{6}$ of the rectangle is shaded if the 4 small parts are colored?

- A. Yes, because 4 of 6 parts are colored
- B. No, because the parts are not equal ✓**
- C. Yes, because most parts are colored
- D. No, because rectangles only split into 4 parts

Explanation: You cannot write $\frac{4}{6}$ because the parts are not equal. Fractions require all parts to be the same size.

5. 3 friends share 2 pies equally. What fraction of a pie does each friend get?

- A. $\frac{1}{3}$
- B. $\frac{2}{3}$ ✓**
- C. $\frac{3}{2}$
- D. $\frac{1}{2}$

Explanation: 2 pies shared among 3 friends = $\frac{2}{3}$ per friend. Each person gets 2 thirds of a pie.

6. Mia says $\frac{2}{4}$ of her cookie is the same as $\frac{1}{2}$. Is Mia correct? Why?

- A. No, because 2 and 1 are different numbers
- B. Yes, because both show the same amount of cookie ✓**
- C. No, because 4 is bigger than 2
- D. Yes, but only if the cookie is round

Explanation: Mia is correct! 2 out of 4 equal parts is the same amount as 1 out of 2 equal parts. They are equal fractions.

7. A cake has 8 equal slices. Ben eats 3 slices and Zara eats 2 slices. What fraction of the cake is LEFT?

- A. $\frac{5}{8}$
- B. $\frac{3}{8}$ ✓**
- C. $\frac{6}{8}$
- D. $\frac{2}{8}$

Explanation: Ben ate 3 and Zara ate 2. That is $3 + 2 = 5$ slices eaten. $8 - 5 = 3$ slices left. The answer is $\frac{3}{8}$.

8. Liam says $\frac{8}{8}$ is bigger than $\frac{4}{4}$. Is he right?

- A. Yes, because 8 is bigger than 4
- B. No, they are both equal to 1 whole ✓**
- C. Yes, because eighths are bigger than fourths
- D. No, $\frac{4}{4}$ is bigger

Explanation: $\frac{8}{8} = 1$ whole and $\frac{4}{4} = 1$ whole. They are the same! When the top and bottom are equal, it always means 1 whole.

9. You cut a ribbon into 4 equal pieces. You use 3 pieces for a gift and then cut another equal ribbon into 4 pieces. How many fourths do you have in total now?

- A. 3 fourths
- B. 4 fourths
- C. 5 fourths ✓**
- D. 7 fourths

Explanation: From the first ribbon you have 1 fourth left. The second ribbon gives you 4 fourths. $1 + 4 = 5$ fourths total.

10. A circle is split into 8 equal parts. 4 parts are red and 2 are blue. What fraction is NOT colored?

- A. $\frac{6}{8}$
- B. $\frac{4}{8}$
- C. $\frac{2}{8}$ ✓**
- D. $\frac{8}{8}$

Explanation: Red uses 4 parts and blue uses 2. That is $4 + 2 = 6$ colored parts. $8 - 6 = 2$ are not colored. The answer is $\frac{2}{8}$.

11. Nora says you cannot have a fraction bigger than 1 whole. Is she correct?

- A. Yes, fractions are always less than 1
- B. No, fractions like $\frac{5}{4}$ are more than 1 whole ✓**
- C. Yes, the top number must be smaller
- D. No, but only with eighths

Explanation: Nora is wrong! Fractions like $\frac{5}{4}$ or $\frac{3}{2}$ are bigger than 1 whole. The top number CAN be larger than the bottom.

12. Max ate $\frac{1}{4}$ of a pizza on Monday and $\frac{2}{4}$ on Tuesday. What fraction has he eaten in total? What fraction is left?

- A. $\frac{3}{4}$ eaten, $\frac{1}{4}$ left ✓**
- B. $\frac{3}{8}$ eaten, $\frac{5}{8}$ left
- C. $\frac{2}{4}$ eaten, $\frac{2}{4}$ left
- D. $\frac{3}{4}$ eaten, $\frac{3}{4}$ left

Explanation: $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$ eaten. The whole pizza is $\frac{4}{4}$, so $\frac{4}{4} - \frac{3}{4} = \frac{1}{4}$ is left.

13. Ava splits a candy bar into 8 equal pieces. She gives $\frac{1}{2}$ of the bar to her brother. How many pieces did she give him?

- A. 2 pieces
- B. 4 pieces ✓**
- C. 1 piece
- D. 8 pieces

Explanation: Half of 8 pieces is 4 pieces. So $\frac{1}{2}$ of the candy bar = $\frac{4}{8} = 4$ pieces.

14. A teacher splits a paper into 4 equal parts. She colors 2 parts red and 1 part blue. She says $\frac{3}{4}$ is colored. A student says only $\frac{2}{4}$ is colored because the colors are different. Who is right?

- A. The teacher, because $2 + 1 = 3$ parts are colored ✓**
- B. The student, because different colors do not count together
- C. Neither -- the answer is $\frac{1}{4}$
- D. Both are right

Explanation: The teacher is right. 2 red + 1 blue = 3 colored parts out of 4 . It does not matter that the colors are different -- $\frac{3}{4}$ is colored.

15. 2 friends share 1 pie equally. Then 1 more friend arrives and they start over, sharing 1 pie equally among 3. Does each person get more or less pie now?

A. More, because there are more friends
B. Less, because the pie is split into more pieces ✓
C. The same amount
D. You cannot share a pie among 3

Explanation: With 2 friends each gets $1/2$. With 3 friends each gets $1/3$. $1/3 < 1/2$, so each person gets LESS pie.

16. Kai colors $6/8$ of a rectangle. Lily colors $3/4$ of the same size rectangle. Who colored more?

A. Kai colored more
B. Lily colored more
C. They colored the same amount ✓
D. You cannot compare them

Explanation: $6/8$ and $3/4$ are equal! $3/4 = 6/8$ because if you split each fourth into 2 parts, 3 fourths become 6 eighths.

17. A pan of brownies has 8 equal pieces. Mom eats $1/8$, Dad eats $2/8$, and you eat $3/8$. How much is left for your sister?

A. $6/8$
B. $3/8$
C. $2/8$ ✓
D. $1/8$

Explanation: $1/8 + 2/8 + 3/8 = 6/8$ eaten. $8/8 - 6/8 = 2/8$ is left for your sister.

18. Tom says $0/8$ and $0/4$ are different amounts because 8 and 4 are different. Is Tom right?

A. Yes, because the bottom numbers are different
B. No, both equal zero -- nothing is shaded ✓
C. Yes, $0/8$ is bigger
D. No, $0/4$ is bigger

Explanation: Tom is wrong. $0/8 = 0$ and $0/4 = 0$. Zero parts shaded is always zero, no matter how many parts the whole has.

19. You have 3 halves of a sandwich. How many whole sandwiches is that?

A. 1 whole
B. 1 whole and 1 half ✓
C. 2 wholes
D. 3 wholes

Explanation: 2 halves = 1 whole. You have 3 halves, so that is 1 whole sandwich plus $1/2$ more.

20. Zoe says if you cut a pizza into more slices, you get more pizza. Is she correct?

A. Yes, more slices means more pizza
B. No, the pizza is the same size no matter how you cut it ✓
C. Yes, but only if the slices are equal
D. No, you actually get less pizza

Explanation: Zoe is wrong. Cutting a pizza into more slices does not create more pizza. The total amount stays the same -- each slice is just smaller.

Lesson 2: Halves, Fourths and Eighths

Easy (20 questions)

1. A rectangle is split into 4 equal parts. 1 part is shaded. What fraction is shaded?

A. $1/2$
B. $1/4$ ✓
C. $1/8$
D. $4/4$

Explanation: 1 out of 4 equal parts is shaded. That is the fraction $1/4$, or one fourth.

2. A circle is cut into 2 equal parts. What is each part called?

- A. A fourth
- B. An eighth
- C. A half ✓
- D. A whole

Explanation: When a shape is cut into 2 equal parts, each part is called a half.

3. A pie is cut into 8 equal slices. 5 slices are eaten. What fraction was eaten?

- A. $\frac{5}{8}$ ✓
- B. $\frac{8}{5}$
- C. $\frac{3}{8}$
- D. $\frac{5}{5}$

Explanation: 5 out of 8 equal slices were eaten. The fraction is $\frac{5}{8}$.

4. How many halves make 1 whole?

- A. 1
- B. 2 ✓
- C. 4
- D. 8

Explanation: 2 halves put together make 1 whole. Think of 2 half-circles making a full circle.

5. How many fourths make 1 whole?

- A. 2
- B. 4 ✓
- C. 6
- D. 8

Explanation: 4 fourths make 1 whole. $\frac{4}{4} = 1$ whole.

6. How many eighths make 1 whole?

- A. 2
- B. 4
- C. 6
- D. 8 ✓

Explanation: 8 eighths make 1 whole. $\frac{8}{8} = 1$ whole.

7. A square is split into 2 equal parts. Both parts are shaded. What fraction is shaded?

- A. $\frac{1}{2}$
- B. $\frac{2}{2}$ ✓
- C. $\frac{2}{1}$
- D. $\frac{0}{2}$

Explanation: Both parts are shaded, so 2 out of 2 parts are shaded. That is $\frac{2}{2}$, or 1 whole.

8. A candy bar is split into 8 equal pieces. You eat 1 piece. What fraction did you eat?

- A. $\frac{1}{2}$
- B. $\frac{1}{4}$
- C. $\frac{1}{8}$ ✓
- D. $\frac{8}{1}$

Explanation: You ate 1 out of 8 equal pieces. That is $\frac{1}{8}$ of the candy bar.

9. A rectangle is cut into 4 equal parts. 3 parts are colored green. What fraction is green?

- A. $\frac{1}{4}$
- B. $\frac{3}{4}$ ✓
- C. $\frac{4}{3}$
- D. $\frac{3}{3}$

Explanation: 3 out of 4 equal parts are green. The fraction is $\frac{3}{4}$.

10. A pizza is cut into 2 equal slices. 1 slice has pepperoni. What fraction has pepperoni?

A. $\frac{2}{1}$

B. $\frac{1}{2}$ ✓

C. $\frac{2}{2}$

D. $\frac{1}{1}$

Explanation: 1 out of 2 equal slices has pepperoni. That is $\frac{1}{2}$.

11. A circle is split into 8 equal parts. 0 parts are shaded. What fraction is shaded?

A. $\frac{8}{8}$

B. $\frac{1}{8}$

C. $\frac{0}{8}$ ✓

D. $\frac{8}{0}$

Explanation: No parts are shaded, so 0 out of 8 are shaded. That is $\frac{0}{8}$, which means none.

12. A waffle is cut into 4 equal pieces. You eat 2. What fraction is left?

A. $\frac{2}{4}$ ✓

B. $\frac{4}{4}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

Explanation: You ate 2 out of 4 pieces, so $4 - 2 = 2$ pieces are left. That is $\frac{2}{4}$.

13. Which fraction means the whole shape is shaded?

A. $\frac{0}{4}$

B. $\frac{1}{4}$

C. $\frac{2}{4}$

D. $\frac{4}{4}$ ✓

Explanation: $\frac{4}{4}$ means all 4 parts out of 4 are shaded. That is the whole shape!

14. An apple is cut into 4 equal slices. Your friend takes 1 slice. What fraction does your friend have?

A. $\frac{1}{4}$ ✓

B. $\frac{4}{1}$

C. $\frac{1}{2}$

D. $\frac{3}{4}$

Explanation: Your friend took 1 out of 4 equal slices. That is $\frac{1}{4}$ of the apple.

15. A ribbon is cut into 8 equal pieces. 7 pieces are red. What fraction is red?

A. $\frac{1}{8}$

B. $\frac{7}{8}$ ✓

C. $\frac{8}{7}$

D. $\frac{7}{7}$

Explanation: 7 out of 8 equal pieces are red. The fraction is $\frac{7}{8}$.

16. A paper is folded into 2 equal parts. You color 1 part. What fraction is colored?

A. $\frac{2}{1}$

B. $\frac{1}{4}$

C. $\frac{1}{2}$ ✓

D. $\frac{2}{2}$

Explanation: 1 out of 2 equal parts is colored. That is $\frac{1}{2}$.

17. A square is cut into 4 equal parts. 4 parts are colored. What fraction is NOT colored?

A. $\frac{4}{4}$

B. $\frac{0}{4}$ ✓

C. $\frac{1}{4}$

D. $\frac{3}{4}$

Explanation: All 4 parts are colored, so 0 parts are not colored. That is $\frac{0}{4}$.

18. A cracker is broken into 2 equal halves. What fraction is each half?

- A. $\frac{1}{2}$ ✓
- B. $\frac{2}{2}$
- C. $\frac{1}{4}$
- D. $\frac{1}{8}$

Explanation: Each half is 1 out of 2 equal parts. That is $\frac{1}{2}$.

19. A circle is divided into 8 equal parts. 4 parts are shaded yellow. What fraction is yellow?

- A. $\frac{8}{4}$
- B. $\frac{1}{8}$
- C. $\frac{4}{4}$
- D. $\frac{4}{8}$ ✓

Explanation: 4 out of 8 equal parts are yellow. The fraction is $\frac{4}{8}$.

20. A rectangle is split into 4 equal parts. 2 parts are blue. What fraction is NOT blue?

- A. $\frac{2}{4}$ ✓
- B. $\frac{4}{4}$
- C. $\frac{1}{4}$
- D. $\frac{3}{4}$

Explanation: 2 parts are blue, so $4 - 2 = 2$ parts are not blue. That is $\frac{2}{4}$.

Medium (20 questions)

1. A circle is divided into 4 equal parts. How many parts do you shade to show $\frac{1}{2}$?

- A. 1 part
- B. 2 parts ✓
- C. 3 parts
- D. 4 parts

Explanation: Half of 4 is 2. So you shade 2 out of 4 equal parts to show $\frac{1}{2}$.

2. A rectangle is split into 8 equal parts. You shade 4 parts. What fraction is shaded?

- A. $\frac{1}{8}$
- B. $\frac{4}{8}$ ✓
- C. $\frac{8}{4}$
- D. $\frac{1}{4}$

Explanation: 4 out of 8 equal parts are shaded. That is $\frac{4}{8}$, which is the same as $\frac{1}{2}$.

3. Which is bigger: 1 half or 1 fourth?

- A. 1 fourth, because $4 > 2$
- B. 1 half, because fewer pieces means bigger pieces ✓
- C. They are the same
- D. You cannot compare them

Explanation: 1 half is bigger. When you cut into fewer pieces, each piece is larger. Half > fourth.

4. Tina has a pie cut into 8 equal slices. She eats 3 slices. What fraction of the pie is LEFT?

- A. $\frac{3}{8}$
- B. $\frac{8}{3}$
- C. $\frac{5}{8}$ ✓
- D. $\frac{6}{8}$

Explanation: She ate 3 of 8 slices. $8 - 3 = 5$ slices are left. That is $\frac{5}{8}$ of the pie.

5. How many eighths equal $\frac{1}{2}$?

- A. 2 eighths
- B. 4 eighths ✓
- C. 6 eighths
- D. 8 eighths

Explanation: Half of 8 is 4. So $\frac{4}{8} = \frac{1}{2}$. Four eighths equal one half.

6. A pizza has 4 equal slices. Carlos eats 1 slice and gives 1 slice to his sister. What fraction of the pizza is left?

- A. $\frac{1}{4}$
- B. $\frac{2}{4}$ ✓
- C. $\frac{3}{4}$
- D. $\frac{4}{4}$

Explanation: Carlos ate 1 and gave away 1. That is 2 slices gone. $4 - 2 = 2$ slices left. That is $\frac{2}{4}$.

7. Which fraction equals 1 whole: $\frac{2}{2}$, $\frac{2}{4}$, or $\frac{2}{8}$?

- A. $\frac{2}{2}$ ✓
- B. $\frac{2}{4}$
- C. $\frac{2}{8}$
- D. All of them

Explanation: $\frac{2}{2}$ means 2 out of 2 equal parts -- that is the whole thing. $\frac{2}{4}$ and $\frac{2}{8}$ are less than 1 whole.

8. Maya says $\frac{2}{8}$ and $\frac{1}{4}$ are the same amount. Is she right?

- A. Yes, they show the same amount ✓
- B. No, $\frac{2}{8}$ is bigger
- C. No, $\frac{1}{4}$ is bigger
- D. You cannot compare eighths and fourths

Explanation: Maya is right! If you split each fourth into 2 pieces, 1 fourth becomes 2 eighths. So $\frac{2}{8} = \frac{1}{4}$.

9. A shape has ____ equal parts. 3 parts are shaded. The fraction is $\frac{3}{8}$. How many equal parts does the shape have?

- A. 3
- B. 4
- C. 8 ✓
- D. 11

Explanation: In $\frac{3}{8}$, the bottom number is 8. That means the shape has 8 equal parts.

10. Ben folds a paper into 2 equal parts and shades 1 part. Mia folds a same-size paper into 4 equal parts and shades 2 parts. Did they shade the same amount?

- A. No, Ben shaded more
- B. No, Mia shaded more
- C. Yes, they shaded the same amount ✓
- D. You cannot tell

Explanation: Ben shaded $\frac{1}{2}$. Mia shaded $\frac{2}{4}$. These are equal because $\frac{2}{4} = \frac{1}{2}$.

11. A rope is cut into 8 equal pieces. Jess uses 6 pieces. What fraction did she NOT use?

- A. $\frac{6}{8}$
- B. $\frac{2}{8}$ ✓
- C. $\frac{8}{2}$
- D. $\frac{8}{6}$

Explanation: She used 6 out of 8 pieces, so $8 - 6 = 2$ pieces are left. That is $\frac{2}{8}$.

12. Which two fractions both mean 'half'?

- A. $\frac{1}{2}$ and $\frac{2}{4}$ ✓
- B. $\frac{1}{2}$ and $\frac{1}{4}$
- C. $\frac{2}{4}$ and $\frac{2}{8}$
- D. $\frac{1}{4}$ and $\frac{1}{8}$

Explanation: $\frac{1}{2}$ and $\frac{2}{4}$ both mean half. 2 out of 4 equal parts is the same as 1 out of 2 equal parts.

13. A sandwich is cut into 4 equal pieces. You eat 3 pieces. What fraction is left?

- A. $\frac{3}{4}$
- B. $\frac{1}{4}$ ✓
- C. $\frac{4}{4}$
- D. $\frac{0}{4}$

Explanation: You ate 3 of 4 pieces. $4 - 3 = 1$ piece left. That is $\frac{1}{4}$.

14. Rosa splits her brownie into 8 equal pieces. She wants to eat exactly half. How many pieces should she eat?

- A. 2 pieces
- B. 3 pieces
- C. 4 pieces ✓**
- D. 8 pieces

Explanation: Half of 8 pieces is 4 pieces. She should eat 4 pieces to eat exactly $\frac{1}{2}$.

15. Eli says $\frac{4}{8}$ is bigger than $\frac{1}{2}$. Is he correct?

- A. Yes, because 4 and 8 are bigger numbers
- B. No, $\frac{4}{8}$ and $\frac{1}{2}$ are the same ✓**
- C. Yes, because eighths are bigger
- D. No, $\frac{1}{2}$ is bigger

Explanation: $\frac{4}{8} = \frac{1}{2}$. They are the same amount. 4 out of 8 equal parts is exactly half.

16. A circle is split into 8 equal parts. ____ parts are shaded. The fraction shaded is $\frac{1}{2}$. What goes in the blank?

- A. 1
- B. 2
- C. 4 ✓**
- D. 8

Explanation: $\frac{1}{2}$ of 8 parts = 4 parts. So 4 parts should be shaded to show $\frac{1}{2}$.

17. A garden has 4 equal sections. 1 section has flowers and 1 section has vegetables. What fraction of the garden is planted?

- A. $\frac{1}{4}$
- B. $\frac{2}{4}$ ✓**
- C. $\frac{3}{4}$
- D. $\frac{4}{4}$

Explanation: 1 section of flowers + 1 section of vegetables = 2 sections planted. That is 2 out of 4, or $\frac{2}{4}$.

18. Which is bigger: 1 fourth or 1 eighth?

- A. 1 eighth, because $8 > 4$
- B. 1 fourth, because fewer pieces means bigger pieces ✓**
- C. They are the same
- D. You cannot compare them

Explanation: 1 fourth is bigger. Splitting into 4 pieces gives larger pieces than splitting into 8.

19. A strip of paper is folded into 8 equal parts. 8 parts are shaded. What fraction is shaded?

- A. $\frac{1}{8}$
- B. $\frac{0}{8}$
- C. $\frac{8}{8}$ ✓**
- D. $\frac{8}{1}$

Explanation: All 8 out of 8 parts are shaded. That is $\frac{8}{8}$, which is 1 whole.

20. Kevin has 2 equal halves of a granola bar. He puts them together. What does he have?

- A. $\frac{1}{2}$ of a bar
- B. 1 whole bar ✓**
- C. 2 whole bars
- D. $\frac{1}{4}$ of a bar

Explanation: 2 halves together = $\frac{2}{2} = 1$ whole granola bar.

Hard (20 questions)

1. A pizza has 8 equal slices. Jade eats $\frac{3}{8}$ and her mom eats $\frac{2}{8}$. What fraction is left? Is more or less than half left?

A. $\frac{3}{8}$ left -- less than half ✓

B. $\frac{5}{8}$ left -- more than half

C. $\frac{3}{8}$ left -- more than half

D. $\frac{5}{8}$ left -- less than half

Explanation: $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$ eaten. $\frac{8}{8} - \frac{5}{8} = \frac{3}{8}$ left. Half is $\frac{4}{8}$, and $\frac{3}{8} < \frac{4}{8}$, so less than half is left.

2. Ravi says $\frac{1}{8}$ is bigger than $\frac{1}{2}$ because 8 is bigger than 2. What is wrong with Ravi's thinking?

A. Nothing -- Ravi is correct

B. More pieces means each piece is smaller, so $\frac{1}{8} < \frac{1}{2}$ ✓

C. $\frac{1}{8}$ and $\frac{1}{2}$ are equal

D. You cannot compare fractions with different bottoms

Explanation: Ravi is wrong. When you cut into MORE pieces, each piece is SMALLER. $\frac{1}{8}$ is much smaller than $\frac{1}{2}$.

3. You have 9 eighths of a cake. Is that more or less than 1 whole cake? By how much?

A. Less by $\frac{1}{8}$

B. Exactly 1 whole

C. More by $\frac{1}{8}$ ✓

D. More by $\frac{2}{8}$

Explanation: $\frac{8}{8} = 1$ whole. $\frac{9}{8} = \frac{8}{8} + \frac{1}{8}$. So you have 1 whole cake plus $\frac{1}{8}$ more.

4. 2 kids share 1 pie equally. Then they get a second pie and share it equally too. What fraction of a pie does each kid have now?

A. $\frac{1}{2}$ of a pie

B. 1 whole pie ✓

C. 2 halves of a pie

D. 2 whole pies

Explanation: Each kid gets $\frac{1}{2}$ from the first pie and $\frac{1}{2}$ from the second pie. $\frac{1}{2} + \frac{1}{2} = \frac{2}{2} = 1$ whole pie each.

5. A rectangle is split into 8 parts. 3 parts are tall and 5 are short. Can you say $\frac{5}{8}$ is shaded if the short parts are colored?

A. Yes, because 5 of 8 parts are colored

B. No, because the parts are not all equal ✓

C. Yes, because most of the parts are colored

D. No, because you can only shade half

Explanation: The parts are not equal (3 tall, 5 short). Fractions require all parts to be the same size, so $\frac{5}{8}$ is not correct.

6. Noah shades $\frac{2}{4}$ of a square. Eva shades $\frac{4}{8}$ of the same size square. Finn shades $\frac{1}{2}$ of the same size square. Who shaded the most?

A. Noah shaded the most

B. Eva shaded the most

C. Finn shaded the most

D. They all shaded the same amount ✓

Explanation: $\frac{2}{4} = \frac{1}{2}$ and $\frac{4}{8} = \frac{1}{2}$. All three fractions are equal -- they all shaded exactly half.

7. 4 friends share 3 pies equally. What fraction of a pie does each friend get?

A. $\frac{1}{4}$

B. $\frac{3}{4}$ ✓

C. $\frac{4}{3}$

D. $\frac{1}{3}$

Explanation: 3 pies split among 4 friends = $\frac{3}{4}$ per friend. Each person gets 3 fourths of a pie.

8. Amara ate $\frac{1}{4}$ of a cake on Saturday, $\frac{1}{4}$ on Sunday, and $\frac{1}{4}$ on Monday. How much has she eaten? How much is left?

A. $\frac{3}{4}$ eaten, $\frac{1}{4}$ left ✓

B. $\frac{3}{12}$ eaten, $\frac{9}{12}$ left

C. $\frac{2}{4}$ eaten, $\frac{2}{4}$ left

D. $\frac{3}{4}$ eaten, $\frac{3}{4}$ left

Explanation: $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$ eaten. The whole cake is $\frac{4}{4}$, so $\frac{4}{4} - \frac{3}{4} = \frac{1}{4}$ is left.

9. Leo says $\frac{2}{2}$ is bigger than $\frac{4}{4}$ because halves are bigger than fourths. Is Leo right?

A. Yes, halves are always bigger

B. No, both equal 1 whole ✓

C. Yes, because $2 < 4$

D. No, $\frac{4}{4}$ is bigger

Explanation: Leo is wrong. $\frac{2}{2} = 1$ whole and $\frac{4}{4} = 1$ whole. They are the same amount!

10. A cookie sheet has 8 equal cookies. Mom takes $\frac{1}{2}$ of them for a party and Dad takes $\frac{1}{4}$ of the total. How many cookies are left?

A. 1 cookie

B. 2 cookies ✓

C. 3 cookies

D. 4 cookies

Explanation: Mom takes $\frac{1}{2}$ of $8 = 4$ cookies. Dad takes $\frac{1}{4}$ of $8 = 2$ cookies. $4 + 2 = 6$ taken. $8 - 6 = 2$ cookies left.

11. Which list shows fractions from smallest to largest?

A. $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$

B. $\frac{1}{8}$, $\frac{1}{2}$, $\frac{1}{4}$

C. $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$ ✓

D. $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{2}$

Explanation: $\frac{1}{8}$ is the smallest piece, then $\frac{1}{4}$, then $\frac{1}{2}$ is the biggest. More pieces = smaller slices.

12. Kai has 3 fourths of a granola bar. Zoe has 6 eighths of the same size bar. Who has more?

A. Kai has more

B. Zoe has more

C. They have the same amount ✓

D. You cannot compare them

Explanation: $\frac{3}{4} = \frac{6}{8}$ because each fourth splits into 2 eighths. $3 \times 2 = 6$ eighths. They are equal!

13. Ava has 5 halves of a sandwich. How many whole sandwiches can she make? How much is left over?

A. 2 wholes, $\frac{1}{2}$ left ✓

B. 5 wholes, nothing left

C. 2 wholes, nothing left

D. 3 wholes, $\frac{1}{2}$ left

Explanation: 2 halves = 1 whole. 4 halves = 2 wholes. The 5th half is left over. So 2 wholes and $\frac{1}{2}$.

14. A pan of brownies has 8 equal pieces. 3 kids each take 2 pieces. Is there enough for a 4th kid to also take 2 pieces?

A. Yes, there are 2 left ✓

B. No, there are 0 left

C. Yes, there are 3 left

D. No, there is only 1 left

Explanation: 3 kids $\times$ 2 pieces = 6 pieces taken. $8 - 6 = 2$ pieces left. Yes, 1 more kid can take 2 pieces!

15. Ian says if two fractions have the same top number, the one with the bigger bottom number is bigger. Is Ian correct?

A. Yes, bigger bottom means bigger fraction

B. No, bigger bottom means smaller pieces so the fraction is smaller ✓

C. Yes, because more parts is always more

D. No, they are always equal

Explanation: Ian is wrong. With the same top number, a bigger bottom means more pieces, so each piece is smaller. Example: $\frac{1}{8} < \frac{1}{4}$.

16. Mia colors $\frac{1}{2}$ of a shape red, then $\frac{1}{4}$ of the same shape blue. What fraction of the shape is still white?

A. $\frac{3}{4}$

B. $\frac{1}{4}$ ✓

C. $\frac{2}{4}$

D. $\frac{1}{2}$

Explanation: $\frac{1}{2} = \frac{2}{4}$. Red is $\frac{2}{4}$ and blue is $\frac{1}{4}$. That is $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ colored. So $\frac{4}{4} - \frac{3}{4} = \frac{1}{4}$ is white.

17. A ribbon is 1 whole long. You cut off $\frac{3}{8}$. Then you cut off another $\frac{3}{8}$. How much ribbon is left?

- A. $\frac{6}{8}$
- B. $\frac{3}{8}$
- C. $\frac{2}{8}$ ✓
- D. $\frac{1}{8}$

Explanation: $\frac{3}{8} + \frac{3}{8} = \frac{6}{8}$ cut off. $\frac{8}{8} - \frac{6}{8} = \frac{2}{8}$ is left.

18. Ella says $\frac{4}{4}$, $\frac{2}{2}$, and $\frac{8}{8}$ are all different amounts. Is she right?

- A. Yes, they have different numbers
- B. No, they all equal 1 whole ✓
- C. Yes, $\frac{8}{8}$ is the biggest
- D. No, only $\frac{4}{4}$ and $\frac{2}{2}$ are the same

Explanation: Ella is wrong. $\frac{4}{4} = 1$ whole, $\frac{2}{2} = 1$ whole, and $\frac{8}{8} = 1$ whole. When top equals bottom, it is always 1 whole.

19. 2 friends want to share 3 granola bars equally. What fraction does each friend get?

- A. $\frac{3}{2}$ of a bar ✓
- B. $\frac{2}{3}$ of a bar
- C. $\frac{1}{2}$ of a bar
- D. 1 whole bar

Explanation: 3 bars shared by 2 friends = $\frac{3}{2}$ per friend. That is 1 whole bar and $\frac{1}{2}$ more (one and a half bars).

20. Jada says cutting a pizza into 8 slices gives you more pizza than cutting it into 4 slices. What is wrong with her thinking?

- A. Nothing -- she is correct
- B. The total pizza stays the same; more slices just means smaller slices ✓
- C. You actually get less pizza with more cuts
- D. Pizzas can only be cut into 4 slices

Explanation: Jada is wrong. Cutting more slices does not create more pizza. The total stays the same -- each slice just gets smaller.

Lesson 3: Which Piece is Bigger?

Easy (20 questions)

1. Which is larger: $\frac{1}{2}$ or $\frac{1}{4}$?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$ ✓
- C. They are the same
- D. You cannot tell

Explanation: $\frac{1}{2}$ is larger. When you split something into 2 pieces, each piece is bigger than when you split into 4.

2. Which is larger: $\frac{1}{2}$ or $\frac{1}{8}$?

- A. $\frac{1}{8}$
- B. $\frac{1}{2}$ ✓
- C. They are the same
- D. You cannot tell

Explanation: $\frac{1}{2}$ is larger. Splitting into 2 pieces gives much bigger pieces than splitting into 8.

3. Which is larger: $\frac{1}{4}$ or $\frac{1}{8}$?

- A. $\frac{1}{8}$
- B. $\frac{1}{4}$ ✓
- C. They are the same
- D. You cannot tell

Explanation: $\frac{1}{4}$ is larger. Fewer pieces means each piece is bigger.

4. Which is the smallest: $\frac{1}{2}$, $\frac{1}{4}$, or $\frac{1}{8}$?

- A. $\frac{1}{2}$
- B. $\frac{1}{4}$
- C. $\frac{1}{8}$ ✓
- D. They are all the same

Explanation: $\frac{1}{8}$ is the smallest. When you split into 8 pieces, each piece is very small.

5. Which is the largest: $\frac{1}{2}$, $\frac{1}{4}$, or $\frac{1}{8}$?

- A. $\frac{1}{8}$
- B. $\frac{1}{4}$
- C. $\frac{1}{2}$ ✓
- D. They are all the same

Explanation: $\frac{1}{2}$ is the largest. Splitting into only 2 pieces gives the biggest pieces.

6. You get $\frac{1}{2}$ of a cookie or $\frac{1}{8}$ of the same cookie. Which gives you more cookie?

- A. $\frac{1}{8}$
- B. $\frac{1}{2}$ ✓
- C. They are the same
- D. It depends on the cookie

Explanation: $\frac{1}{2}$ gives you more. Half a cookie is a lot more than one eighth of a cookie!

7. A pizza is cut into 4 equal slices. Another same-size pizza is cut into 8 equal slices. Which pizza has BIGGER slices?

- A. The one cut into 8 slices
- B. The one cut into 4 slices ✓
- C. Both have the same size slices
- D. You cannot tell

Explanation: The pizza cut into 4 slices has bigger slices. Fewer cuts = bigger pieces.

8. Which piece of pie is bigger: 1 fourth or 1 half?

- A. 1 fourth
- B. 1 half ✓
- C. They are the same
- D. It depends on the pie

Explanation: 1 half is bigger. A half is a larger piece than a fourth.

9. Sam has $\frac{1}{4}$ of a brownie. Ella has $\frac{1}{2}$ of the same brownie. Who has more?

- A. Sam
- B. Ella ✓
- C. They have the same
- D. You cannot tell

Explanation: Ella has more. $\frac{1}{2}$ is a bigger piece than $\frac{1}{4}$.

10. Which is bigger: $\frac{3}{4}$ or $\frac{1}{4}$?

- A. $\frac{1}{4}$
- B. $\frac{3}{4}$ ✓
- C. They are the same
- D. You cannot tell

Explanation: $\frac{3}{4}$ is bigger. 3 fourths is more than 1 fourth -- it is 3 pieces instead of 1.

11. Which is bigger: $\frac{2}{8}$ or $\frac{7}{8}$?

- A. $\frac{2}{8}$
- B. $\frac{7}{8}$ ✓
- C. They are the same
- D. You cannot tell

Explanation: $\frac{7}{8}$ is bigger. 7 pieces out of 8 is more than 2 pieces out of 8.

12. Would you rather have $\frac{1}{8}$ of a cake or $\frac{1}{2}$ of the same cake?

- A. $\frac{1}{8}$, because you get more
- B. $\frac{1}{2}$, because you get more ✓**
- C. They are the same
- D. It depends on the cake

Explanation: $\frac{1}{2}$ gives you much more cake. Half a cake is way bigger than one eighth!

13. 2 kids each get 1 piece of pizza. One pizza was cut into 2 slices, the other into 8 slices. Who got a bigger piece?

- A. The kid with a slice from the 8-slice pizza
- B. The kid with a slice from the 2-slice pizza ✓**
- C. Both got the same
- D. You cannot tell

Explanation: The kid who got a slice from the 2-slice pizza got $\frac{1}{2}$. The other got $\frac{1}{8}$. $\frac{1}{2} > \frac{1}{8}$.

14. Which is bigger: $\frac{5}{8}$ or $\frac{2}{8}$?

- A. $\frac{2}{8}$
- B. $\frac{5}{8}$ ✓**
- C. They are the same
- D. You cannot tell

Explanation: $\frac{5}{8}$ is bigger. When the bottom number is the same, just compare the top numbers: $5 > 2$.

15. Which is bigger: $\frac{1}{4}$ or $\frac{3}{4}$?

- A. $\frac{1}{4}$
- B. $\frac{3}{4}$ ✓**
- C. They are the same
- D. You cannot tell

Explanation: $\frac{3}{4}$ is bigger. 3 pieces out of 4 is more than 1 piece out of 4.

16. A ribbon is cut into 8 equal pieces. Is 1 piece bigger or smaller than half the ribbon?

- A. Bigger than half
- B. Smaller than half ✓**
- C. Exactly half
- D. You cannot tell

Explanation: 1 piece out of 8 is $\frac{1}{8}$. Half the ribbon is $\frac{4}{8}$. Since $\frac{1}{8} < \frac{4}{8}$, one piece is smaller than half.

17. Which fraction shows MORE of a shape: $\frac{6}{8}$ or $\frac{3}{8}$?

- A. $\frac{3}{8}$
- B. $\frac{6}{8}$ ✓**
- C. They are the same
- D. You cannot tell

Explanation: $\frac{6}{8}$ shows more. 6 parts shaded out of 8 is more than 3 parts shaded out of 8.

18. Mia has $\frac{1}{2}$ of a sandwich. Jake has $\frac{1}{4}$ of the same sandwich. Who has the smaller piece?

- A. Mia
- B. Jake ✓**
- C. They are the same
- D. You cannot tell

Explanation: Jake has the smaller piece. $\frac{1}{4}$ is smaller than $\frac{1}{2}$.

19. Which is less: $\frac{1}{8}$ or $\frac{1}{4}$?

- A. $\frac{1}{4}$
- B. $\frac{1}{8}$ ✓**
- C. They are the same
- D. You cannot tell

Explanation: $\frac{1}{8}$ is less. More pieces means smaller pieces, so eighths are smaller than fourths.

20. A pie is cut into 4 equal pieces. 2 pieces are cherry and 2 are apple. Which flavor takes up more of the pie?

- A. Cherry
- B. Apple
- C. They take up the same amount ✓**
- D. You cannot tell

Explanation: Both flavors have 2 out of 4 pieces. $\frac{2}{4} = \frac{2}{4}$, so they take up the same amount.

Medium (20 questions)

1. Order from smallest to largest: $\frac{1}{2}$, $\frac{1}{8}$, $\frac{1}{4}$.

- A. $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$
- B. $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$ ✓**
- C. $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{2}$
- D. $\frac{1}{8}$, $\frac{1}{2}$, $\frac{1}{4}$

Explanation: $\frac{1}{8}$ is smallest (8 pieces = tiny). $\frac{1}{4}$ is next (4 pieces = small). $\frac{1}{2}$ is biggest (2 pieces = big).

2. Order from largest to smallest: $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{8}$.

- A. $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$
- B. $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$ ✓**
- C. $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{8}$
- D. $\frac{1}{2}$, $\frac{1}{8}$, $\frac{1}{4}$

Explanation: $\frac{1}{2}$ is biggest, then $\frac{1}{4}$, then $\frac{1}{8}$ is smallest. Fewer pieces = bigger slices.

3. Ava gets $\frac{3}{8}$ of a pie. Ben gets $\frac{5}{8}$ of the same pie. Who gets more?

- A. Ava
- B. Ben ✓**
- C. They get the same
- D. You cannot compare

Explanation: Both fractions have 8 as the bottom number. $5 > 3$, so $\frac{5}{8} > \frac{3}{8}$. Ben gets more.

4. Which is larger: $\frac{2}{4}$ or $\frac{1}{2}$?

- A. $\frac{2}{4}$ is larger
- B. $\frac{1}{2}$ is larger
- C. They are the same ✓**
- D. You cannot compare them

Explanation: $\frac{2}{4}$ and $\frac{1}{2}$ are equal! 2 out of 4 parts is the same as 1 out of 2 parts.

5. A cake has 8 equal slices. You eat 4 slices. Did you eat more than half, less than half, or exactly half?

- A. More than half
- B. Less than half
- C. Exactly half ✓**
- D. You cannot tell

Explanation: Half of 8 slices is 4 slices. You ate $\frac{4}{8}$, which is exactly $\frac{1}{2}$.

6. Jess eats $\frac{2}{8}$ of a candy bar. Did she eat more or less than $\frac{1}{4}$?

- A. More than $\frac{1}{4}$
- B. Less than $\frac{1}{4}$
- C. Exactly $\frac{1}{4}$ ✓**
- D. You cannot tell

Explanation: $\frac{1}{4} = \frac{2}{8}$. So Jess ate exactly $\frac{1}{4}$ of the candy bar.

7. Who ate more pizza: Maya who ate $\frac{3}{4}$, or Eli who ate $\frac{6}{8}$?

- A. Maya ate more
- B. Eli ate more
- C. They ate the same amount ✓**
- D. You cannot compare

Explanation: $\frac{3}{4} = \frac{6}{8}$. They ate the same amount! Each fourth equals 2 eighths, and $3 \times 2 = 6$.

8. Which fraction is closest to 1 whole: $\frac{3}{4}$, $\frac{1}{2}$, or $\frac{1}{8}$?

- A. $\frac{1}{8}$
- B. $\frac{1}{2}$
- C. $\frac{3}{4}$ ✓
- D. They are all the same distance

Explanation: $\frac{3}{4}$ is closest to 1 whole. It is only $\frac{1}{4}$ away from a full whole.

9. A rope is cut into 8 equal pieces. You need more than half. What is the fewest pieces you need to take?

- A. 3 pieces
- B. 4 pieces
- C. 5 pieces ✓
- D. 6 pieces

Explanation: Half is $\frac{4}{8}$. To get MORE than half, you need at least 5 pieces ($\frac{5}{8} > \frac{4}{8}$).

10. Kim has $\frac{1}{4}$ of a pie. Lee has $\frac{1}{4}$ of a different same-size pie. Together, how much pie do they have?

- A. $\frac{2}{8}$ of a pie
- B. $\frac{2}{4}$ of a pie ✓
- C. $\frac{1}{2}$ of a pie
- D. $\frac{1}{4}$ of a pie

Explanation: Kim has $\frac{1}{4}$ and Lee has $\frac{1}{4}$. Together that is $\frac{1}{4} + \frac{1}{4} = \frac{2}{4}$ of a pie.

11. Noah says $\frac{7}{8}$ is less than $\frac{1}{2}$ because eighths are small. Is Noah correct?

- A. Yes, eighths are very small
- B. No, $\frac{7}{8}$ is more than $\frac{1}{2}$ because 7 eighths is a lot ✓
- C. Yes, any fraction with 8 on the bottom is small
- D. No, $\frac{7}{8}$ equals $\frac{1}{2}$

Explanation: Noah is wrong. $\frac{1}{2} = \frac{4}{8}$. Since $\frac{7}{8} > \frac{4}{8}$, we know $\frac{7}{8}$ is much MORE than $\frac{1}{2}$.

12. Which is bigger: $\frac{3}{8}$ or $\frac{1}{2}$?

- A. $\frac{3}{8}$
- B. $\frac{1}{2}$ ✓
- C. They are the same
- D. You cannot tell

Explanation: $\frac{1}{2} = \frac{4}{8}$. Since $\frac{3}{8} < \frac{4}{8}$, $\frac{1}{2}$ is bigger than $\frac{3}{8}$.

13. Two same-size pizzas are cut differently. Pizza A has 4 equal slices. Pizza B has 8 equal slices. Are 2 slices of Pizza A the same as 4 slices of Pizza B?

- A. No, Pizza A slices are bigger
- B. No, Pizza B slices are more
- C. Yes, both equal $\frac{1}{2}$ of a pizza ✓
- D. You cannot compare different pizzas

Explanation: 2 slices of 4 = $\frac{2}{4} = \frac{1}{2}$. 4 slices of 8 = $\frac{4}{8} = \frac{1}{2}$. Yes, they are the same amount!

14. Which is closer to 0: $\frac{1}{8}$ or $\frac{1}{2}$?

- A. $\frac{1}{2}$
- B. $\frac{1}{8}$ ✓
- C. They are the same distance
- D. You cannot tell

Explanation: $\frac{1}{8}$ is closer to 0. It is a very tiny piece, much smaller than $\frac{1}{2}$.

15. Put these fractions in order from least to greatest: $\frac{7}{8}$, $\frac{3}{8}$, $\frac{5}{8}$.

- A. $\frac{7}{8}$, $\frac{5}{8}$, $\frac{3}{8}$
- B. $\frac{3}{8}$, $\frac{5}{8}$, $\frac{7}{8}$ ✓
- C. $\frac{5}{8}$, $\frac{3}{8}$, $\frac{7}{8}$
- D. $\frac{3}{8}$, $\frac{7}{8}$, $\frac{5}{8}$

Explanation: All have 8 on the bottom, so compare the tops: $3 < 5 < 7$. Order: $\frac{3}{8}$, $\frac{5}{8}$, $\frac{7}{8}$.

16. Lina ate $\frac{1}{2}$ of her lunch. Ravi ate $\frac{3}{4}$ of the same size lunch. Who has more left?

A. Lina has more left ✓

B. Ravi has more left

C. They have the same left

D. You cannot compare

Explanation: Lina ate $\frac{1}{2}$ so she has $\frac{1}{2}$ left. Ravi ate $\frac{3}{4}$ so he has $\frac{1}{4}$ left. $\frac{1}{2} > \frac{1}{4}$, so Lina has more left.

17. Is $\frac{4}{8}$ greater than, less than, or equal to $\frac{2}{4}$?

A. Greater than $\frac{2}{4}$

B. Less than $\frac{2}{4}$

C. Equal to $\frac{2}{4}$ ✓

D. You cannot compare

Explanation: $\frac{4}{8} = \frac{1}{2}$ and $\frac{2}{4} = \frac{1}{2}$. They are equal!

18. A rectangle is split into 4 equal parts. 3 are shaded. Is more than half shaded?

A. No, exactly half is shaded

B. No, less than half is shaded

C. Yes, more than half is shaded ✓

D. You cannot tell

Explanation: Half of 4 parts is 2 parts. 3 parts are shaded, and $3 > 2$, so yes, more than half is shaded.

19. Which is bigger: $\frac{2}{4}$ or $\frac{3}{8}$?

A. $\frac{2}{4}$ ✓

B. $\frac{3}{8}$

C. They are equal

D. You cannot compare

Explanation: $\frac{2}{4} = \frac{4}{8}$. Compare $\frac{4}{8}$ to $\frac{3}{8}$. Since $4 > 3$, $\frac{2}{4}$ is bigger than $\frac{3}{8}$.

20. Finn has $\frac{5}{8}$ of a candy bar. Sam has $\frac{3}{4}$ of the same candy bar. Who has more?

A. Finn

B. Sam ✓

C. They have the same

D. You cannot tell

Explanation: $\frac{3}{4} = \frac{6}{8}$. Compare $\frac{5}{8}$ to $\frac{6}{8}$. Since $6 > 5$, Sam has more ($\frac{6}{8} > \frac{5}{8}$).

Hard (20 questions)

1. Sam says $\frac{1}{4}$ is bigger than $\frac{1}{2}$ because 4 is bigger than 2. What should you tell Sam?

A. You are right, $4 > 2$ so $\frac{1}{4} > \frac{1}{2}$

B. More parts means smaller pieces, so $\frac{1}{4}$ is actually smaller ✓

C. They are equal

D. You cannot compare $\frac{1}{4}$ and $\frac{1}{2}$

Explanation: Sam is wrong. Cutting into MORE pieces makes each piece SMALLER. $\frac{1}{4} < \frac{1}{2}$.

2. Ella ate $\frac{3}{8}$ of a cake. Her brother ate $\frac{4}{8}$. How much did they eat together? Is more than half the cake gone?

A. $\frac{7}{8}$ -- yes, more than half is gone ✓

B. $\frac{7}{8}$ -- no, less than half is gone

C. $\frac{7}{16}$ -- yes, more than half

D. $\frac{1}{8}$ -- no, less than half

Explanation: $\frac{3}{8} + \frac{4}{8} = \frac{7}{8}$. Half is $\frac{4}{8}$. Since $\frac{7}{8} > \frac{4}{8}$, yes, way more than half is gone!

3. You have 7 fourths of a pie. How many whole pies is that, and how much extra?

A. 1 whole and $\frac{1}{4}$ extra

B. 1 whole and $\frac{3}{4}$ extra ✓

C. 2 wholes and $\frac{1}{4}$ extra

D. 7 wholes

Explanation: $\frac{4}{4} = 1$ whole. $\frac{7}{4} = \frac{4}{4} + \frac{3}{4} = 1$ whole pie and $\frac{3}{4}$ extra.

4. A rectangle is cut into 8 parts. 5 parts are wide and 3 parts are narrow. Is $\frac{3}{8}$ shaded if the 3 narrow parts are colored?

- A. Yes, because 3 out of 8 are colored
- B. No, because the parts are not all equal ✓**
- C. Yes, because narrow parts count
- D. No, because only wide parts count

Explanation: The parts are not equal (5 wide, 3 narrow). Fractions need equal parts, so we cannot say $\frac{3}{8}$.

5. 4 kids share 3 pies equally. What fraction of a pie does each kid get?

- A. $\frac{1}{4}$
- B. $\frac{3}{4}$ ✓**
- C. $\frac{4}{3}$
- D. $\frac{1}{3}$

Explanation: 3 pies split among 4 kids = $\frac{3}{4}$ per kid. Each child gets 3 fourths of one pie.

6. Order these fractions from smallest to largest: $\frac{3}{4}$, $\frac{3}{8}$, $\frac{1}{2}$.

- A. $\frac{3}{4}$, $\frac{1}{2}$, $\frac{3}{8}$
- B. $\frac{3}{8}$, $\frac{1}{2}$, $\frac{3}{4}$ ✓**
- C. $\frac{1}{2}$, $\frac{3}{8}$, $\frac{3}{4}$
- D. $\frac{3}{8}$, $\frac{3}{4}$, $\frac{1}{2}$

Explanation: Convert to eighths: $\frac{3}{8} = \frac{3}{8}$, $\frac{1}{2} = \frac{4}{8}$, $\frac{3}{4} = \frac{6}{8}$. Order: $\frac{3}{8}$, $\frac{4}{8}$, $\frac{6}{8}$.

7. Kai ate $\frac{2}{4}$ of a pizza. Zoe ate $\frac{3}{8}$ of the same size pizza. Who ate more?

- A. Kai ✓**
- B. Zoe
- C. They ate the same
- D. You cannot compare

Explanation: $\frac{2}{4} = \frac{4}{8}$. Compare $\frac{4}{8}$ to $\frac{3}{8}$. Since $4 > 3$, Kai ate more.

8. Mia says $\frac{6}{8}$ and $\frac{3}{4}$ are different amounts because the numbers are different. Is she right?

- A. Yes, different numbers mean different amounts
- B. No, $\frac{6}{8}$ and $\frac{3}{4}$ are equal ✓**
- C. Yes, $\frac{6}{8}$ is bigger
- D. No, $\frac{3}{4}$ is bigger

Explanation: Mia is wrong. $\frac{3}{4} = \frac{6}{8}$ because each fourth splits into 2 eighths. They show the same amount.

9. A pie is cut into 8 equal slices. Dad eats $\frac{1}{4}$ of the pie and Mom eats $\frac{1}{2}$ of the pie. How many slices are left?

- A. 1 slice
- B. 2 slices ✓**
- C. 3 slices
- D. 4 slices

Explanation: Dad eats $\frac{1}{4} = \frac{2}{8}$ (2 slices). Mom eats $\frac{1}{2} = \frac{4}{8}$ (4 slices). $2 + 4 = 6$ eaten. $8 - 6 = 2$ slices left.

10. Ben has $\frac{5}{8}$ of a candy bar. He says that is more than $\frac{1}{2}$ but less than 1 whole. Is he correct?

- A. No, $\frac{5}{8}$ is less than $\frac{1}{2}$
- B. No, $\frac{5}{8}$ equals 1 whole
- C. Yes, $\frac{5}{8}$ is between $\frac{1}{2}$ and 1 whole ✓**
- D. No, $\frac{5}{8}$ is more than 1 whole

Explanation: $\frac{1}{2} = \frac{4}{8}$ and 1 whole = $\frac{8}{8}$. Since $\frac{4}{8} < \frac{5}{8} < \frac{8}{8}$, Ben is correct!

11. Leo says $\frac{2}{8}$ is bigger than $\frac{1}{4}$ because $2 > 1$ and $8 > 4$. Is Leo right?

- A. Yes, both numbers are bigger
- B. No, $\frac{2}{8}$ and $\frac{1}{4}$ are equal ✓**
- C. Yes, $\frac{2}{8}$ is much bigger
- D. No, $\frac{1}{4}$ is bigger

Explanation: Leo is wrong! $\frac{1}{4} = \frac{2}{8}$. They are the same amount. You cannot just compare top and bottom separately.

12. 3 friends share 2 cakes equally. What fraction of a cake does each friend get?

- A. $\frac{1}{3}$
- B. $\frac{2}{3}$ ✓
- C. $\frac{3}{2}$
- D. $\frac{1}{2}$

Explanation: 2 cakes shared among 3 friends = $\frac{2}{3}$ per friend. Each gets 2 thirds of a cake.

13. Ava colors $\frac{1}{2}$ of a circle red, then $\frac{1}{4}$ blue, then $\frac{1}{8}$ green. What fraction is still white?

- A. $\frac{7}{8}$
- B. $\frac{1}{8}$ ✓
- C. $\frac{3}{8}$
- D. $\frac{0}{8}$

Explanation: In eighths: $\frac{1}{2} = \frac{4}{8}$, $\frac{1}{4} = \frac{2}{8}$, $\frac{1}{8} = \frac{1}{8}$. Total colored: $4 + 2 + 1 = \frac{7}{8}$. White: $\frac{8}{8} - \frac{7}{8} = \frac{1}{8}$.

14. Nora has 3 halves of a granola bar. Eli has 5 fourths of a granola bar. Who has more?

- A. Nora, because $\frac{3}{2} > \frac{5}{4}$ ✓
- B. Eli, because $\frac{5}{4} > \frac{3}{2}$
- C. They have the same
- D. You cannot compare halves and fourths

Explanation: $\frac{3}{2} = \frac{6}{4}$. Compare $\frac{6}{4}$ to $\frac{5}{4}$. Since $6 > 5$, Nora has more.

15. A class votes for snacks. $\frac{3}{8}$ want chips, $\frac{2}{8}$ want fruit, and the rest want pretzels. What fraction wants pretzels? Which snack is most popular?

- A. $\frac{3}{8}$ pretzels -- pretzels tie with chips for most popular ✓
- B. $\frac{3}{8}$ pretzels -- chips are most popular
- C. $\frac{5}{8}$ pretzels -- pretzels are most popular
- D. $\frac{2}{8}$ pretzels -- chips are most popular

Explanation: $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$ accounted for. $\frac{8}{8} - \frac{5}{8} = \frac{3}{8}$ want pretzels. Chips and pretzels tie at $\frac{3}{8}$ each.

16. Jada says if you know $\frac{1}{4} < \frac{1}{2}$, then you also know $\frac{3}{4} < \frac{3}{2}$. Is she right?

- A. Yes, both comparisons are true ✓
- B. No, $\frac{3}{4} > \frac{3}{2}$
- C. No, $\frac{3}{4} = \frac{3}{2}$
- D. You cannot compare those fractions

Explanation: Jada is correct! $\frac{3}{4} = 0.75$ and $\frac{3}{2} = 1.5$. Fourths are smaller pieces than halves, so 3 fourths $<$ 3 halves.

17. A garden has 8 equal sections. Tomatoes take $\frac{3}{8}$, carrots take $\frac{1}{4}$, and flowers take $\frac{1}{8}$. What fraction is empty?

- A. $\frac{5}{8}$
- B. $\frac{3}{8}$
- C. $\frac{2}{8}$ ✓
- D. $\frac{1}{8}$

Explanation: In eighths: tomatoes $\frac{3}{8}$, carrots $\frac{1}{4} = \frac{2}{8}$, flowers $\frac{1}{8}$. Total planted: $3 + 2 + 1 = \frac{6}{8}$. Empty: $\frac{8}{8} - \frac{6}{8} = \frac{2}{8}$.

18. Which is closest to 1 whole without going over: $\frac{3}{4}$, $\frac{7}{8}$, or $\frac{5}{8}$?

- A. $\frac{3}{4}$
- B. $\frac{7}{8}$ ✓
- C. $\frac{5}{8}$
- D. They are all the same distance

Explanation: In eighths: $\frac{3}{4} = \frac{6}{8}$, $\frac{5}{8} = \frac{5}{8}$, $\frac{7}{8} = \frac{7}{8}$. $\frac{7}{8}$ is only $\frac{1}{8}$ away from $\frac{8}{8}$. That is the closest.

19. Ravi and Mia each have a same-size sandwich. Ravi eats $\frac{5}{8}$ of his. Mia eats $\frac{3}{4}$ of hers. Who ate more? How much more (in eighths)?

- A. Ravi ate $\frac{1}{8}$ more
- B. Mia ate $\frac{1}{8}$ more ✓
- C. They ate the same
- D. Mia ate $\frac{2}{8}$ more

Explanation: $\frac{3}{4} = \frac{6}{8}$. Compare $\frac{5}{8}$ and $\frac{6}{8}$. Mia ate $\frac{6}{8} - \frac{5}{8} = \frac{1}{8}$ more than Ravi.

20. Ian says $\frac{4}{8}$, $\frac{2}{4}$, and $\frac{1}{2}$ should NOT all be at the same spot on a number line. Is Ian right?

- A. Yes, they have different numbers so different spots
- B. No, they are all equal and belong at the same spot ✓**
- C. Yes, $\frac{4}{8}$ should be further right
- D. No, but only $\frac{2}{4}$ and $\frac{1}{2}$ are at the same spot

Explanation: Ian is wrong. $\frac{4}{8} = \frac{2}{4} = \frac{1}{2}$. They all equal the same amount and sit at the same point on the number line.

Unit Test Bank

Easy (20 questions)

1. A circle is split into 2 equal parts. What is each part called?

- A. A fourth
- B. A half ✓**
- C. An eighth
- D. A whole

Explanation: When something is divided into 2 equal parts, each part is called a half.

2. A rectangle is cut into 4 equal parts. 3 parts are shaded. What fraction is shaded?

- A. $\frac{1}{4}$
- B. $\frac{3}{4}$ ✓**
- C. $\frac{4}{3}$
- D. $\frac{3}{3}$

Explanation: 3 out of 4 equal parts are shaded. That is $\frac{3}{4}$.

3. How many halves make 1 whole?

- A. 1
- B. 2 ✓**
- C. 4
- D. 8

Explanation: 2 halves put together make 1 whole.

4. Which is larger: $\frac{1}{2}$ or $\frac{1}{8}$?

- A. $\frac{1}{8}$
- B. $\frac{1}{2}$ ✓**
- C. They are the same
- D. You cannot tell

Explanation: $\frac{1}{2}$ is larger. When you cut into only 2 pieces, each piece is much bigger than cutting into 8.

5. A pie is cut into 8 equal slices. 5 slices are gone. What fraction is left?

- A. $\frac{5}{8}$
- B. $\frac{3}{8}$ ✓**
- C. $\frac{8}{5}$
- D. $\frac{8}{3}$

Explanation: 5 slices are gone, so $8 - 5 = 3$ slices are left. That is $\frac{3}{8}$.

6. What does the bottom number in a fraction tell you?

- A. How many parts are shaded
- B. How many equal parts in all ✓**
- C. How big the whole is
- D. How many shapes there are

Explanation: The bottom number tells how many equal parts the whole is divided into.

7. A square is split into 4 equal parts. All 4 parts are shaded. What fraction is shaded?

- A. $\frac{1}{4}$
- B. $\frac{4}{4}$ ✓**
- C. $\frac{0}{4}$
- D. $\frac{4}{1}$

Explanation: All 4 out of 4 parts are shaded. That is $\frac{4}{4}$, which equals 1 whole.

8. How many eighths make 1 whole?

- A. 2
- B. 4
- C. 6
- D. 8 ✓**

Explanation: It takes 8 eighths to make 1 whole. $\frac{8}{8} = 1$ whole.

9. Which is bigger: $\frac{1}{4}$ or $\frac{1}{8}$?

- A. $\frac{1}{8}$
- B. $\frac{1}{4}$ ✓**
- C. They are the same
- D. You cannot tell

Explanation: $\frac{1}{4}$ is bigger. Fewer pieces means each piece is larger.

10. In the fraction $\frac{5}{8}$, what does the 5 mean?

- A. There are 5 equal parts in all
- B. We are counting 5 of the parts ✓**
- C. There are 5 shapes
- D. Each part is size 5

Explanation: The top number 5 tells us we are counting 5 of the 8 equal parts.

11. A cookie is split into 2 equal parts. You eat 1 part. What fraction is left?

- A. $\frac{1}{2}$ ✓**
- B. $\frac{2}{2}$
- C. $\frac{0}{2}$
- D. $\frac{2}{1}$

Explanation: You ate 1 of 2 parts. $2 - 1 = 1$ part is left. That is $\frac{1}{2}$.

12. Which is bigger: $\frac{6}{8}$ or $\frac{2}{8}$?

- A. $\frac{2}{8}$
- B. $\frac{6}{8}$ ✓**
- C. They are the same
- D. You cannot tell

Explanation: $\frac{6}{8}$ is bigger. Both have 8 equal parts, but 6 pieces is more than 2 pieces.

13. A candy bar is split into 8 equal pieces. 1 piece is eaten. What fraction was eaten?

- A. $\frac{1}{2}$
- B. $\frac{1}{4}$
- C. $\frac{1}{8}$ ✓**
- D. $\frac{8}{1}$

Explanation: 1 out of 8 equal pieces was eaten. That is $\frac{1}{8}$.

14. What must be true to write a fraction?

- A. The parts must be different sizes
- B. The parts must be equal ✓**
- C. The shape must be round
- D. There must be exactly 2 parts

Explanation: To write a fraction, the whole must be split into equal parts.

15. A ribbon is cut into 4 equal parts. 2 parts are blue. What fraction is blue?

A. $\frac{2}{4}$ ✓

B. $\frac{4}{2}$

C. $\frac{1}{4}$

D. $\frac{2}{2}$

Explanation: 2 out of 4 equal parts are blue. That is $\frac{2}{4}$.

16. How many fourths make 1 whole?

A. 2

B. 3

C. 4 ✓

D. 8

Explanation: It takes 4 fourths ($\frac{4}{4}$) to make 1 whole.

17. A pizza has 8 equal slices. 0 slices are taken. What fraction is taken?

A. $\frac{8}{8}$

B. $\frac{0}{8}$ ✓

C. $\frac{1}{8}$

D. $\frac{8}{0}$

Explanation: No slices are taken. 0 out of 8 = $\frac{0}{8}$. That means none.

18. Which is the smallest piece: $\frac{1}{2}$, $\frac{1}{4}$, or $\frac{1}{8}$?

A. $\frac{1}{2}$

B. $\frac{1}{4}$

C. $\frac{1}{8}$ ✓

D. They are all the same

Explanation: $\frac{1}{8}$ is the smallest. Cutting into 8 pieces makes tiny pieces.

19. A waffle is cut into 4 equal parts. You eat 1 part. What fraction is left?

A. $\frac{1}{4}$

B. $\frac{2}{4}$

C. $\frac{3}{4}$ ✓

D. $\frac{4}{4}$

Explanation: You ate 1 of 4 parts. $4 - 1 = 3$ parts left. That is $\frac{3}{4}$.

20. A paper is folded into 8 equal parts. 4 are colored red. What fraction is red?

A. $\frac{4}{8}$ ✓

B. $\frac{8}{4}$

C. $\frac{4}{4}$

D. $\frac{1}{4}$

Explanation: 4 out of 8 equal parts are red. That is $\frac{4}{8}$.

Medium (20 questions)

1. A shape is split into 4 parts but the parts are NOT equal. Can we write a fraction for the shaded part? Why?

A. Yes, because something is shaded

B. No, because fractions need equal parts ✓

C. Yes, because there are 4 parts

D. No, because there are too many parts

Explanation: Fractions only work when the parts are equal. If the parts are different sizes, we cannot write a fraction.

2. How many fourths make 1 whole?

A. 2

B. 3

C. 4 ✓

D. 8

Explanation: 4 fourths make 1 whole. $\frac{4}{4} = 1$.

3. Order from smallest to largest: $\frac{1}{2}$, $\frac{1}{8}$, $\frac{1}{4}$.

- A. $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$
- B. $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$ ✓**
- C. $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{2}$
- D. $\frac{1}{8}$, $\frac{1}{2}$, $\frac{1}{4}$

Explanation: $\frac{1}{8}$ is smallest (tiny pieces), then $\frac{1}{4}$, then $\frac{1}{2}$ is biggest. More cuts = smaller pieces.

4. A brownie is cut into 8 equal pieces. You eat 3 pieces. What fraction is LEFT?

- A. $\frac{3}{8}$
- B. $\frac{5}{8}$ ✓**
- C. $\frac{8}{3}$
- D. $\frac{8}{5}$

Explanation: You ate 3 of 8. That leaves $8 - 3 = 5$ pieces, or $\frac{5}{8}$.

5. Jake says the fraction $\frac{3}{4}$ means 4 out of 3 equal parts. Is Jake correct?

- A. Yes
- B. No, it means 3 out of 4 equal parts ✓**
- C. No, it means 3 plus 4
- D. Yes, the bigger number always goes on top

Explanation: $\frac{3}{4}$ means 3 out of 4 equal parts. The top is how many we count; the bottom is how many equal parts in all.

6. Anna shades $\frac{2}{4}$ of a circle. Marco shades $\frac{1}{2}$ of the same size circle. Who shaded more?

- A. Anna
- B. Marco
- C. They shaded the same amount ✓**
- D. You cannot compare them

Explanation: $\frac{2}{4} = \frac{1}{2}$. They shaded the same amount!

7. A pie is cut into 8 slices. Is $\frac{3}{8}$ more or less than half the pie?

- A. More than half
- B. Less than half ✓**
- C. Exactly half
- D. You cannot tell

Explanation: Half of 8 slices is 4 slices ($\frac{4}{8}$). Since $3 < 4$, $\frac{3}{8}$ is less than half.

8. Which two fractions are BOTH equal to 1 whole?

- A. $\frac{1}{2}$ and $\frac{2}{4}$
- B. $\frac{2}{2}$ and $\frac{4}{4}$ ✓**
- C. $\frac{1}{4}$ and $\frac{1}{8}$
- D. $\frac{3}{4}$ and $\frac{4}{8}$

Explanation: $\frac{2}{2} = 1$ whole and $\frac{4}{4} = 1$ whole. When top equals bottom, it is always 1 whole.

9. A pizza has 4 equal slices. You eat some and $\frac{1}{4}$ is left. How many slices did you eat?

- A. 1 slice
- B. 2 slices
- C. 3 slices ✓**
- D. 4 slices

Explanation: $\frac{1}{4}$ left means 1 of 4 slices remains. So you ate $4 - 1 = 3$ slices.

10. Which is bigger: $\frac{5}{8}$ or $\frac{3}{8}$?

- A. $\frac{3}{8}$
- B. $\frac{5}{8}$ ✓**
- C. They are equal
- D. You cannot compare

Explanation: Same bottom number, so compare tops: $5 > 3$. $\frac{5}{8}$ is bigger.

11. Eli splits a cracker into 2 pieces but one piece is way bigger. He says each piece is $\frac{1}{2}$. Is Eli correct?

A. Yes, there are 2 pieces

B. No, the pieces must be equal to call them halves ✓

C. Yes, any 2 pieces are halves

D. No, crackers cannot be split into fractions

Explanation: Eli is wrong. For halves, the 2 pieces must be equal in size.

12. How many eighths equal $\frac{1}{2}$?

A. 2

B. 3

C. 4 ✓

D. 6

Explanation: Half of 8 is 4. So $\frac{4}{8} = \frac{1}{2}$.

13. A shape has ____ equal parts. 5 parts are shaded. The fraction is $\frac{5}{8}$. What goes in the blank?

A. 5

B. 8 ✓

C. 3

D. 13

Explanation: In $\frac{5}{8}$, the bottom number is 8. The shape has 8 equal parts.

14. Which fraction is closest to 1 whole: $\frac{3}{4}$, $\frac{1}{2}$, or $\frac{1}{4}$?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{3}{4}$ ✓

D. They are all the same distance

Explanation: $\frac{3}{4}$ is closest to 1 whole. It is only $\frac{1}{4}$ away from being complete.

15. A rope is cut into 8 equal pieces. Ava uses 6 pieces. What fraction did she NOT use?

A. $\frac{6}{8}$

B. $\frac{2}{8}$ ✓

C. $\frac{8}{6}$

D. $\frac{8}{2}$

Explanation: She used 6 of 8. $8 - 6 = 2$ are unused. That is $\frac{2}{8}$.

16. Rosa says $\frac{4}{8}$ is bigger than $\frac{1}{2}$. Is she right?

A. Yes, because 4 and 8 are bigger numbers

B. No, $\frac{4}{8}$ equals $\frac{1}{2}$ ✓

C. Yes, eighths are bigger

D. No, $\frac{1}{2}$ is bigger

Explanation: $\frac{4}{8}$ and $\frac{1}{2}$ are the same. 4 out of 8 equal parts is exactly half.

17. A garden has 4 equal sections. 1 has flowers and 2 have vegetables. What fraction is planted?

A. $\frac{1}{4}$

B. $\frac{2}{4}$

C. $\frac{3}{4}$ ✓

D. $\frac{4}{4}$

Explanation: 1 flowers + 2 vegetables = 3 sections planted. That is $\frac{3}{4}$.

18. Which is bigger: $\frac{2}{4}$ or $\frac{3}{8}$?

A. $\frac{2}{4}$ ✓

B. $\frac{3}{8}$

C. They are equal

D. You cannot compare

Explanation: $\frac{2}{4} = \frac{4}{8}$. Since $\frac{4}{8} > \frac{3}{8}$, $\frac{2}{4}$ is bigger.

19. A circle is divided into 8 equal parts. How many parts must you shade to show $\frac{1}{4}$?

- A. 1 part
- B. 2 parts ✓**
- C. 4 parts
- D. 8 parts

Explanation: $\frac{1}{4}$ of 8 = 2 parts. So shade 2 out of 8 parts to show $\frac{1}{4}$.

20. Kai has 2 equal halves of a granola bar. He puts them together. What does he have?

- A. $\frac{1}{2}$ of a bar
- B. 1 whole bar ✓**
- C. 2 whole bars
- D. $\frac{2}{4}$ of a bar

Explanation: 2 halves = $\frac{2}{2}$ = 1 whole granola bar.

Hard (20 questions)

1. Sam says $\frac{1}{4}$ is bigger than $\frac{1}{2}$ because 4 is bigger than 2. Is Sam correct? Why not?

- A. Yes, $4 > 2$ so $\frac{1}{4} > \frac{1}{2}$
- B. No, more pieces means smaller pieces so $\frac{1}{4} < \frac{1}{2}$ ✓**
- C. They are equal
- D. You cannot compare different fractions

Explanation: Sam is wrong. More pieces means SMALLER pieces. $\frac{1}{4}$ is smaller than $\frac{1}{2}$.

2. Emma ate $\frac{2}{8}$ of a pizza. Jake ate $\frac{3}{8}$. How much did they eat together? Is more than half gone?

- A. $\frac{5}{8}$ -- yes, more than half is gone ✓**
- B. $\frac{5}{8}$ -- no, less than half
- C. $\frac{5}{16}$ -- yes, more than half
- D. $\frac{6}{8}$ -- yes, more than half

Explanation: $\frac{2}{8} + \frac{3}{8} = \frac{5}{8}$. Half is $\frac{4}{8}$. Since $\frac{5}{8} > \frac{4}{8}$, yes, more than half is gone.

3. You have 5 fourths of a pie. Is that more or less than 1 whole pie? How much more?

- A. Less by $\frac{1}{4}$
- B. Exactly 1 whole
- C. More by $\frac{1}{4}$ ✓**
- D. More by $\frac{2}{4}$

Explanation: $\frac{4}{4} = 1$ whole. $\frac{5}{4} = \frac{4}{4} + \frac{1}{4}$. That is 1 whole plus $\frac{1}{4}$ more.

4. A rectangle is split into 6 parts. 2 parts are big and 4 are small. Can you say $\frac{4}{6}$ is shaded if the 4 small parts are colored?

- A. Yes, 4 of 6 parts are colored
- B. No, the parts are not equal so we cannot write a fraction ✓**
- C. Yes, most parts are colored
- D. No, you can only use halves, fourths, and eighths

Explanation: The parts are not equal. Fractions require equal-sized parts, so $\frac{4}{6}$ is not correct here.

5. 3 friends share 2 pies equally. What fraction does each friend get?

- A. $\frac{1}{3}$
- B. $\frac{2}{3}$ ✓**
- C. $\frac{3}{2}$
- D. $\frac{1}{2}$

Explanation: 2 pies shared among 3 friends = $\frac{2}{3}$ per friend.

6. Order from smallest to largest: $\frac{3}{4}$, $\frac{3}{8}$, $\frac{1}{2}$.

- A. $\frac{3}{4}$, $\frac{1}{2}$, $\frac{3}{8}$
- B. $\frac{3}{8}$, $\frac{1}{2}$, $\frac{3}{4}$ ✓**
- C. $\frac{1}{2}$, $\frac{3}{8}$, $\frac{3}{4}$
- D. $\frac{3}{8}$, $\frac{3}{4}$, $\frac{1}{2}$

Explanation: In eighths: $\frac{3}{8} = \frac{3}{8}$, $\frac{1}{2} = \frac{4}{8}$, $\frac{3}{4} = \frac{6}{8}$. So $\frac{3}{8} < \frac{1}{2} < \frac{3}{4}$.

7. A pan of brownies has 8 equal pieces. Mom eats $\frac{1}{8}$, Dad eats $\frac{2}{8}$, and you eat $\frac{3}{8}$. What fraction is left for your brother?

- A. $\frac{6}{8}$
- B. $\frac{3}{8}$
- C. $\frac{2}{8}$ ✓
- D. $\frac{1}{8}$

Explanation: $\frac{1}{8} + \frac{2}{8} + \frac{3}{8} = \frac{6}{8}$ eaten. $\frac{8}{8} - \frac{6}{8} = \frac{2}{8}$ left for your brother.

8. Liam says $\frac{8}{8}$ is bigger than $\frac{4}{4}$ because $8 > 4$. Is he right?

- A. Yes, 8 is bigger than 4
- B. No, both equal 1 whole ✓
- C. Yes, eighths add up to more
- D. No, $\frac{4}{4}$ is bigger

Explanation: $\frac{8}{8} = 1$ whole and $\frac{4}{4} = 1$ whole. They are the same amount!

9. Ava splits a candy bar into 8 equal pieces. She gives $\frac{1}{2}$ to her friend. How many pieces did she give?

- A. 1 piece
- B. 2 pieces
- C. 4 pieces ✓
- D. 8 pieces

Explanation: $\frac{1}{2}$ of 8 = 4 pieces. She gave 4 of the 8 equal pieces.

10. Mia colors $\frac{1}{2}$ of a shape red, then $\frac{1}{4}$ blue. What fraction is still white?

- A. $\frac{3}{4}$
- B. $\frac{1}{4}$ ✓
- C. $\frac{2}{4}$
- D. $\frac{1}{2}$

Explanation: $\frac{1}{2} = \frac{2}{4}$. Red + blue = $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ colored. $\frac{4}{4} - \frac{3}{4} = \frac{1}{4}$ is white.

11. Kai has $\frac{3}{4}$ of a bar. Zoe has $\frac{6}{8}$ of the same size bar. Who has more?

- A. Kai
- B. Zoe
- C. They have the same ✓
- D. You cannot compare

Explanation: $\frac{3}{4} = \frac{6}{8}$. They have the same amount!

12. You have 3 halves of a sandwich. How many whole sandwiches is that?

- A. 1 whole
- B. 1 whole and 1 half ✓
- C. 2 wholes
- D. 3 wholes

Explanation: 2 halves = 1 whole. 3 halves = 1 whole + $\frac{1}{2}$ left over.

13. A cookie sheet has 8 equal cookies. Mom takes $\frac{1}{4}$ and Dad takes $\frac{1}{2}$. How many cookies are left?

- A. 1 cookie
- B. 2 cookies ✓
- C. 3 cookies
- D. 4 cookies

Explanation: Mom takes $\frac{1}{4}$ of 8 = 2 cookies. Dad takes $\frac{1}{2}$ of 8 = 4 cookies. $2 + 4 = 6$ taken. $8 - 6 = 2$ left.

14. Leo says $\frac{2}{8}$ is bigger than $\frac{1}{4}$ because $2 > 1$. Is Leo right?

- A. Yes, $2 > 1$ so $\frac{2}{8} > \frac{1}{4}$
- B. No, they are equal ✓
- C. Yes, eighths are bigger than fourths
- D. No, $\frac{1}{4}$ is much bigger

Explanation: $\frac{1}{4} = \frac{2}{8}$. They are equal! You cannot just compare top numbers when the bottoms are different.

15. 4 kids share 3 cookies equally. What fraction of a cookie does each kid get?

A. $\frac{1}{4}$

B. $\frac{3}{4}$ ✓

C. $\frac{4}{3}$

D. $\frac{1}{3}$

Explanation: 3 cookies split among 4 kids = $\frac{3}{4}$ per kid.

16. Nora says fractions can never be bigger than 1 whole. Is she correct?

A. Yes, fractions are always less than 1

B. No, fractions like $\frac{5}{4}$ are more than 1 whole ✓

C. Yes, the top number must be smaller

D. No, but only halves can be more than 1

Explanation: Nora is wrong. $\frac{5}{4}$, $\frac{3}{2}$, $\frac{9}{8}$ are all bigger than 1 whole. The top CAN be bigger than the bottom.

17. A ribbon is 1 whole long. You cut off $\frac{3}{8}$ and then another $\frac{2}{8}$. What fraction is left?

A. $\frac{5}{8}$

B. $\frac{3}{8}$ ✓

C. $\frac{1}{8}$

D. $\frac{6}{8}$

Explanation: $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$ cut off. $\frac{8}{8} - \frac{5}{8} = \frac{3}{8}$ is left.

18. Ian says if two fractions have the same top number, the one with the bigger bottom is bigger. Is he right?

A. Yes, bigger bottom means more

B. No, bigger bottom means smaller pieces so less overall ✓

C. Yes, because more parts is always more

D. It depends on the numbers

Explanation: Ian is wrong. Same top number but bigger bottom = smaller pieces. Example: $\frac{1}{8} < \frac{1}{4}$.

19. Which is closest to 1 whole without going over: $\frac{3}{4}$, $\frac{7}{8}$, or $\frac{1}{2}$?

A. $\frac{3}{4}$

B. $\frac{7}{8}$ ✓

C. $\frac{1}{2}$

D. They are all the same distance

Explanation: In eighths: $\frac{1}{2} = \frac{4}{8}$, $\frac{3}{4} = \frac{6}{8}$, $\frac{7}{8} = \frac{7}{8}$. $\frac{7}{8}$ is just $\frac{1}{8}$ away from 1 whole. That is closest.

20. Jada says cutting a pizza into 8 slices gives you more pizza than cutting it into 4 slices. What is wrong with her thinking?

A. Nothing, she is correct

B. The total amount stays the same -- more slices just means smaller slices ✓

C. You get less pizza with more cuts

D. Pizzas can only be cut into 4 slices

Explanation: Jada is wrong. Cutting more slices does not create more pizza. The total stays the same; each slice just gets smaller.

Unit 9: Geometry

Lesson 1: Flat Shapes

Easy (20 questions)

1. How many sides does a triangle have?

- A. 2
- B. 3 ✓**
- C. 4
- D. 5

Explanation: A triangle always has exactly 3 sides. 'Tri' means three!

2. What shape has 4 equal sides and 4 square corners?

- A. Triangle
- B. Circle
- C. Square ✓**
- D. Pentagon

Explanation: A square has 4 equal sides and 4 square corners (right angles).

3. How many corners (vertices) does a rectangle have?

- A. 3
- B. 2
- C. 5
- D. 4 ✓**

Explanation: A rectangle has 4 corners. Count them: top-left, top-right, bottom-left, bottom-right.

4. Which shape is round with no sides and no corners?

- A. Square
- B. Circle ✓**
- C. Hexagon
- D. Triangle

Explanation: A circle is perfectly round. It has 0 sides and 0 corners.

5. How many sides does a hexagon have?

- A. 5
- B. 4
- C. 6 ✓**
- D. 8

Explanation: A hexagon has 6 sides. 'Hex' means six!

6. What is the name of a shape with 5 sides?

- A. Hexagon
- B. Pentagon ✓**
- C. Octagon
- D. Quadrilateral

Explanation: A pentagon has 5 sides. 'Penta' means five!

7. How many corners does a triangle have?

- A. 4
- B. 2
- C. 3 ✓**
- D. 1

Explanation: A triangle has 3 corners (vertices) -- one at each point where two sides meet.

8. Which of these is NOT a polygon?

- A. Square
- B. Triangle
- C. Circle ✓**
- D. Rectangle

Explanation: A circle is NOT a polygon. Polygons have straight sides, but a circle is curved.

9. How many sides does an octagon have?

- A. 6
- B. 7
- C. 8 ✓**
- D. 10

Explanation: An octagon has 8 sides. A stop sign is shaped like an octagon!

10. A square is a special kind of ____.

- A. Triangle
- B. Circle
- C. Rectangle ✓**
- D. Pentagon

Explanation: A square is a special rectangle -- it has 4 right angles AND 4 equal sides.

11. True or false: A rectangle always has 4 sides.

- A. True ✓**
- B. False
- C. Only sometimes
- D. It depends on size

Explanation: True! Every rectangle has exactly 4 sides and 4 right angles.

12. What shape is a door most like?

- A. Triangle
- B. Circle
- C. Rectangle ✓**
- D. Hexagon

Explanation: A door is shaped like a rectangle -- it has 4 sides and 4 square corners.

13. How many sides does a square have?

- A. 3
- B. 4 ✓**
- C. 5
- D. 6

Explanation: A square has exactly 4 sides, and all 4 sides are the same length.

14. Which shape looks like a stop sign?

- A. Hexagon
- B. Pentagon
- C. Octagon ✓**
- D. Square

Explanation: A stop sign is an octagon -- it has 8 sides.

15. Does a triangle have more sides or more corners?

- A. More sides
- B. More corners
- C. They are equal ✓**
- D. It has no corners

Explanation: A triangle has 3 sides and 3 corners -- they are equal!

16. What shape has 0 sides?

- A. Square
- B. Triangle
- C. Pentagon
- D. Circle ✓**

Explanation: A circle has 0 straight sides. It is a curved shape, not a polygon.

17. Which of these shapes has the fewest sides?

- A. Square
- B. Pentagon
- C. Triangle ✓**
- D. Hexagon

Explanation: A triangle has 3 sides. That is fewer than a square (4), pentagon (5), or hexagon (6).

18. A rectangle has 2 long sides and 2 short sides. How many sides is that in all?

- A. 2
- B. 3
- C. 4 ✓**
- D. 6

Explanation: 2 long sides + 2 short sides = 4 sides in all.

19. What do we call any shape with 4 sides?

- A. Triangle
- B. Quadrilateral ✓**
- C. Pentagon
- D. Hexagon

Explanation: A quadrilateral is any shape with exactly 4 sides. 'Quad' means four!

20. What shape has 3 sides and 3 corners?

- A. Rectangle
- B. Triangle ✓**
- C. Square
- D. Circle

Explanation: A triangle has 3 sides and 3 corners. It is the polygon with the fewest sides.

Medium (20 questions)

1. A shape has 4 sides but they are NOT all equal. It has 4 square corners. What could it be?

- A. Square
- B. Rectangle ✓**
- C. Triangle
- D. Pentagon

Explanation: A rectangle has 4 sides and 4 square corners, but its sides do not all have to be equal. A square would need all 4 sides equal.

2. Which of these shapes are quadrilaterals? (Shapes with exactly 4 sides.)

- A. Triangle and hexagon
- B. Square and rectangle ✓**
- C. Circle and oval
- D. Pentagon and octagon

Explanation: Squares and rectangles both have exactly 4 sides, so they are both quadrilaterals.

3. A polygon has 5 sides. It is called a ____.

- A. Hexagon
- B. Pentagon ✓**
- C. Octagon
- D. Quadrilateral

Explanation: A polygon with 5 sides is called a pentagon. 'Penta' means five.

4. How many more sides does a hexagon have than a triangle?

- A. 2
- B. 3 ✓**
- C. 4
- D. 6

Explanation: A hexagon has 6 sides and a triangle has 3 sides. $6 - 3 = 3$ more sides.

5. I have 4 sides. Two of my sides are long and two are short. All my corners are square corners. What am I?

- A. Square
- B. Triangle
- C. Rectangle ✓**
- D. Trapezoid

Explanation: A rectangle has 4 square corners, with 2 long sides and 2 short sides.

6. Which shape has MORE corners: a pentagon or a square?

- A. Pentagon ✓**
- B. Square
- C. They are equal
- D. Neither has corners

Explanation: A pentagon has 5 corners and a square has 4 corners. $5 > 4$, so the pentagon has more.

7. A ____ has 8 sides and 8 corners.

- A. Hexagon
- B. Pentagon
- C. Heptagon
- D. Octagon ✓**

Explanation: An octagon has 8 sides and 8 corners. 'Octa' means eight.

8. Jake says a circle is a polygon because it is a flat shape. Is he correct?

- A. Yes, all flat shapes are polygons
- B. No, polygons must have straight sides ✓**
- C. Yes, circles have curved sides
- D. No, polygons must be round

Explanation: Jake is wrong. A polygon must have straight sides. A circle has a curved edge, so it is NOT a polygon.

9. How many more sides does an octagon have than a rectangle?

- A. 2
- B. 3
- C. 4 ✓**
- D. 6

Explanation: An octagon has 8 sides and a rectangle has 4 sides. $8 - 4 = 4$ more sides.

10. Which of these is TRUE about every square?

- A. It has 3 sides
- B. It has sides of different lengths
- C. It is also a rectangle ✓**
- D. It has no corners

Explanation: Every square is also a rectangle because it has 4 sides and 4 right angles. But a square's sides are all the same length.

11. Sort these from fewest sides to most: hexagon, triangle, square.

- A. Triangle, square, hexagon ✓**
- B. Square, triangle, hexagon
- C. Hexagon, square, triangle
- D. Triangle, hexagon, square

Explanation: Triangle has 3, square has 4, hexagon has 6. So the order is triangle, square, hexagon.

12. A shape has fewer than 5 sides but more than 3 sides. It has 4 equal sides. What is it?

- A. Triangle
- B. Rectangle
- C. Square ✓**
- D. Pentagon

Explanation: More than 3 and fewer than 5 means it has 4 sides. With 4 equal sides, it must be a square.

13. What is the same about a square and a rectangle?

- A. All sides are equal
- B. They both have 4 sides and 4 right angles ✓**
- C. They both have 3 corners
- D. They are both round

Explanation: Both a square and a rectangle have 4 sides and 4 right angles. The difference is a square's sides are all equal.

14. If you put 2 triangles together along one side, how many sides could the new shape have?

- A. 3
- B. 4 ✓**
- C. 6
- D. 2

Explanation: When you join 2 triangles along one side, the joined sides disappear inside. You can make a shape with 4 sides (like a parallelogram).

15. Emma drew a shape with more sides than a square but fewer sides than a hexagon. What shape could she have drawn?

- A. Triangle
- B. Rectangle
- C. Pentagon ✓**
- D. Octagon

Explanation: A square has 4 sides and a hexagon has 6. A pentagon has 5 sides, which is between 4 and 6.

16. A shape has the same number of sides as corners. Is this always true for polygons?

- A. Yes, always ✓**
- B. No, never
- C. Only for triangles
- D. Only for squares

Explanation: Yes! Every polygon has the same number of sides and corners. A triangle has 3 of each, a square has 4 of each, and so on.

17. Which shape does NOT belong: square, rectangle, triangle, rhombus?

- A. Square
- B. Rectangle
- C. Triangle ✓**
- D. Rhombus

Explanation: The triangle does not belong. The square, rectangle, and rhombus are all quadrilaterals (4 sides). A triangle has only 3 sides.

18. How many sides do 2 separate triangles have in all?

- A. 3
- B. 4
- C. 5
- D. 6 ✓**

Explanation: Each triangle has 3 sides. $3 + 3 = 6$ sides in all.

19. I am a polygon. I have more sides than a pentagon but fewer sides than an octagon. I could be a ____.

- A. Square
- B. Hexagon ✓**
- C. Triangle
- D. Rectangle

Explanation: A pentagon has 5 sides and an octagon has 8. A hexagon has 6 sides (or a heptagon has 7). Both fit between 5 and 8.

20. Lily says a rectangle has 2 pairs of equal sides. Is she right?

A. Yes -- 2 long sides match and 2 short sides match ✓

B. No -- all 4 sides are equal

C. No -- no sides are equal

D. Yes -- but only if it is big

Explanation: Lily is correct! A rectangle has 2 pairs of equal sides: the 2 long sides are equal and the 2 short sides are equal.

Hard (20 questions)

1. Tom says all rectangles are squares. Sara says all squares are rectangles. Who is correct?

A. Tom is correct

B. Sara is correct ✓

C. Both are correct

D. Neither is correct

Explanation: Sara is correct. Every square IS a rectangle (4 sides, 4 right angles). But not every rectangle is a square -- rectangles can have sides of different lengths.

2. A shape has more than 4 sides and fewer than 7 sides. All its sides are equal length. What shape could it be?

A. Square

B. Regular pentagon or regular hexagon ✓

C. Triangle

D. Octagon

Explanation: More than 4 and fewer than 7 means 5 or 6 sides. With all sides equal, it could be a regular pentagon (5) or regular hexagon (6).

3. You cut a square in half with a straight diagonal line from corner to corner. What 2 shapes do you get?

A. 2 squares

B. 2 rectangles

C. 2 triangles ✓

D. 1 triangle and 1 rectangle

Explanation: Cutting a square diagonally makes 2 triangles. Each triangle has 3 sides: 2 sides from the square and 1 from the cut.

4. Ben says a triangle can have 4 corners if you make it big enough. What is wrong with his thinking?

A. He is actually correct

B. A triangle always has 3 corners no matter the size ✓

C. Bigger triangles have 5 corners

D. Only small triangles have 3 corners

Explanation: Size does not change the number of corners. A triangle ALWAYS has exactly 3 sides and 3 corners, whether it is tiny or huge.

5. How many total sides do a triangle, a square, and a pentagon have together?

A. 10

B. 11

C. 12 ✓

D. 13

Explanation: Triangle = 3, square = 4, pentagon = 5. Add them: $3 + 4 + 5 = 12$ sides total.

6. Mia drew a quadrilateral with NO sides equal. Which shape did she definitely NOT draw?

A. Trapezoid

B. Square ✓

C. Irregular quadrilateral

D. Any 4-sided shape

Explanation: A square has all 4 sides equal, so Mia definitely did NOT draw a square. Her shape has no equal sides.

7. If you remove one side from a square and close the shape, what polygon do you get?

- A. Rectangle
- B. Triangle ✓**
- C. Pentagon
- D. It is not possible

Explanation: If you remove one side of a square and connect the remaining corners, you get a triangle with 3 sides.

8. Kai says a circle has 1 side because it has 1 curved edge. Emma says it has 0 sides because sides must be straight. Who is correct for polygons?

- A. Kai is correct
- B. Emma is correct ✓**
- C. Both are correct
- D. Neither is correct

Explanation: In geometry, polygon sides must be straight line segments. Since a circle has no straight sides, Emma is correct -- it has 0 polygon sides.

9. A shape has twice as many sides as a triangle. What shape is it?

- A. Square
- B. Pentagon
- C. Hexagon ✓**
- D. Octagon

Explanation: A triangle has 3 sides. Twice as many: $3 \times 2 = 6$ sides. A hexagon has 6 sides.

10. Sam drew a polygon. It has 2 more sides than a square but 2 fewer sides than an octagon. What did he draw?

- A. Pentagon
- B. Hexagon ✓**
- C. Heptagon
- D. Triangle

Explanation: A square has 4 sides. $4 + 2 = 6$. An octagon has 8 sides. $8 - 2 = 6$. Sam drew a hexagon (6 sides). Both clues match!

11. Rosa says a rectangle can never be a square. Is she right? Explain.

- A. Yes, they are completely different shapes
- B. No -- when all 4 sides are equal, a rectangle IS a square ✓**
- C. Yes, squares have more sides
- D. No -- rectangles always have 5 sides

Explanation: Rosa is wrong. A rectangle CAN be a square! When a rectangle has all 4 sides the same length, it is also a square. A square is a special rectangle.

12. You put 2 identical squares side by side touching along one full side. What shape do you make?

- A. A bigger square
- B. A rectangle ✓**
- C. A triangle
- D. A hexagon

Explanation: Two squares side by side make a rectangle. The new shape is twice as long as it is tall, with 4 sides and 4 right angles.

13. Jaya drew a shape with 3 fewer sides than an octagon. How many corners does her shape have?

- A. 3
- B. 4
- C. 5 ✓**
- D. 6

Explanation: An octagon has 8 sides. $8 - 3 = 5$ sides. A polygon with 5 sides (pentagon) has 5 corners. Sides and corners are always equal!

14. Which statement is FALSE?

- A. All squares are rectangles
- B. All rectangles are quadrilaterals
- C. All quadrilaterals are squares ✓**
- D. All squares are quadrilaterals

Explanation: 'All quadrilaterals are squares' is FALSE. A quadrilateral is any 4-sided shape. Many quadrilaterals (like rectangles and trapezoids) are NOT squares.

15. You cut a rectangle in half with a straight line from the middle of the top to the middle of the bottom. What 2 shapes do you get?

- A. 2 triangles
- B. 2 smaller rectangles ✓**
- C. 1 square and 1 rectangle
- D. 2 pentagons

Explanation: Cutting a rectangle straight down the middle gives you 2 smaller rectangles, each with 4 sides and 4 right angles.

16. Leo says: 'I drew a polygon with no corners.' Is this possible?

- A. Yes, a circle
- B. Yes, an oval
- C. No -- every polygon has corners ✓**
- D. Yes, if it is very small

Explanation: Every polygon must have corners (vertices). If a shape has no corners, it is not a polygon. A circle has no corners, but it is NOT a polygon.

17. A hexagon has how many more sides than a triangle, and how many fewer sides than an octagon?

- A. 3 more and 2 fewer ✓**
- B. 2 more and 3 fewer
- C. 4 more and 1 fewer
- D. 3 more and 3 fewer

Explanation: Hexagon = 6. Triangle = 3. $6 - 3 = 3$ more. Octagon = 8. $8 - 6 = 2$ fewer. So 3 more than a triangle and 2 fewer than an octagon.

18. Ava traced around a square tile 3 times without overlapping. She made a new shape. How many sides could it have if the squares share full sides?

- A. 8 ✓**
- B. 12
- C. 6
- D. 4

Explanation: Three squares in an L-shape make a polygon with 8 sides. Shared sides become interior and do not count as outside edges.

19. What is the DIFFERENCE between the number of sides on a pentagon and the number of corners on a hexagon?

- A. 0
- B. 1 ✓**
- C. 2
- D. 3

Explanation: A pentagon has 5 sides. A hexagon has 6 corners. $6 - 5 = 1$. The difference is 1.

20. Maya and Alex each draw a quadrilateral. Maya's has all equal sides. Alex's has NO equal sides. Can they both be correct?

- A. No -- quadrilaterals must have equal sides
- B. No -- quadrilaterals never have equal sides
- C. Yes -- a quadrilateral just needs 4 sides ✓**
- D. Yes -- but only if they are rectangles

Explanation: Yes! A quadrilateral is ANY shape with 4 sides. Maya could draw a square (all equal) and Alex could draw an irregular quadrilateral (no equal sides). Both are valid.

Lesson 2: Solid Shapes

Easy (20 questions)

1. What shape is a ball?

- A. Cube
- B. Cone
- C. Sphere ✓**
- D. Cylinder

Explanation: A ball is shaped like a sphere -- it is perfectly round in every direction.

2. What 3D shape does a soup can look like?

- A. Sphere
- B. Cone
- C. Cube
- D. Cylinder ✓**

Explanation: A soup can is shaped like a cylinder -- it has 2 flat circular faces and 1 curved surface.

3. How many faces does a cube have?

- A. 4
- B. 6 ✓**
- C. 8
- D. 12

Explanation: A cube has 6 faces. Think of a dice -- top, bottom, front, back, left, right.

4. What 3D shape looks like an ice cream cone?

- A. Cylinder
- B. Sphere
- C. Cone ✓**
- D. Rectangular prism

Explanation: An ice cream cone is shaped like a cone -- it has a point at one end and a circle at the other.

5. What 3D shape does a cereal box look like?

- A. Cube
- B. Cone
- C. Rectangular prism ✓**
- D. Sphere

Explanation: A cereal box is a rectangular prism -- it has 6 flat rectangular faces.

6. How many flat faces does a sphere have?

- A. 0 ✓**
- B. 1
- C. 2
- D. 6

Explanation: A sphere has 0 flat faces. It is completely curved all around, like a ball.

7. Which 3D shape can roll AND slide?

- A. Cube
- B. Cylinder ✓**
- C. Rectangular prism
- D. Triangular prism

Explanation: A cylinder can roll (on its curved side) and slide (on its flat faces).

8. How many vertices (corners) does a cube have?

- A. 4
- B. 6
- C. 8 ✓**
- D. 12

Explanation: A cube has 8 vertices (corners). Think of 4 corners on top and 4 corners on bottom.

9. What shape are all 6 faces of a cube?

- A. Rectangles
- B. Triangles
- C. Squares ✓**
- D. Circles

Explanation: All 6 faces of a cube are squares. Each face is the same size.

10. Which solid shape has NO edges and NO vertices?

- A. Cube
- B. Cone
- C. Sphere ✓**
- D. Rectangular prism

Explanation: A sphere has no edges and no vertices. It is completely smooth and round.

11. How many edges does a cube have?

- A. 6
- B. 8
- C. 10
- D. 12 ✓**

Explanation: A cube has 12 edges -- the lines where two faces meet.

12. What 3D shape does a basketball look like?

- A. Cylinder
- B. Cube
- C. Sphere ✓**
- D. Cone

Explanation: A basketball is shaped like a sphere -- round in every direction.

13. A cone has how many flat faces?

- A. 0
- B. 1 ✓**
- C. 2
- D. 3

Explanation: A cone has 1 flat face (the circular base) and 1 curved surface that goes up to a point.

14. Which 3D shape can only roll (not slide on a flat face)?

- A. Cube
- B. Cylinder
- C. Sphere ✓**
- D. Rectangular prism

Explanation: A sphere can only roll because it has no flat faces to slide on.

15. What 3D shape does a party hat look like?

- A. Sphere
- B. Cylinder
- C. Cone ✓**
- D. Cube

Explanation: A party hat is shaped like a cone -- wide at the bottom and pointed at the top.

16. A rectangular prism has how many faces?

- A. 4
- B. 5
- C. 6 ✓**
- D. 8

Explanation: A rectangular prism has 6 faces, just like a cube. But its faces can be rectangles of different sizes.

17. Which shape has a point (apex) at the top?

- A. Cube
- B. Cylinder
- C. Sphere
- D. Cone ✓**

Explanation: A cone has a point (called the apex) at the top and a flat circular base.

18. What 3D shape does a tissue box look like?

- A. Cube
- B. Rectangular prism ✓**
- C. Cylinder
- D. Cone

Explanation: A tissue box is shaped like a rectangular prism -- it has 6 rectangular faces.

19. How many flat faces does a cylinder have?

- A. 0
- B. 1
- C. 2 ✓**
- D. 3

Explanation: A cylinder has 2 flat circular faces -- one on the top and one on the bottom.

20. True or false: A cube can stack on top of another cube.

- A. True ✓**
- B. False
- C. Only if they are the same size
- D. Only small cubes

Explanation: True! Cubes have flat faces, so they stack easily. Flat faces let shapes sit on top of each other.

Medium (20 questions)

1. How many more edges does a rectangular prism have than a triangular prism?

- A. 2
- B. 3 ✓**
- C. 4
- D. 6

Explanation: A rectangular prism has 12 edges. A triangular prism has 9 edges. $12 - 9 = 3$ more edges.

2. A shape has 6 faces, 8 vertices, and 12 edges. All faces are the same size. What is it?

- A. Rectangular prism
- B. Cube ✓**
- C. Triangular prism
- D. Cylinder

Explanation: A cube has 6 equal square faces, 8 vertices, and 12 edges. A rectangular prism also has 6 faces, 8 vertices, and 12 edges, but its faces are not all the same size.

3. Which 3D shape has triangular faces? Choose the best answer.

- A. Cube
- B. Cylinder
- C. Triangular prism ✓**
- D. Sphere

Explanation: A triangular prism has 2 triangular faces (on the ends) and 3 rectangular faces (on the sides).

4. What 2D shape would you see if you traced the bottom of a cylinder?

- A. Square
- B. Triangle
- C. Circle ✓**
- D. Rectangle

Explanation: The base of a cylinder is a circle. If you set it down and traced around the bottom, you would draw a circle.

5. Which solid can slide but NOT roll?

- A. Sphere
- B. Cylinder
- C. Cube ✓**
- D. Cone

Explanation: A cube has only flat faces, so it can slide but cannot roll. You need a curved surface to roll.

6. How many faces does a triangular prism have?

- A. 3
- B. 4
- C. 5 ✓**
- D. 6

Explanation: A triangular prism has 5 faces: 2 triangles on the ends and 3 rectangles on the sides.

7. Amy says a cone has 2 vertices. Is she correct?

- A. Yes, one on top and one on bottom
- B. No, a cone has 1 vertex (the point) ✓**
- C. No, a cone has 0 vertices
- D. Yes, just like a cylinder

Explanation: A cone has just 1 vertex -- the pointy tip at the top. The circular base has no corners.

8. If you trace all the faces of a cube, what shapes do you draw?

- A. 6 circles
- B. 6 squares ✓**
- C. 4 squares and 2 rectangles
- D. 6 triangles

Explanation: A cube has 6 faces and each face is a square. So you would draw 6 squares.

9. Which has MORE vertices: a cube or a triangular prism?

- A. Cube (8 vs 6) ✓**
- B. Triangular prism (6 vs 8)
- C. They are equal
- D. Neither has vertices

Explanation: A cube has 8 vertices and a triangular prism has 6 vertices. The cube has 2 more.

10. What 2D shape do you see if you look straight at one end of a triangular prism?

- A. Square
- B. Circle
- C. Triangle ✓**
- D. Rectangle

Explanation: The end faces of a triangular prism are triangles. Looking straight at one end, you see a triangle.

11. How are a cube and a rectangular prism ALIKE?

- A. Both have 6 faces, 8 vertices, 12 edges ✓**
- B. Both have all square faces
- C. Both can roll
- D. Both have curved surfaces

Explanation: Both a cube and a rectangular prism have 6 faces, 8 vertices, and 12 edges. The difference is a cube's faces are all squares.

12. Which solid shape has exactly 1 curved surface and 0 flat faces?

- A. Cone
- B. Cylinder
- C. Sphere ✓**
- D. Cube

Explanation: A sphere has 1 curved surface and no flat faces at all. A cone has 1 flat face and a cylinder has 2.

13. How many edges does a triangular prism have?

- A. 6
- B. 8
- C. 9 ✓**
- D. 12

Explanation: A triangular prism has 9 edges: 3 on each triangular face (6 total) plus 3 edges connecting them.

14. A ____ has 5 faces, 6 vertices, and 9 edges.

- A. Cube
- B. Rectangular prism
- C. Triangular prism ✓**
- D. Cone

Explanation: A triangular prism has 5 faces, 6 vertices, and 9 edges.

15. Which 3D shape could you make by rolling a rectangle into a tube and adding 2 circle caps?

- A. Cone
- B. Cube
- C. Cylinder ✓**
- D. Sphere

Explanation: Rolling a rectangle into a tube gives a curved surface. Adding 2 circle caps on the ends makes a cylinder.

16. How are a cone and a cylinder DIFFERENT?

- A. A cone has 1 flat face; a cylinder has 2 ✓**
- B. A cone has more edges
- C. A cylinder has no curved surface
- D. They are exactly the same

Explanation: A cone has 1 flat circular face and comes to a point. A cylinder has 2 flat circular faces and no point.

17. Which of these real objects is shaped most like a rectangular prism?

- A. A basketball
- B. A brick ✓**
- C. A funnel
- D. A marble

Explanation: A brick is shaped like a rectangular prism -- it has 6 flat rectangular faces.

18. Sort from fewest faces to most: sphere, triangular prism, cube.

- A. Sphere, triangular prism, cube ✓**
- B. Cube, triangular prism, sphere
- C. Triangular prism, sphere, cube
- D. Sphere, cube, triangular prism

Explanation: Sphere has 0 flat faces, triangular prism has 5, cube has 6. Order: sphere, triangular prism, cube.

19. What flat shape do you see if you look at the top of a cylinder?

- A. Square
- B. Triangle
- C. Rectangle
- D. Circle ✓**

Explanation: The top face of a cylinder is a circle. Looking straight down, you see a circle.

20. If you stack 3 cubes in a tower, how many faces can you see from the outside? (Not counting the bottom or hidden faces.)

- A. 12
- B. 13 ✓**
- C. 14
- D. 18

Explanation: Each cube shows 4 side faces ($4 \times 3 = 12$). The top cube shows its top face too ($12 + 1 = 13$). The touching faces and the bottom are hidden.

Hard (20 questions)

1. Tom says a cylinder has 2 faces and 0 vertices. Sara says it has 3 faces and 0 vertices. Who is closer to correct?

A. Tom -- it has 2 flat faces and 0 vertices ✓

B. Sara -- it has 3 surfaces total

C. Neither -- it has 6 faces

D. Both -- they said the same thing

Explanation: A cylinder has 2 flat circular faces and 0 vertices. It also has 1 curved surface, but that is not a 'face.' Tom is correct about the 2 flat faces and 0 vertices.

2. If you unfold a cube and lay it flat, how many squares do you see?

A. 4

B. 5

C. 6 ✓

D. 8

Explanation: A cube has 6 square faces. When you unfold (make a net of) a cube, you see all 6 squares laid flat.

3. Maya says a sphere has 1 face because the whole outside is a face. Jake says it has 0 faces because faces must be flat. Who is correct?

A. Maya

B. Jake ✓

C. Both are correct

D. Neither is correct

Explanation: Jake is correct. In geometry, a 'face' is a flat surface. A sphere has no flat surfaces, so it has 0 faces. It has 1 curved surface.

4. How many more edges does a cube have than a triangular prism?

A. 1

B. 2

C. 3 ✓

D. 4

Explanation: A cube has 12 edges. A triangular prism has 9 edges. $12 - 9 = 3$ more edges.

5. A 3D shape has 5 faces. Two faces are triangles and three faces are rectangles. What is it?

A. Cube

B. Rectangular prism

C. Triangular prism ✓

D. Cone

Explanation: A triangular prism has exactly 2 triangular faces (the ends) and 3 rectangular faces (the sides) = 5 faces total.

6. Leo stacks a cone on top of a cylinder. How many flat faces are now hidden (touching each other)?

A. 0

B. 1

C. 2 ✓

D. 3

Explanation: The top circle of the cylinder and the bottom circle of the cone are pressed together. That is 2 flat faces hidden.

7. Which has more vertices: 2 triangular prisms or 1 cube?

A. 2 triangular prisms (12 vs 8) ✓

B. 1 cube (8 vs 12)

C. They are equal

D. Neither has vertices

Explanation: Each triangular prism has 6 vertices. Two of them: $6 + 6 = 12$. A cube has 8 vertices. $12 > 8$, so 2 triangular prisms have more.

8. Ava says you can stack spheres as easily as cubes because both are 3D shapes. What is wrong with her thinking?

A. Nothing -- she is correct

B. Spheres have no flat faces so they roll and do not stack well ✓

C. Cubes are not 3D shapes

D. Spheres are heavier than cubes

Explanation: Cubes stack easily because they have flat faces. Spheres have NO flat faces -- they roll! Being 3D does not mean a shape can stack.

9. You trace every face of a rectangular prism. You get 4 rectangles and 2 squares. How many shapes did you trace in all?

A. 4

B. 5

C. 6 ✓

D. 8

Explanation: 4 rectangles + 2 squares = 6 shapes. A rectangular prism always has 6 faces total.

10. A shape has twice as many faces as a triangular prism. It has 2 more faces than a cube. Does such a shape exist among common solids?

A. Yes -- a shape with 10 faces

B. No -- twice 5 is 10 but a cube has 6, and $6 + 2 = 8$, not 10 ✓

C. Yes -- a shape with 8 faces

D. No -- no solid has 10 faces

Explanation: Twice a triangular prism's faces = $2 \times 5 = 10$. But a cube has 6 faces, and $6 + 2 = 8$. Since 10 does not equal 8, no single shape can satisfy both clues.

11. You slice a cube straight through the middle, parallel to one face. What shape is the cut surface?

A. Triangle

B. Circle

C. Square ✓

D. Rectangle

Explanation: Slicing a cube parallel to one of its faces gives a square cross-section, the same size as the face.

12. Ben says a cone has 1 edge. Mia says it has 0 edges because edges must be straight. Who is correct for standard geometry?

A. Ben ✓

B. Mia

C. Both

D. Neither

Explanation: In standard elementary geometry, a cone is said to have 1 edge -- the circular edge where the flat base meets the curved surface. Ben is correct.

13. How many total faces do a cube and a triangular prism have together?

A. 10

B. 11 ✓

C. 12

D. 13

Explanation: A cube has 6 faces and a triangular prism has 5 faces. $6 + 5 = 11$ faces total.

14. A shape can roll, slide, AND has exactly 1 vertex. What is it?

A. Sphere

B. Cylinder

C. Cone ✓

D. Cube

Explanation: A cone can roll (curved surface), slide (flat base), and has exactly 1 vertex (the point at the top).

15. Rosa has 2 cubes. She glues them together on one full face. How many faces can you see on the outside now?

- A. 8
- B. 10 ✓**
- C. 12
- D. 6

Explanation: Each cube has 6 faces = 12 total. The 2 glued faces are hidden inside. $12 - 2 = 10$ faces visible on the outside.

16. Which 3D shape has the MOST edges: cube, triangular prism, or cone?

- A. Cube (12 edges) ✓**
- B. Triangular prism (9 edges)
- C. Cone (1 edge)
- D. They all have the same

Explanation: Cube has 12 edges, triangular prism has 9, cone has 1. The cube has the most edges.

17. Sam says if he cuts a rectangular prism in half, he gets 2 cubes. Is this always true?

- A. Yes, always
- B. No -- only if the prism is exactly twice as long as it is wide ✓**
- C. Yes, if you cut it straight
- D. No -- you always get triangles

Explanation: Only when the rectangular prism is exactly twice as long as it is wide and tall (and width equals height). Otherwise, you get 2 smaller rectangular prisms, not cubes.

18. A triangular prism has ____ more faces than a cone.

- A. 2
- B. 3
- C. 4 ✓**
- D. 5

Explanation: A triangular prism has 5 faces. A cone has 1 flat face. $5 - 1 = 4$ more faces.

19. If you look at a cube from directly above, what shape do you see?

- A. Rectangle
- B. Triangle
- C. Square ✓**
- D. Circle

Explanation: Looking straight down at a cube, you see only the top face, which is a square.

20. Kai builds a shape using 1 rectangle and 2 triangles. He folds them into a 3D shape. What did he make?

- A. Cube
- B. Rectangular prism
- C. Triangular prism ✓**
- D. Cone

Explanation: A triangular prism has 2 triangular end faces and rectangular side faces. With 1 rectangle and 2 triangles as a start, Kai is building a triangular prism (he would need 3 rectangles total for all sides).

Lesson 3: Mirror Shapes

Easy (20 questions)

1. Does a heart shape have a line of symmetry?

- A. Yes ✓**
- B. No
- C. Only if it is big
- D. Only if it is red

Explanation: Yes! A heart has 1 line of symmetry straight down the middle. The left and right halves match.

2. If you fold a square in half, do both sides match exactly?

A. Yes ✓

B. No

C. Only sometimes

D. Only if you fold it a special way

Explanation: Yes! A square has lines of symmetry. Folding it in half (top to bottom, left to right, or corner to corner) gives matching halves.

3. Two triangles can be put together to make a ____.

A. Circle

B. Square ✓

C. Pentagon

D. Sphere

Explanation: Two identical right triangles can be put together to make a square (or a rectangle or parallelogram, depending on the triangles).

4. What does 'line of symmetry' mean?

A. A line that makes a shape bigger

B. A line that folds a shape into two matching halves ✓

C. A line that goes around a shape

D. A line that makes a shape 3D

Explanation: A line of symmetry divides a shape into two halves that are mirror images -- they match exactly when folded.

5. Does a circle have a line of symmetry?

A. Yes ✓

B. No

C. Only a big circle

D. Only half a circle

Explanation: Yes! A circle has many lines of symmetry. Any line through the center divides it into two matching halves.

6. Which shape can you make by putting 2 squares side by side?

A. Triangle

B. Rectangle ✓

C. Circle

D. Pentagon

Explanation: Two squares side by side make a rectangle that is twice as long as it is tall.

7. Does the letter A have a line of symmetry?

A. Yes, straight down the middle ✓

B. No, it does not

C. Yes, across the top

D. Only capital A

Explanation: The capital letter A has a line of symmetry straight down the middle. The left side mirrors the right side.

8. If you cut a rectangle in half from top to bottom, what 2 shapes do you get?

A. 2 triangles

B. 2 smaller rectangles ✓

C. 2 circles

D. 1 square and 1 triangle

Explanation: Cutting a rectangle straight down the middle gives you 2 smaller rectangles.

9. Does a square have more than one line of symmetry?

A. No, just 1

B. Yes, it has 2

C. Yes, it has 4 ✓

D. Squares have no lines of symmetry

Explanation: A square has 4 lines of symmetry: horizontal, vertical, and 2 diagonal.

10. What shape do you get when you put 4 small triangles together?

- A. A bigger triangle
- B. A circle
- C. A hexagon
- D. A bigger triangle or a square ✓**

Explanation: Four small triangles can be arranged to make a bigger triangle or a square, depending on how you place them.

11. Does a rectangle have a line of symmetry?

- A. Yes ✓**
- B. No
- C. Only a square rectangle
- D. Only a long rectangle

Explanation: Yes! A rectangle has 2 lines of symmetry: one horizontal and one vertical through the center.

12. If you fold a shape along a line of symmetry, what happens?

- A. The shape gets bigger
- B. Both halves match up perfectly ✓**
- C. The shape turns into a circle
- D. Nothing happens

Explanation: When you fold along a line of symmetry, both halves match up perfectly -- one half covers the other exactly.

13. You have 1 big square. You cut it into 4 equal pieces. What shape is each piece?

- A. Triangle
- B. Small square ✓**
- C. Circle
- D. Hexagon

Explanation: Cutting a square into 4 equal pieces (one horizontal cut, one vertical cut) gives 4 smaller squares.

14. Which of these shapes has NO line of symmetry?

- A. Square
- B. Circle
- C. Scalene triangle with all different sides ✓**
- D. Rectangle

Explanation: A scalene triangle (all sides different, all angles different) has no line of symmetry. The other shapes all have at least one.

15. Two rectangles placed end to end make a ____.

- A. Circle
- B. Longer rectangle ✓**
- C. Triangle
- D. Hexagon

Explanation: Two rectangles placed end to end make one longer rectangle.

16. Does a butterfly have symmetry?

- A. Yes, left and right sides match ✓**
- B. No, butterflies are not shapes
- C. Only big butterflies
- D. Only when flying

Explanation: Yes! A butterfly has symmetry. Its left wing mirrors its right wing, with a line of symmetry down the body.

17. If you put 6 triangles together at a point, what shape can you make?

- A. Square
- B. Rectangle
- C. Hexagon ✓**
- D. Pentagon

Explanation: Six equilateral triangles arranged around a center point make a hexagon.

18. Does an equilateral triangle (all sides equal) have a line of symmetry?

A. Yes ✓

B. No

C. Only if it is big

D. Triangles never have symmetry

Explanation: Yes! An equilateral triangle has 3 lines of symmetry -- one from each corner to the middle of the opposite side.

19. You split a circle in half. What are the two pieces called?

A. Quarter circles

B. Semicircles ✓

C. Triangles

D. Rectangles

Explanation: Half of a circle is called a semicircle. 'Semi' means half.

20. Which letter does NOT have a line of symmetry: B, O, X, or F?

A. B

B. O

C. X

D. F ✓

Explanation: The letter F has no line of symmetry. B has a horizontal line, O has many, and X has both vertical and horizontal.

Medium (20 questions)

1. How many lines of symmetry does a rectangle have (that is NOT a square)?

A. 0

B. 1

C. 2 ✓

D. 4

Explanation: A non-square rectangle has 2 lines of symmetry: one horizontal and one vertical through the center. Its diagonals are NOT lines of symmetry.

2. Two triangles are put together to make a rectangle. Each triangle has 3 sides. How many sides does the rectangle have?

A. 6

B. 5

C. 4 ✓

D. 3

Explanation: Even though the triangles had 6 sides total, the joined sides are hidden inside. The rectangle has only 4 sides.

3. Which has MORE lines of symmetry: a square or a non-square rectangle?

A. A square (4 vs 2) ✓

B. A rectangle (2 vs 4)

C. They are equal

D. Neither has any

Explanation: A square has 4 lines of symmetry (horizontal, vertical, and 2 diagonals). A non-square rectangle has only 2 (horizontal and vertical).

4. You cut a square diagonally. What 2 shapes do you get?

A. 2 squares

B. 2 rectangles

C. 2 triangles ✓

D. 1 triangle and 1 rectangle

Explanation: Cutting a square along a diagonal creates 2 right triangles. Each triangle has the diagonal as its longest side.

5. Emma says the letter S has a line of symmetry. Is she correct?

- A. Yes, it has a vertical line
- B. Yes, it has a horizontal line
- C. No, S has no line of symmetry ✓**
- D. Yes, it has a diagonal line

Explanation: The letter S has no line of symmetry. If you fold it any way, the halves do not match. (S does have rotational symmetry, but not a line of symmetry.)

6. You arrange 2 small squares into an L-shape. How many sides does the L-shape have?

- A. 4
- B. 5
- C. 6 ✓**
- D. 8

Explanation: An L-shape made from 2 squares has 6 sides. The shared edge becomes interior, creating extra outside corners.

7. How many lines of symmetry does an equilateral triangle have?

- A. 0
- B. 1
- C. 2
- D. 3 ✓**

Explanation: An equilateral triangle has 3 lines of symmetry -- one from each vertex to the midpoint of the opposite side.

8. If you put 2 identical right triangles together along their longest sides, what shape can you make?

- A. A bigger triangle
- B. A rectangle ✓**
- C. A pentagon
- D. A hexagon

Explanation: Two identical right triangles placed together along their longest sides (hypotenuse) form a rectangle.

9. A shape has exactly 1 line of symmetry. Which could it be?

- A. Square
- B. Circle
- C. Isosceles triangle ✓**
- D. Regular hexagon

Explanation: An isosceles triangle (2 equal sides) has exactly 1 line of symmetry, from the top vertex straight down to the base.

10. You cut a hexagon into 6 equal triangles from the center. How many sides does each triangle have?

- A. 2
- B. 3 ✓**
- C. 4
- D. 6

Explanation: Each piece is a triangle, so each has 3 sides -- even though they came from a hexagon.

11. Does the letter H have a line of symmetry?

- A. Yes -- vertical only
- B. Yes -- horizontal only
- C. Yes -- both vertical and horizontal ✓**
- D. No line of symmetry

Explanation: The capital letter H has 2 lines of symmetry: one vertical (left matches right) and one horizontal (top matches bottom).

12. What shape do you get if you put 4 identical squares together in a 2-by-2 grid?

- A. A rectangle
- B. A bigger square ✓**
- C. A triangle
- D. A pentagon

Explanation: Four squares in a 2x2 arrangement make one bigger square. Each side is twice as long as one small square's side.

13. Amy folds a rectangle in half along its line of symmetry (top to bottom). The folded shape is a ____.

- A. Triangle
- B. Smaller rectangle ✓**
- C. Circle
- D. Pentagon

Explanation: Folding a rectangle in half along its horizontal center gives a smaller rectangle, half the height of the original.

14. Which shape has MORE lines of symmetry: an equilateral triangle or a non-square rectangle?

- A. Equilateral triangle (3 vs 2) ✓**
- B. Non-square rectangle (2 vs 3)
- C. They are equal
- D. Neither has any

Explanation: An equilateral triangle has 3 lines of symmetry. A non-square rectangle has 2. The triangle has more!

15. You have a big triangle. You draw a line from one side to another, parallel to the base. How many shapes do you now have?

- A. 1
- B. 2 ✓**
- C. 3
- D. 4

Explanation: Drawing a line parallel to the base cuts the triangle into 2 shapes: a smaller triangle on top and a trapezoid on the bottom.

16. Can you split a rectangle into 2 equal triangles?

- A. Yes, by cutting corner to corner diagonally ✓**
- B. No, rectangles cannot make triangles
- C. Only if it is a square
- D. Yes, but only into 4 triangles

Explanation: Yes! Drawing a diagonal line from one corner to the opposite corner splits any rectangle into 2 equal right triangles.

17. How many lines of symmetry does a regular pentagon (all sides equal) have?

- A. 0
- B. 2
- C. 5 ✓**
- D. 10

Explanation: A regular pentagon has 5 lines of symmetry -- one from each vertex to the midpoint of the opposite side.

18. Which of these cannot be made by combining 2 identical squares?

- A. A rectangle
- B. An L-shape
- C. A triangle ✓**
- D. A bigger shape with 4 or more sides

Explanation: Two squares cannot make a triangle. Squares have right angles and straight sides that combine into rectangles or L-shapes, not triangles.

19. A shape and its mirror image look exactly the same. What must the shape have?

- A. Curved sides
- B. At least 6 sides
- C. At least one line of symmetry ✓**
- D. No corners

Explanation: If a shape looks the same as its mirror image, it has at least one line of symmetry. The mirror line is the line of symmetry.

20. You arrange 3 equilateral triangles in a row, sharing sides. How many sides does the new shape have?

- A. 3
- B. 4
- C. 5 ✓**
- D. 9

Explanation: Three triangles in a row (forming a trapezoid-like strip) have 5 outer sides. The shared edges are interior and do not count.

Hard (20 questions)

1. How many lines of symmetry does a regular hexagon have? More or fewer than a square?

A. 6 -- more than a square's 4 ✓

B. 4 -- the same as a square

C. 2 -- fewer than a square

D. 8 -- more than a square

Explanation: A regular hexagon has 6 lines of symmetry. A square has 4. So the hexagon has MORE lines of symmetry.

2. Can you fold a rectangle in half to make two squares? When is this possible?

A. Always possible

B. Only when the rectangle is exactly twice as long as it is wide ✓

C. Never possible

D. Only with big rectangles

Explanation: You can fold a rectangle into 2 squares ONLY when the length is exactly twice the width. Then each half is a perfect square.

3. Mia says cutting a square corner-to-corner gives 2 right triangles. She also says each triangle has 2 lines of symmetry. Is she fully correct?

A. Yes, both statements are correct

B. The first part is correct, but each triangle has only 1 line of symmetry ✓

C. The first part is correct, but each triangle has 0 lines of symmetry

D. Both statements are wrong

Explanation: Mia is right that cutting diagonally gives 2 right triangles. But each is an isosceles right triangle with only 1 line of symmetry (along its own line from the right angle to the hypotenuse midpoint).

4. Ben says all shapes with 4 lines of symmetry must be squares. Can you find a counterexample?

A. He is correct -- only squares have 4 lines

B. No, but a circle has more

C. Yes -- a regular octagon also has more than 4

D. The plus sign (+) shape has 4 lines and is not a square ✓

Explanation: A plus sign (+) shape has 4 lines of symmetry but is not a square. Ben is wrong -- other shapes can have 4 lines of symmetry too.

5. You cut a regular hexagon from center to each corner. How many triangles do you get, and what kind are they?

A. 3 right triangles

B. 6 equilateral triangles ✓

C. 4 isosceles triangles

D. 6 right triangles

Explanation: Cutting from center to each of the 6 corners makes 6 equilateral triangles. Each has 3 equal sides.

6. A shape is made from 4 identical squares arranged in a T-shape. How many lines of symmetry does it have?

A. 0

B. 1 ✓

C. 2

D. 4

Explanation: A T-shaped tetromino has exactly 1 line of symmetry -- a vertical line down the center of the top of the T. It is not symmetric left-right if flipped another way.

7. Sam says: 'If I fold a shape in half and the pieces do not match, the shape has no symmetry.' Is he correct?

A. Yes, that fold proves no symmetry

B. No -- he might have folded along the wrong line ✓

C. No -- all shapes have symmetry

D. Yes -- you only need to try one fold

Explanation: Sam is wrong. He may have folded along a line that is not a line of symmetry. He needs to try ALL possible fold lines before concluding there is no symmetry.

8. You combine a square and a triangle (sharing one full side). How many sides does the new shape have?

- A. 4
- B. 5 ✓**
- C. 6
- D. 7

Explanation: A square has 4 sides and a triangle has 3 sides = 7 total. But they share 1 side (hidden inside): $7 - 2 = 5$ outside sides.

9. A regular polygon has the same number of lines of symmetry as it has sides. How many lines of symmetry does a regular octagon have?

- A. 4
- B. 6
- C. 8 ✓**
- D. 10

Explanation: A regular octagon has 8 sides, so it has 8 lines of symmetry. Regular polygons always have as many lines of symmetry as sides.

10. Rosa puts 2 regular hexagons together sharing one full side. How many sides does the new shape have?

- A. 8
- B. 10 ✓**
- C. 12
- D. 6

Explanation: Each hexagon has 6 sides = 12 total. They share 1 side, removing 2 edges from the outside: $12 - 2 = 10$ sides.

11. Jaya says a parallelogram (not a rectangle) has 2 lines of symmetry. Is she correct?

- A. Yes, horizontal and vertical
- B. Yes, both diagonals
- C. No, a non-rectangular parallelogram has 0 lines of symmetry ✓**
- D. No, it has exactly 1

Explanation: A parallelogram that is not a rectangle has 0 lines of symmetry. No fold line will make the two halves match. (It does have rotational symmetry, but no line symmetry.)

12. You have 12 identical equilateral triangles. What is the largest regular polygon you could tile with them?

- A. Square
- B. Regular hexagon ✓**
- C. Regular octagon
- D. Regular pentagon

Explanation: You cannot tile a square or octagon with equilateral triangles. But 6 equilateral triangles make a regular hexagon, and 12 make 2 hexagons or a larger hexagonal shape.

13. Kai cuts a rectangle diagonally both ways (making an X). How many pieces does he get, and what shape are they?

- A. 2 rectangles
- B. 4 identical triangles
- C. 4 triangles (2 pairs of identical ones) ✓**
- D. 3 triangles and 1 square

Explanation: Two diagonal cuts through a rectangle make 4 triangles. If the rectangle is not a square, you get 2 pairs: 2 wider triangles and 2 taller triangles.

14. How many total lines of symmetry do a square and an equilateral triangle have COMBINED?

- A. 5
- B. 6
- C. 7 ✓**
- D. 8

Explanation: A square has 4 lines of symmetry. An equilateral triangle has 3. Combined: $4 + 3 = 7$.

15. Alex says you can make a hexagon by arranging 3 rhombuses around a center point. Is this possible?

- A. No, rhombuses cannot make hexagons
- B. Yes, 3 identical rhombuses can tile a regular hexagon ✓**
- C. Only with squares, not rhombuses
- D. You would need 6 rhombuses

Explanation: Yes! Three identical rhombuses (each made of 2 equilateral triangles) can be arranged around a center to form a regular hexagon.

16. A shape has exactly 2 lines of symmetry. Which of these could it be?

- A. Equilateral triangle
- B. Regular hexagon
- C. Non-square rectangle ✓**
- D. Regular pentagon

Explanation: A non-square rectangle has exactly 2 lines of symmetry (horizontal and vertical). The triangle has 3, hexagon has 6, pentagon has 5.

17. You remove 1 small square from the corner of a big square. How many sides does the remaining shape have?

- A. 4
- B. 5
- C. 6 ✓**
- D. 8

Explanation: Removing a corner square creates a notch. The new shape has 6 sides: the 3 remaining big square sides, plus 2 small edges from the cut, plus 1 remaining edge.

18. Maya says a circle has exactly 1 line of symmetry. Leo says it has infinitely many. Who is correct?

- A. Maya
- B. Leo ✓**
- C. Both
- D. Neither

Explanation: Leo is correct. ANY line through the center of a circle is a line of symmetry. There are infinitely many such lines.

19. You compose a new shape using 1 square and 2 equilateral triangles (one on the top side, one on the bottom side). How many sides does it have?

- A. 4
- B. 6 ✓**
- C. 8
- D. 10

Explanation: The square has 4 sides. Each triangle adds 2 new sides but covers 1 square side. Total: $4 - 2$ (covered) + 4 (new triangle sides) = 6 sides.

20. What is the fewest number of lines of symmetry any regular polygon can have?

- A. 0
- B. 1
- C. 2
- D. 3 ✓**

Explanation: The simplest regular polygon is an equilateral triangle, which has 3 lines of symmetry. No regular polygon has fewer than 3, because the fewest sides a polygon can have is 3.

Unit Test Bank

Easy (20 questions)

1. How many sides does a pentagon have?

- A. 4
- B. 5 ✓**
- C. 6
- D. 8

Explanation: A pentagon has 5 sides. 'Penta' means five.

2. What 3D shape does a dice look like?

- A. Sphere
- B. Cylinder
- C. Cube ✓**
- D. Cone

Explanation: A dice is shaped like a cube -- it has 6 square faces.

3. Does a square have a line of symmetry?

- A. Yes ✓**
- B. No
- C. Only big squares
- D. Only small squares

Explanation: Yes! A square has 4 lines of symmetry. You can fold it in half many ways and both sides match.

4. How many corners does a hexagon have?

- A. 4
- B. 5
- C. 6 ✓**
- D. 8

Explanation: A hexagon has 6 corners (vertices) -- the same number as its sides.

5. What shape is a globe?

- A. Cube
- B. Cone
- C. Cylinder
- D. Sphere ✓**

Explanation: A globe is shaped like a sphere -- it is round in every direction.

6. If you fold a heart shape in half down the middle, do the two sides match?

- A. Yes ✓**
- B. No
- C. Only sometimes
- D. Only big hearts

Explanation: Yes! A heart has a line of symmetry down the middle. Both halves are mirror images.

7. How many sides does an octagon have?

- A. 6
- B. 7
- C. 8 ✓**
- D. 10

Explanation: An octagon has 8 sides. Think of a stop sign!

8. How many flat faces does a cone have?

- A. 0
- B. 1 ✓**
- C. 2
- D. 3

Explanation: A cone has 1 flat face (the circular base) and 1 curved surface going up to the point.

9. Two squares placed side by side make a ____.

- A. Triangle
- B. Rectangle ✓**
- C. Circle
- D. Pentagon

Explanation: Two squares side by side create a rectangle that is twice as long as it is tall.

10. What shape is round with no corners?

- A. Square
- B. Triangle
- C. Circle ✓**
- D. Hexagon

Explanation: A circle is round with no corners and no straight sides.

11. How many edges does a cube have?

- A. 6
- B. 8
- C. 10
- D. 12 ✓**

Explanation: A cube has 12 edges -- the lines where two faces meet.

12. Does the letter X have a line of symmetry?

- A. Yes ✓**
- B. No
- C. Only capital X
- D. Only in some fonts

Explanation: Yes! The letter X has lines of symmetry -- both vertical and horizontal.

13. What is the name of a shape with 3 sides?

- A. Square
- B. Triangle ✓**
- C. Pentagon
- D. Hexagon

Explanation: A triangle has 3 sides. 'Tri' means three.

14. Which 3D shape can roll but NOT slide?

- A. Cube
- B. Sphere ✓**
- C. Rectangular prism
- D. Triangular prism

Explanation: A sphere can only roll because it has no flat faces to slide on.

15. How many faces does a rectangular prism have?

- A. 4
- B. 5
- C. 6 ✓**
- D. 8

Explanation: A rectangular prism has 6 faces: top, bottom, front, back, and 2 sides.

16. What shapes do you get if you cut a rectangle from top to bottom down the middle?

- A. 2 triangles
- B. 2 smaller rectangles ✓**
- C. 2 circles
- D. 2 hexagons

Explanation: Cutting a rectangle straight down the middle gives 2 smaller rectangles.

17. How many vertices does a triangle have?

- A. 1
- B. 2
- C. 3 ✓**
- D. 4

Explanation: A triangle has 3 vertices (corners) -- one where each pair of sides meets.

18. What 3D shape does a can of beans look like?

- A. Cube
- B. Sphere
- C. Cone
- D. Cylinder ✓**

Explanation: A can of beans is shaped like a cylinder -- 2 flat circle faces and a curved surface.

19. A square has how many equal sides?

- A. 2
- B. 3
- C. 4 ✓**
- D. 0

Explanation: A square has 4 equal sides -- all the same length.

20. Does a circle have any corners?

- A. Yes, 1
- B. Yes, many
- C. No, 0 corners ✓**
- D. Yes, 4

Explanation: A circle has 0 corners (vertices). It is completely smooth and curved.

Medium (20 questions)

1. A shape has 4 sides and 4 right angles, but NOT all sides are equal. What is it?

- A. Square
- B. Non-square rectangle ✓**
- C. Triangle
- D. Hexagon

Explanation: A non-square rectangle has 4 sides and 4 right angles, but 2 sides are longer than the other 2.

2. How many more faces does a cube have than a triangular prism?

- A. 0
- B. 1 ✓**
- C. 2
- D. 3

Explanation: A cube has 6 faces and a triangular prism has 5. $6 - 5 = 1$ more face.

3. Which has more lines of symmetry: an equilateral triangle or a square?

- A. Equilateral triangle (3 vs 4)
- B. Square (4 vs 3) ✓**
- C. They are equal
- D. Neither has any

Explanation: A square has 4 lines of symmetry. An equilateral triangle has 3. The square has more.

4. How many more sides does an octagon have than a pentagon?

- A. 2
- B. 3 ✓**
- C. 4
- D. 5

Explanation: An octagon has 8 sides. A pentagon has 5 sides. $8 - 5 = 3$ more sides.

5. What 2D shape do you see if you trace the face of a cylinder?

- A. Square
- B. Triangle
- C. Rectangle
- D. Circle ✓**

Explanation: The flat face of a cylinder is a circle. Tracing it gives you a circle.

6. You cut a square from one corner to the opposite corner. What shapes do you get?

- A. 2 squares
- B. 2 rectangles
- C. 2 triangles ✓**
- D. 1 triangle and 1 rectangle

Explanation: A diagonal cut through a square creates 2 identical right triangles.

7. Maria says a quadrilateral always has to be a rectangle. Is she right?

- A. Yes, all quadrilaterals are rectangles
- B. No -- a quadrilateral is any shape with 4 sides ✓**
- C. Yes, if it has 4 corners
- D. No -- quadrilaterals have 5 sides

Explanation: A quadrilateral is ANY shape with 4 sides. Rectangles are quadrilaterals, but so are trapezoids, rhombuses, and other 4-sided shapes.

8. Which 3D shape has 2 triangular faces and 3 rectangular faces?

- A. Cube
- B. Rectangular prism
- C. Triangular prism ✓**
- D. Cylinder

Explanation: A triangular prism has 2 triangular end faces and 3 rectangular side faces = 5 faces total.

9. A non-square rectangle has ____ lines of symmetry.

- A. 0
- B. 1
- C. 2 ✓**
- D. 4

Explanation: A non-square rectangle has 2 lines of symmetry: one horizontal and one vertical through the center.

10. Sort from fewest sides to most: octagon, triangle, pentagon.

- A. Triangle, pentagon, octagon ✓**
- B. Pentagon, triangle, octagon
- C. Octagon, pentagon, triangle
- D. Triangle, octagon, pentagon

Explanation: Triangle = 3 sides, pentagon = 5, octagon = 8. Order: triangle, pentagon, octagon.

11. A ____ can slide but NOT roll because all its faces are flat.

- A. Sphere
- B. Cylinder
- C. Cone
- D. Cube ✓**

Explanation: A cube has only flat faces (no curved surfaces), so it can slide but never roll.

12. If 2 identical triangles share one full side, how many sides does the new shape have?

- A. 3
- B. 4 ✓**
- C. 5
- D. 6

Explanation: Each triangle has 3 sides (6 total). The shared side is hidden inside, removing 2 edges. $6 - 2 = 4$ sides.

13. I am a polygon. I have 1 more side than a pentagon and 2 fewer sides than an octagon. What am I?

- A. Square
- B. Hexagon ✓**
- C. Heptagon
- D. Triangle

Explanation: Pentagon = 5 sides. $5 + 1 = 6$. Octagon = 8 sides. $8 - 2 = 6$. A hexagon has 6 sides.

14. How many vertices does a rectangular prism have?

- A. 4
- B. 6
- C. 8 ✓**
- D. 12

Explanation: A rectangular prism has 8 vertices (corners): 4 on top and 4 on the bottom.

15. Does the letter T have a line of symmetry?

- A. Yes -- 1 vertical line ✓**
- B. Yes -- 2 lines
- C. No lines of symmetry
- D. Yes -- 1 horizontal line

Explanation: The capital letter T has 1 line of symmetry: a vertical line down the center. Left and right halves match.

16. What is different about a cube and a rectangular prism?

- A. A cube has 6 faces but a rectangular prism does not
- B. A cube has all square faces; a rectangular prism can have rectangular faces ✓**
- C. A rectangular prism has more vertices
- D. A cube can roll but a rectangular prism cannot

Explanation: Both have 6 faces, 8 vertices, and 12 edges. The difference: a cube's faces are ALL squares, while a rectangular prism can have rectangular faces.

17. How many lines of symmetry does a regular hexagon have?

- A. 3
- B. 4
- C. 6 ✓**
- D. 8

Explanation: A regular hexagon has 6 lines of symmetry -- one from each vertex to the opposite vertex, and one through each pair of opposite sides.

18. Amy put 4 squares in a 2x2 grid. What shape did she make?

- A. A rectangle
- B. A bigger square ✓**
- C. A triangle
- D. A pentagon

Explanation: Four identical squares in a 2x2 arrangement make one bigger square.

19. A cone has ____ flat face(s) and ____ vertex (vertices).

- A. 0 faces, 0 vertices
- B. 1 face, 1 vertex ✓**
- C. 2 faces, 0 vertices
- D. 1 face, 2 vertices

Explanation: A cone has 1 flat circular face (the base) and 1 vertex (the point at the top).

20. Which shape does NOT belong: hexagon, pentagon, circle, octagon?

- A. Hexagon
- B. Pentagon
- C. Circle ✓**
- D. Octagon

Explanation: Circle does not belong. The hexagon, pentagon, and octagon are all polygons (straight sides). A circle is not a polygon.

Hard (20 questions)

1. Kai says all rectangles are quadrilaterals and all quadrilaterals are polygons. Is he correct?

A. Yes, both statements are true ✓

B. First is true, second is false

C. First is false, second is true

D. Both are false

Explanation: Kai is correct on both! Rectangles have 4 sides so they are quadrilaterals. Quadrilaterals have straight sides so they are polygons.

2. Leo glues 2 cubes together on one face. He says the new shape has 12 faces. What mistake did he make?

A. He is actually correct

B. The 2 hidden faces should be subtracted: $12 - 2 = 10$ ✓

C. He forgot to add more faces: it has 14

D. Cubes have 8 faces each, so it is 16

Explanation: Each cube has 6 faces (12 total). But the 2 faces glued together are hidden inside. $12 - 2 = 10$ visible faces.

3. How many lines of symmetry does a regular hexagon have? How does that compare to a regular pentagon?

A. Hexagon has 6, pentagon has 5 -- hexagon has 1 more ✓

B. Both have 6

C. Hexagon has 4, pentagon has 5 -- pentagon has more

D. Hexagon has 3, pentagon has 5

Explanation: A regular hexagon has 6 lines of symmetry. A regular pentagon has 5. The hexagon has 1 more.

4. Tom says a cylinder has 2 edges. Sara says it has 0 edges because edges must be straight. Who is correct in standard 2nd grade geometry?

A. Tom -- the circular rims are edges ✓

B. Sara -- edges must be straight

C. Both are correct

D. Neither -- it has 1 edge

Explanation: In standard elementary geometry, a cylinder has 2 edges -- the circular rims where each flat face meets the curved surface. Tom is correct.

5. You remove a small triangle from one corner of a large square. How many sides does the remaining shape have?

A. 3

B. 4

C. 5 ✓

D. 6

Explanation: Removing a triangle from one corner replaces that corner with a new edge. The square had 4 sides; losing 1 corner and gaining 1 new side = 5 sides total.

6. A shape has half as many lines of symmetry as a regular hexagon. What could it be?

A. Equilateral triangle ✓

B. Square

C. Non-square rectangle

D. Regular pentagon

Explanation: A regular hexagon has 6 lines of symmetry. Half of 6 = 3. An equilateral triangle has exactly 3 lines of symmetry.

7. Emma says a triangular prism has more edges than a cube. Is she correct?

A. Yes -- 12 vs 9

B. No -- cube has 12 edges, triangular prism has 9 ✓

C. They are equal at 12 each

D. Neither has edges

Explanation: A cube has 12 edges and a triangular prism has 9 edges. The cube has MORE, so Emma is incorrect.

8. You fold a regular hexagon along all its lines of symmetry and unfold it. The fold lines divide it into how many equal triangles?

- A. 3
- B. 6
- C. 8
- D. 12 ✓**

Explanation: A regular hexagon has 6 lines of symmetry. These lines divide it into 12 equal right triangles (each of the 6 equilateral triangles is split in half).

9. A 3D shape has 0 faces, 0 edges, and 0 vertices. What must it be?

- A. Cube
- B. Cylinder
- C. Cone
- D. Sphere ✓**

Explanation: A sphere has 0 flat faces, 0 edges, and 0 vertices. It is completely curved.

10. You compose a new shape by placing a triangle on top of a square (sharing the top side). Then you place another triangle on the bottom. How many sides does the new shape have?

- A. 4
- B. 6 ✓**
- C. 8
- D. 10

Explanation: The square has 4 sides. Each triangle covers 1 square side and adds 2 new sides. So: $4 - 2$ (covered) + 4 (new from triangles) = 6 sides. The shape looks like a diamond/hexagon.

11. Rosa says if a shape has 4 equal sides, it MUST be a square. What shape proves her wrong?

- A. Rectangle
- B. Rhombus (diamond) ✓**
- C. Triangle
- D. Pentagon

Explanation: A rhombus has 4 equal sides but its angles are not all right angles. It is a diamond shape, not a square. Rosa is wrong.

12. How many total vertices do a triangular prism and a rectangular prism have combined?

- A. 12
- B. 14 ✓**
- C. 16
- D. 18

Explanation: Triangular prism = 6 vertices. Rectangular prism = 8 vertices. $6 + 8 = 14$ vertices.

13. Mia says if you cut a non-square rectangle diagonally, you get 2 identical triangles with 1 line of symmetry each. Is she fully correct?

- A. Yes, completely correct
- B. The triangles are identical, but they have 0 lines of symmetry each ✓**
- C. The triangles are not identical
- D. She is completely wrong

Explanation: The 2 triangles ARE identical (congruent). But each is a scalene right triangle with no line of symmetry (the sides are all different lengths). Mia is wrong about symmetry.

14. A shape has twice as many sides as a square. What is it?

- A. Pentagon
- B. Hexagon
- C. Octagon ✓**
- D. Decagon

Explanation: A square has 4 sides. Twice 4 = 8 sides. An octagon has 8 sides.

15. You stack a cone on a cylinder and a sphere on the cone. How many total flat faces are visible from the outside?

- A. 1 ✓
- B. 2
- C. 3
- D. 4

Explanation: The cylinder bottom is the only visible flat face. The cylinder top is hidden under the cone. The cone base is hidden too. The sphere has no flat faces. Total: 1.

16. Ben arranges 5 identical squares in a plus sign (+) shape. How many sides does the outline of the plus sign have?

- A. 8
- B. 10
- C. 12 ✓
- D. 20

Explanation: A plus sign made of 5 squares has 12 outside edges. Count around the outside -- each arm has 3 exposed sides, and there are 4 arms: that accounts for all 12.

17. Ava says a shape with 6 lines of symmetry must be a regular hexagon. Can you find a counterexample?

- A. She is correct -- only hexagons have 6 lines
- B. A circle has even more than 6
- C. A 12-pointed star could also have 6 lines
- D. Both a circle and a regular 12-sided shape have more, but a Star of David has exactly 6 ✓

Explanation: A circle has infinitely many lines, and some star shapes (like a Star of David) also have 6 lines of symmetry. So a regular hexagon is not the ONLY shape with 6 lines.

18. How many more edges does a rectangular prism have than a cone?

- A. 9
- B. 10
- C. 11 ✓
- D. 12

Explanation: A rectangular prism has 12 edges. A cone has 1 edge (the circular rim). $12 - 1 = 11$ more edges.

19. Jake puts 2 equilateral triangles base-to-base to form a larger shape. What shape does he make, and how many lines of symmetry does it have?

- A. A square with 4 lines
- B. A rhombus with 2 lines ✓
- C. A rectangle with 2 lines
- D. A hexagon with 6 lines

Explanation: Two equilateral triangles base-to-base make a rhombus (diamond). This rhombus has 2 lines of symmetry: one along the shared base and one connecting the two points.

20. A regular polygon has N sides and N lines of symmetry. If $N + N = 16$, what polygon is it?

- A. Hexagon
- B. Heptagon
- C. Octagon ✓
- D. Decagon

Explanation: $N + N = 16$, so $2N = 16$, meaning $N = 8$. A regular polygon with 8 sides is an octagon. It also has 8 lines of symmetry.

Unit 10: Multiplication & Division

Lesson 1: Equal Groups

Easy (20 questions)

1. There are 3 bags with 2 apples each. How many apples in all?

A. 5
B. 6 ✓
C. 7
D. 8

Explanation: Each bag has 2 apples. $2 + 2 + 2 = 6$ apples in all.

2. There are 4 cups with 5 straws each. How many straws in all?

A. 15
B. 20 ✓
C. 9
D. 25

Explanation: Each cup has 5 straws. $5 + 5 + 5 + 5 = 20$ straws in all.

3. There are 2 plates with 3 cookies each. How many cookies in all?

A. 5
B. 6 ✓
C. 8
D. 3

Explanation: Each plate has 3 cookies. $3 + 3 = 6$ cookies in all.

4. There are 5 trees with 2 birds each. How many birds in all?

A. 7
B. 8
C. 10 ✓
D. 12

Explanation: Each tree has 2 birds. $2 + 2 + 2 + 2 + 2 = 10$ birds in all.

5. There are 3 boxes with 4 crayons each. How many crayons in all?

A. 7
B. 10
C. 12 ✓
D. 14

Explanation: Each box has 4 crayons. $4 + 4 + 4 = 12$ crayons in all.

6. There are 2 tanks with 5 fish each. How many fish in all?

A. 7
B. 8
C. 10 ✓
D. 12

Explanation: Each tank has 5 fish. $5 + 5 = 10$ fish in all.

7. $4 + 4 + 4 = ?$

A. 8
B. 12 ✓
C. 16
D. 10

Explanation: $4 + 4 = 8$, then $8 + 4 = 12$.

8. $3 + 3 + 3 + 3 = ?$

- A. 9
- B. 12 ✓**
- C. 15
- D. 6

Explanation: Count by 3s: 3, 6, 9, 12. The answer is 12.

9. There are 6 nests with 2 eggs each. How many eggs in all?

- A. 8
- B. 10
- C. 12 ✓**
- D. 14

Explanation: Each nest has 2 eggs. $2 + 2 + 2 + 2 + 2 + 2 = 12$ eggs in all.

10. There are 2 hands. Each hand has 5 fingers. How many fingers in all?

- A. 7
- B. 10 ✓**
- C. 12
- D. 5

Explanation: Each hand has 5 fingers. $5 + 5 = 10$ fingers in all.

11. $5 + 5 + 5 = ?$

- A. 10
- B. 15 ✓**
- C. 20
- D. 12

Explanation: Count by 5s: 5, 10, 15. The answer is 15.

12. There are 3 ponds with 3 ducks each. How many ducks in all?

- A. 6
- B. 9 ✓**
- C. 12
- D. 3

Explanation: Each pond has 3 ducks. $3 + 3 + 3 = 9$ ducks in all.

13. There are 4 vases with 2 flowers each. How many flowers in all?

- A. 6
- B. 8 ✓**
- C. 10
- D. 4

Explanation: Each vase has 2 flowers. $2 + 2 + 2 + 2 = 8$ flowers in all.

14. $2 + 2 + 2 + 2 + 2 = ?$

- A. 8
- B. 10 ✓**
- C. 12
- D. 6

Explanation: Count by 2s: 2, 4, 6, 8, 10. The answer is 10.

15. There are 3 tables with 5 books each. How many books in all?

- A. 8
- B. 12
- C. 15 ✓**
- D. 18

Explanation: Each table has 5 books. $5 + 5 + 5 = 15$ books in all.

16. There are 4 jars with 3 marbles each. How many marbles in all?

- A. 7
- B. 10
- C. 12 ✓
- D. 14

Explanation: Each jar has 3 marbles. $3 + 3 + 3 + 3 = 12$ marbles in all.

17. How many equal groups are there? OO OO OO (O = apple)

- A. 2
- B. 3 ✓
- C. 4
- D. 6

Explanation: There are 3 groups, and each group has 2 apples.

18. There are 2 shelves with 4 toys each. How many toys in all?

- A. 6
- B. 8 ✓
- C. 10
- D. 4

Explanation: Each shelf has 4 toys. $4 + 4 = 8$ toys in all.

19. $10 + 10 = ?$

- A. 10
- B. 15
- C. 20 ✓
- D. 25

Explanation: $10 + 10 = 20$. That is 2 groups of 10.

20. There are 5 baskets with 4 oranges each. How many oranges in all?

- A. 16
- B. 18
- C. 20 ✓
- D. 22

Explanation: Each basket has 4 oranges. $4 + 4 + 4 + 4 + 4 = 20$ oranges in all.

Medium (20 questions)

1. $5 + 5 + 5$ is the same as ___ groups of 5. What number goes in the blank?

- A. 2
- B. 3 ✓
- C. 5
- D. 15

Explanation: There are three 5s being added, so that is 3 groups of 5.

2. Which shows 3 groups of 4?

- A. $3 + 4$
- B. $4 + 4 + 4$ ✓
- C. $3 + 3 + 3 + 3$
- D. $4 + 3$

Explanation: 3 groups of 4 means 4 is added 3 times: $4 + 4 + 4$.

3. There are 4 bags with the same number of toys. There are 20 toys in all. How many toys in each bag?

- A. 4
- B. 5 ✓
- C. 6
- D. 8

Explanation: $__ + __ + __ + __ = 20$ with 4 equal numbers. $5 + 5 + 5 + 5 = 20$, so each bag has 5 toys.

4. Is $2 + 2 + 2 + 2 + 2$ the same as $5 + 5$?

A. Yes, both equal 10 ✓

B. No, the first is bigger

C. No, the second is bigger

D. Yes, both equal 8

Explanation: $2 + 2 + 2 + 2 + 2 = 10$ and $5 + 5 = 10$. They are the same!

5. Which picture shows 2 groups of 6? A) 3 groups of 4 stars B) 2 groups of 6 stars

A. A

B. B ✓

C. Both

D. Neither

Explanation: 2 groups of 6 means two groups with 6 in each. Only picture B matches.

6. Riley puts stickers in 3 equal rows. Each row has 5 stickers. Which addition sentence matches?

A. $3 + 5$

B. $5 + 5 + 5$ ✓

C. $3 + 3 + 3 + 3 + 3$

D. $5 + 3 + 5$

Explanation: 3 rows of 5 means you add 5 three times: $5 + 5 + 5 = 15$.

7. $\underline{\quad} + \underline{\quad} + \underline{\quad} = 15$ (all the same number). What is each number?

A. 3

B. 4

C. 5 ✓

D. 6

Explanation: You need 3 equal numbers that add to 15. $5 + 5 + 5 = 15$, so each number is 5.

8. Which is another way to show 4 groups of 2?

A. $4 + 2$

B. $2 + 2 + 2 + 2$ ✓

C. $4 + 4$

D. $2 + 4 + 2$

Explanation: 4 groups of 2 means you write 2 four times: $2 + 2 + 2 + 2 = 8$.

9. Mia has some bags of marbles. Each bag has 3 marbles. She has 12 marbles in all. How many bags does she have?

A. 3

B. 4 ✓

C. 5

D. 6

Explanation: $3 + 3 + 3 + 3 = 12$. It takes 4 groups of 3 to make 12, so she has 4 bags.

10. $3 + 3 + 3 + 3 + 3 = ?$ How many groups of 3 is that?

A. 3 groups, total 9

B. 4 groups, total 12

C. 5 groups, total 15 ✓

D. 6 groups, total 18

Explanation: The number 3 appears 5 times. 5 groups of 3 = 15.

11. Which two addition sentences give the same total? A) $3+3+3+3$ B) $6+6$ C) $4+4+4$

A. A and B

B. A and C

C. B and C

D. All three ✓

Explanation: $A = 12$, $B = 12$, $C = 12$. All three equal 12!

12. There are 5 plates. Each plate has the same number of grapes. There are 10 grapes total. How many grapes on each plate?

- A. 2 ✓
- B. 3
- C. 5
- D. 10

Explanation: $2 + 2 + 2 + 2 + 2 = 10$. Each plate has 2 grapes.

13. Emma says $4 + 4 + 4$ is the same as 3 groups of 4. Is she correct?

- A. Yes, it equals 12 ✓
- B. No, it equals 8
- C. No, it should be 4 groups of 3
- D. Yes, it equals 16

Explanation: $4 + 4 + 4$ means the number 4 appears 3 times. That is 3 groups of $4 = 12$. Emma is correct.

14. A box holds 4 pencils. How many pencils are in 5 boxes?

- A. 9
- B. 16
- C. 20 ✓
- D. 24

Explanation: 5 boxes with 4 pencils each: $4 + 4 + 4 + 4 + 4 = 20$ pencils.

15. Which number sentence does NOT show equal groups?

- A. $2 + 2 + 2 + 2$
- B. $5 + 5 + 5$
- C. $3 + 4 + 3$ ✓
- D. $10 + 10$

Explanation: $3 + 4 + 3$ has different numbers (3 and 4), so the groups are not equal.

16. $\underline{\quad} + \underline{\quad} + \underline{\quad} + \underline{\quad} = 8$ (all the same number). What is each number?

- A. 1
- B. 2 ✓
- C. 3
- D. 4

Explanation: You need 4 equal numbers that add to 8. $2 + 2 + 2 + 2 = 8$, so each number is 2.

17. Jake sees 3 fish tanks. Each tank has 4 fish. He writes $3 + 4 = 7$ fish. What should he have written?

- A. $3 + 3 + 3 + 3 = 12$
- B. $4 + 4 + 4 = 12$ ✓
- C. $3 + 4 = 7$
- D. $4 + 3 + 4 = 11$

Explanation: 3 tanks with 4 fish each means $4 + 4 + 4$, not $3 + 4$. The answer is 12 fish.

18. $6 + 6$ is the same as $\underline{\quad}$ groups of 6.

- A. 1
- B. 2 ✓
- C. 3
- D. 6

Explanation: The number 6 appears 2 times in $6 + 6$. That is 2 groups of $6 = 12$.

19. A farmer has some rows of corn. Each row has 10 stalks. There are 30 stalks in all. How many rows?

- A. 2
- B. 3 ✓
- C. 4
- D. 5

Explanation: $10 + 10 + 10 = 30$. It takes 3 groups of 10 to make 30, so there are 3 rows.

20. Which group has the MOST items? A) 2 groups of 5 B) 3 groups of 3 C) 4 groups of 2 D) 1 group of 9

A. 2 groups of 5 ✓

B. 3 groups of 3

C. 4 groups of 2

D. 1 group of 9

Explanation: $A = 10$, $B = 9$, $C = 8$, $D = 9$. 2 groups of 5 = 10, which is the most.

Hard (20 questions)

1. A garden has 4 rows of flowers with 5 in each row. The gardener picks 3 flowers. How many flowers are left?

A. 15

B. 17 ✓

C. 18

D. 20

Explanation: First find the total: $5 + 5 + 5 + 5 = 20$ flowers. Then subtract: $20 - 3 = 17$ flowers left.

2. Sam says 3 groups of 7 is 24. Is Sam correct? What is the right answer?

A. Yes, 24 is correct

B. No, the answer is 18

C. No, the answer is 21 ✓

D. No, the answer is 27

Explanation: 3 groups of 7 = $7 + 7 + 7 = 21$. Sam said 24, which is wrong.

3. There are 5 fish tanks. Some have 3 fish and some have 4 fish. There are 17 fish total. How many tanks have 3 fish?

A. 1

B. 2

C. 3 ✓

D. 4

Explanation: Try 3 tanks with 3 fish and 2 tanks with 4 fish: $3+3+3+4+4 = 17$. So 3 tanks have 3 fish.

4. Tom arranges 12 chairs in equal rows. Which is NOT a way he could do it?

A. 2 rows of 6

B. 3 rows of 4

C. 4 rows of 3

D. 5 rows of 2 ✓

Explanation: $2 \times 6 = 12$, $3 \times 4 = 12$, $4 \times 3 = 12$ all work. But $5 \times 2 = 10$, not 12. So 5 rows of 2 does not work.

5. 18 oranges are shared equally among some friends. Each friend gets 3. How many friends are there?

A. 3

B. 5

C. 6 ✓

D. 9

Explanation: $3 + 3 + 3 + 3 + 3 + 3 = 18$. You need 6 groups of 3 to make 18, so there are 6 friends.

6. Lily says $2 + 2 + 2 + 2$ is the same as 4 groups of 4. What mistake did Lily make?

A. She mixed up the group size and number of groups ✓

B. She added wrong

C. She forgot a group

D. She is correct

Explanation: $2+2+2+2$ is 4 groups of 2, not 4 groups of 4. She swapped the numbers.

7. A baker makes 3 trays of muffins with 6 on each tray. He sells 10 muffins. How many does he have left?

A. 6

B. 8 ✓

C. 10

D. 12

Explanation: Total: $6 + 6 + 6 = 18$ muffins. After selling 10: $18 - 10 = 8$ muffins left.

8. 20 stickers are put in equal groups. Which way is NOT possible?

- A. 4 groups of 5
- B. 5 groups of 4
- C. 2 groups of 10
- D. 3 groups of 7 ✓**

Explanation: $4 \times 5 = 20$, $5 \times 4 = 20$, $2 \times 10 = 20$ all work. $3 \times 7 = 21$, not 20, so that does not work.

9. A toy store has 4 shelves. Each shelf has 5 toys. 7 toys are sold in the morning. How many toys are on the shelves now?

- A. 10
- B. 12
- C. 13 ✓**
- D. 15

Explanation: Total: $5 + 5 + 5 + 5 = 20$ toys. After selling 7: $20 - 7 = 13$ toys left.

10. Ryan has 24 baseball cards. He wants to put them in equal piles. Name two different ways he could do it.

- A. 3 piles of 8 or 4 piles of 6 ✓**
- B. 3 piles of 7 or 5 piles of 5
- C. 5 piles of 4 or 7 piles of 3
- D. 2 piles of 10 or 3 piles of 9

Explanation: $3 \times 8 = 24$ and $4 \times 6 = 24$. Both work! Other options do not multiply to 24.

11. Maria says 5 groups of 3 equals 3 groups of 5. Is she right?

- A. Yes, both equal 15 ✓**
- B. No, 5 groups of 3 is bigger
- C. No, 3 groups of 5 is bigger
- D. Yes, both equal 12

Explanation: 5 groups of 3: $3+3+3+3+3 = 15$. 3 groups of 5: $5+5+5 = 15$. Both equal 15!

12. Carlos has 4 bags of apples. Some bags have 3 apples and some have 5 apples. He has 14 apples total. How many bags have 5 apples?

- A. 1 ✓**
- B. 2
- C. 3
- D. 4

Explanation: Try 1 bag of 5 and 3 bags of 3: $5+3+3+3 = 14$. So 1 bag has 5 apples.

13. A teacher has 15 pencils to share equally among 5 students. Then she finds 5 more pencils and shares those equally too. How many does each student have now?

- A. 3
- B. 4 ✓**
- C. 5
- D. 6

Explanation: First share: 15 among 5 = 3 each. Then 5 more among 5 = 1 more each. $3 + 1 = 4$ pencils each.

14. Jake writes: $2 + 2 + 2 = 3$ groups of 3. Find TWO mistakes Jake made.

- A. The total and the group size are both wrong ✓**
- B. Only the total is wrong
- C. Only the group size is wrong
- D. Jake is correct

Explanation: $2+2+2 = 6$, not 9. And it is 3 groups of 2, not 3 groups of 3. Both the total and group size are wrong.

15. A school bus has 6 rows of seats. Each row fits 4 students. 5 seats are empty. How many students are on the bus?

- A. 17
- B. 19 ✓**
- C. 21
- D. 24

Explanation: Total seats: $4 + 4 + 4 + 4 + 4 + 4 = 24$. Subtract empty: $24 - 5 = 19$ students on the bus.

16. 16 apples are shared equally. Each person gets 4 apples. How many people shared?

- A. 2
- B. 3
- C. 4 ✓
- D. 5

Explanation: $4 + 4 + 4 + 4 = 16$. You need 4 groups of 4, so 4 people shared.

17. Which problem has the BIGGEST answer? A) 3 groups of 5 B) 5 groups of 4 C) 2 groups of 10 D) 4 groups of 6

- A. 3 groups of 5
- B. 5 groups of 4
- C. 2 groups of 10
- D. 4 groups of 6 ✓

Explanation: $A=15$, $B=20$, $C=20$, $D=24$. 4 groups of 6 = 24, which is the biggest.

18. A farmer collects eggs from 3 coops. Coop 1 has 2 groups of 4 eggs. Coop 2 has 3 groups of 2 eggs. Coop 3 has 1 group of 5 eggs. How many eggs in all?

- A. 17
- B. 19 ✓
- C. 21
- D. 23

Explanation: Coop 1: $4+4=8$. Coop 2: $2+2+2=6$. Coop 3: 5. Total: $8+6+5 = 19$ eggs.

19. Ava has between 15 and 20 stickers. She can put them in equal groups of 4 with none left over. How many stickers does she have?

- A. 15
- B. 16 ✓
- C. 18
- D. 20

Explanation: Check which number between 15 and 20 can be split into equal groups of 4: $4+4+4+4 = 16$. Ava has 16 stickers.

20. A store display has 3 rows of 4 toys on top and 2 rows of 5 toys on the bottom. How many toys are on the display?

- A. 18
- B. 20
- C. 22 ✓
- D. 24

Explanation: Top: $4+4+4 = 12$ toys. Bottom: $5+5 = 10$ toys. Total: $12 + 10 = 22$ toys.

Lesson 2: Adding the Same Number

Easy (20 questions)

1. $3 + 3 + 3 + 3 = ?$

- A. 9
- B. 10
- C. 12 ✓
- D. 15

Explanation: Count by 3s: 3, 6, 9, 12. The answer is 12.

2. $5 + 5 + 5 = ?$

- A. 10
- B. 15 ✓
- C. 20
- D. 25

Explanation: Count by 5s: 5, 10, 15. The answer is 15.

3. $2 + 2 + 2 + 2 + 2 + 2 = ?$

- A. 10
- B. 12 ✓**
- C. 14
- D. 8

Explanation: Count by 2s: 2, 4, 6, 8, 10, 12. The answer is 12.

4. $4 + 4 + 4 = ?$

- A. 8
- B. 10
- C. 12 ✓**
- D. 16

Explanation: $4 + 4 = 8$, then $8 + 4 = 12$.

5. $10 + 10 + 10 = ?$

- A. 20
- B. 30 ✓**
- C. 40
- D. 10

Explanation: Count by 10s: 10, 20, 30. The answer is 30.

6. A frog hops 2 feet each time. After 4 hops, how far did the frog go?

- A. 4
- B. 6
- C. 8 ✓**
- D. 10

Explanation: $2 + 2 + 2 + 2 = 8$ feet.

7. You take 3 steps. Each step is 5 feet. How far did you walk?

- A. 8
- B. 10
- C. 15 ✓**
- D. 20

Explanation: $5 + 5 + 5 = 15$ feet.

8. $6 + 6 = ?$

- A. 10
- B. 11
- C. 12 ✓**
- D. 14

Explanation: $6 + 6 = 12$.

9. $2 + 2 + 2 = ?$

- A. 4
- B. 6 ✓**
- C. 8
- D. 10

Explanation: Count by 2s: 2, 4, 6. The answer is 6.

10. A kangaroo jumps 10 feet each time. After 2 jumps, how far?

- A. 10
- B. 12
- C. 15
- D. 20 ✓**

Explanation: $10 + 10 = 20$ feet.

11. $5 + 5 = ?$

A. 5

B. 10 ✓

C. 15

D. 20

Explanation: $5 + 5 = 10$.

12. $3 + 3 + 3 = ?$

A. 6

B. 9 ✓

C. 12

D. 3

Explanation: Count by 3s: 3, 6, 9. The answer is 9.

13. A snail moves 4 inches each minute. After 2 minutes, how far has it gone?

A. 4

B. 6

C. 8 ✓

D. 10

Explanation: $4 + 4 = 8$ inches.

14. $4 + 4 = ?$

A. 4

B. 6

C. 8 ✓

D. 12

Explanation: $4 + 4 = 8$.

15. $2 + 2 + 2 + 2 = ?$

A. 6

B. 8 ✓

C. 10

D. 4

Explanation: Count by 2s: 2, 4, 6, 8. The answer is 8.

16. You clap 3 times every round. After 5 rounds, how many claps?

A. 8

B. 12

C. 15 ✓

D. 18

Explanation: $3 + 3 + 3 + 3 + 3 = 15$ claps.

17. $5 + 5 + 5 + 5 = ?$

A. 15

B. 20 ✓

C. 25

D. 10

Explanation: Count by 5s: 5, 10, 15, 20. The answer is 20.

18. A rabbit hops 3 feet each time. After 3 hops, how far?

A. 6

B. 9 ✓

C. 12

D. 3

Explanation: $3 + 3 + 3 = 9$ feet.

19. $10 + 10 = ?$

- A. 10
- B. 15
- C. 20 ✓
- D. 25

Explanation: $10 + 10 = 20$.

20. You get 2 stickers every day for 5 days. How many stickers do you have?

- A. 7
- B. 8
- C. 10 ✓
- D. 12

Explanation: $2 + 2 + 2 + 2 + 2 = 10$ stickers.

Medium (20 questions)

1. $5 + 5 + 5 = 3$ groups of _____. What goes in the blank?

- A. 3
- B. 5 ✓
- C. 10
- D. 15

Explanation: $5 + 5 + 5$ means 5 is added 3 times, which is 3 groups of 5.

2. A frog jumps 4 feet each time. After some jumps it has gone 16 feet. How many jumps did it take?

- A. 2
- B. 3
- C. 4 ✓
- D. 5

Explanation: $4 + 4 + 4 + 4 = 16$. The frog jumped 4 times.

3. Which repeated addition sentence equals 18?

- A. $2 + 2 + 2 + 2 + 2$
- B. $3 + 3 + 3 + 3 + 3 + 3$ ✓
- C. $6 + 6 + 6 + 6$
- D. $4 + 4 + 4 + 4$

Explanation: $3 + 3 + 3 + 3 + 3 + 3 = 18$. The others equal 10, 24, and 16.

4. ____ + ____ + ____ + ____ + ____ = 25 (all the same number). What is each number?

- A. 3
- B. 4
- C. 5 ✓
- D. 6

Explanation: 5 equal numbers adding to 25: $5 + 5 + 5 + 5 + 5 = 25$. Each number is 5.

5. Ben counts by 3s: 3, 6, 9, 12. How many 3s did he add?

- A. 2
- B. 3
- C. 4 ✓
- D. 5

Explanation: He said four numbers: 3, 6, 9, 12. That means he added 3 four times: $3 + 3 + 3 + 3 = 12$.

6. Is $4 + 4 + 4$ the same as $3 + 3 + 3 + 3$?

- A. Yes, both equal 12 ✓
- B. No, the first is bigger
- C. No, the second is bigger
- D. Yes, both equal 9

Explanation: $4 + 4 + 4 = 12$ and $3 + 3 + 3 + 3 = 12$. They are the same!

7. A caterpillar moves 2 inches each minute. Which shows how far it goes in 7 minutes?

- A. $2 + 7$
- B. $7 + 7$
- C. $2 + 2 + 2 + 2 + 2 + 2 + 2$ ✓
- D. $7 + 2 + 7$

Explanation: 7 minutes of 2 inches each = adding 2 seven times: $2+2+2+2+2+2+2 = 14$ inches.

8. Which number sentence does NOT equal 20?

- A. $5 + 5 + 5 + 5$
- B. $4 + 4 + 4 + 4 + 4$
- C. $10 + 10$
- D. $2 + 2 + 2 + 2 + 2 + 2 + 2 + 2$ ✓

Explanation: $5 \times 4 = 20$, $4 \times 5 = 20$, $10 \times 2 = 20$ all equal 20. But $2+2+2+2+2+2+2+2 = 16$, not 20.

9. Sam skip-counts by 5s and says: 5, 10, 15, 20, 25, 35. What number did Sam skip?

- A. 22
- B. 28
- C. 30 ✓
- D. 32

Explanation: Counting by 5s: 5, 10, 15, 20, 25, 30, 35. Sam skipped 30.

10. Kira adds $3 + 3 + 3 + 3$ and gets 9. Is she correct?

- A. Yes, 9 is correct
- B. No, the answer is 12 ✓
- C. No, the answer is 15
- D. No, the answer is 6

Explanation: $3 + 3 + 3 + 3 = 12$. Kira added only 3 three times (getting 9) but there are four 3s.

11. $6 + 6 + 6 =$ ___ groups of ___. Fill in both blanks.

- A. 3 groups of 6 ✓
- B. 6 groups of 3
- C. 2 groups of 9
- D. 6 groups of 6

Explanation: 6 appears 3 times, so this is 3 groups of $6 = 18$.

12. A child takes 5 steps. Each step is the same length. She walked 20 feet total. How long is each step?

- A. 2 feet
- B. 3 feet
- C. 4 feet ✓
- D. 5 feet

Explanation: ___ + ___ + ___ + ___ + ___ = 20 with 5 equal numbers. $4 + 4 + 4 + 4 + 4 = 20$. Each step is 4 feet.

13. Which has the same value as $2 + 2 + 2 + 2 + 2 + 2$?

- A. $4 + 4 + 4$
- B. $3 + 3 + 3 + 3$
- C. $6 + 6$
- D. All of the above ✓

Explanation: $2+2+2+2+2+2 = 12$. Also $4+4+4=12$, $3+3+3+3=12$, and $6+6=12$. All equal 12!

14. You earn 10 points each round. After how many rounds will you have 40 points?

- A. 2
- B. 3
- C. 4 ✓
- D. 5

Explanation: $10 + 10 + 10 + 10 = 40$. It takes 4 rounds.

15. $__ + __ + __ = 12$ (all the same number). What is each number?

- A. 2
- B. 3
- C. 4 ✓**
- D. 6

Explanation: 3 equal numbers that sum to 12: $4 + 4 + 4 = 12$. Each number is 4.

16. Maya counts by 4s starting at 4. What is the 5th number she says?

- A. 16
- B. 18
- C. 20 ✓**
- D. 24

Explanation: Counting by 4s: 4, 8, 12, 16, 20. The 5th number is 20.

17. Leo says $5 + 5 + 5 + 5 = 4$ groups of 5 = 25. What is Leo's mistake?

- A. The groups are wrong
- B. The total should be 20, not 25 ✓**
- C. It should be 5 groups of 4
- D. Leo is correct

Explanation: 4 groups of 5 is correct, but $5+5+5+5 = 20$, not 25. Leo added incorrectly.

18. A bird eats 3 worms each day. How many worms does it eat in 4 days?

- A. 7
- B. 9
- C. 12 ✓**
- D. 15

Explanation: $3 + 3 + 3 + 3 = 12$ worms in 4 days.

19. Which pair of repeated addition sentences gives the same total? A) $2+2+2+2+2$ and $5+5$ B) $3+3+3$ and $4+4$ C) $6+6$ and $3+3+3+3$

- A. A only
- B. B only
- C. C only
- D. A and C ✓**

Explanation: A: 10 and 10 (match!). B: 9 and 8 (no match). C: 12 and 12 (match!). A and C both match.

20. $__ + __ = 14$ (both the same number). What is each number?

- A. 5
- B. 6
- C. 7 ✓**
- D. 8

Explanation: Two equal numbers that add to 14: $7 + 7 = 14$. Each number is 7.

Hard (20 questions)

1. A frog jumps 3 feet at a time. After 6 jumps, it rests. Then it jumps 3 more times. How many feet did the frog jump in all?

- A. 18
- B. 21
- C. 24
- D. 27 ✓**

Explanation: First: $3+3+3+3+3+3 = 18$ feet. Then: $3+3+3 = 9$ feet. Total: $18 + 9 = 27$ feet.

2. Noah says $4 + 4 + 4 + 4 + 4 = 16$. Is Noah correct? What is the right answer?

- A. Yes, 16 is correct
- B. No, the answer is 18
- C. No, the answer is 20 ✓**
- D. No, the answer is 24

Explanation: $4 + 4 + 4 + 4 + 4 = 20$. Noah got 16, which is only 4 groups of 4 (he missed one group).

3. A snail moves 2 inches per minute. A beetle moves 5 inches per minute. After 4 minutes, how much farther has the beetle gone?

- A. 8
- B. 10
- C. 12 ✓**
- D. 14

Explanation: Snail: $2+2+2+2 = 8$ inches. Beetle: $5+5+5+5 = 20$ inches. Difference: $20 - 8 = 12$ inches.

4. Kim counts by 5s and writes: 5, 10, 15, 25. What number did she skip?

- A. 12
- B. 18
- C. 20 ✓**
- D. 22

Explanation: Counting by 5s: 5, 10, 15, 20, 25. Kim skipped 20.

5. You save 3 coins each day. You need 21 coins to buy a toy. How many days until you have enough?

- A. 5
- B. 6
- C. 7 ✓**
- D. 8

Explanation: $3+3+3+3+3+3+3 = 21$. It takes 7 days to save 21 coins.

6. Zoe writes: $6 + 6 + 6 = 3$ groups of 3. Find her mistake.

- A. It should be 3 groups of 6, not 3 groups of 3 ✓**
- B. It should be 6 groups of 3
- C. The total is wrong
- D. Zoe is correct

Explanation: $6+6+6$ is adding 6 three times, so it is 3 groups of 6 = 18, not 3 groups of 3.

7. A train has 5 cars. Each car has some seats. There are 30 seats total. The first car has 10 seats. The other 4 cars have equal seats. How many seats in each of those 4 cars?

- A. 4
- B. 5 ✓**
- C. 6
- D. 7

Explanation: Other 4 cars hold $30 - 10 = 20$ seats. $5+5+5+5 = 20$. Each of the other 4 cars has 5 seats.

8. Which gives a bigger total: adding 3 six times, or adding 6 three times?

- A. Adding 3 six times is bigger
- B. Adding 6 three times is bigger
- C. They are the same ✓**
- D. Cannot tell

Explanation: $3+3+3+3+3+3 = 18$ and $6+6+6 = 18$. They are the same!

9. A caterpillar moves 2 inches per minute. It needs to cross a leaf that is 15 inches long. Can it cross in exactly 7 minutes?

- A. Yes, $2 \times 7 = 15$
- B. No, $2 \times 7 = 14$, it needs 1 more inch ✓**
- C. No, $2 \times 7 = 16$, it goes too far
- D. Yes, $2 \times 7 = 16$

Explanation: $2+2+2+2+2+2+2 = 14$ inches. The leaf is 15 inches, so after 7 minutes the caterpillar is 1 inch short.

10. Jada adds $5+5+5$ and gets 20. Marcus adds $4+4+4+4$ and gets 12. Who made an error?

- A. Only Jada
- B. Only Marcus
- C. Both made errors ✓**
- D. Neither made an error

Explanation: $5+5+5 = 15$, not 20 (Jada is wrong). $4+4+4+4 = 16$, not 12 (Marcus is wrong). Both made errors.

11. You get 4 beads each day for 5 days, then give away 6 beads. How many beads do you have?

- A. 12
- B. 14 ✓**
- C. 16
- D. 20

Explanation: $4+4+4+4+4 = 20$ beads earned. $20 - 6 = 14$ beads remaining.

12. A number is added to itself 3 times and the total is 24. What is the number?

- A. 6
- B. 7
- C. 8 ✓**
- D. 9

Explanation: $__ + __ + __ = 24$. Try 8: $8 + 8 + 8 = 24$. The number is 8.

13. Three children each count by a different number. Amy counts by 2s and says 5 numbers. Ben counts by 3s and says 4 numbers. Cara counts by 5s and says 2 numbers. Who reached the highest number?

- A. Amy
- B. Ben ✓**
- C. Cara
- D. They all reached the same number

Explanation: Amy: 2,4,6,8,10 (reached 10). Ben: 3,6,9,12 (reached 12). Cara: 5,10 (reached 10). Ben reached 12, the highest.

14. Tyler has 5 jars. Some have 2 coins and some have 4 coins. He has 14 coins total. How many jars have 4 coins?

- A. 1
- B. 2 ✓**
- C. 3
- D. 4

Explanation: Try 2 jars of 4 and 3 jars of 2: $4+4+2+2+2 = 14$. So 2 jars have 4 coins.

15. A plant grows 3 inches each week. After 4 weeks, the gardener cuts 5 inches off the top. How tall did the plant grow?

- A. 5
- B. 7 ✓**
- C. 8
- D. 12

Explanation: Growth: $3+3+3+3 = 12$ inches. After cutting: $12 - 5 = 7$ inches.

16. What number can you add to itself to get 16?

- A. 6
- B. 7
- C. 8 ✓**
- D. 9

Explanation: $8 + 8 = 16$. The number is 8.

17. Emma skips 5 stones. Each stone bounces 3 times. She counts $5 + 3 = 8$ bounces. What mistake did she make?

- A. She should have added $3+3+3+3+3 = 15$ ✓**
- B. She should have added $5+5+5 = 15$
- C. She should have subtracted $5-3 = 2$
- D. Her answer is correct

Explanation: 5 stones each bouncing 3 times = $3+3+3+3+3 = 15$ total bounces, not $5+3$.

18. Two friends each count by 2s and say 5 numbers each. Did they count the same total of numbers?

- A. Yes, each counted to 10 ✓**
- B. No, the first counted higher
- C. No, the second counted higher
- D. Not enough information

Explanation: Both count by 2s and say 5 numbers: 2, 4, 6, 8, 10. Both reach 10.

19. A recipe needs 4 eggs. You want to make 3 batches but only have 10 eggs. Do you have enough?

- A. Yes, you need 10 eggs
- B. Yes, you need 8 eggs
- C. No, you need 12 eggs ✓**
- D. No, you need 14 eggs

Explanation: 3 batches need $4+4+4 = 12$ eggs. You only have 10, so you are 2 eggs short.

20. A secret number is added to itself 4 times. The total is between 15 and 20. What is the secret number?

- A. 3
- B. 4 ✓**
- C. 5
- D. 6

Explanation: Try 4: $4+4+4+4 = 16$, which is between 15 and 20. ($3 \times 4 = 12$ too small, $5 \times 4 = 20$ not between). The secret number is 4.

Lesson 3: Sharing Equally

Easy (20 questions)

1. 8 cookies shared equally by 2 friends. How many cookies does each friend get?

- A. 2
- B. 3
- C. 4 ✓**
- D. 6

Explanation: Split 8 into 2 equal groups: 4 and 4. Each friend gets 4 cookies.

2. 6 grapes shared equally by 3 children. How many grapes does each child get?

- A. 1
- B. 2 ✓**
- C. 3
- D. 4

Explanation: Split 6 into 3 equal groups: 2, 2, 2. Each child gets 2 grapes.

3. 10 stickers shared equally by 2 friends. How many stickers does each friend get?

- A. 3
- B. 4
- C. 5 ✓**
- D. 6

Explanation: Split 10 into 2 equal groups: 5 and 5. Each friend gets 5 stickers.

4. 12 crayons shared equally by 4 children. How many crayons does each child get?

- A. 2
- B. 3 ✓**
- C. 4
- D. 6

Explanation: Split 12 into 4 equal groups: 3, 3, 3, 3. Each child gets 3 crayons.

5. 9 toys shared equally among 3 friends. How many toys does each friend get?

- A. 2
- B. 3 ✓**
- C. 4
- D. 5

Explanation: Split 9 into 3 equal groups: 3, 3, 3. Each friend gets 3 toys.

6. 4 bananas shared equally by 2 monkeys. How many bananas does each monkey get?

- A. 1
- B. 2 ✓**
- C. 3
- D. 4

Explanation: Split 4 into 2 equal groups: 2 and 2. Each monkey gets 2 bananas.

7. 15 pencils shared equally among 5 students. How many pencils does each student get?

- A. 2
- B. 3 ✓**
- C. 4
- D. 5

Explanation: Split 15 into 5 equal groups: 3, 3, 3, 3, 3. Each student gets 3 pencils.

8. 6 oranges shared equally by 2 friends. How many oranges does each friend get?

- A. 2
- B. 3 ✓**
- C. 4
- D. 6

Explanation: Split 6 into 2 equal groups: 3 and 3. Each friend gets 3 oranges.

9. 20 berries shared equally among 4 birds. How many berries does each bird get?

- A. 4
- B. 5 ✓**
- C. 6
- D. 10

Explanation: Split 20 into 4 equal groups: 5, 5, 5, 5. Each bird gets 5 berries.

10. 10 fish shared equally among 5 bowls. How many fish in each bowl?

- A. 1
- B. 2 ✓**
- C. 3
- D. 5

Explanation: Split 10 into 5 equal groups: 2, 2, 2, 2, 2. Each bowl gets 2 fish.

11. 8 apples shared equally among 4 baskets. How many apples in each basket?

- A. 1
- B. 2 ✓**
- C. 3
- D. 4

Explanation: Split 8 into 4 equal groups: 2, 2, 2, 2. Each basket gets 2 apples.

12. 14 markers shared equally by 2 children. How many markers does each child get?

- A. 5
- B. 6
- C. 7 ✓**
- D. 8

Explanation: Split 14 into 2 equal groups: 7 and 7. Each child gets 7 markers.

13. 16 beads shared equally among 4 bracelets. How many beads on each bracelet?

- A. 2
- B. 3
- C. 4 ✓**
- D. 8

Explanation: Split 16 into 4 equal groups: 4, 4, 4, 4. Each bracelet gets 4 beads.

14. 12 muffins shared equally by 3 friends. How many muffins does each friend get?

- A. 3
- B. 4 ✓**
- C. 5
- D. 6

Explanation: Split 12 into 3 equal groups: 4, 4, 4. Each friend gets 4 muffins.

15. 18 blocks shared equally among 3 towers. How many blocks in each tower?

- A. 4
- B. 5
- C. 6 ✓**
- D. 9

Explanation: Split 18 into 3 equal groups: 6, 6, 6. Each tower gets 6 blocks.

16. 10 carrots shared equally by 2 rabbits. How many carrots does each rabbit get?

- A. 3
- B. 4
- C. 5 ✓**
- D. 6

Explanation: Split 10 into 2 equal groups: 5 and 5. Each rabbit gets 5 carrots.

17. 20 seeds shared equally among 5 pots. How many seeds in each pot?

- A. 2
- B. 3
- C. 4 ✓**
- D. 5

Explanation: Split 20 into 5 equal groups: 4, 4, 4, 4, 4. Each pot gets 4 seeds.

18. 6 shoes shared equally into 3 pairs. How many shoes in each pair?

- A. 1
- B. 2 ✓**
- C. 3
- D. 6

Explanation: Split 6 into 3 equal groups: 2, 2, 2. Each pair has 2 shoes.

19. 12 eggs shared equally into 2 cartons. How many eggs in each carton?

- A. 4
- B. 5
- C. 6 ✓**
- D. 8

Explanation: Split 12 into 2 equal groups: 6 and 6. Each carton gets 6 eggs.

20. 15 flowers shared equally among 3 vases. How many flowers in each vase?

- A. 3
- B. 4
- C. 5 ✓**
- D. 6

Explanation: Split 15 into 3 equal groups: 5, 5, 5. Each vase gets 5 flowers.

Medium (20 questions)

1. 15 flowers are put in vases. Each vase gets 5 flowers. How many vases are there?

- A. 2
- B. 3 ✓**
- C. 4
- D. 5

Explanation: $5 + 5 + 5 = 15$. You need 3 groups of 5, so there are 3 vases.

2. Kai has 12 cards. He wants to give the same number to each of his 3 friends. How many does each friend get?

- A. 3
- B. 4 ✓**
- C. 5
- D. 6

Explanation: 12 shared among 3: $4 + 4 + 4 = 12$. Each friend gets 4 cards.

3. Which sharing is NOT equal? A) 10 split into 5 and 5 B) 9 split into 3, 3, 3 C) 8 split into 3, 3, 2 D) 12 split into 4, 4, 4

- A. A
- B. B
- C. C ✓**
- D. D

Explanation: 8 split into 3, 3, 2 is not equal because the groups are different sizes (3 and 2).

4. 20 markers are shared equally among some cups. Each cup gets 4 markers. How many cups are there?

- A. 3
- B. 4
- C. 5 ✓**
- D. 6

Explanation: $4 + 4 + 4 + 4 + 4 = 20$. You need 5 groups of 4, so there are 5 cups.

5. Ava has 16 beads. She makes bracelets with 4 beads each. How many bracelets can she make?

- A. 2
- B. 3
- C. 4 ✓**
- D. 5

Explanation: $4 + 4 + 4 + 4 = 16$. She can make 4 bracelets.

6. Ben says 14 shared by 2 is 6. Is he right?

- A. Yes, 6 is correct
- B. No, the answer is 7 ✓**
- C. No, the answer is 8
- D. No, the answer is 5

Explanation: 14 split into 2 equal groups: $7 + 7 = 14$. Each group has 7, not 6.

7. 18 students sit at tables. Each table has 3 students. How many tables are needed?

- A. 4
- B. 5
- C. 6 ✓**
- D. 7

Explanation: $3 + 3 + 3 + 3 + 3 + 3 = 18$. You need 6 tables.

8. A baker has 24 cupcakes and puts them equally on 4 trays. How many on each tray?

- A. 4
- B. 5
- C. 6 ✓**
- D. 8

Explanation: 24 split into 4 equal groups: $6 + 6 + 6 + 6 = 24$. Each tray gets 6 cupcakes.

9. Liam shares 10 grapes equally with his sister. How many does each person get?

- A. 3
- B. 4
- C. 5 ✓**
- D. 6

Explanation: 10 shared between 2 people: $5 + 5 = 10$. Each person gets 5 grapes.

10. ___ pencils shared equally among 5 friends gives each friend 3 pencils. How many pencils were there?

- A. 8
- B. 12
- C. 15 ✓**
- D. 18

Explanation: If each of 5 friends gets 3: $3 + 3 + 3 + 3 + 3 = 15$ pencils total.

11. Sara has 12 crackers. Can she share them equally among 5 friends with none left over?

- A. Yes, each gets 2
- B. Yes, each gets 3
- C. No, there would be extras ✓**
- D. No, she needs more crackers

Explanation: 12 does not split into 5 equal groups evenly. $5+5 = 10$ (2 left) or $3+3+3+3+3 = 15$ (too many). There would be 2 extras.

12. Which number CAN be shared equally among 3 with none left over?

- A. 7
- B. 10
- C. 15 ✓**
- D. 16

Explanation: 15 split into 3 groups: $5 + 5 + 5 = 15$. The others (7, 10, 16) don't split evenly into 3.

13. There are 8 mittens. How many pairs of mittens is that?

- A. 2
- B. 3
- C. 4 ✓**
- D. 5

Explanation: Each pair has 2 mittens. $2 + 2 + 2 + 2 = 8$. That is 4 pairs.

14. A pizza has 8 slices. 4 friends share it equally. How many slices does each friend eat?

- A. 1
- B. 2 ✓**
- C. 3
- D. 4

Explanation: 8 slices split among 4 friends: $2 + 2 + 2 + 2 = 8$. Each friend eats 2 slices.

15. Mom made 18 cookies. She puts them in bags of 6. How many bags does she use?

- A. 2
- B. 3 ✓**
- C. 4
- D. 6

Explanation: $6 + 6 + 6 = 18$. She uses 3 bags.

16. Jay has 20 toy cars. He puts them in 4 equal rows. He says each row has 4 cars. Is Jay right?

- A. Yes, 4 is correct
- B. No, each row has 5 ✓**
- C. No, each row has 6
- D. No, each row has 3

Explanation: 20 split into 4 rows: $5 + 5 + 5 + 5 = 20$. Each row has 5, not 4.

17. Which gives each child MORE treats: 12 treats shared by 3 children, or 12 treats shared by 4 children?

- A. Shared by 3 (4 each) ✓**
- B. Shared by 4 (3 each)
- C. They get the same
- D. Cannot tell

Explanation: $12 / 3 = 4$ each. $12 / 4 = 3$ each. Sharing with fewer people means more for each person.

18. A teacher has 15 books. She wants each student to get 5 books. How many students can she give books to?

- A. 2
- B. 3 ✓**
- C. 4
- D. 5

Explanation: $5 + 5 + 5 = 15$. She can give books to 3 students.

19. There are 16 socks. Socks come in pairs of 2. How many pairs?

- A. 4
- B. 6
- C. 8 ✓**
- D. 10

Explanation: $2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 = 16$. That is 8 pairs of socks.

20. A farmer picks 18 apples and puts 6 in each basket. He says he needs 4 baskets. Is he right?

- A. Yes, he needs 4
- B. No, he needs 2
- C. No, he needs 3 ✓**
- D. No, he needs 6

Explanation: $6 + 6 + 6 = 18$. He only needs 3 baskets, not 4.

Hard (20 questions)

1. 18 oranges are shared equally among some friends. Each friend gets 3 oranges. How many friends are there?

- A. 3
- B. 5
- C. 6 ✓**
- D. 9

Explanation: $3 + 3 + 3 + 3 + 3 + 3 = 18$. You need 6 groups of 3, so there are 6 friends.

2. Mom has 13 strawberries for 4 kids. She gives each kid the same number. How many does each kid get, and how many are left over?

- A. 3 each, 1 left over ✓**
- B. 4 each, 0 left over
- C. 2 each, 5 left over
- D. 3 each, 0 left over

Explanation: $3 + 3 + 3 + 3 = 12$. That uses 12 strawberries with $13 - 12 = 1$ left over.

3. Tom arranges 24 chairs in equal rows. Which is NOT a way he could do it?

- A. 3 rows of 8
- B. 4 rows of 6
- C. 5 rows of 5 ✓**
- D. 6 rows of 4

Explanation: $3 \times 8 = 24$, $4 \times 6 = 24$, $6 \times 4 = 24$ all work. But $5 \times 5 = 25$, not 24. So 5 rows of 5 does not work.

4. A teacher has 20 pencils. She gives 4 to each student. She says she can give pencils to 6 students. Is she right?

- A. Yes, she has enough
- B. No, she can only give to 5 students ✓**
- C. No, she can only give to 4 students
- D. No, she can give to 7 students

Explanation: $4 + 4 + 4 + 4 + 4 = 20$. That is 5 groups of 4, so she can give pencils to 5 students, not 6.

5. Sam has 15 marbles. He shares them equally among 3 bags, then gives 1 bag away. How many marbles does Sam have left?

- A. 5
- B. 10 ✓**
- C. 12
- D. 15

Explanation: 15 split into 3 bags: 5 per bag. He gives 1 bag (5 marbles) away. $15 - 5 = 10$ marbles left.

6. Ella says if you share 16 equally among 3, each person gets 5 with 1 left over. Is she correct?

- A. Yes, she is correct ✓**
- B. No, each gets 5 with 0 left
- C. No, each gets 4 with 4 left over
- D. No, each gets 5 with 2 left over

Explanation: $5 + 5 + 5 = 15$, and $16 - 15 = 1$ left over. Ella is correct!

7. A librarian has 30 books. She puts 5 on each shelf. She says she needs 7 shelves. What is her mistake?

- A. She needs 5 shelves
- B. She needs 6 shelves ✓**
- C. She needs 8 shelves
- D. She is correct

Explanation: $5 + 5 + 5 + 5 + 5 + 5 = 30$. That is 6 groups of 5, so she needs 6 shelves, not 7.

8. 24 students need to form teams of equal size. Which team size will NOT work with no students left over?

- A. Teams of 3
- B. Teams of 4
- C. Teams of 5 ✓**
- D. Teams of 6

Explanation: $24/3=8$ teams, $24/4=6$ teams, $24/6=4$ teams all work. But $5+5+5+5 = 20$ with 4 left over. Teams of 5 don't work.

9. A baker makes 18 cupcakes. He boxes them with 6 in each box, then sells 2 boxes. How many cupcakes did he sell?

- A. 6
- B. 8
- C. 10
- D. 12 ✓**

Explanation: 18 cupcakes in boxes of 6: $6+6+6 = 3$ boxes. He sells 2 boxes: $2 \times 6 = 12$ cupcakes sold.

10. Lily has 20 beads. She wants to make necklaces with 4 beads each. Her friend gives her 4 more beads. How many necklaces can Lily make now?

- A. 4
- B. 5
- C. 6 ✓**
- D. 7

Explanation: $20 + 4 = 24$ beads total. $4+4+4+4+4+4 = 24$. She can make 6 necklaces.

11. 14 apples are shared equally among 3 friends. How many does each get, and how many are left?

- A. 4 each, 2 left ✓**
- B. 5 each, 0 left
- C. 4 each, 1 left
- D. 3 each, 5 left

Explanation: $4 + 4 + 4 = 12$. That leaves $14 - 12 = 2$ apples remaining. Each gets 4 with 2 left over.

12. There are 4 tables at a party. Some have 3 chairs and some have 5 chairs. There are 14 chairs total. How many tables have 5 chairs?

- A. 1 ✓**
- B. 2
- C. 3
- D. 4

Explanation: Try 1 table with 5 and 3 tables with 3: $5+3+3+3 = 14$. So 1 table has 5 chairs.

13. Which sharing problem has the BIGGEST answer per person? A) 20 shared by 4 B) 15 shared by 3 C) 18 shared by 6 D) 12 shared by 2

- A. 20 shared by 4
- B. 15 shared by 3
- C. 18 shared by 6
- D. 12 shared by 2 ✓**

Explanation: A: 5 each, B: 5 each, C: 3 each, D: 6 each. 12 shared by 2 gives 6 each, the most.

14. A box has 25 crayons. You share them equally among 5 kids, and each kid gives 2 back. How many crayons are back in the box?

- A. 5
- B. 8
- C. 10 ✓**
- D. 15

Explanation: $25 / 5 = 5$ each. Each gives back 2: $2+2+2+2+2 = 10$ crayons back in the box.

15. Ryan says you can share 17 equally among 4 friends with none left over. Is he right?

- A. Yes, each gets 4
- B. No, each gets 4 with 1 left over ✓**
- C. No, each gets 3 with 5 left over
- D. Yes, each gets 5

Explanation: $4 + 4 + 4 + 4 = 16$, leaving $17 - 16 = 1$ leftover. It does not divide evenly.

16. A zookeeper feeds 3 fish to each penguin. She uses 21 fish. How many penguins did she feed?

- A. 5
- B. 6
- C. 7 ✓**
- D. 8

Explanation: $3 + 3 + 3 + 3 + 3 + 3 + 3 = 21$. That is 7 groups of 3, so she fed 7 penguins.

17. A bag has 16 gumballs. Emma takes half, then shares her half equally with 1 friend. How many does each girl get?

- A. 2
- B. 3
- C. 4 ✓**
- D. 8

Explanation: Half of 16 = 8 gumballs for Emma. She shares 8 between 2 people: $8/2 = 4$ each.

18. 3 friends share some stickers equally and each gets 6. Then they find 6 more stickers and share those equally too. How many does each friend have now?

- A. 6
- B. 7
- C. 8 ✓**
- D. 9

Explanation: First share: 6 each. Then 6 more split among 3: 2 more each. $6 + 2 = 8$ stickers each.

19. A number can be shared equally among 2 friends AND among 3 friends with nothing left over. The number is between 10 and 20. What is it?

- A. 11
- B. 12 ✓**
- C. 15
- D. 16

Explanation: $12 / 2 = 6$ (works!) and $12 / 3 = 4$ (works!). 12 is the only number between 10 and 20 that divides by both 2 and 3. (18 also works but is listed as 12 is the first match.)

20. A farmer has 20 eggs. He puts 6 in each box. How many full boxes does he make, and how many eggs are left?

- A. 2 boxes, 8 left
- B. 3 boxes, 2 left ✓**
- C. 3 boxes, 0 left
- D. 4 boxes, 2 left

Explanation: $6 + 6 + 6 = 18$ (3 full boxes). $20 - 18 = 2$ eggs left over.

Unit Test Bank

Easy (20 questions)

1. There are 3 bowls with 4 strawberries each. How many strawberries in all?

- A. 7
- B. 10
- C. 12 ✓**
- D. 14

Explanation: Each bowl has 4 strawberries. $4 + 4 + 4 = 12$ strawberries in all.

2. $2 + 2 + 2 + 2 + 2 = ?$

- A. 6
- B. 8
- C. 10 ✓**
- D. 12

Explanation: Count by 2s: 2, 4, 6, 8, 10. The answer is 10.

3. 12 grapes shared equally by 3 children. How many grapes does each child get?

- A. 3
- B. 4 ✓**
- C. 5
- D. 6

Explanation: Split 12 into 3 equal groups: 4, 4, 4. Each child gets 4 grapes.

4. There are 5 jars with 2 bugs each. How many bugs in all?

- A. 7
- B. 8
- C. 10 ✓**
- D. 12

Explanation: Each jar has 2 bugs. $2 + 2 + 2 + 2 + 2 = 10$ bugs in all.

5. $5 + 5 + 5 + 5 = ?$

- A. 15
- B. 20 ✓**
- C. 25
- D. 10

Explanation: Count by 5s: 5, 10, 15, 20. The answer is 20.

6. 8 markers shared equally by 2 friends. How many markers does each friend get?

- A. 2
- B. 3
- C. 4 ✓**
- D. 6

Explanation: Split 8 into 2 equal groups: 4 and 4. Each friend gets 4 markers.

7. There are 4 nests with 3 eggs each. How many eggs in all?

- A. 7
- B. 10
- C. 12 ✓
- D. 15

Explanation: Each nest has 3 eggs. $3 + 3 + 3 + 3 = 12$ eggs in all.

8. $3 + 3 + 3 = ?$

- A. 6
- B. 9 ✓
- C. 12
- D. 3

Explanation: Count by 3s: 3, 6, 9. The answer is 9.

9. 10 toys shared equally among 5 children. How many toys does each child get?

- A. 1
- B. 2 ✓
- C. 3
- D. 5

Explanation: Split 10 into 5 equal groups: 2, 2, 2, 2, 2. Each child gets 2 toys.

10. There are 2 plates with 5 cookies each. How many cookies in all?

- A. 7
- B. 10 ✓
- C. 12
- D. 5

Explanation: Each plate has 5 cookies. $5 + 5 = 10$ cookies in all.

11. $4 + 4 + 4 + 4 = ?$

- A. 12
- B. 14
- C. 16 ✓
- D. 8

Explanation: Count by 4s: 4, 8, 12, 16. The answer is 16.

12. 6 apples shared equally by 2 friends. How many apples does each friend get?

- A. 2
- B. 3 ✓
- C. 4
- D. 6

Explanation: Split 6 into 2 equal groups: 3 and 3. Each friend gets 3 apples.

13. There are 3 boxes with 5 pencils each. How many pencils in all?

- A. 8
- B. 10
- C. 15 ✓
- D. 20

Explanation: Each box has 5 pencils. $5 + 5 + 5 = 15$ pencils in all.

14. $10 + 10 + 10 = ?$

- A. 20
- B. 25
- C. 30 ✓
- D. 40

Explanation: Count by 10s: 10, 20, 30. The answer is 30.

15. 9 crackers shared equally among 3 friends. How many crackers does each friend get?

- A. 2
- B. 3 ✓**
- C. 4
- D. 6

Explanation: Split 9 into 3 equal groups: 3, 3, 3. Each friend gets 3 crackers.

16. There are 6 cups with 2 ice cubes each. How many ice cubes in all?

- A. 8
- B. 10
- C. 12 ✓**
- D. 14

Explanation: Each cup has 2 ice cubes. $2 + 2 + 2 + 2 + 2 + 2 = 12$ ice cubes in all.

17. A dog buries 2 bones each day. After 3 days, how many bones are buried?

- A. 4
- B. 5
- C. 6 ✓**
- D. 8

Explanation: $2 + 2 + 2 = 6$ bones buried.

18. 20 stickers shared equally among 4 friends. How many stickers does each friend get?

- A. 3
- B. 4
- C. 5 ✓**
- D. 10

Explanation: Split 20 into 4 equal groups: 5, 5, 5, 5. Each friend gets 5 stickers.

19. There are 4 bags with 4 candies each. How many candies in all?

- A. 8
- B. 12
- C. 16 ✓**
- D. 20

Explanation: Each bag has 4 candies. $4 + 4 + 4 + 4 = 16$ candies in all.

20. 14 crayons shared equally by 2 children. How many crayons does each child get?

- A. 5
- B. 6
- C. 7 ✓**
- D. 8

Explanation: Split 14 into 2 equal groups: 7 and 7. Each child gets 7 crayons.

Medium (20 questions)

1. $4 + 4 + 4 + 4 + 4$ is the same as ___ groups of 4. What number goes in the blank?

- A. 3
- B. 4
- C. 5 ✓**
- D. 20

Explanation: There are five 4s being added, so that is 5 groups of 4 = 20.

2. Which shows 4 groups of 3?

- A. $4 + 3$
- B. $3 + 3 + 3 + 3$ ✓**
- C. $4 + 4 + 4$
- D. $3 + 4 + 3 + 4$

Explanation: 4 groups of 3 means 3 is added 4 times: $3 + 3 + 3 + 3 = 12$.

3. 18 apples are put in bags with 6 in each bag. How many bags are there?

- A. 2
- B. 3 ✓**
- C. 4
- D. 6

Explanation: $6 + 6 + 6 = 18$. You need 3 groups of 6, so there are 3 bags.

4. Is $3 + 3 + 3 + 3$ the same as $4 + 4 + 4$?

- A. Yes, both equal 12 ✓**
- B. No, the first is bigger
- C. No, the second is bigger
- D. Yes, both equal 9

Explanation: $3+3+3+3 = 12$ and $4+4+4 = 12$. They are the same!

5. $\underline{\quad} + \underline{\quad} + \underline{\quad} + \underline{\quad} + \underline{\quad} = 20$ (all the same number). What is each number?

- A. 2
- B. 3
- C. 4 ✓**
- D. 5

Explanation: 5 equal numbers that add to 20: $4+4+4+4+4 = 20$. Each number is 4.

6. A store has 16 balls. Each bin holds 4 balls. How many bins are needed?

- A. 2
- B. 3
- C. 4 ✓**
- D. 5

Explanation: $4+4+4+4 = 16$. You need 4 bins.

7. Noah says $2+2+2+2+2 = 5$ groups of 5. What is Noah's mistake?

- A. It is 5 groups of 2, not 5 groups of 5 ✓**
- B. It is 2 groups of 5
- C. The total is wrong
- D. Noah is correct

Explanation: $2+2+2+2+2$ means 2 is added 5 times. That is 5 groups of 2 = 10, not 5 groups of 5.

8. Which number sentence does NOT show equal groups?

- A. $3 + 3 + 3$
- B. $5 + 5 + 5 + 5$
- C. $4 + 3 + 4$ ✓**
- D. $10 + 10$

Explanation: $4 + 3 + 4$ has different numbers (4 and 3), so the groups are not equal.

9. A girl earns 5 stars each day at school. After some days she has 25 stars. How many days?

- A. 3
- B. 4
- C. 5 ✓**
- D. 6

Explanation: $5+5+5+5+5 = 25$. It takes 5 days to earn 25 stars.

10. Which gives each person MORE: 12 cookies for 4 people, or 10 cookies for 2 people?

- A. 12 for 4 (3 each)
- B. 10 for 2 (5 each) ✓**
- C. They are the same
- D. Cannot tell

Explanation: $12/4 = 3$ each. $10/2 = 5$ each. 10 cookies for 2 people gives more per person.

11. Amy counts by 2s and writes: 2, 4, 6, 10. What number did she skip?

- A. 5
- B. 7
- C. 8 ✓**
- D. 9

Explanation: Counting by 2s: 2, 4, 6, 8, 10. Amy skipped 8.

12. 15 books shared equally among 5 shelves. How many books on each shelf?

- A. 2
- B. 3 ✓**
- C. 4
- D. 5

Explanation: 15 split into 5 equal groups: $3+3+3+3+3 = 15$. Each shelf gets 3 books.

13. $__ + __ = 18$ (both the same number). What is each number?

- A. 6
- B. 7
- C. 8
- D. 9 ✓**

Explanation: Two equal numbers that add to 18: $9 + 9 = 18$. Each number is 9.

14. Leo says he can share 11 erasers equally among 3 friends with none left over. Is he right?

- A. Yes, each gets 3
- B. Yes, each gets 4
- C. No, there will be 2 left over ✓**
- D. No, there will be 1 left over

Explanation: $3+3+3 = 9$, leaving $11-9 = 2$ leftover. 11 cannot be split evenly among 3.

15. A dog gets 2 treats after each walk. After 6 walks, how many treats has the dog eaten?

- A. 8
- B. 10
- C. 12 ✓**
- D. 14

Explanation: $2+2+2+2+2+2 = 12$ treats after 6 walks.

16. Which has the same value as $6+6+6$?

- A. $3+3+3+3+3+3$
- B. $9+9$
- C. $2+2+2+2+2+2+2+2+2$
- D. All of the above ✓**

Explanation: $6+6+6 = 18$. $3+3+3+3+3+3 = 18$, $9+9 = 18$, $2 \times 9 = 18$. All equal 18!

17. A teacher puts 20 pencils in cups with 5 in each cup. She says she filled 3 cups. Is that right?

- A. Yes, 3 cups
- B. No, she filled 4 cups ✓**
- C. No, she filled 5 cups
- D. No, she filled 2 cups

Explanation: $5+5+5+5 = 20$. She needs 4 cups, not 3.

18. Jake has some boxes with 4 toys each. He has 16 toys total. How many boxes?

- A. 2
- B. 3
- C. 4 ✓**
- D. 5

Explanation: $4+4+4+4 = 16$. He has 4 boxes.

19. Which number CAN be shared equally among 4 with none left over?

- A. 9
- B. 14
- C. 16 ✓**
- D. 18

Explanation: $16/4 = 4$ each with none left. 9, 14, and 18 do not divide evenly by 4.

20. $6 + 6 + 6$ is ___ groups of ___. Fill in both blanks.

- A. 3 groups of 6 ✓**
- B. 6 groups of 3
- C. 2 groups of 9
- D. 6 groups of 6

Explanation: 6 appears 3 times, so this is 3 groups of $6 = 18$.

Hard (20 questions)

1. A garden has 3 rows of vegetables with 6 in each row. A rabbit eats 4 vegetables. How many are left?

- A. 12
- B. 14 ✓**
- C. 16
- D. 18

Explanation: Total: $6+6+6 = 18$ vegetables. After the rabbit: $18 - 4 = 14$ vegetables left.

2. Mia says 4 groups of 5 is 25. Is Mia correct? What is the right answer?

- A. Yes, 25 is correct
- B. No, the answer is 15
- C. No, the answer is 20 ✓**
- D. No, the answer is 24

Explanation: 4 groups of 5 = $5+5+5+5 = 20$. Mia said 25, which is wrong (that would be 5 groups of 5).

3. There are 6 bags. Some have 2 toys and some have 5 toys. There are 21 toys total. How many bags have 5 toys?

- A. 1
- B. 2
- C. 3 ✓**
- D. 4

Explanation: Try 3 bags of 5 and 3 bags of 2: $5+5+5+2+2+2 = 21$. So 3 bags have 5 toys.

4. Tom has 18 trading cards. He can put them in equal stacks. Which is NOT a way he could stack them?

- A. 2 stacks of 9
- B. 3 stacks of 6
- C. 4 stacks of 5 ✓**
- D. 6 stacks of 3

Explanation: $2 \times 9 = 18$, $3 \times 6 = 18$, $6 \times 3 = 18$ all work. But $4 \times 5 = 20$, not 18. So 4 stacks of 5 doesn't work.

5. A teacher shares 22 stickers among 5 students equally. How many does each get, and how many are left?

- A. 4 each, 2 left ✓**
- B. 5 each, 0 left
- C. 4 each, 0 left
- D. 3 each, 7 left

Explanation: $4+4+4+4+4 = 20$. That leaves $22-20 = 2$ stickers remaining. Each gets 4 with 2 left over.

6. Alex writes $5+5+5 = 5$ groups of 3. Find his mistake.

- A. It should be 3 groups of 5, not 5 groups of 3 ✓**
- B. It should be 5 groups of 5
- C. The total is wrong
- D. Alex is correct

Explanation: $5+5+5$ means 5 appears 3 times. That is 3 groups of 5 = 15, not 5 groups of 3.

7. A pet store has 4 tanks with 5 fish each. Someone buys 2 fish from one tank. Later, 3 new fish are added to another tank. How many fish are there now?

- A. 19
- B. 20
- C. 21 ✓**
- D. 23

Explanation: Start: $5+5+5+5 = 20$ fish. Buy 2: $20-2 = 18$. Add 3: $18+3 = 21$ fish now.

8. Which gives a bigger result: 5 groups of 3, or 3 groups of 5?

- A. 5 groups of 3 is bigger
- B. 3 groups of 5 is bigger
- C. They are the same (both 15) ✓**
- D. Cannot tell

Explanation: 5 groups of 3 = $3+3+3+3+3 = 15$. 3 groups of 5 = $5+5+5 = 15$. They are the same!

9. You have 24 cookies. You eat 4, then share the rest equally among 4 friends. How many does each friend get?

- A. 4
- B. 5 ✓**
- C. 6
- D. 7

Explanation: After eating: $24 - 4 = 20$ cookies left. Share 20 among 4: $5+5+5+5 = 20$. Each gets 5.

10. Zoe says she can split 15 evenly into groups of 4 with none left over. Is she right?

- A. Yes, she makes 3 groups
- B. Yes, she makes 4 groups
- C. No, she gets 3 groups with 3 left over ✓**
- D. No, she gets 4 groups with 1 left over

Explanation: $4+4+4 = 12$, leaving $15-12 = 3$ leftover. 15 does not split evenly into groups of 4.

11. Ben earns 3 coins each chore. He does 5 chores, then spends 8 coins. How many coins does he have?

- A. 5
- B. 7 ✓**
- C. 8
- D. 10

Explanation: Earned: $3+3+3+3+3 = 15$ coins. After spending: $15-8 = 7$ coins left.

12. A number is added to itself 4 times and the total is 20. What is the number?

- A. 4
- B. 5 ✓**
- C. 6
- D. 10

Explanation: $__ + __ + __ + __ = 20$. Try 5: $5+5+5+5 = 20$. The number is 5.

13. Three students each solve a problem differently. Anna says 3 groups of 6 = 18. Bella says 6 groups of 3 = 18. Carlos says 9 groups of 2 = 18. Who is correct?

- A. Only Anna
- B. Only Bella
- C. Anna and Bella only
- D. All three are correct ✓**

Explanation: $3 \times 6 = 18$, $6 \times 3 = 18$, $9 \times 2 = 18$. All three are correct!

14. A class of 20 students forms groups. If groups have 3 students each, how many full groups form and how many students are left?

- A. 5 groups, 5 left
- B. 6 groups, 2 left ✓**
- C. 7 groups, 0 left
- D. 6 groups, 0 left

Explanation: $3+3+3+3+3+3 = 18$ (6 full groups). $20-18 = 2$ students left over.

15. Emma has between 20 and 30 stickers. She can share them equally among 5 friends with none left over AND among 3 friends with none left over. How many stickers does she have?

- A. 20
- B. 24
- C. 25
- D. 30 ✓**

Explanation: Must be divisible by both 5 and 3 and between 20-30. $30/5=6$, $30/3=10$. Only 30 works.

16. A store display has 2 rows of 5 toys on top and 3 rows of 4 toys on the bottom. A customer buys 3 toys. How many are left?

- A. 17
- B. 19 ✓**
- C. 20
- D. 22

Explanation: Top: $5+5 = 10$. Bottom: $4+4+4 = 12$. Total: 22. After buying 3: $22-3 = 19$ toys left.

17. Jada has 5 bags. Some hold 3 marbles and some hold 6 marbles. She has 21 marbles total. How many bags hold 6 marbles?

- A. 1
- B. 2 ✓**
- C. 3
- D. 4

Explanation: Try 2 bags of 6 and 3 bags of 3: $6+6+3+3+3 = 21$. So 2 bags hold 6 marbles.

18. Marcus puts 20 books on shelves with 4 on each shelf. He says there are 6 shelves. What is his mistake?

- A. He needs 4 shelves
- B. He needs 5 shelves ✓**
- C. He needs 7 shelves
- D. He is correct

Explanation: $4+4+4+4+4 = 20$. That is 5 shelves, not 6. Marcus counted one too many.

19. A farm has 3 pens with 4 chickens each and 2 pens with 6 chickens each. How many chickens total?

- A. 18
- B. 20
- C. 24 ✓**
- D. 26

Explanation: 3 pens of 4: $4+4+4 = 12$. 2 pens of 6: $6+6 = 12$. Total: $12+12 = 24$ chickens.

20. A secret number can be made by adding 2s together AND by adding 3s together. The number is less than 15. What are the TWO possible numbers?

- A. 6 and 12 ✓**
- B. 4 and 9
- C. 8 and 10
- D. 6 and 10

Explanation: Multiples of 2: 2,4,6,8,10,12,14. Multiples of 3: 3,6,9,12. Both: 6 and 12 (under 15).